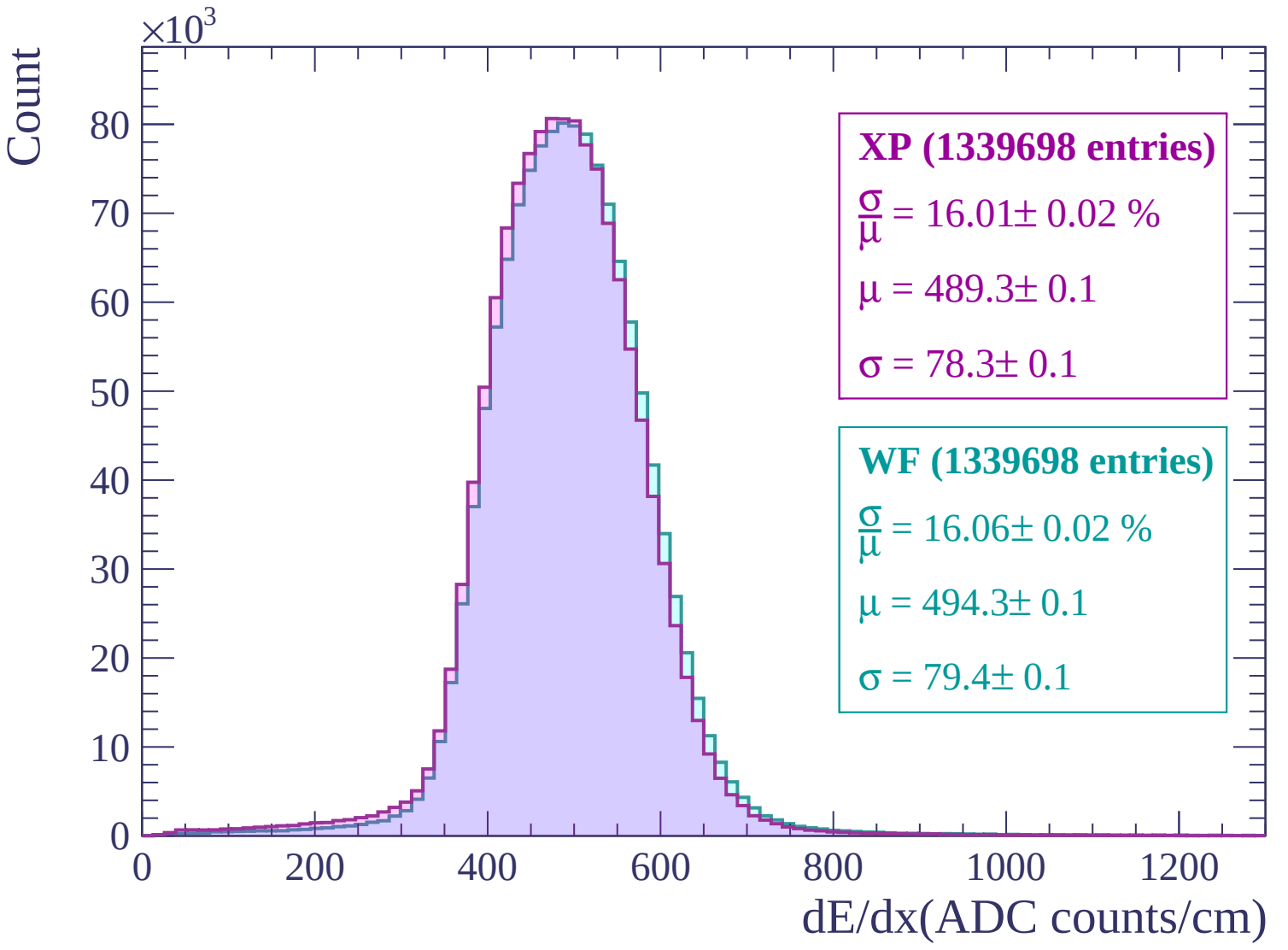
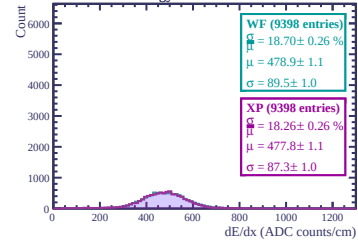
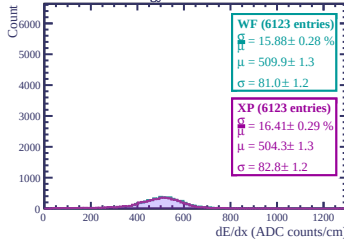




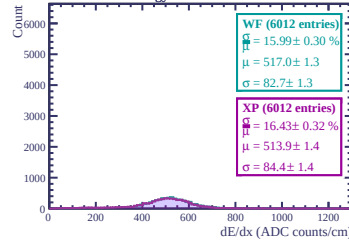
Energy loss in ERAM 24



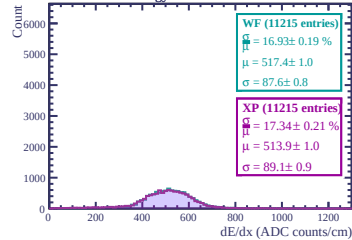
Energy loss in ERAM 30



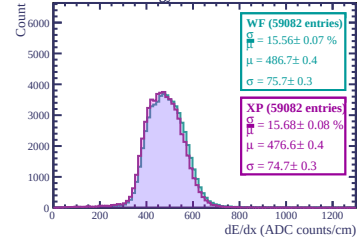
Energy loss in ERAM 36



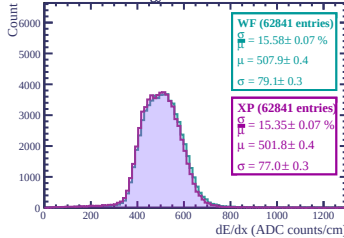
Energy loss in ERAM 42



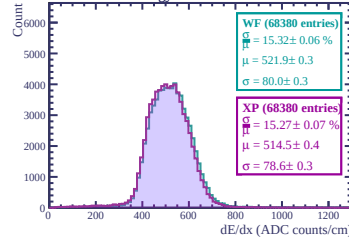
Energy loss in ERAM 21



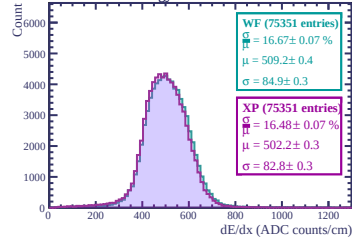
Energy loss in ERAM 13



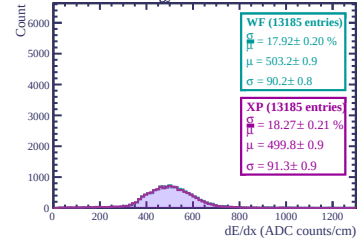
Energy loss in ERAM 9



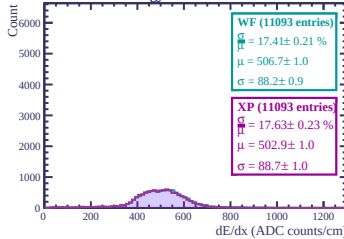
Energy loss in ERAM 2



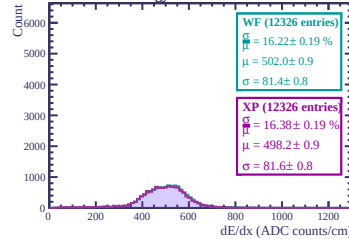
Energy loss in ERAM 26



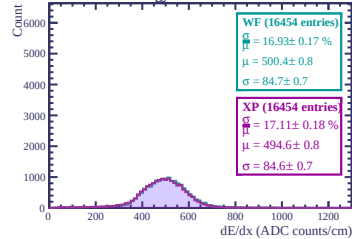
Energy loss in ERAM 17



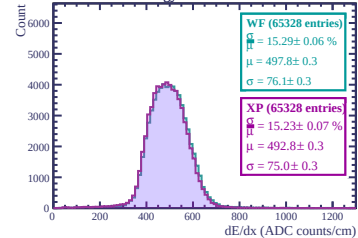
Energy loss in ERAM 23



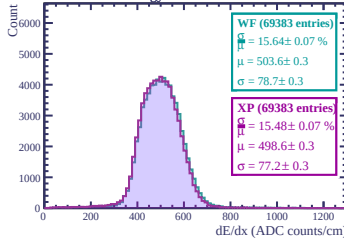
Energy loss in ERAM 29



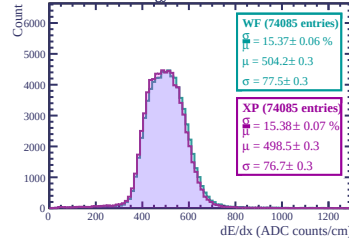
Energy loss in ERAM 1



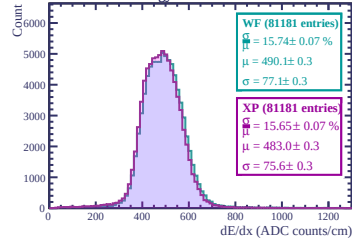
Energy loss in ERAM 10

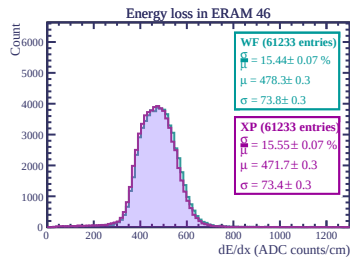
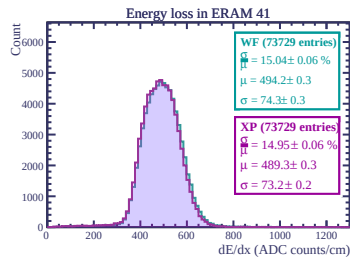
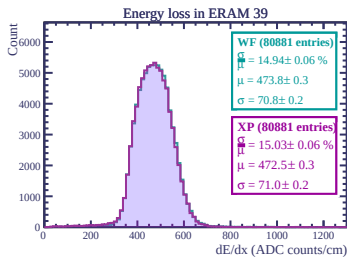
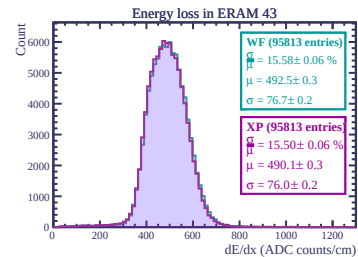
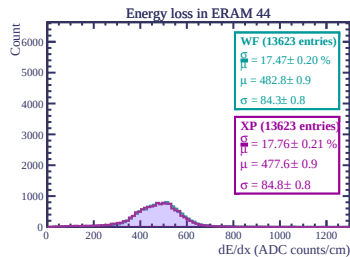
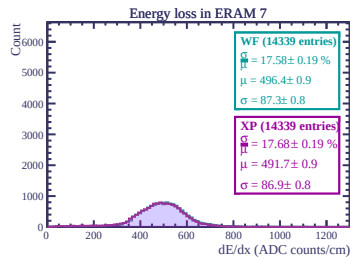
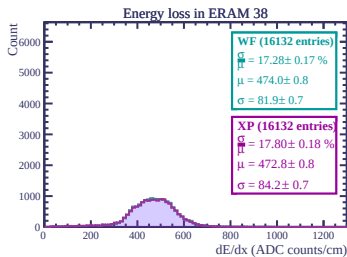
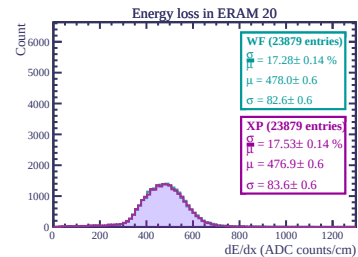
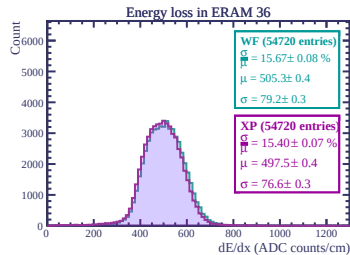
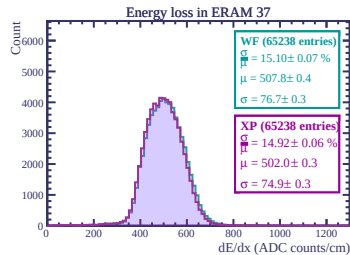
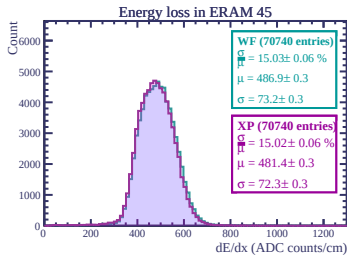
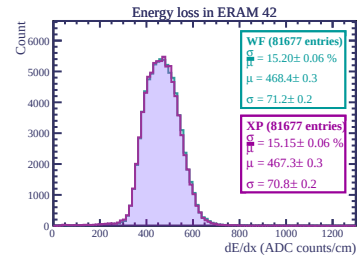
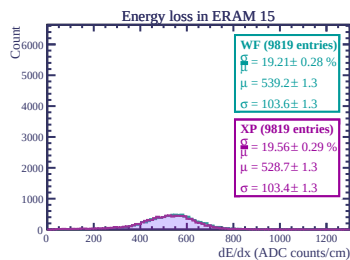
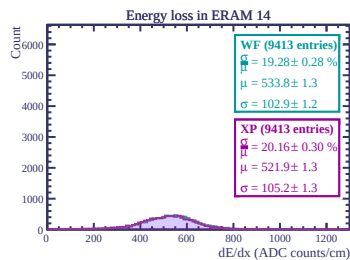
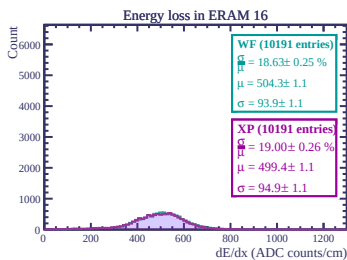
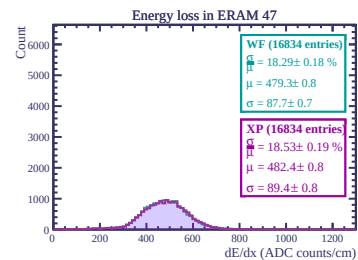


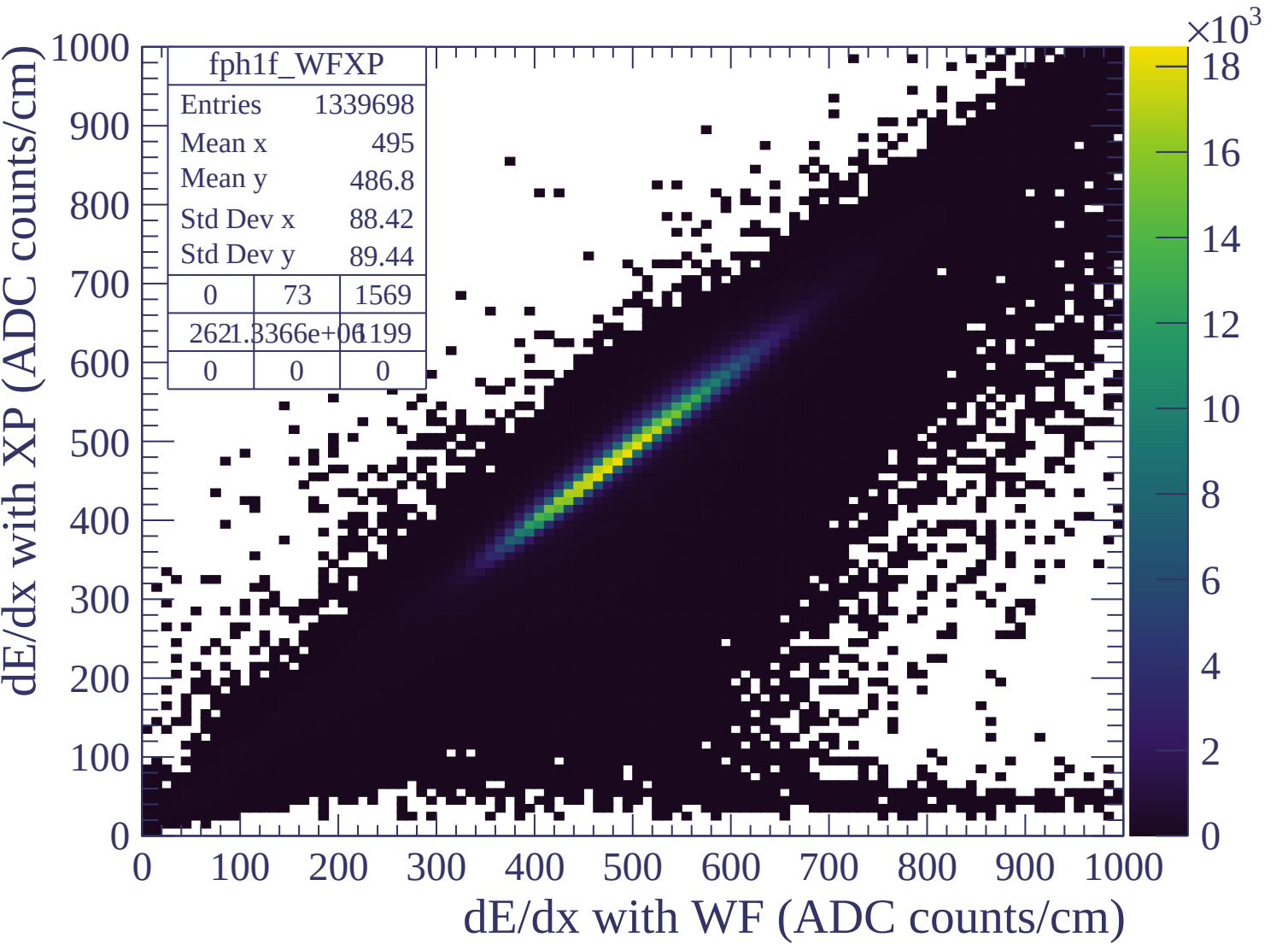
Energy loss in ERAM 11



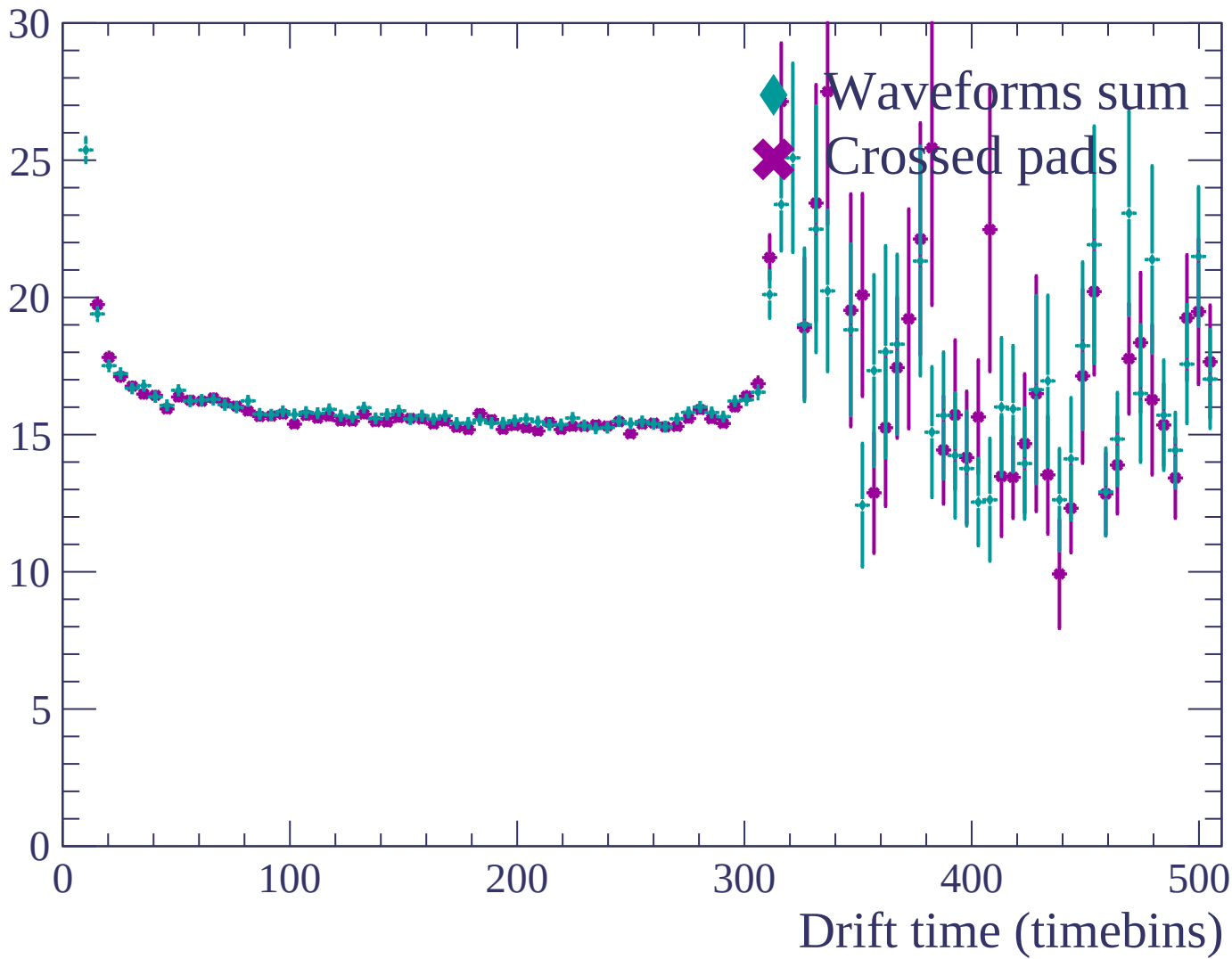
Energy loss in ERAM 3

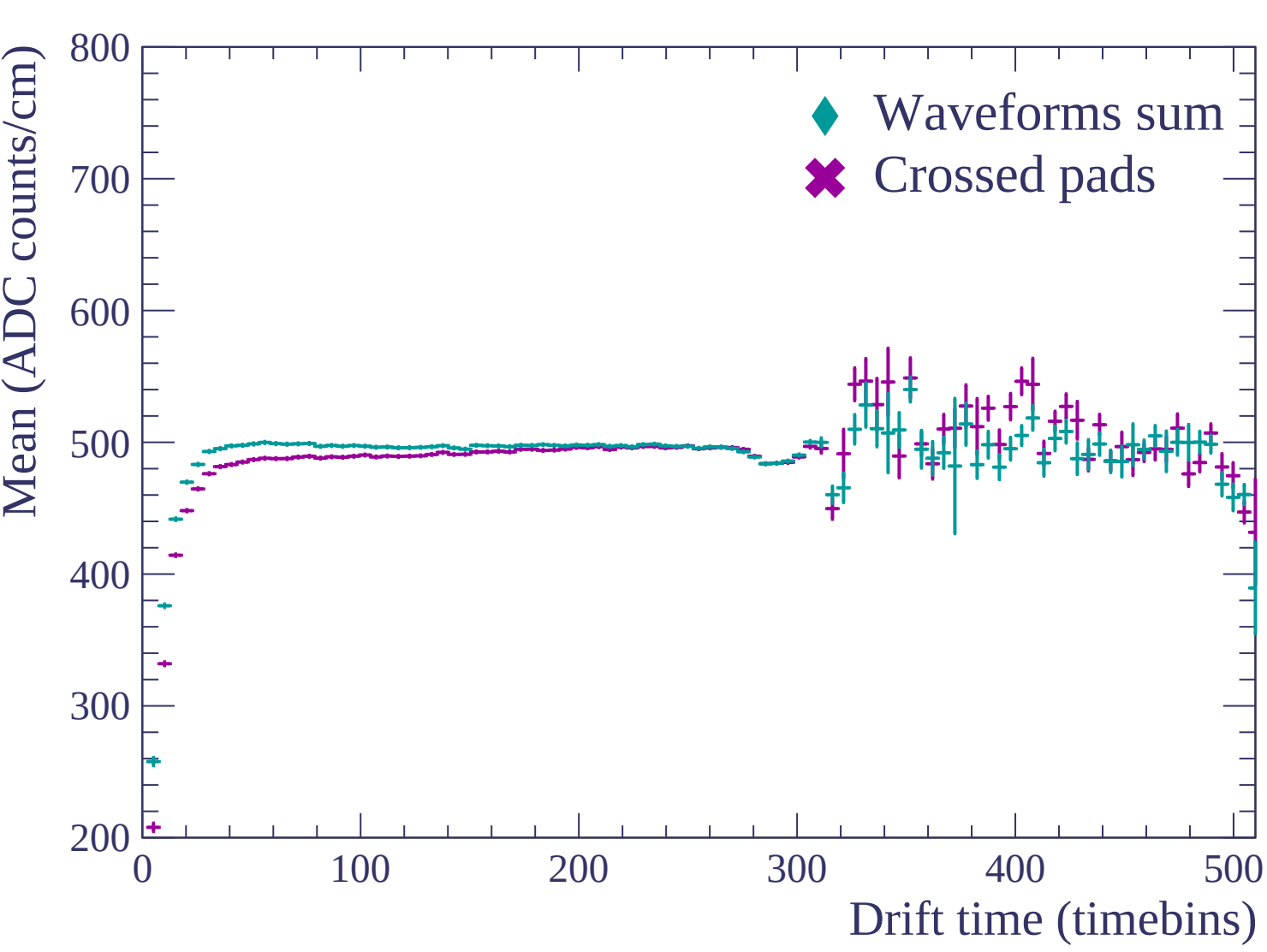


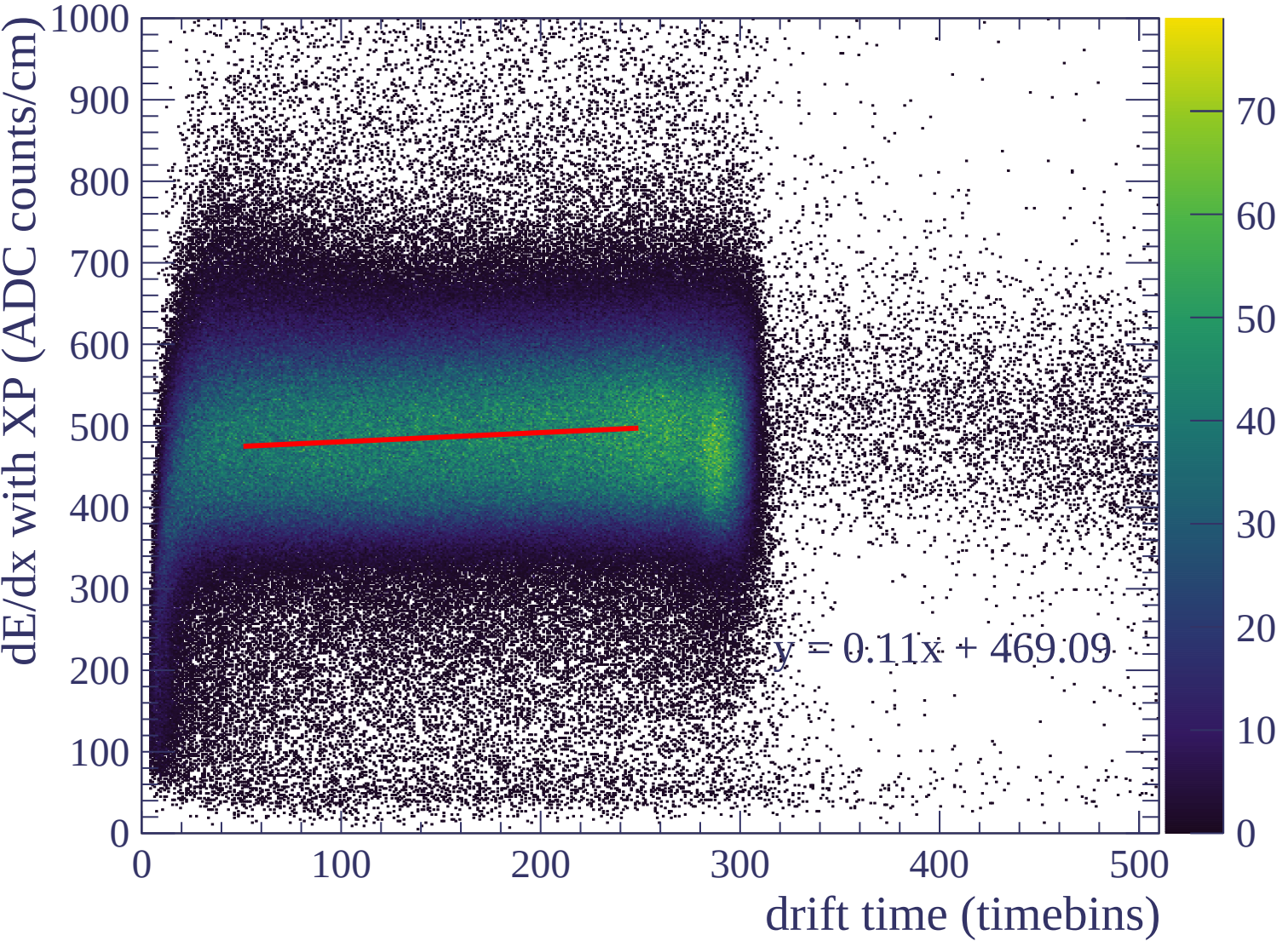


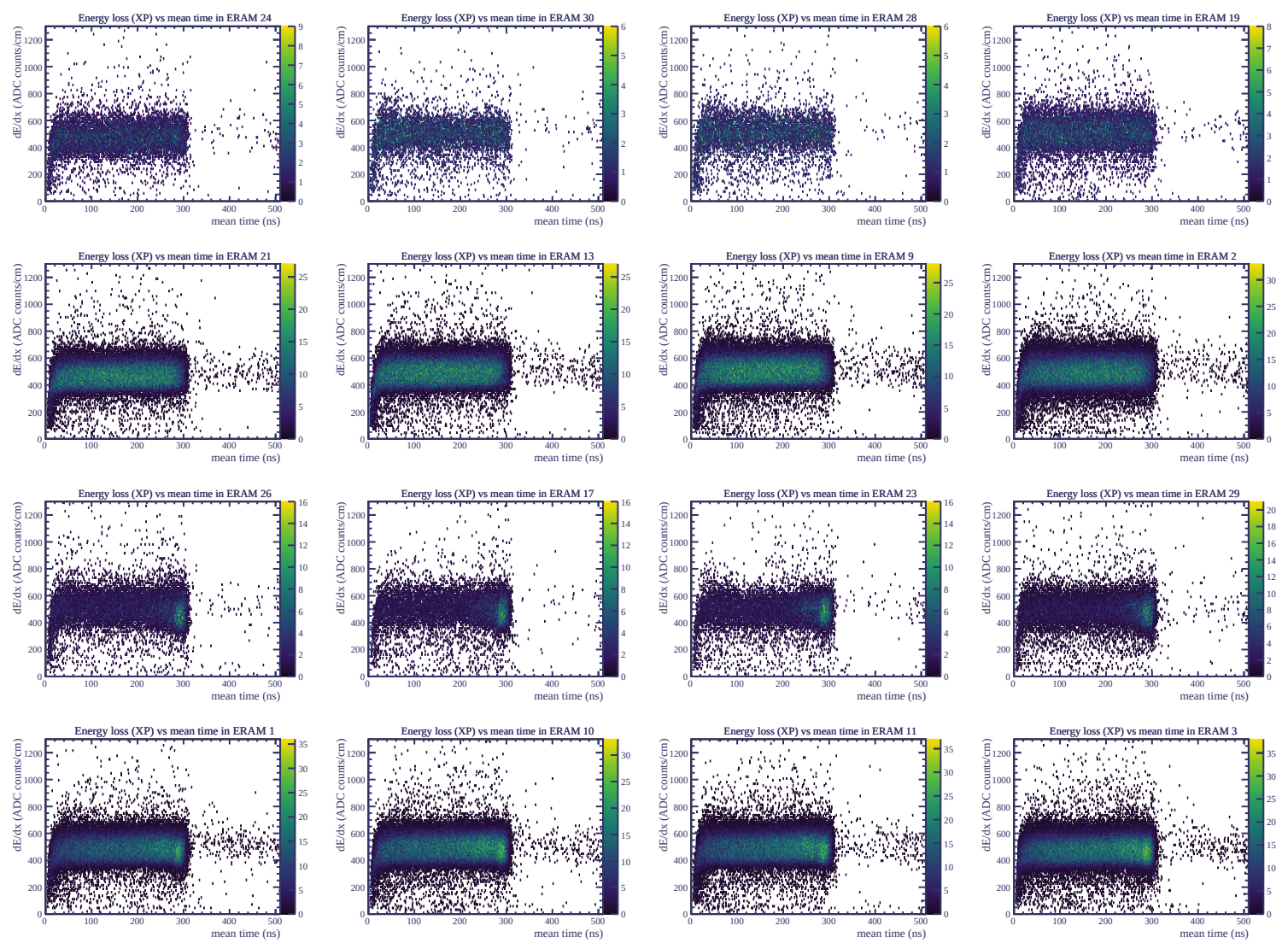


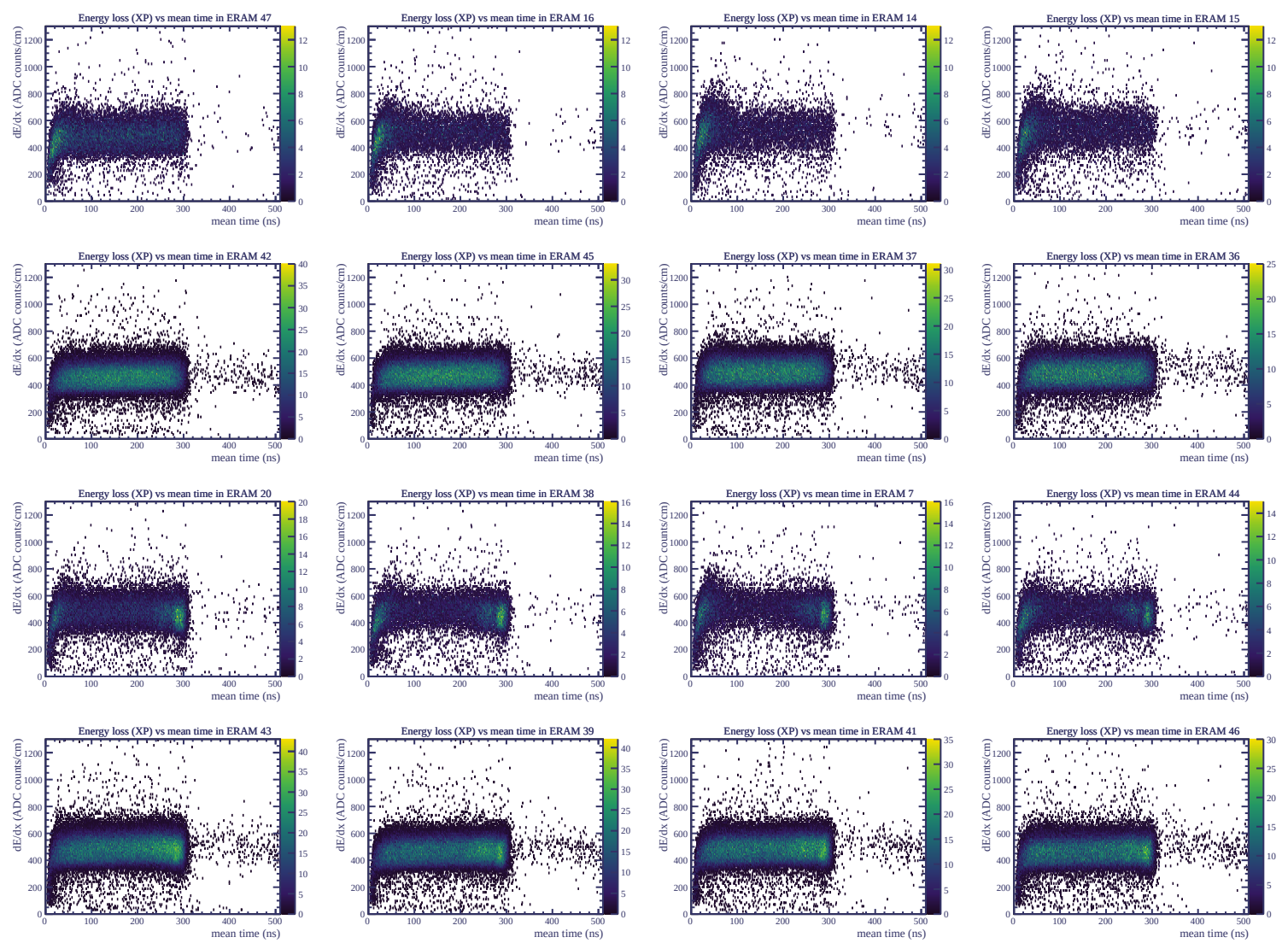
Resolution (%)

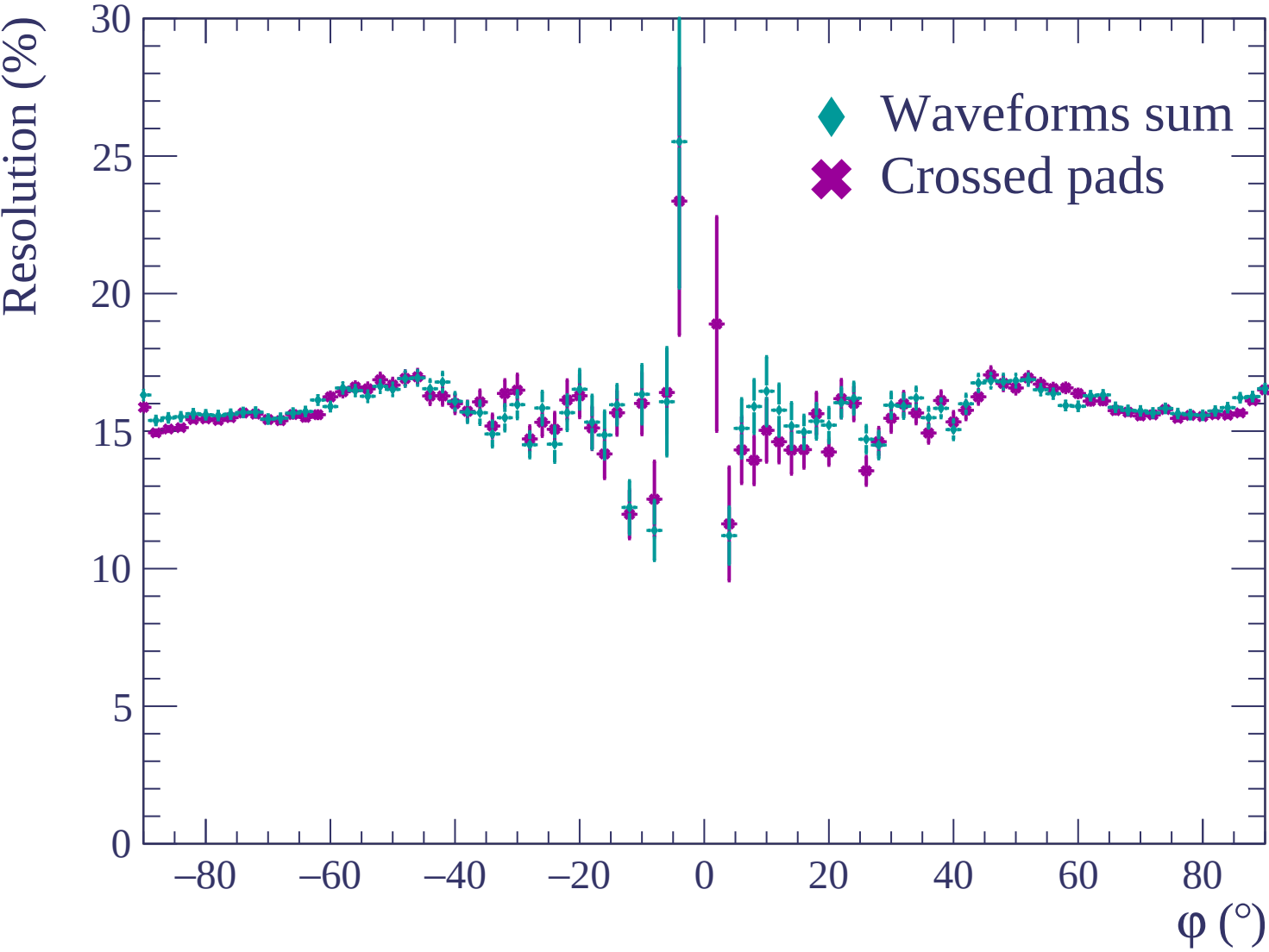


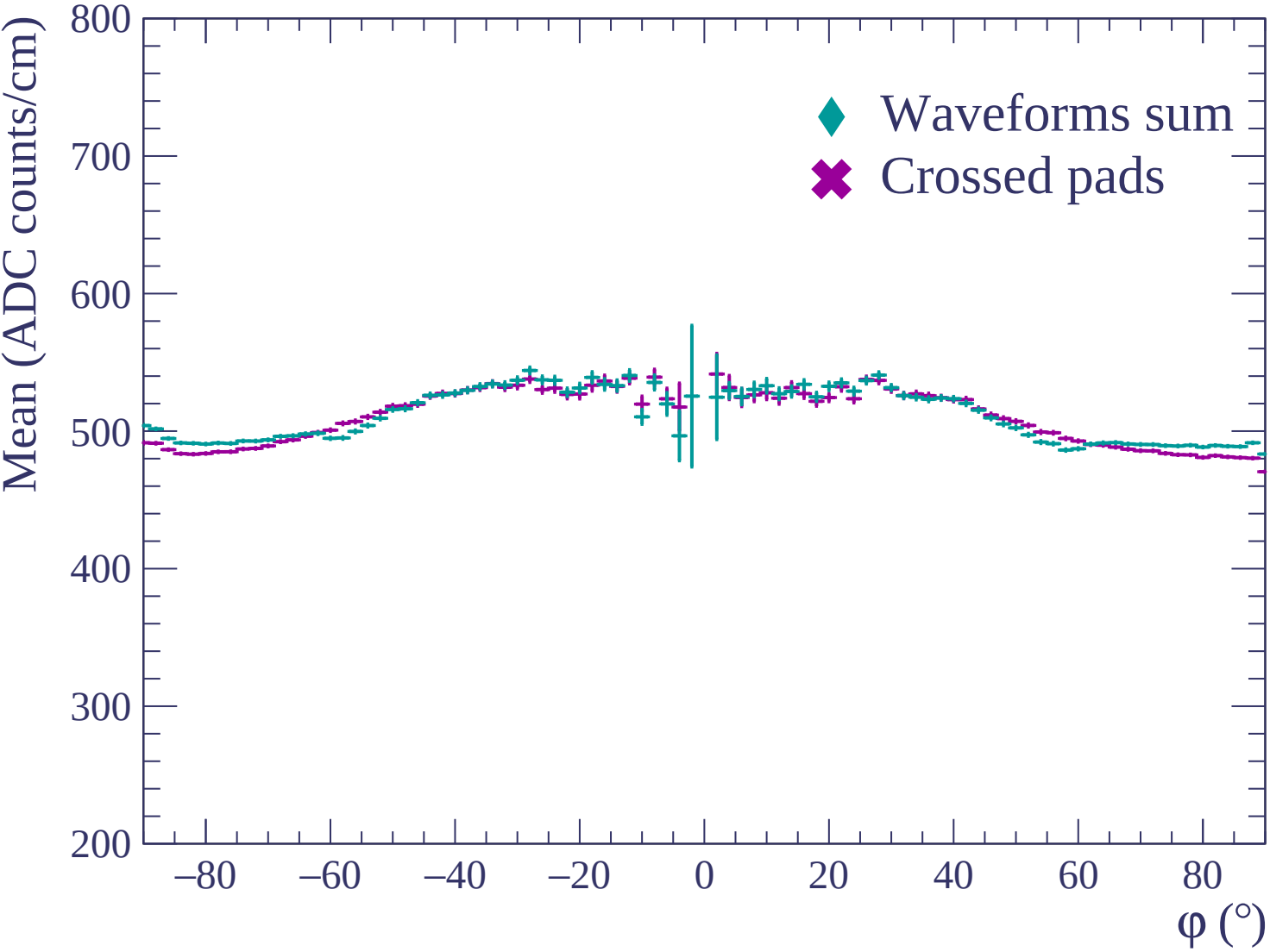


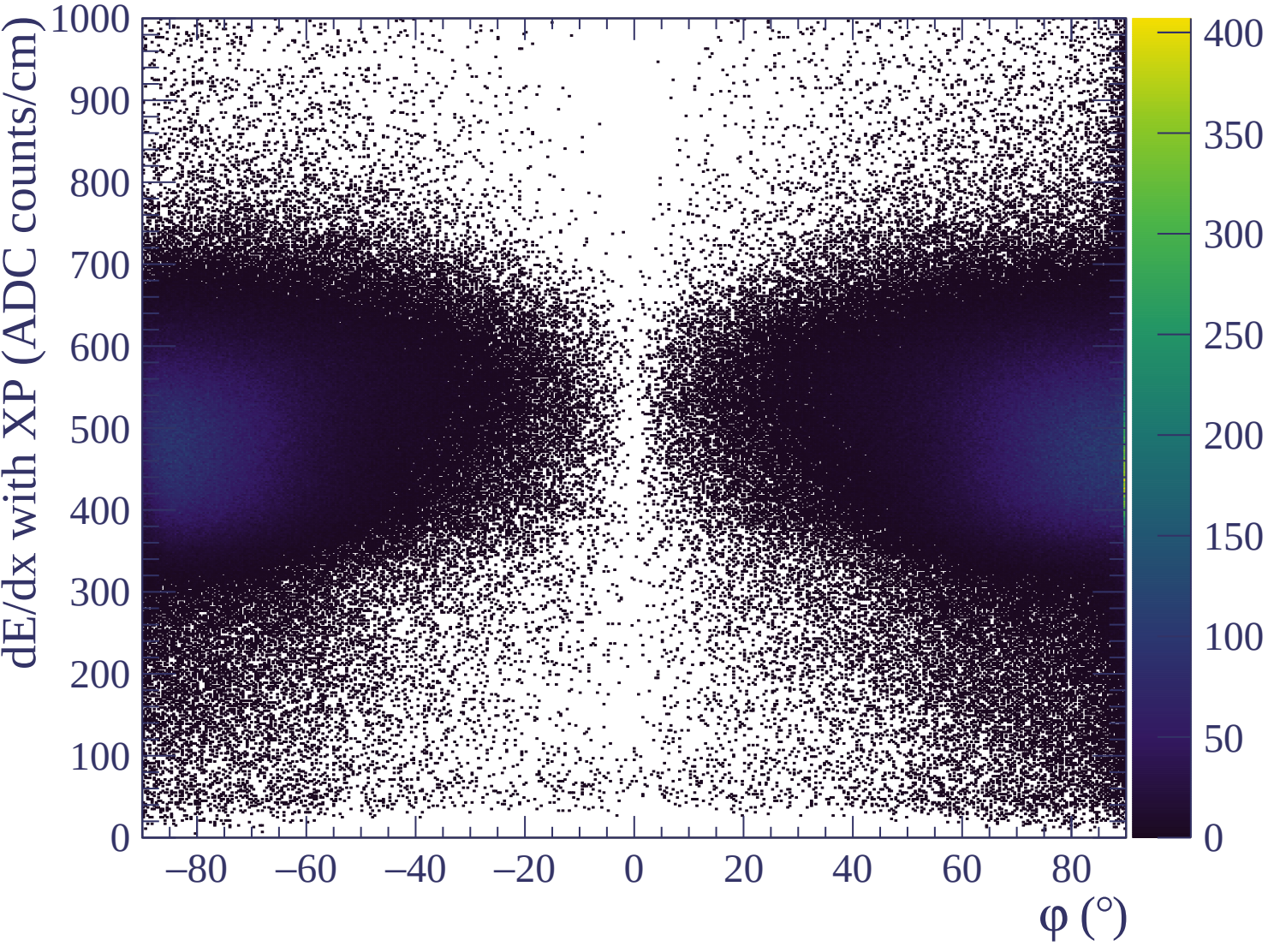


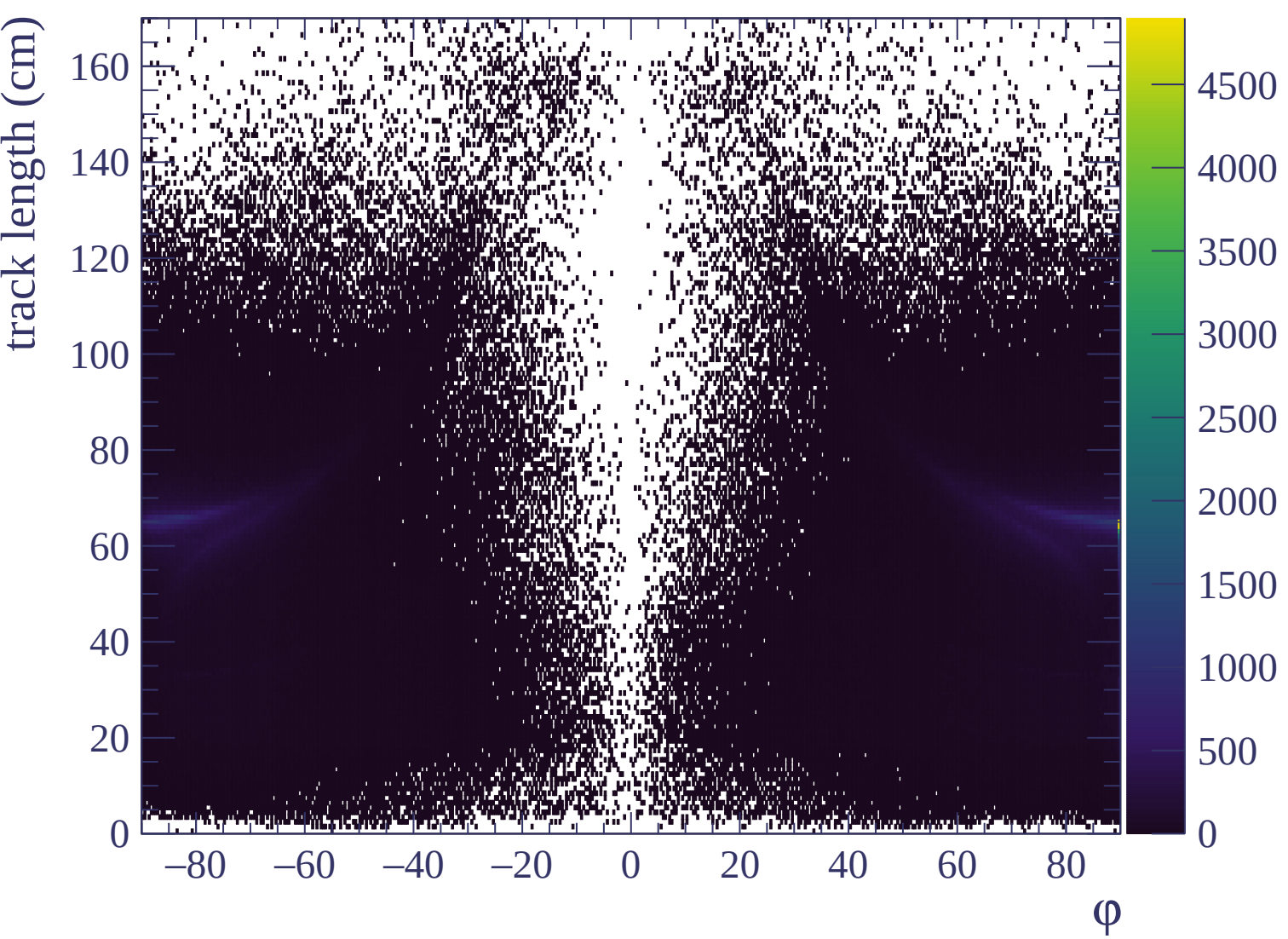


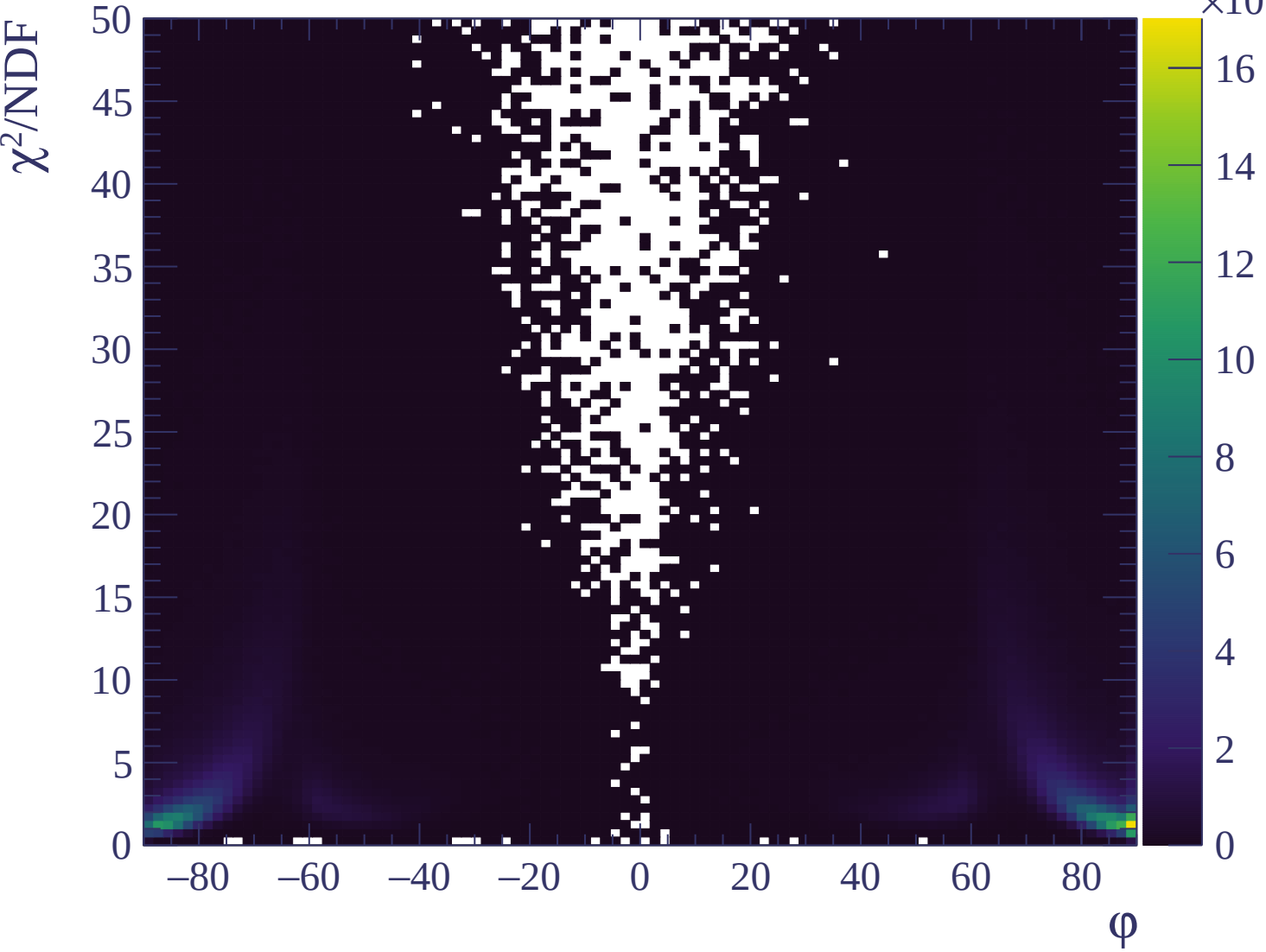


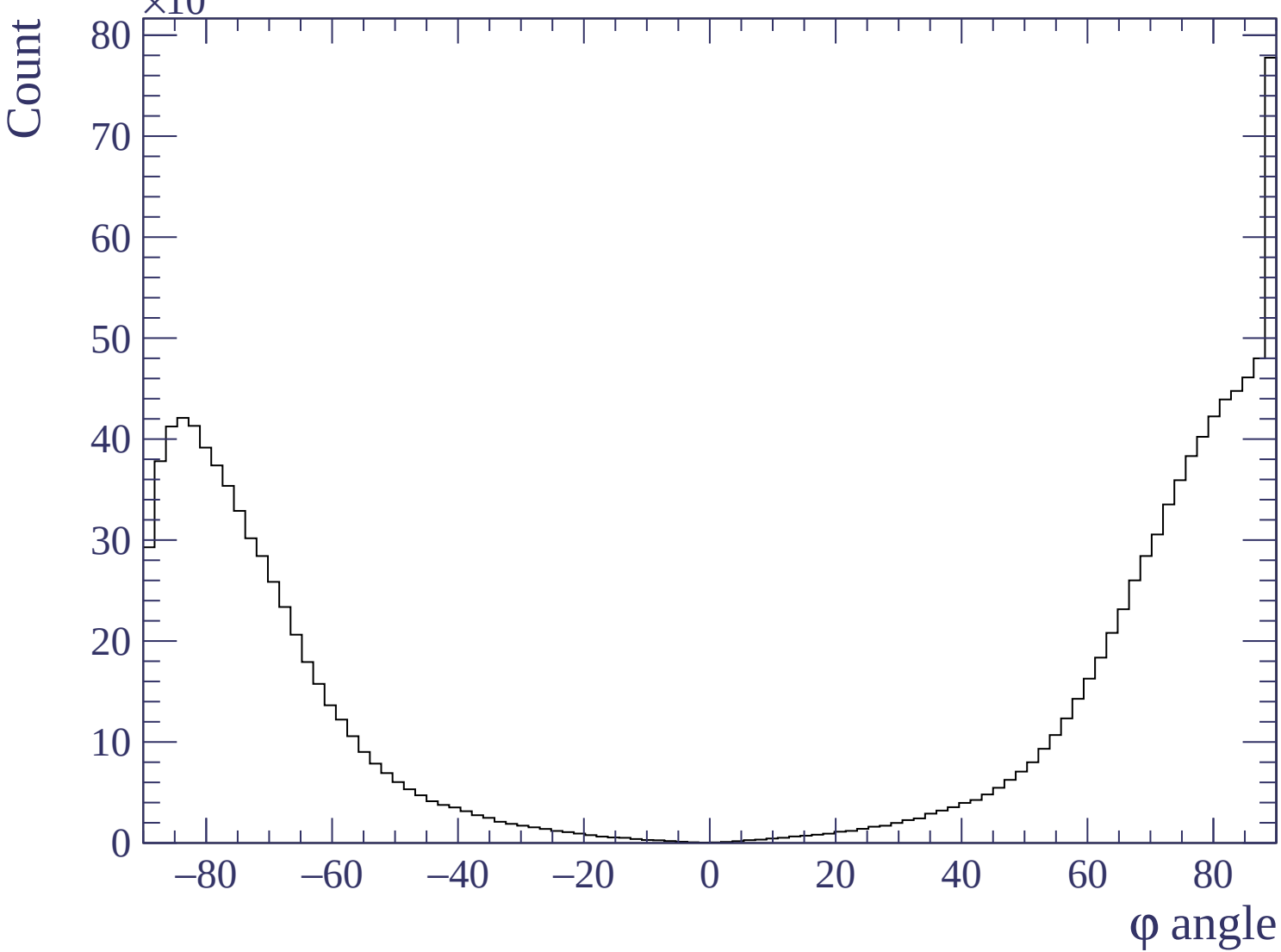


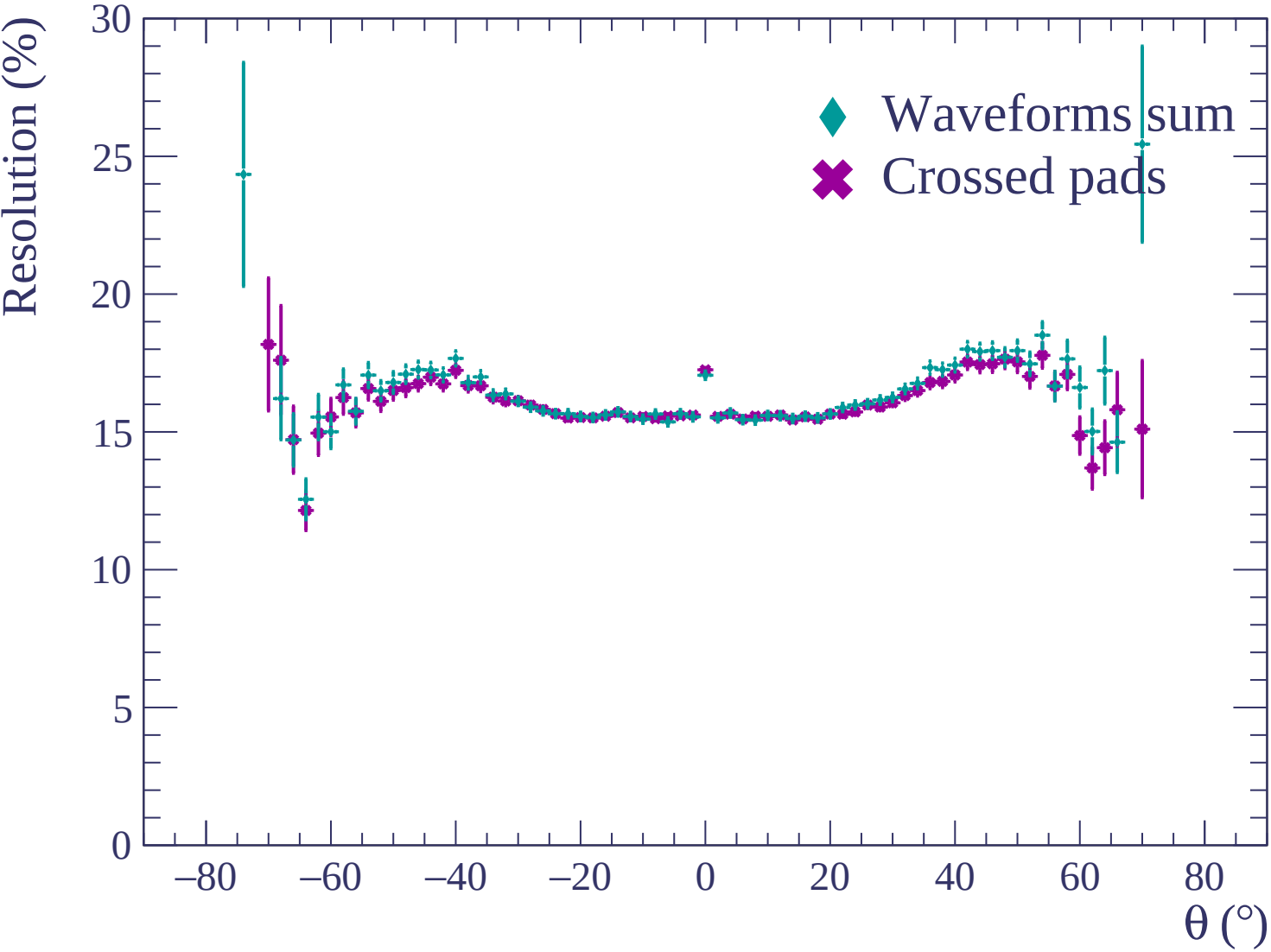


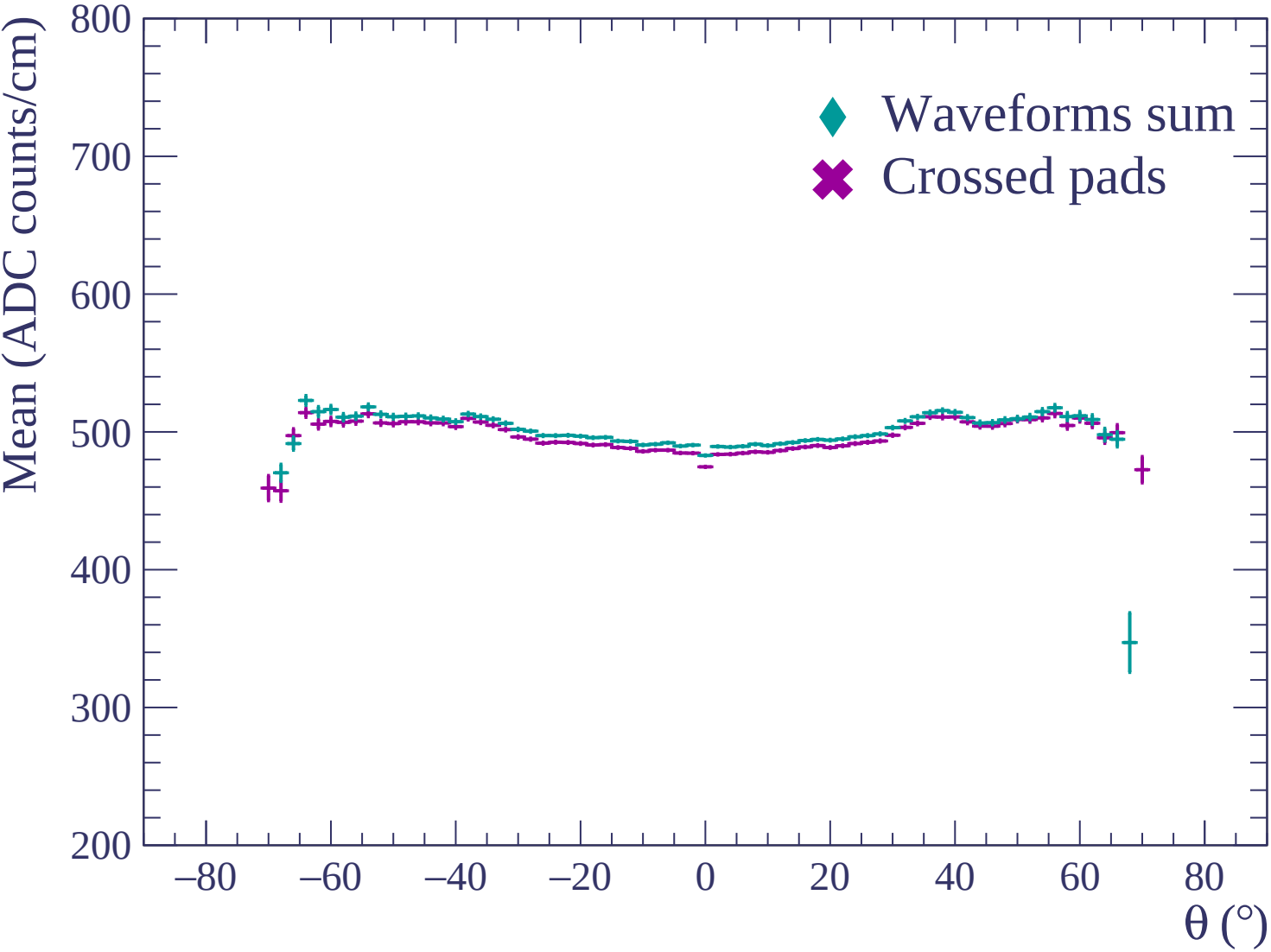


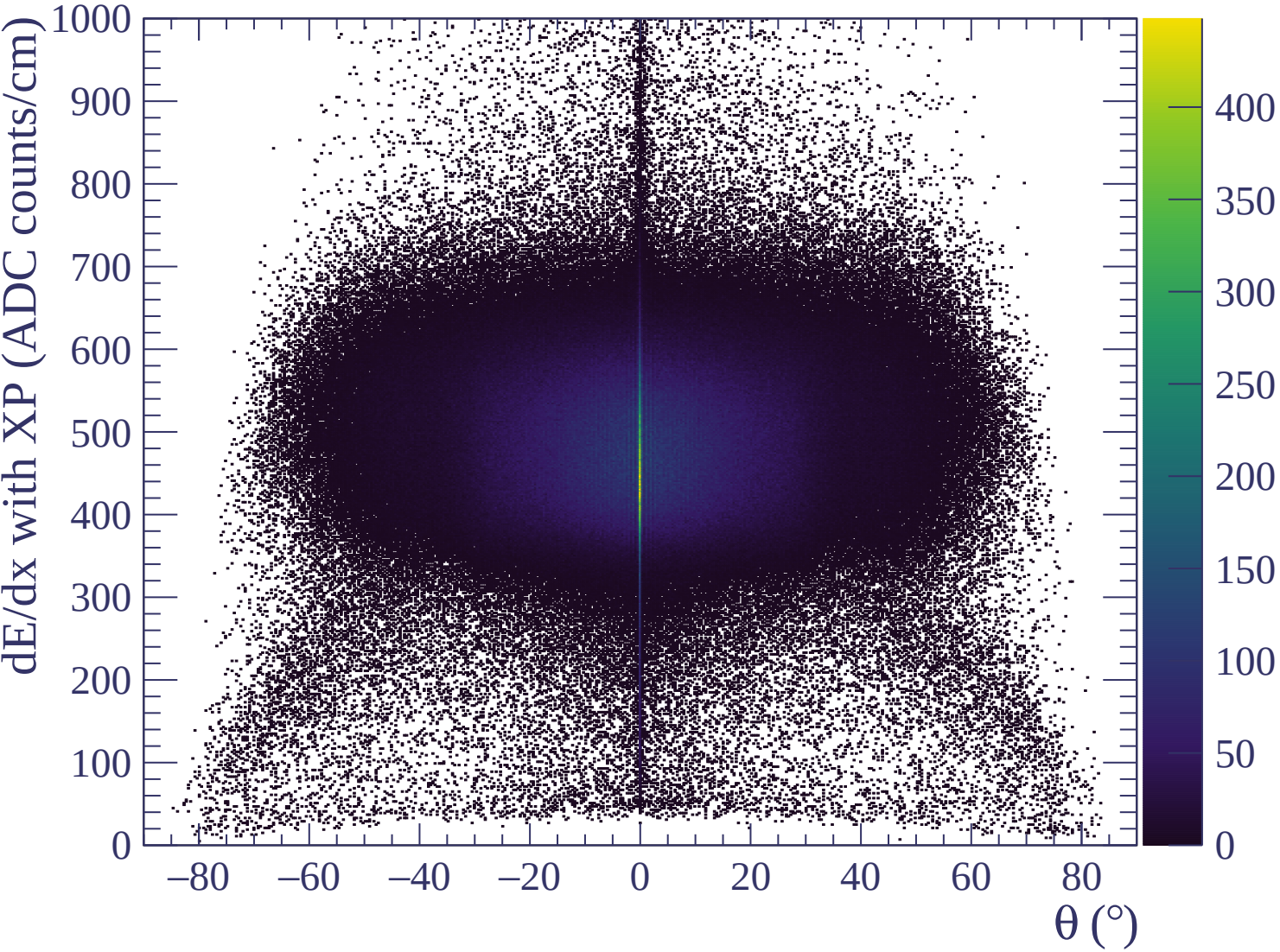


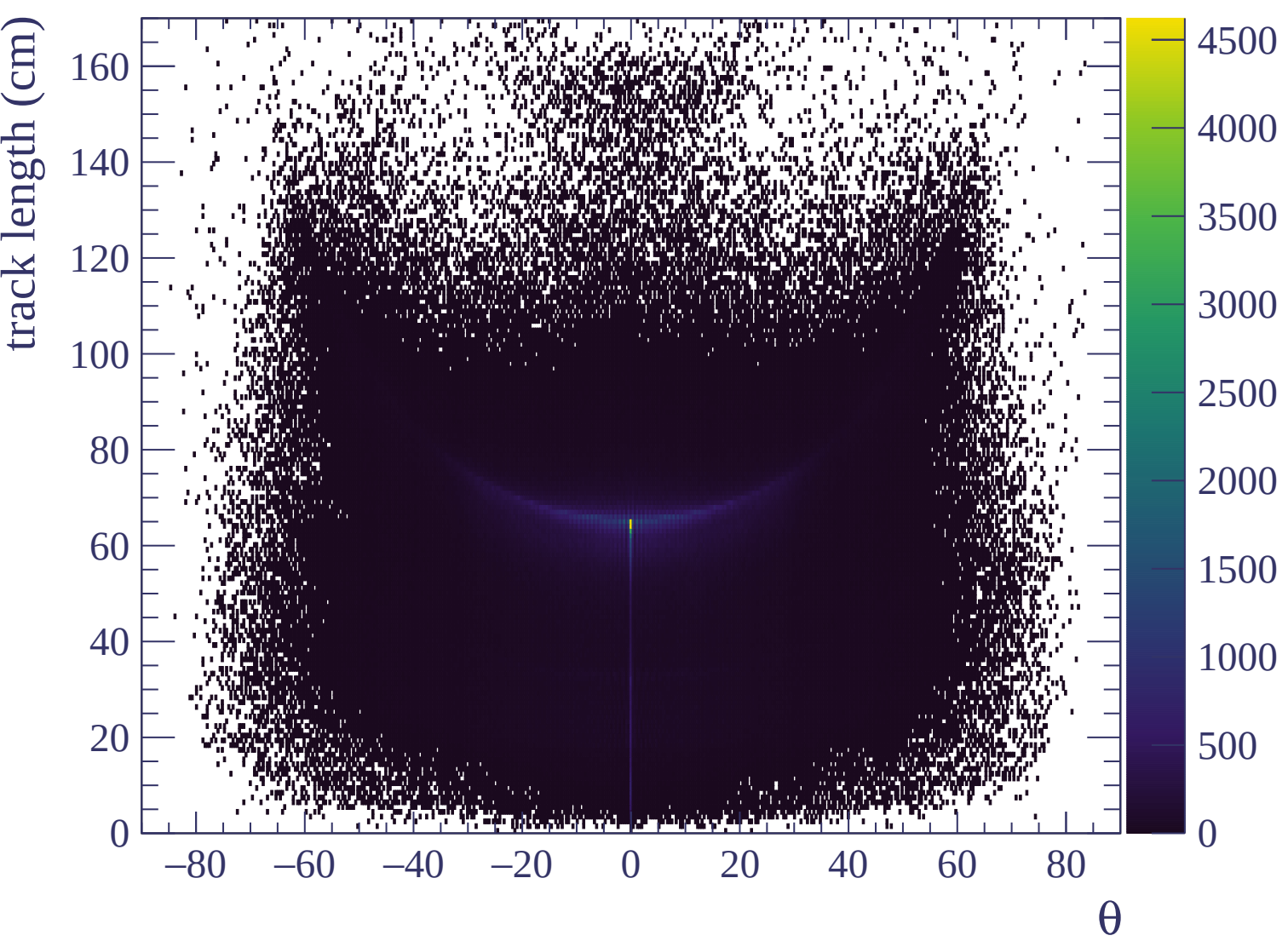


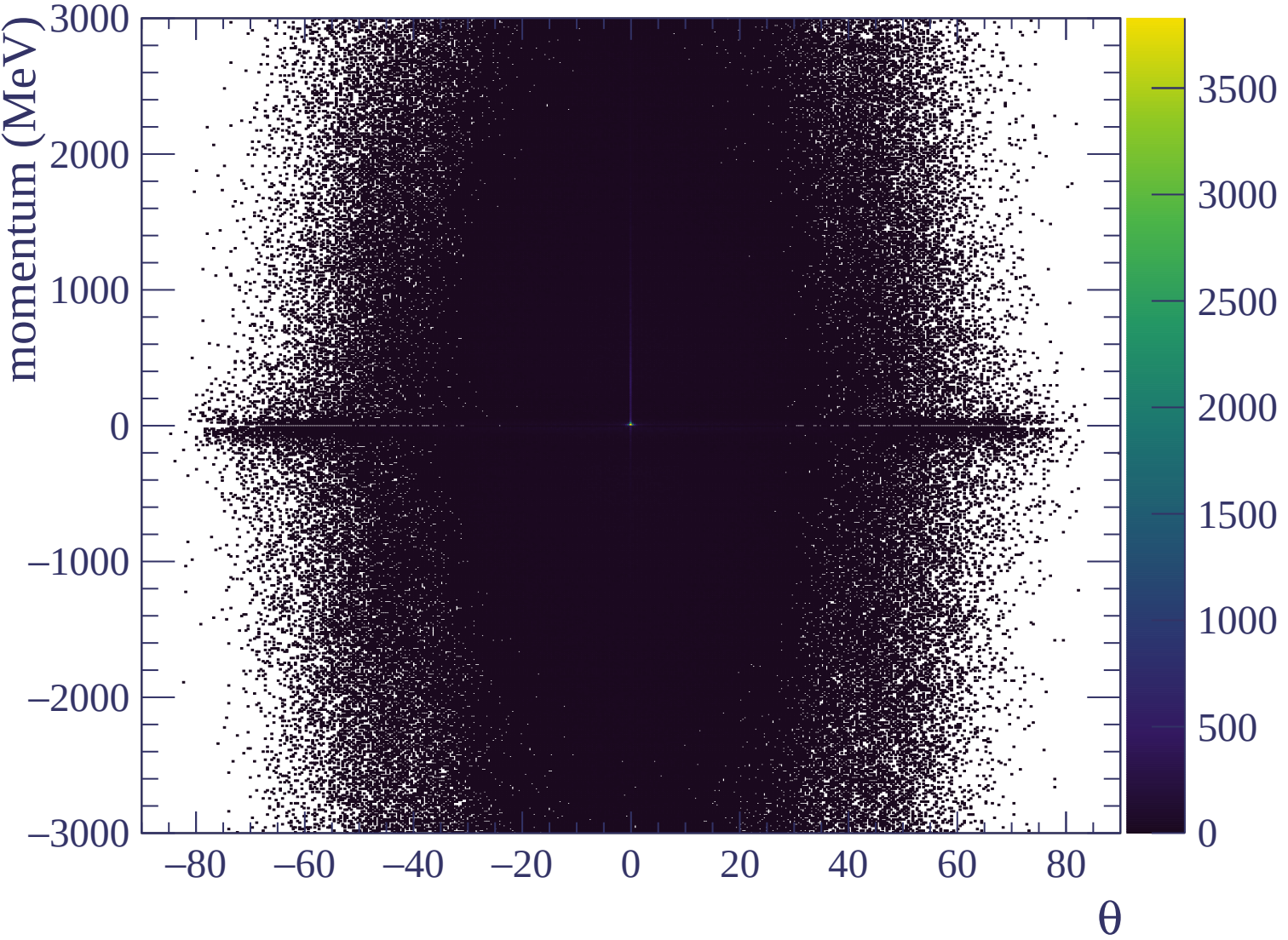


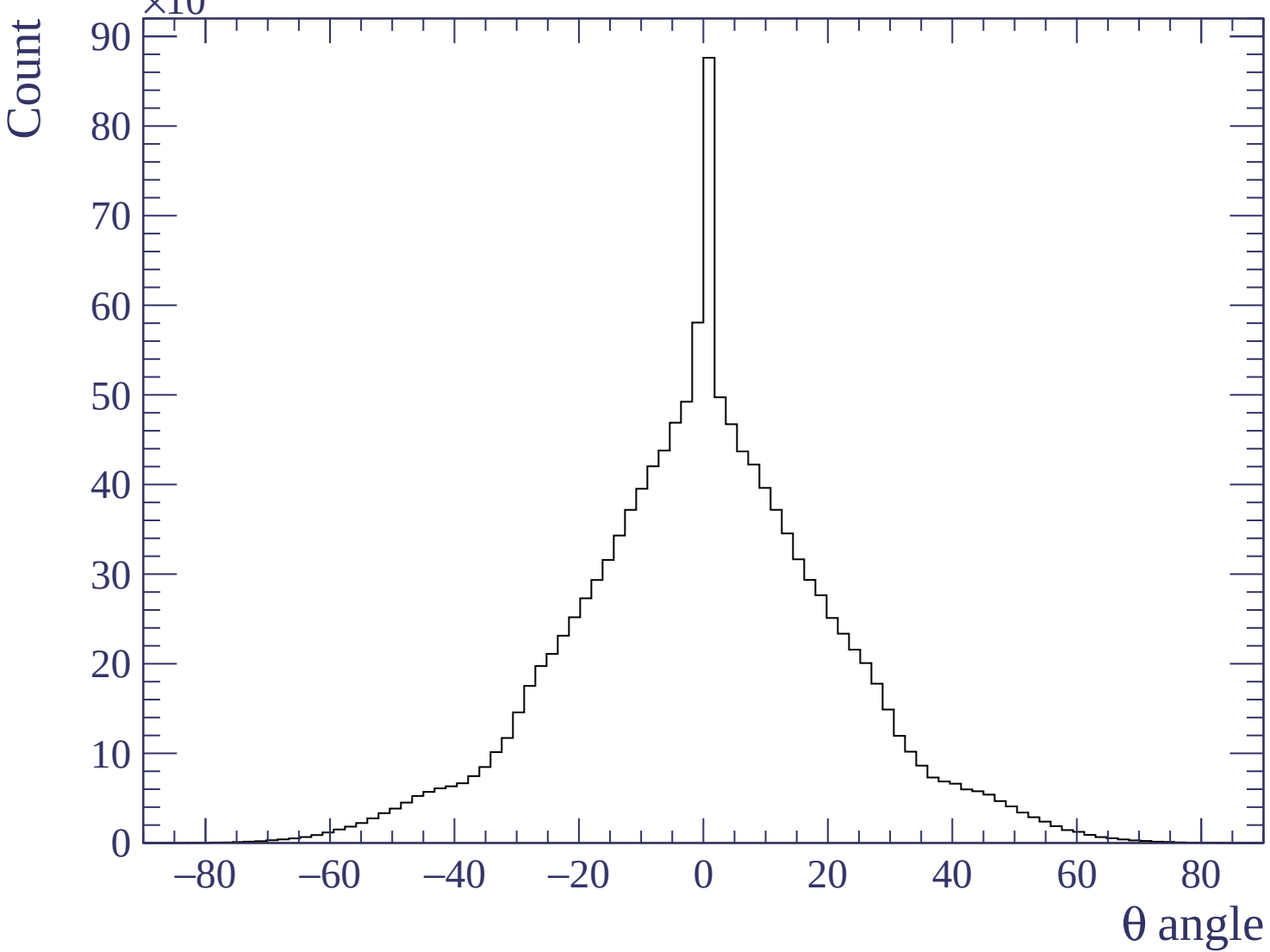


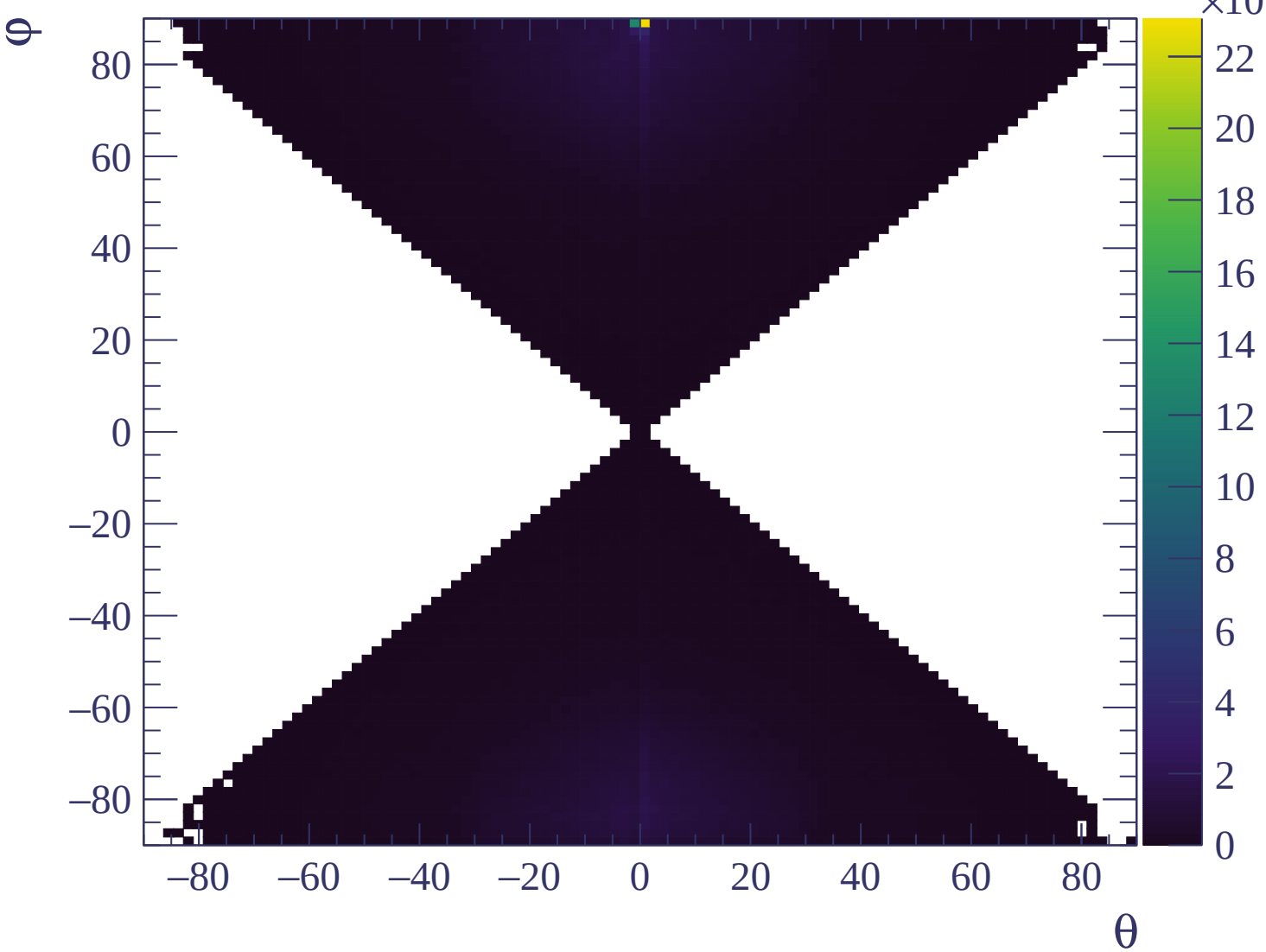


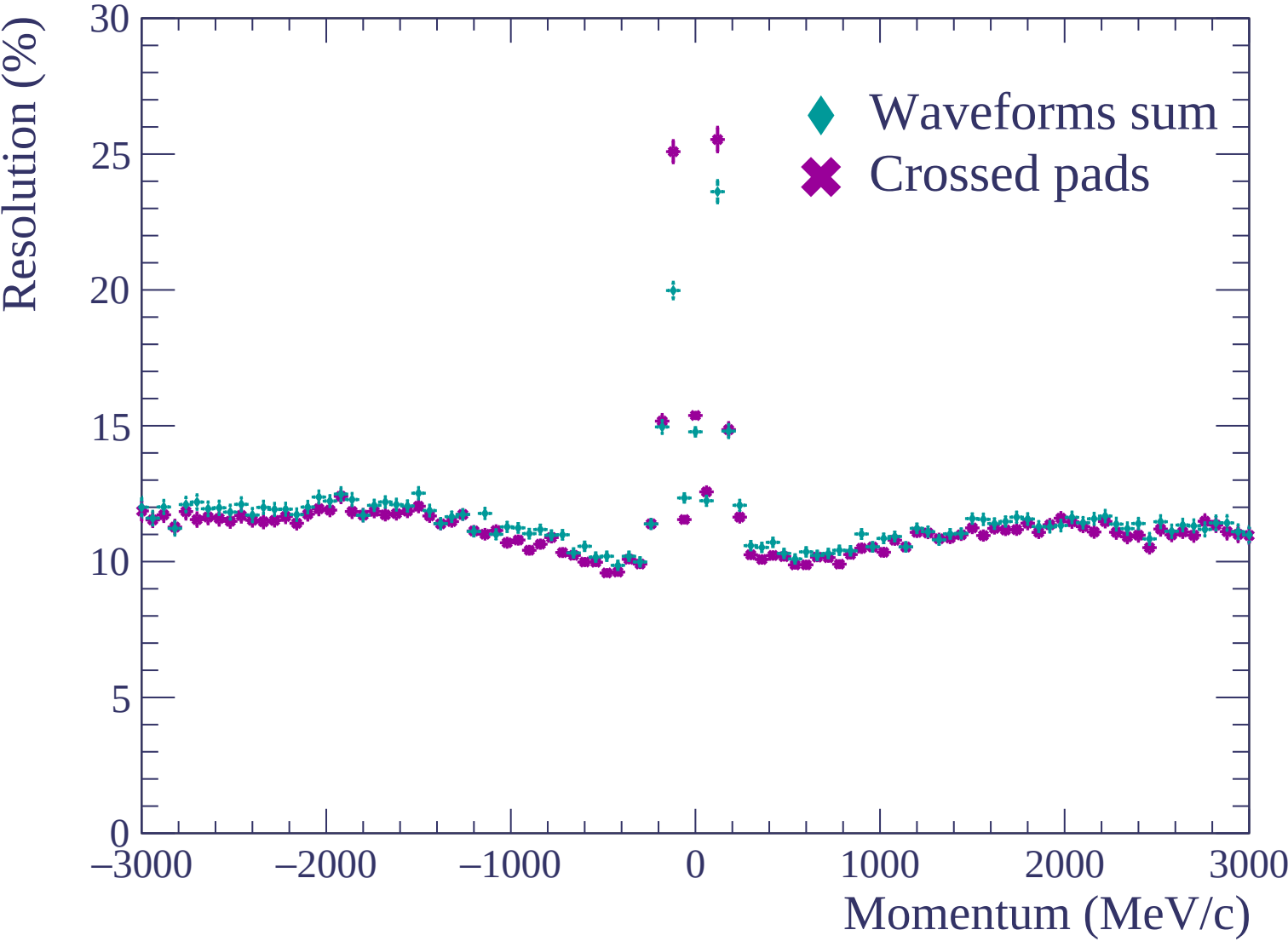


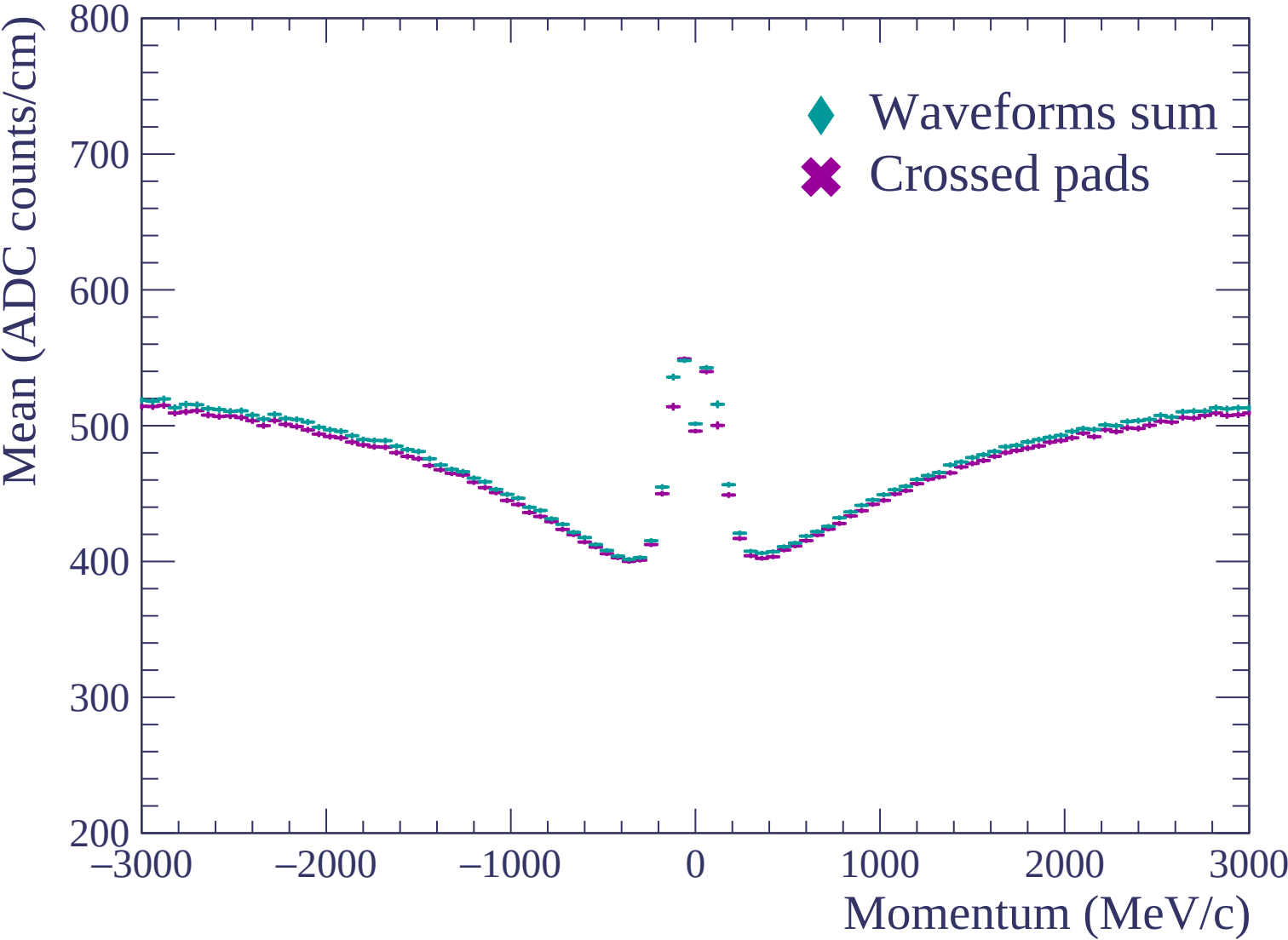


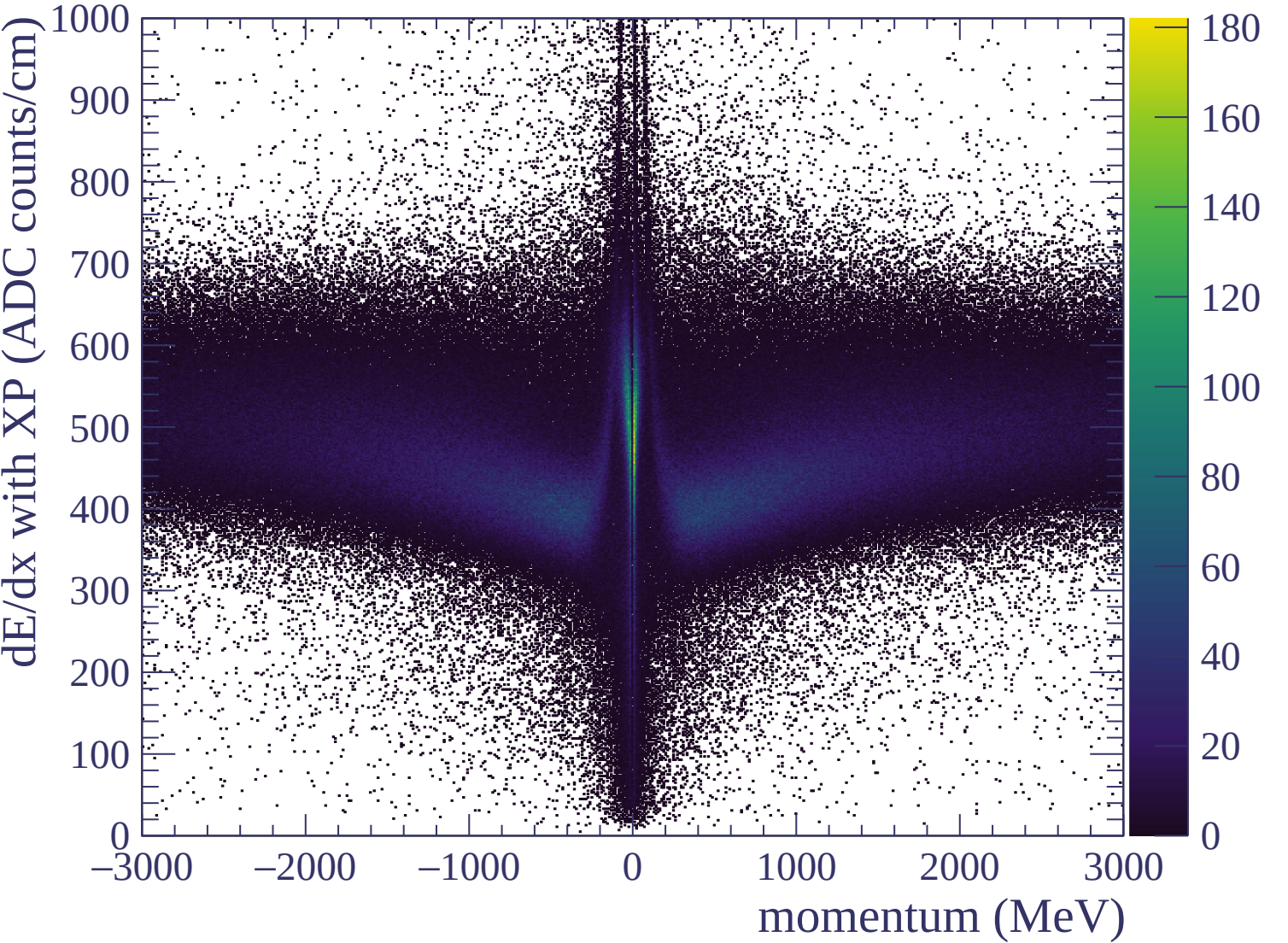


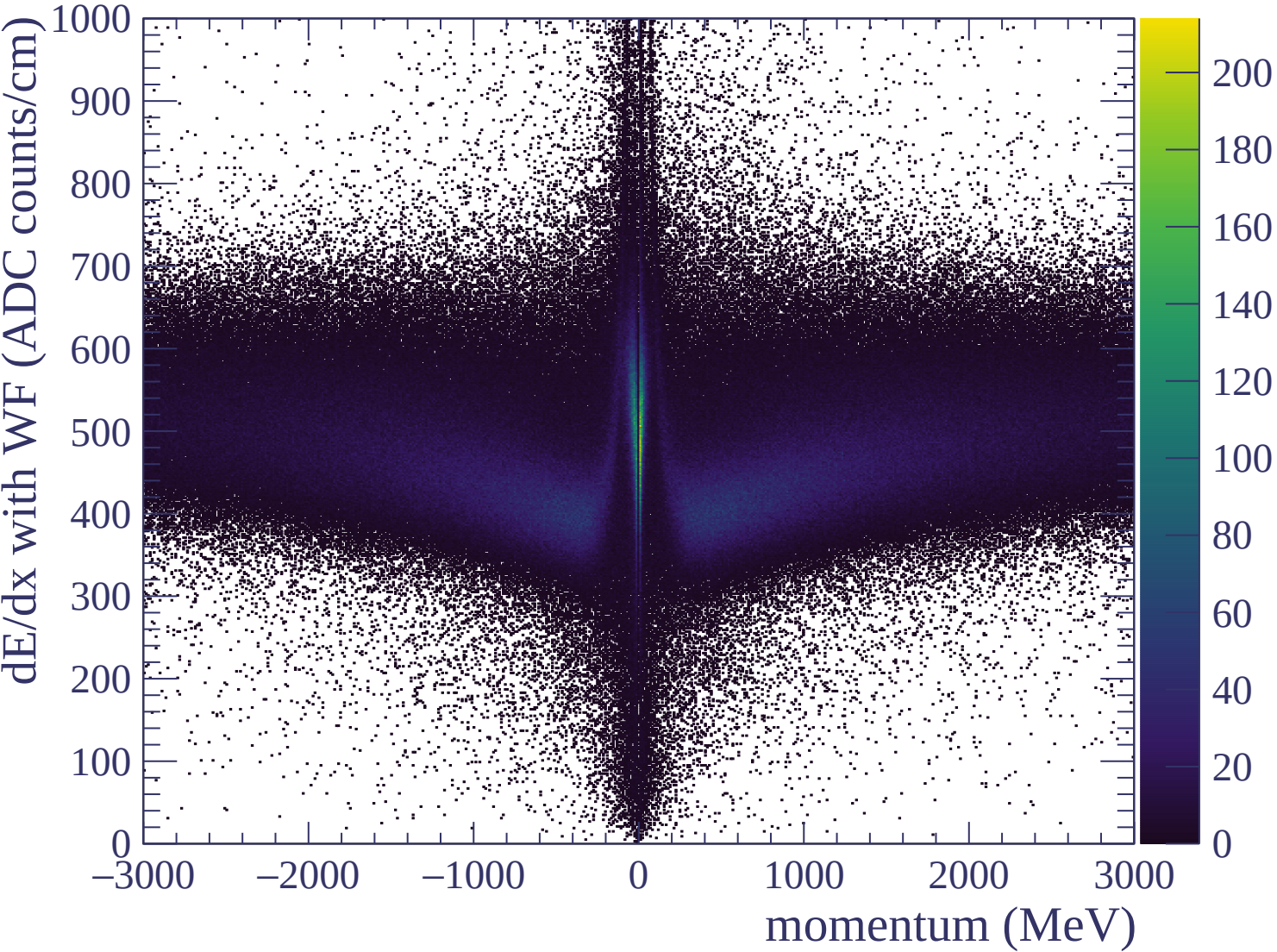


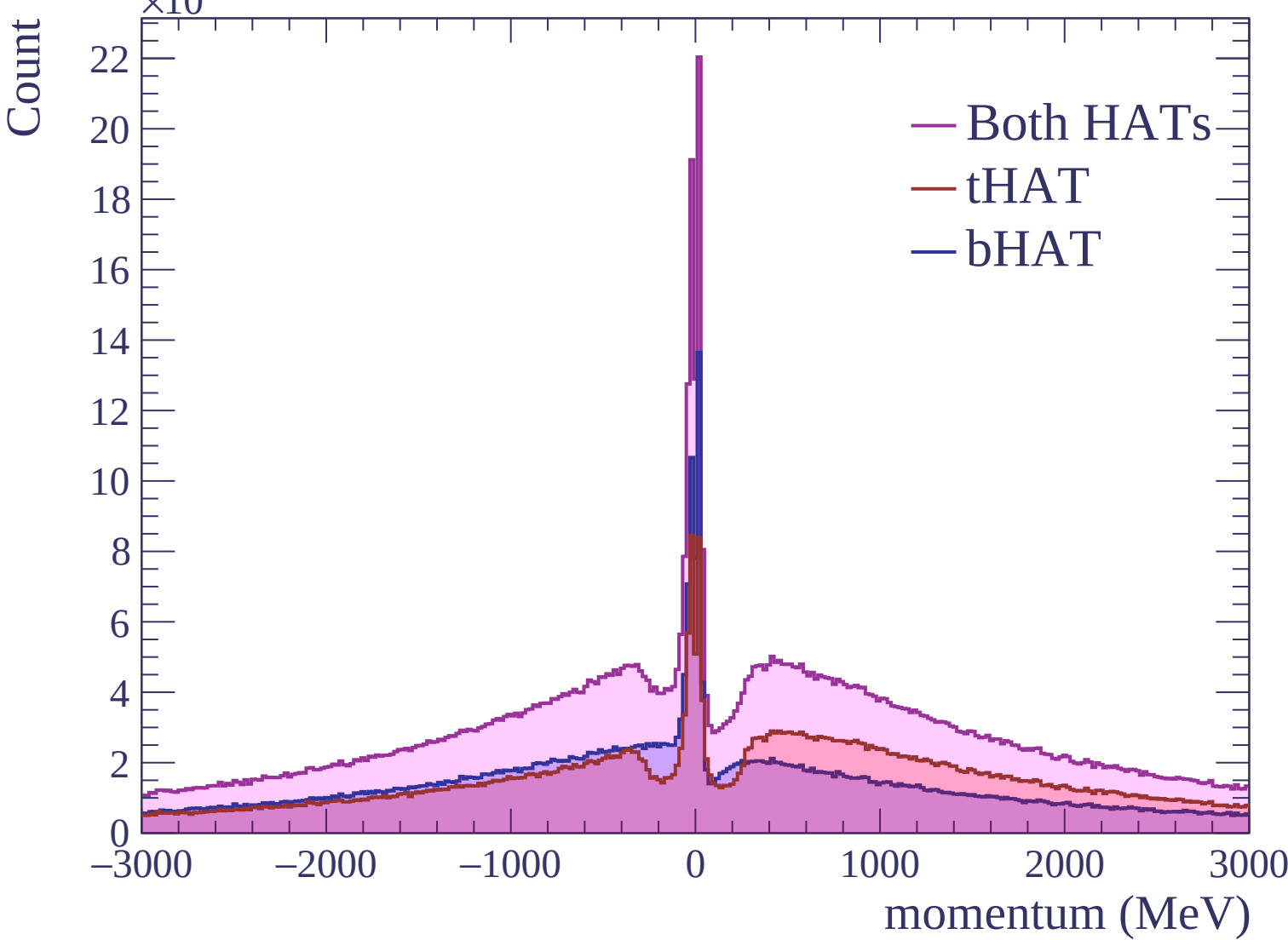


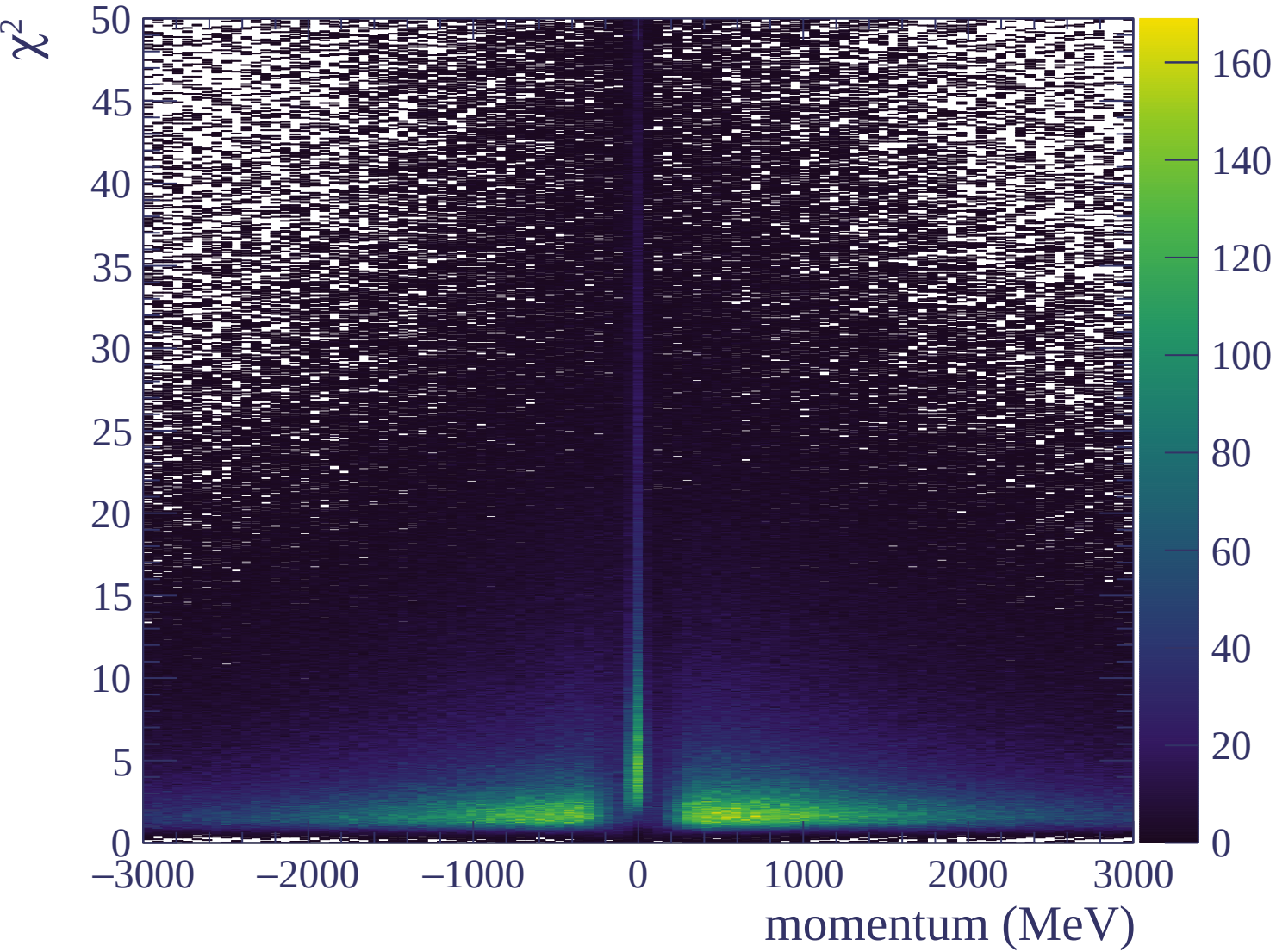


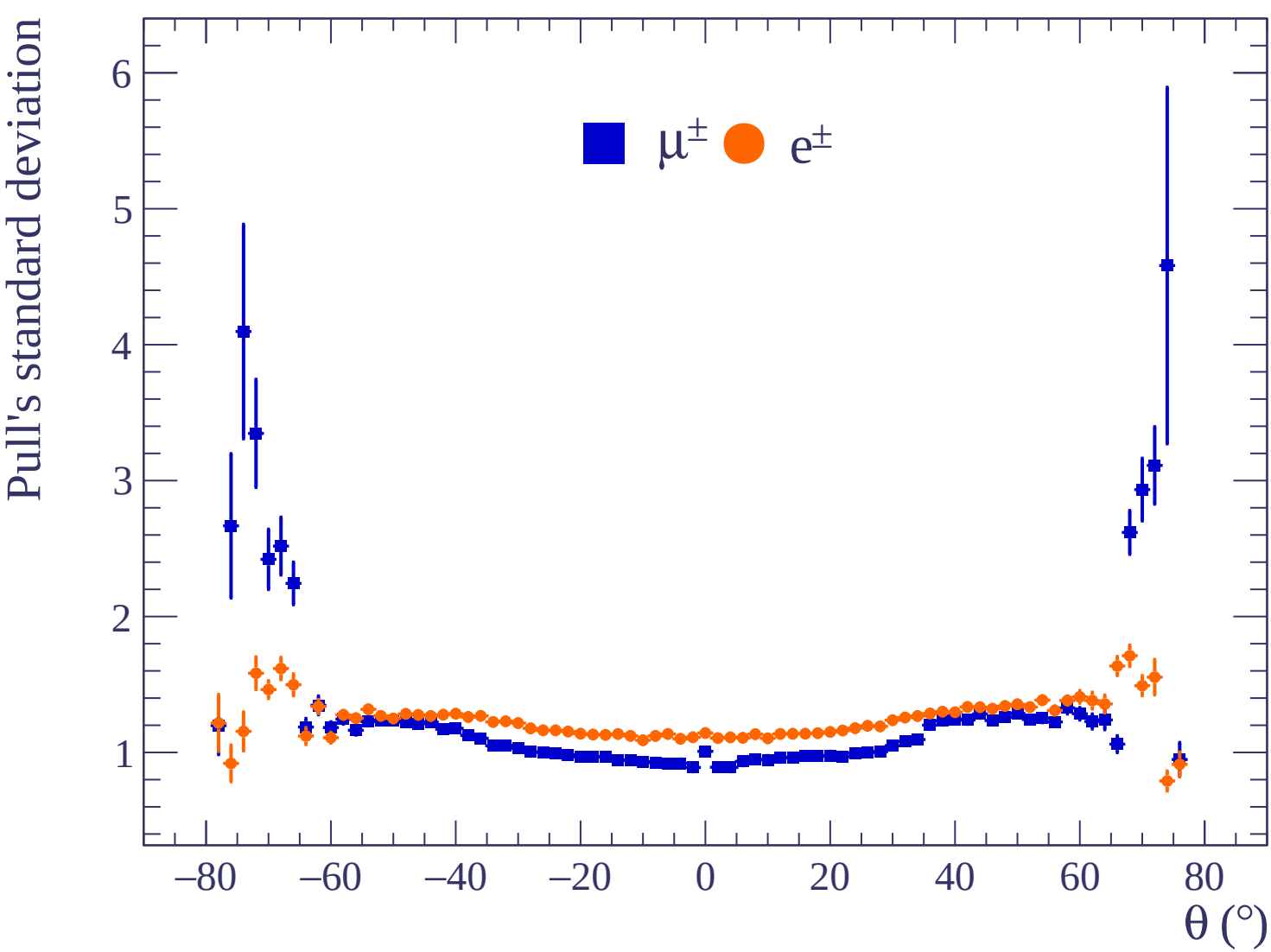


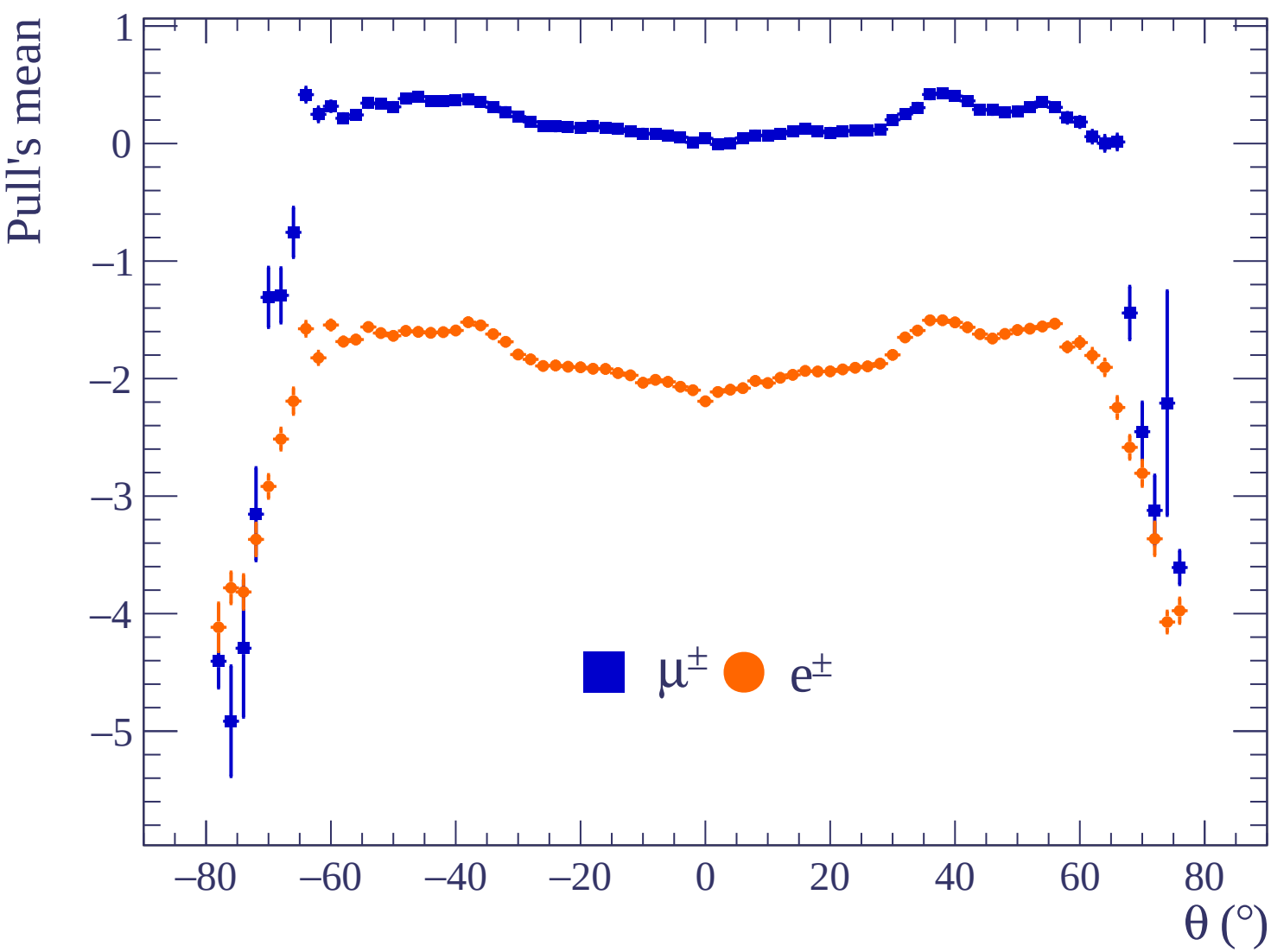


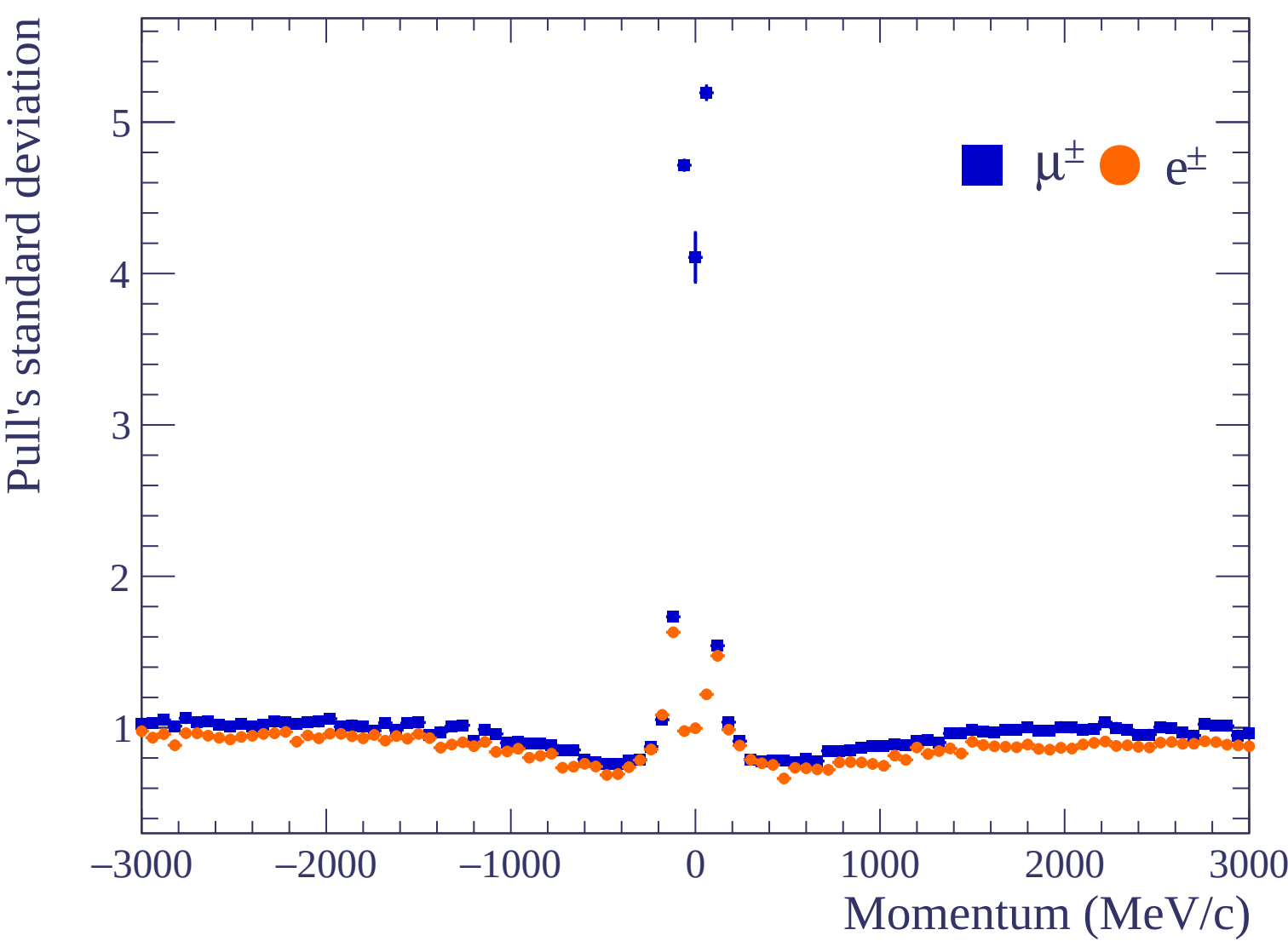


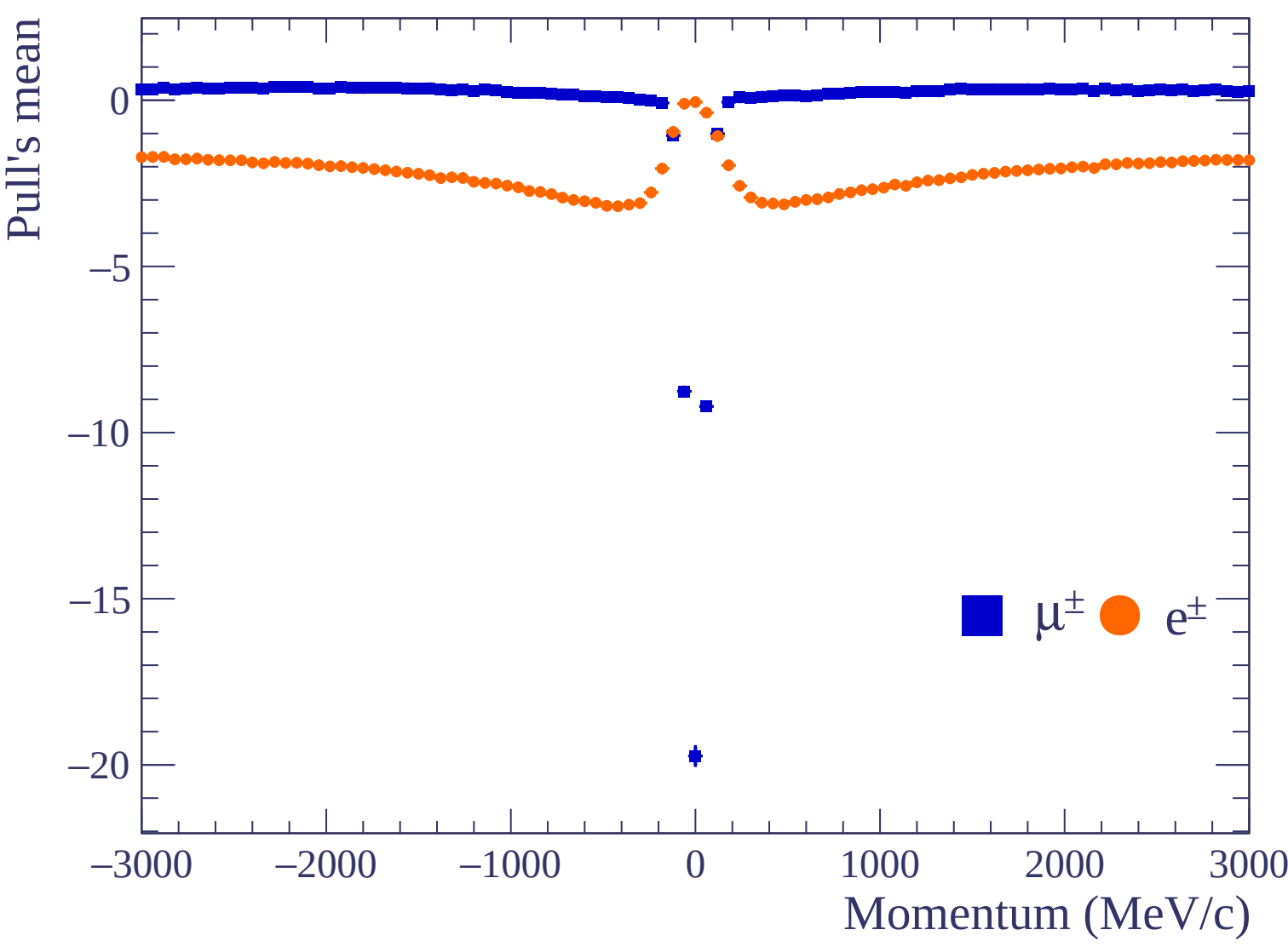


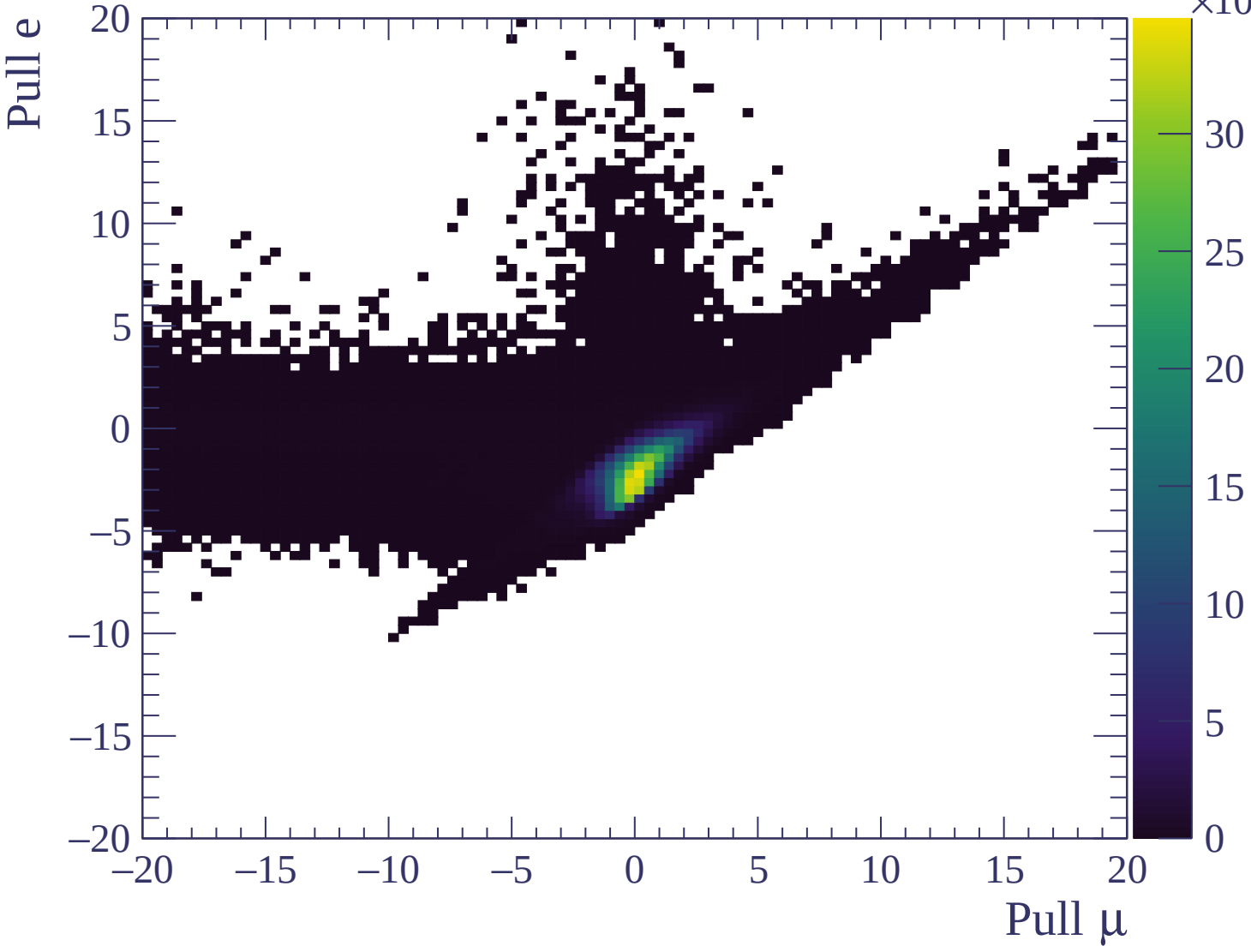


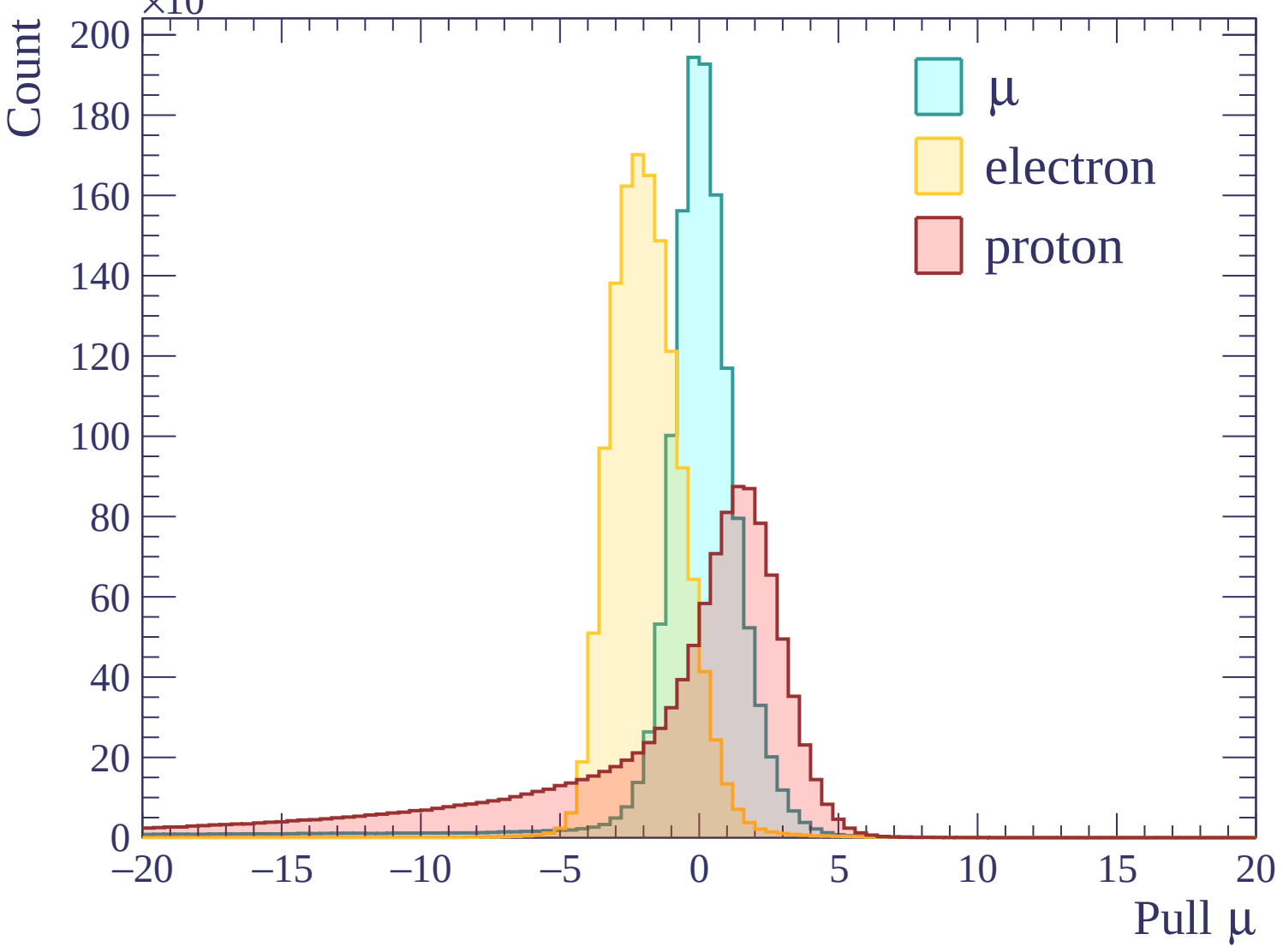


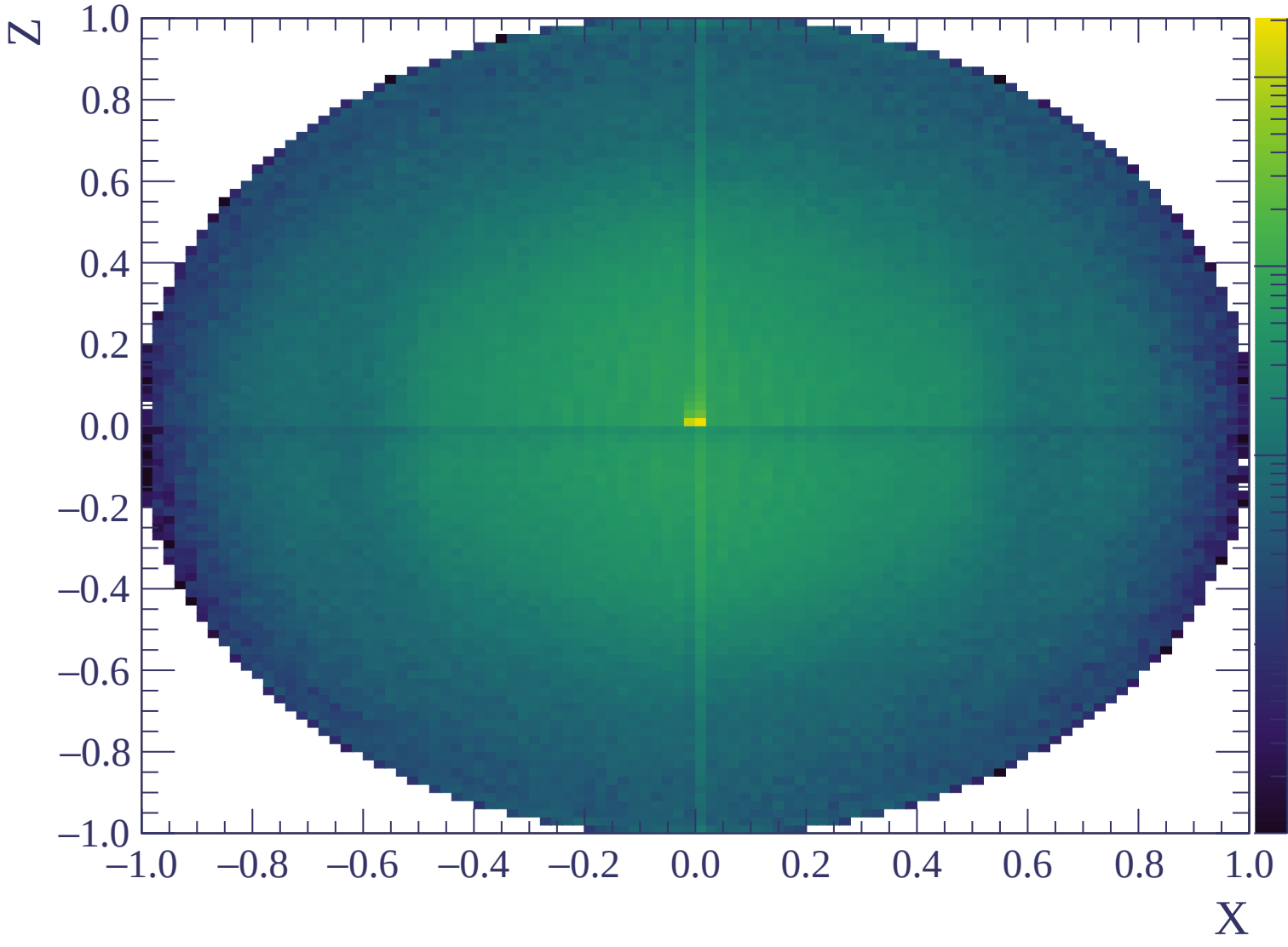


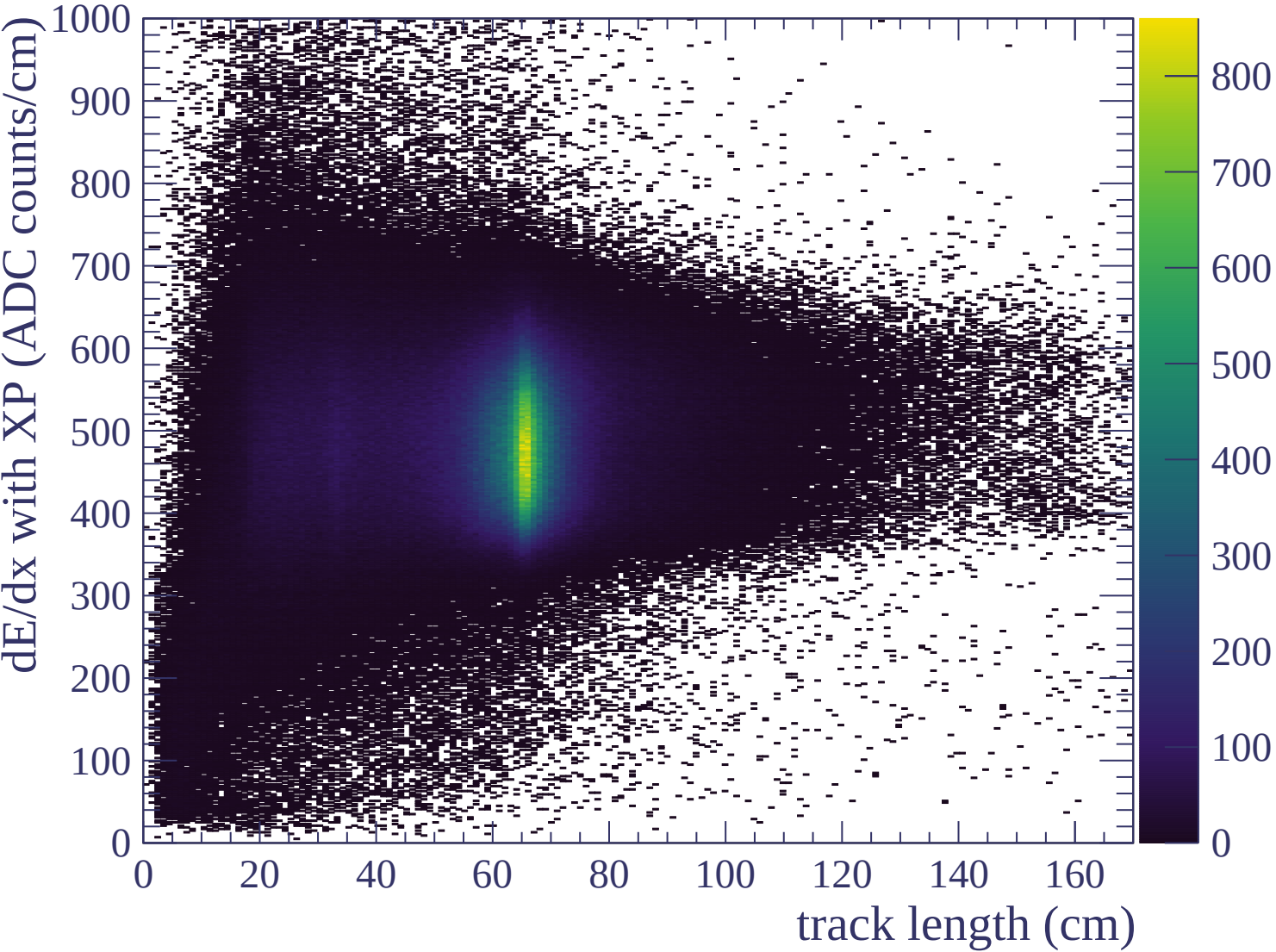


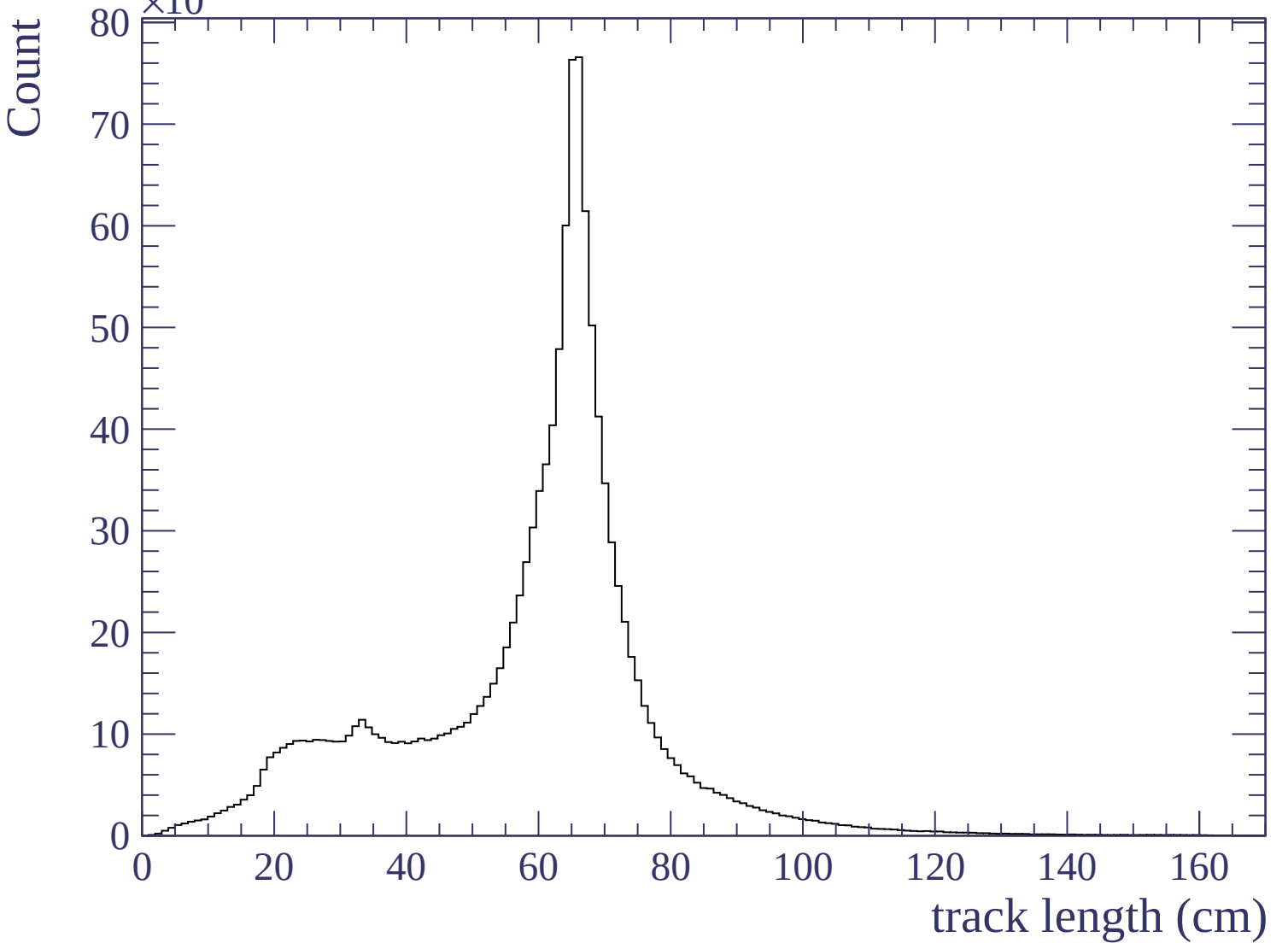


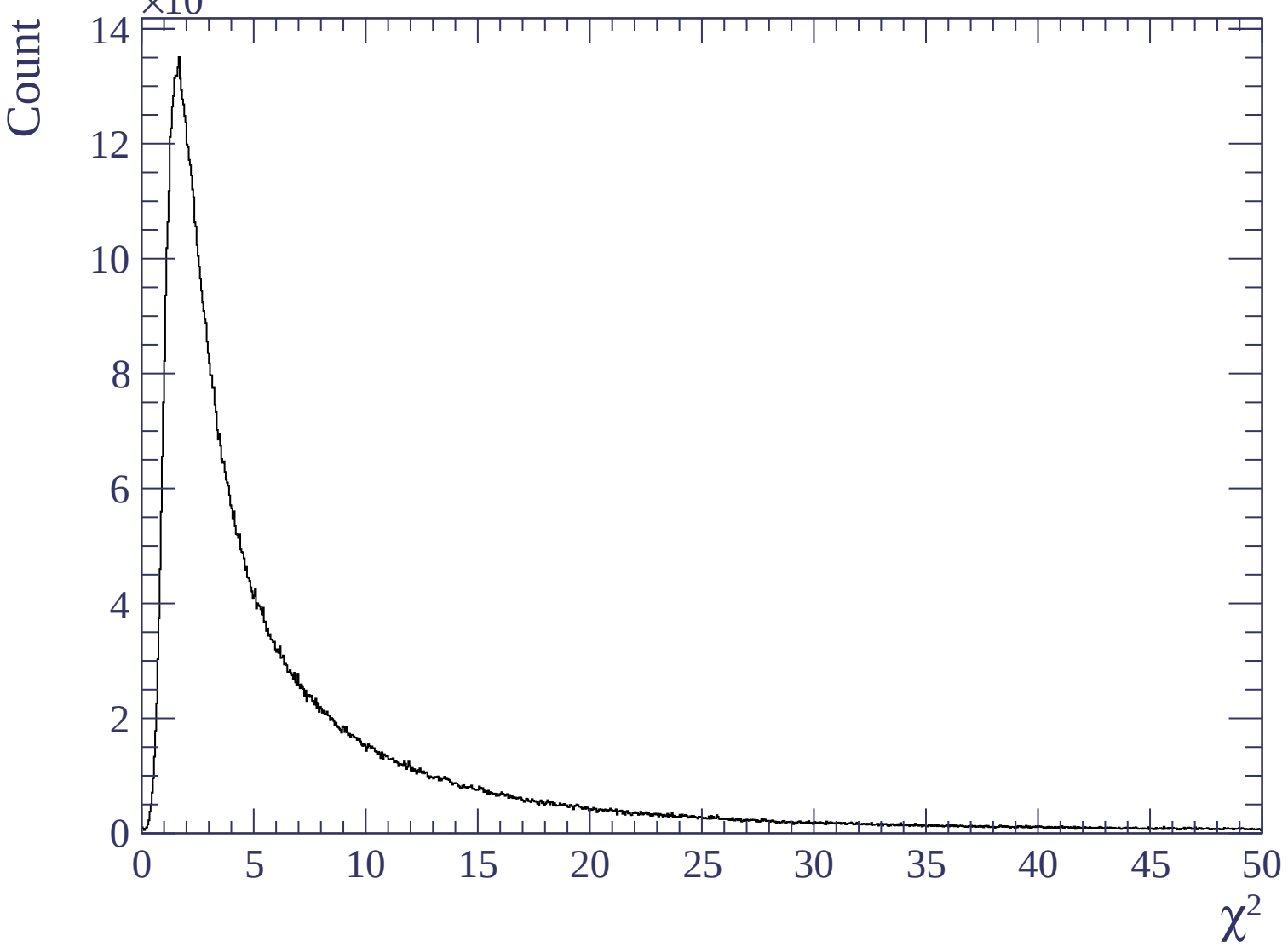


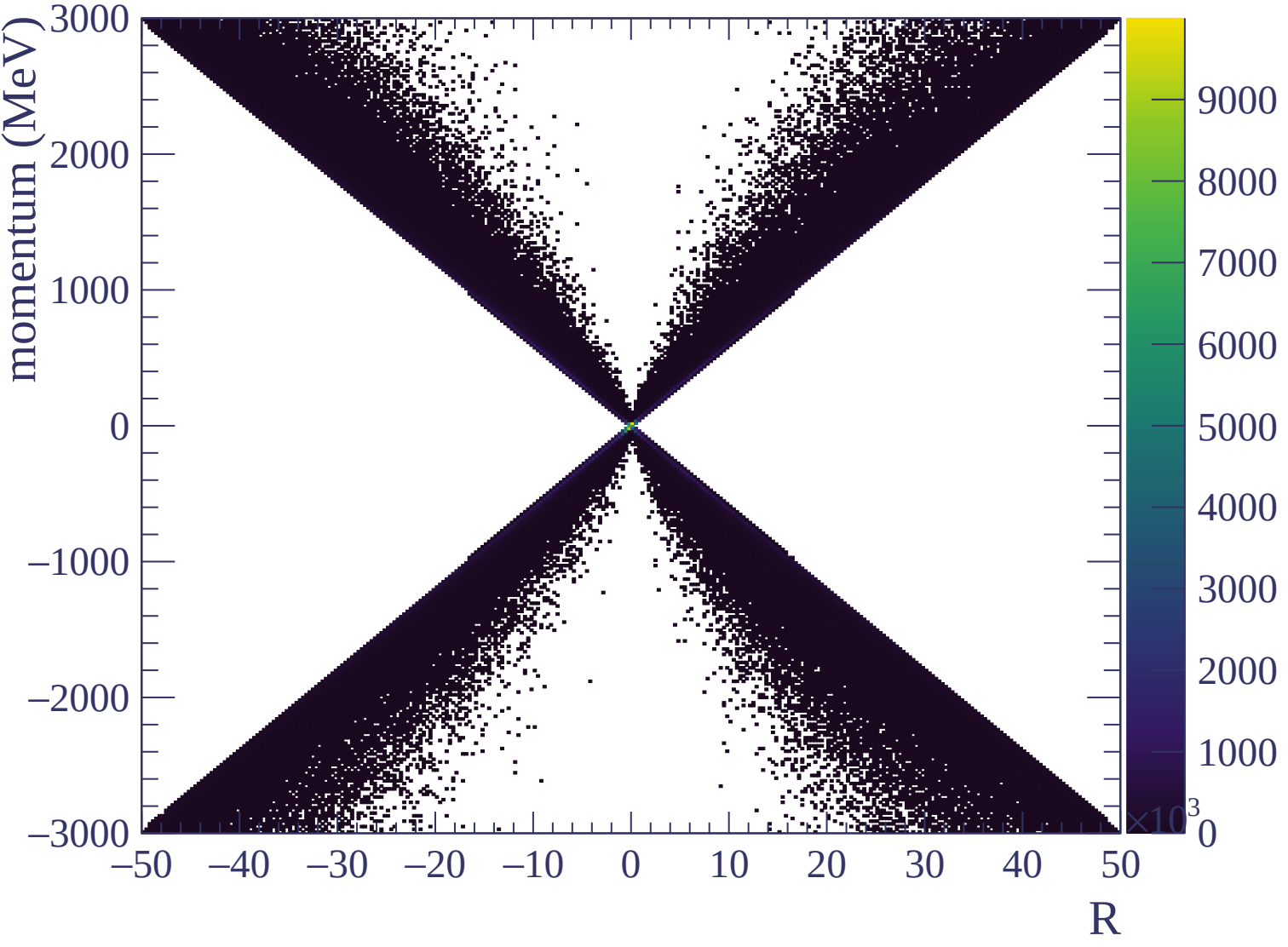


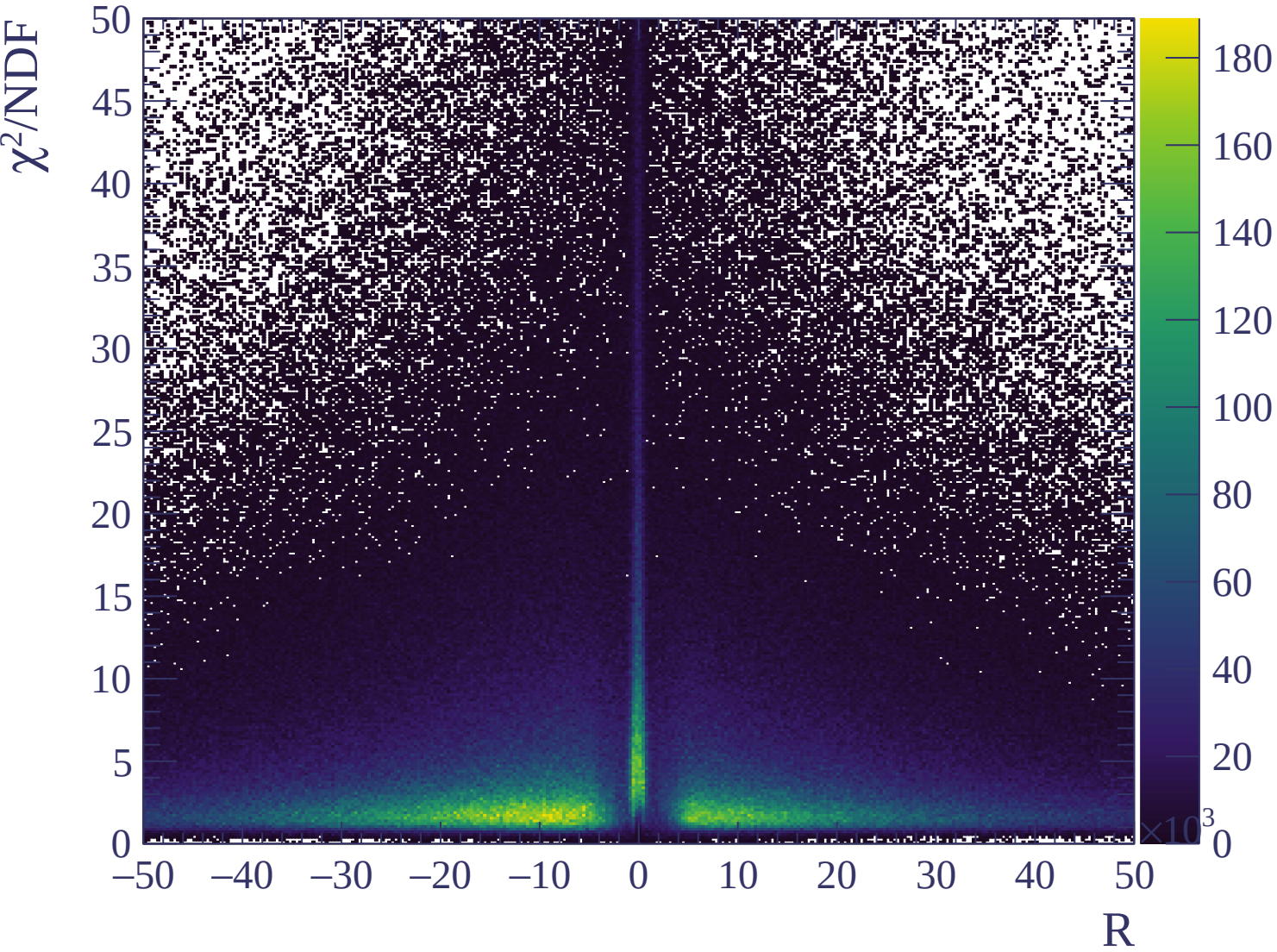






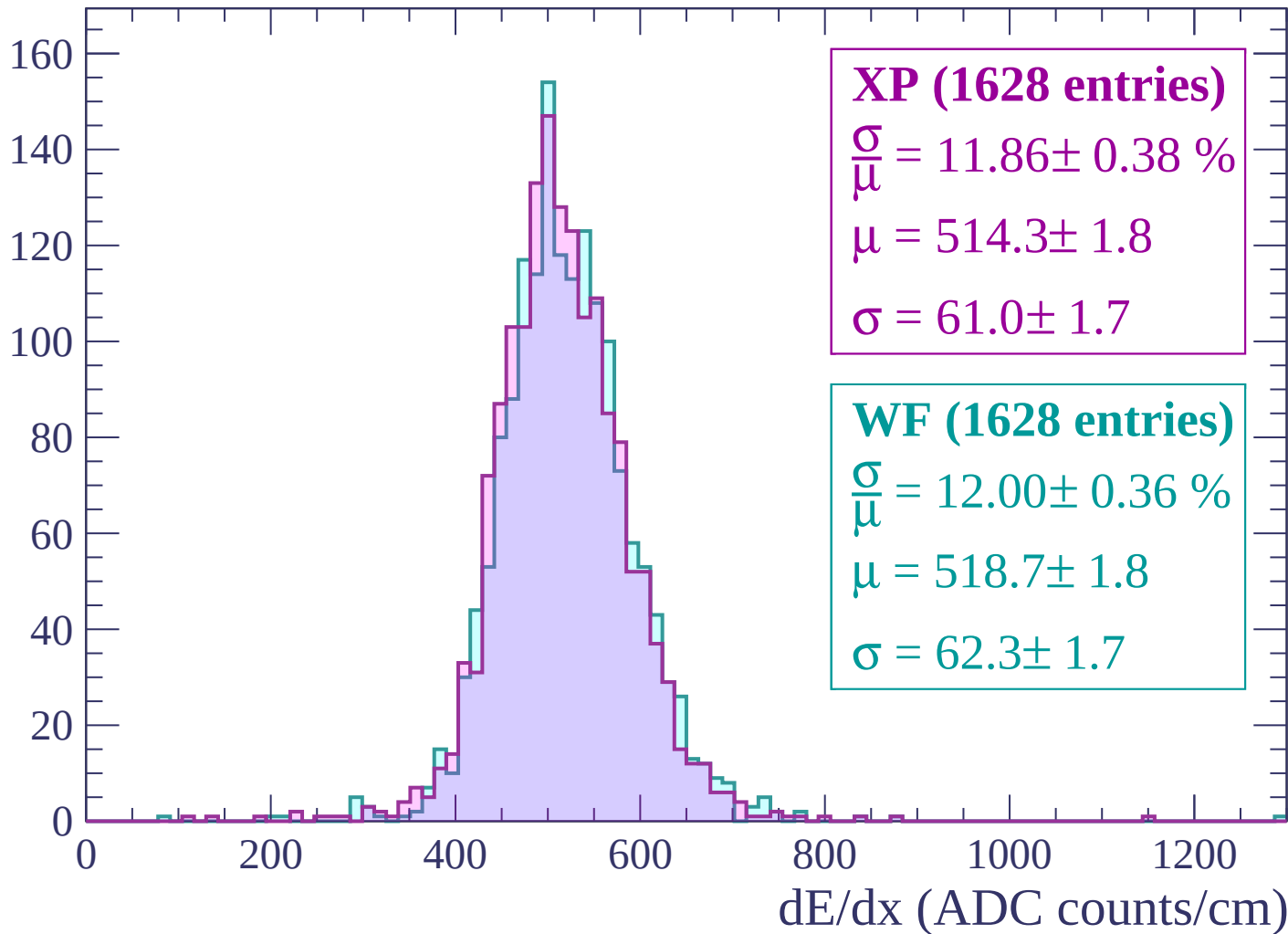






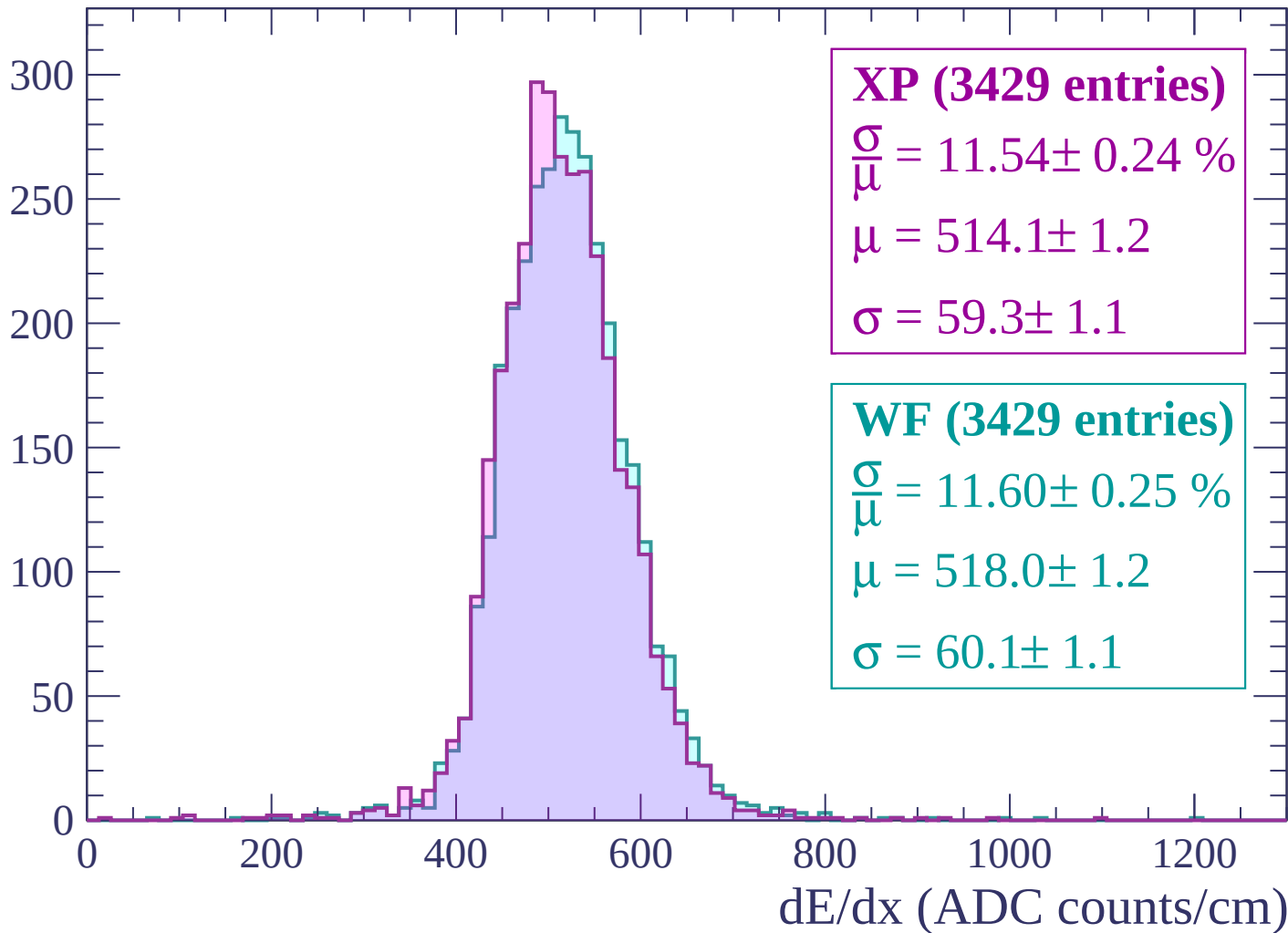
Energy loss | -3000 < p < -2940

Count



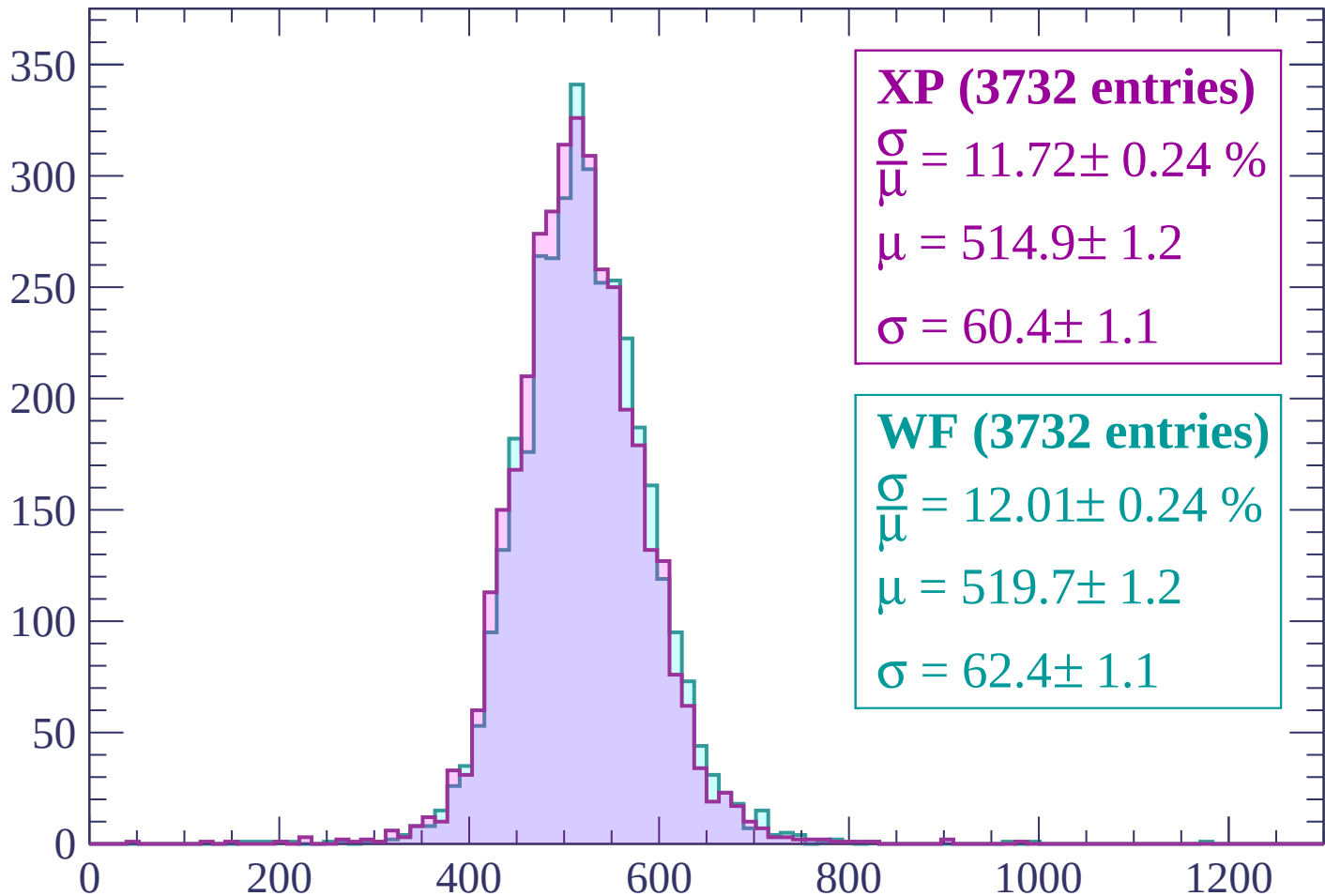
Energy loss | $-2940 < p < -2880$

Count



Energy loss | $-2880 < p < -2820$

Count



XP (3732 entries)

$$\frac{\sigma}{\mu} = 11.72 \pm 0.24 \%$$

$$\mu = 514.9 \pm 1.2$$

$$\sigma = 60.4 \pm 1.1$$

WF (3732 entries)

$$\frac{\sigma}{\mu} = 12.01 \pm 0.24 \%$$

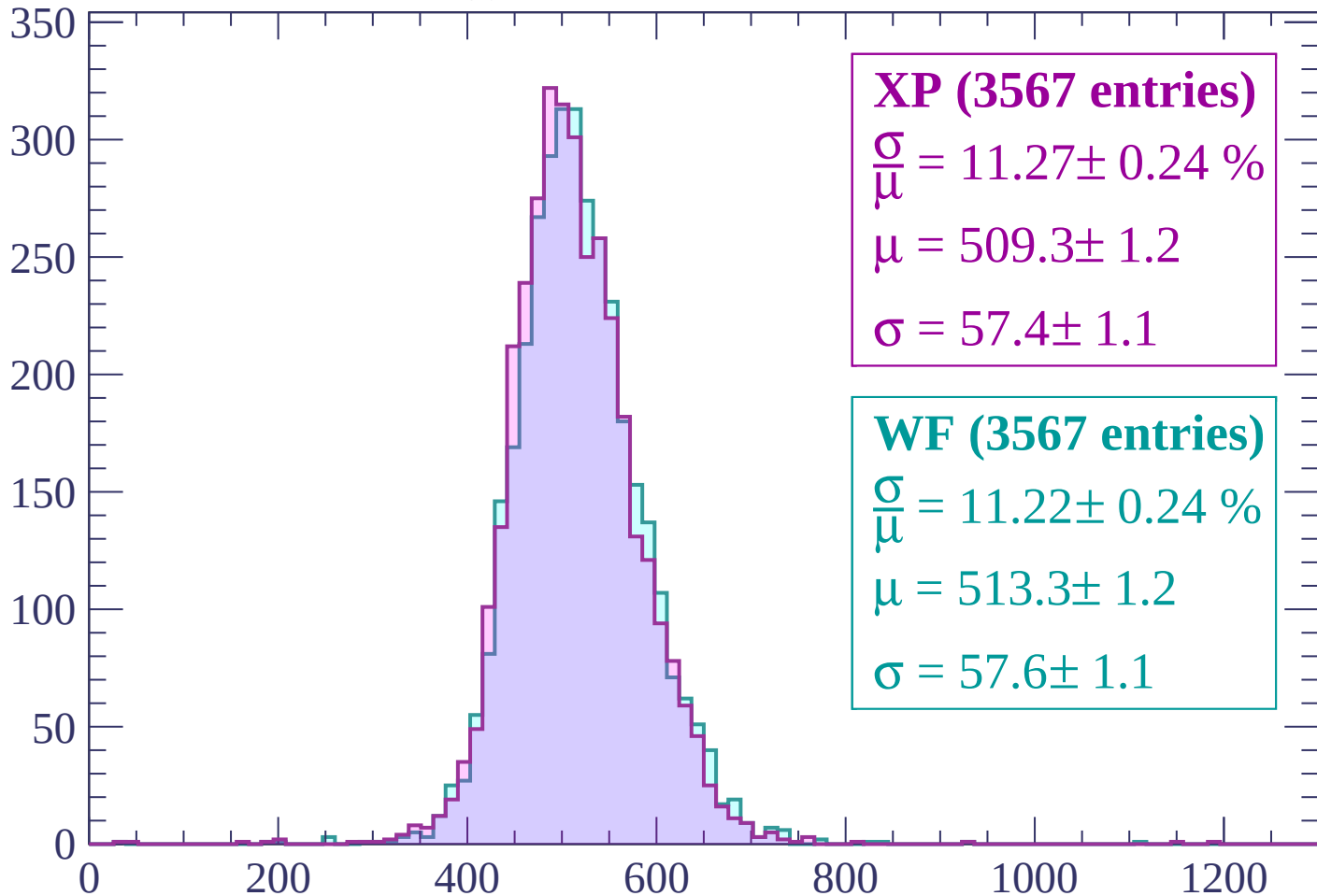
$$\mu = 519.7 \pm 1.2$$

$$\sigma = 62.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-2820 < p < -2760$

Count



XP (3567 entries)

$$\frac{\sigma}{\mu} = 11.27 \pm 0.24 \%$$

$$\mu = 509.3 \pm 1.2$$

$$\sigma = 57.4 \pm 1.1$$

WF (3567 entries)

$$\frac{\sigma}{\mu} = 11.22 \pm 0.24 \%$$

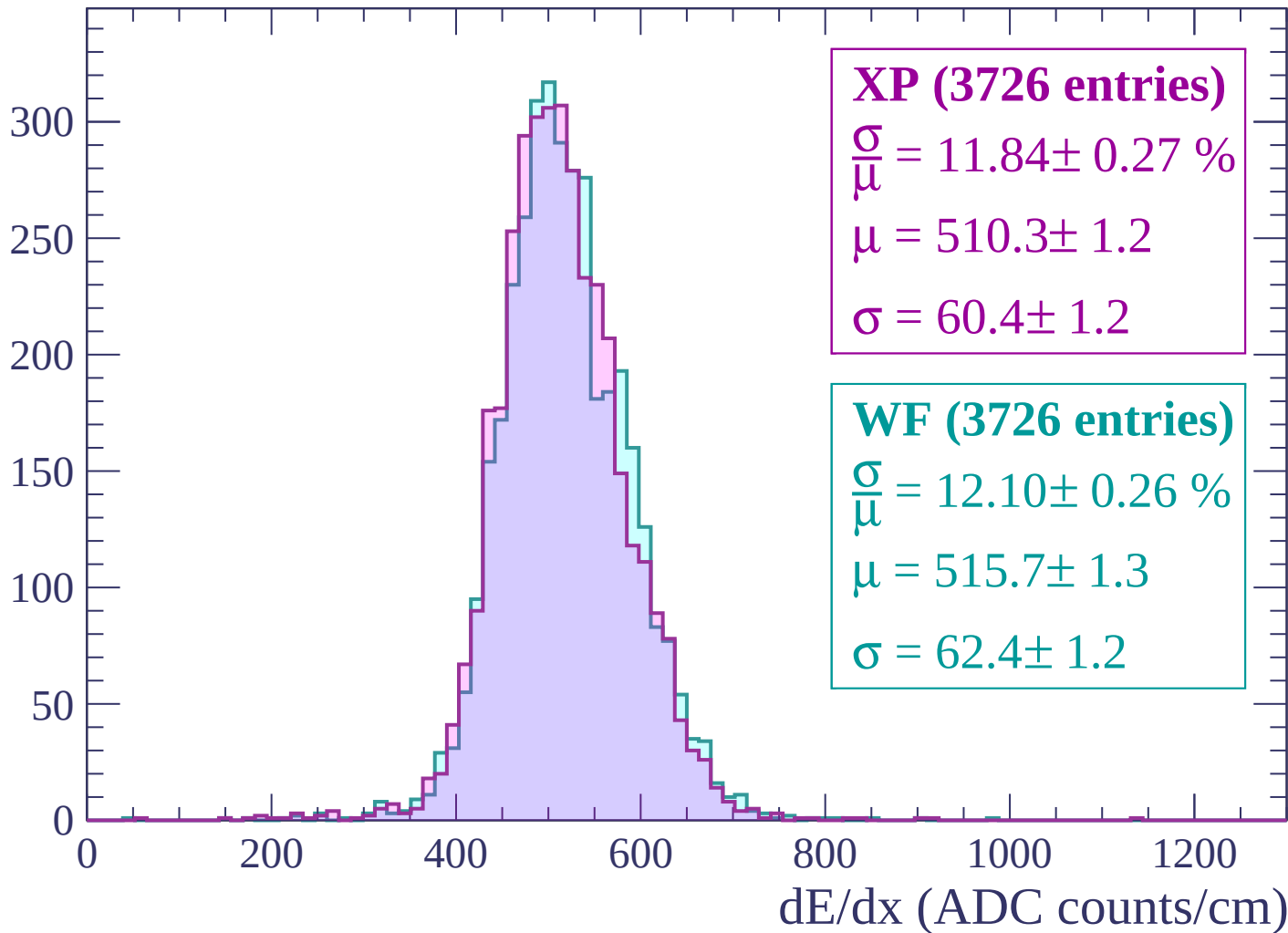
$$\mu = 513.3 \pm 1.2$$

$$\sigma = 57.6 \pm 1.1$$

dE/dx (ADC counts/cm)

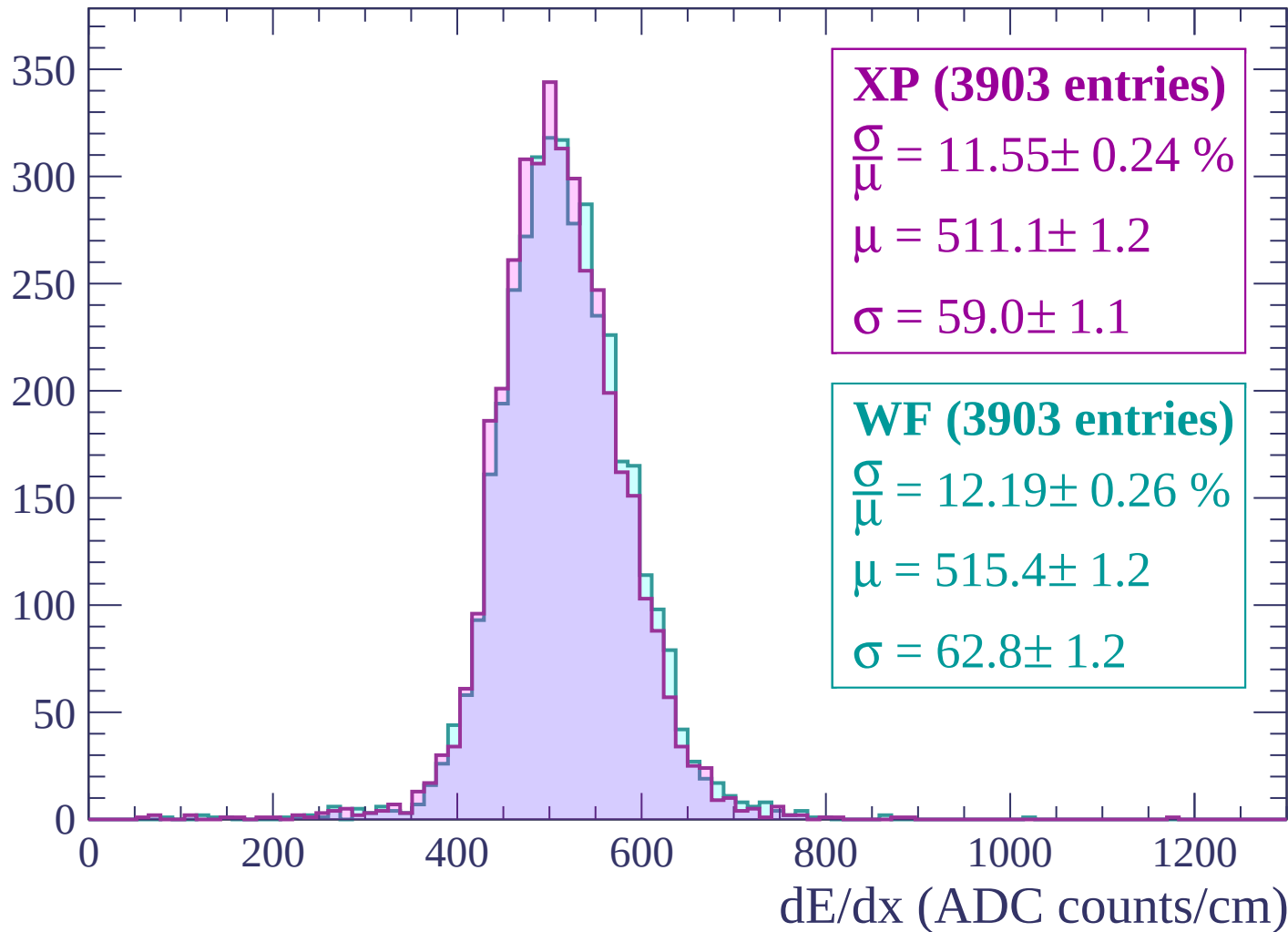
Energy loss | $-2760 < p < -2700$

Count



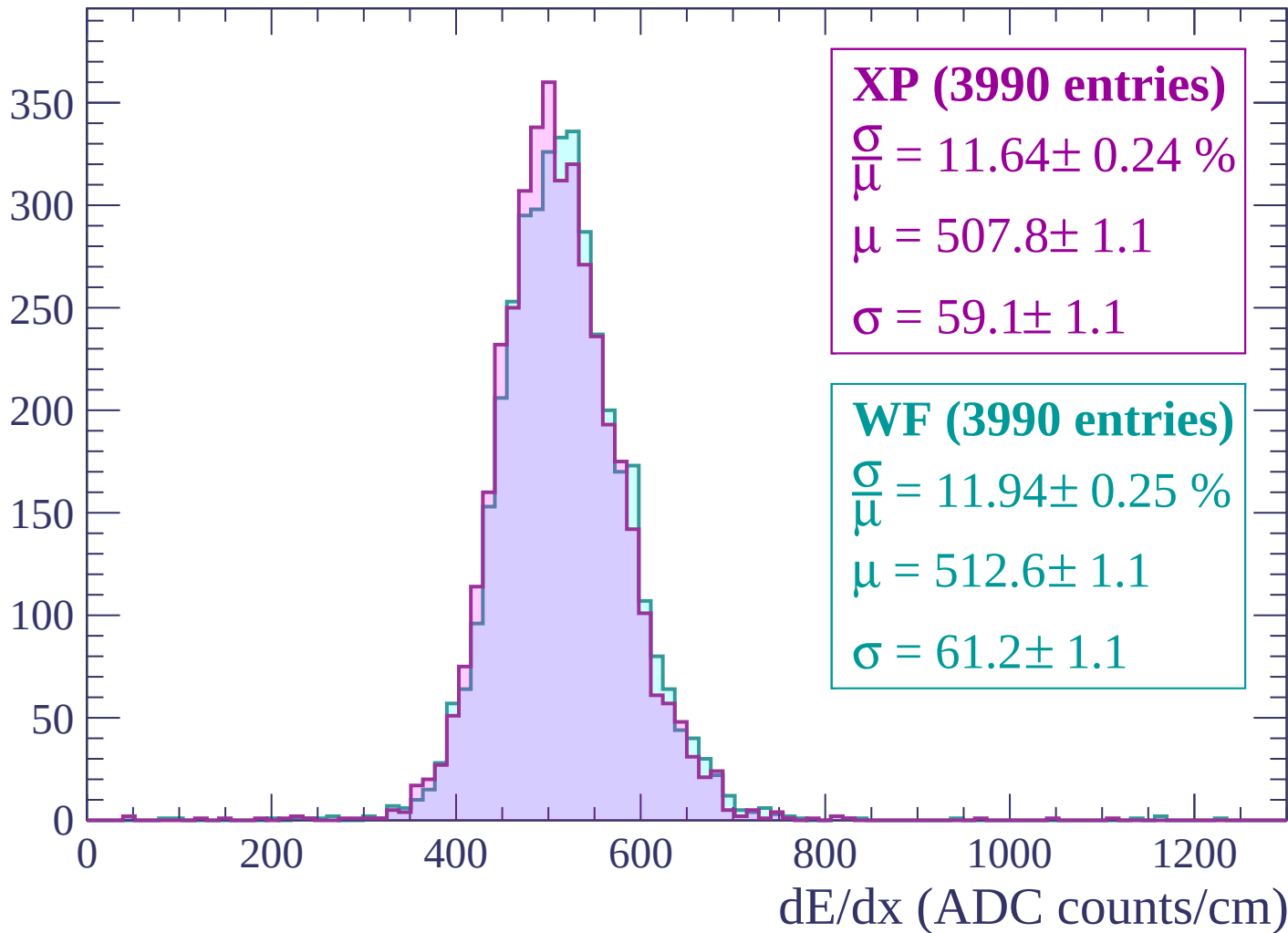
Energy loss | -2700 < p < -2640

Count



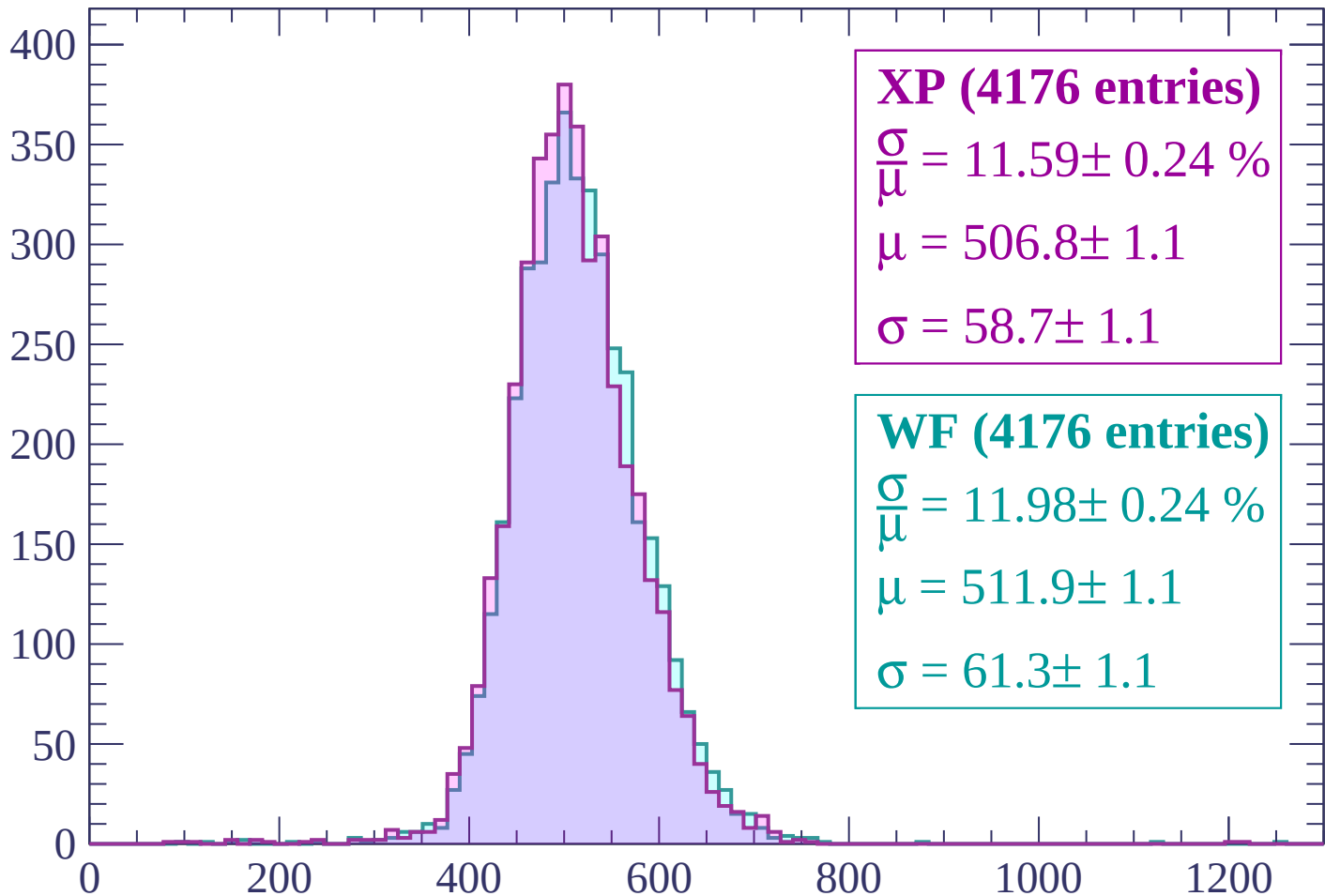
Energy loss | $-2640 < p < -2580$

Count



Energy loss | $-2580 < p < -2520$

Count



XP (4176 entries)

$\frac{\sigma}{\mu} = 11.59 \pm 0.24 \%$

$\mu = 506.8 \pm 1.1$

$\sigma = 58.7 \pm 1.1$

WF (4176 entries)

$\frac{\sigma}{\mu} = 11.98 \pm 0.24 \%$

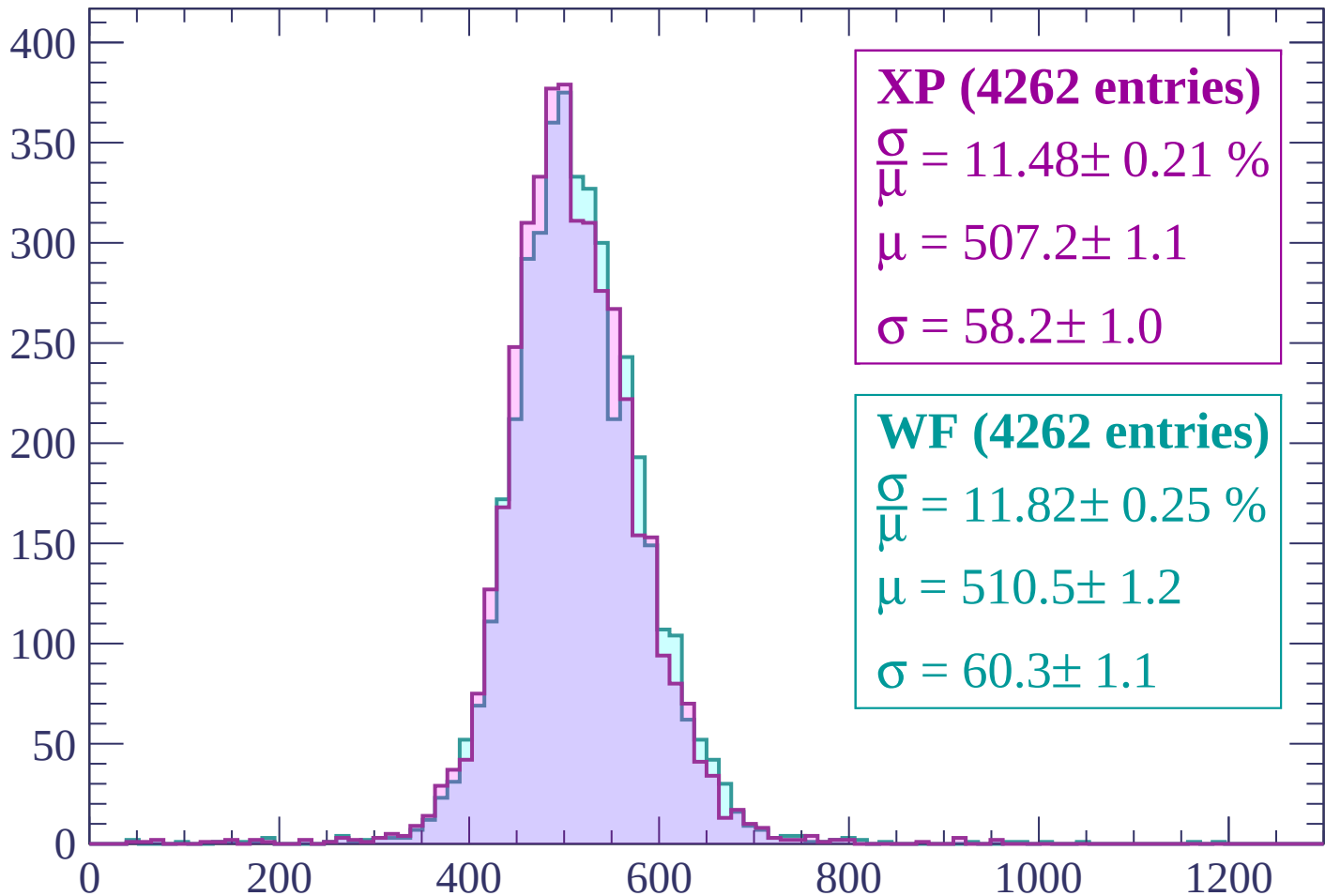
$\mu = 511.9 \pm 1.1$

$\sigma = 61.3 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | -2520 < p < -2460

Count



XP (4262 entries)

$$\frac{\sigma}{\mu} = 11.48 \pm 0.21 \%$$

$$\mu = 507.2 \pm 1.1$$

$$\sigma = 58.2 \pm 1.0$$

WF (4262 entries)

$$\frac{\sigma}{\mu} = 11.82 \pm 0.25 \%$$

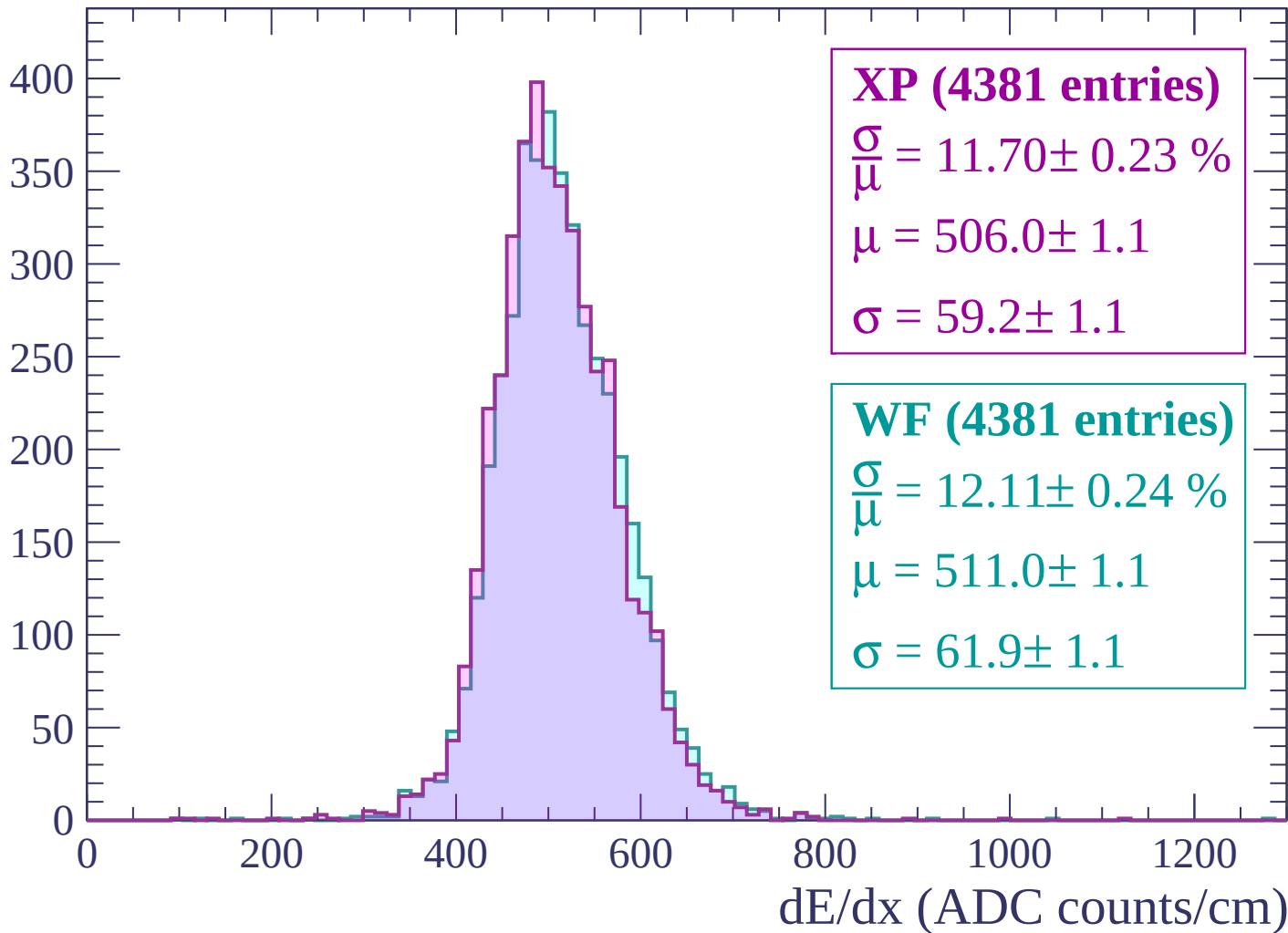
$$\mu = 510.5 \pm 1.2$$

$$\sigma = 60.3 \pm 1.1$$

dE/dx (ADC counts/cm)

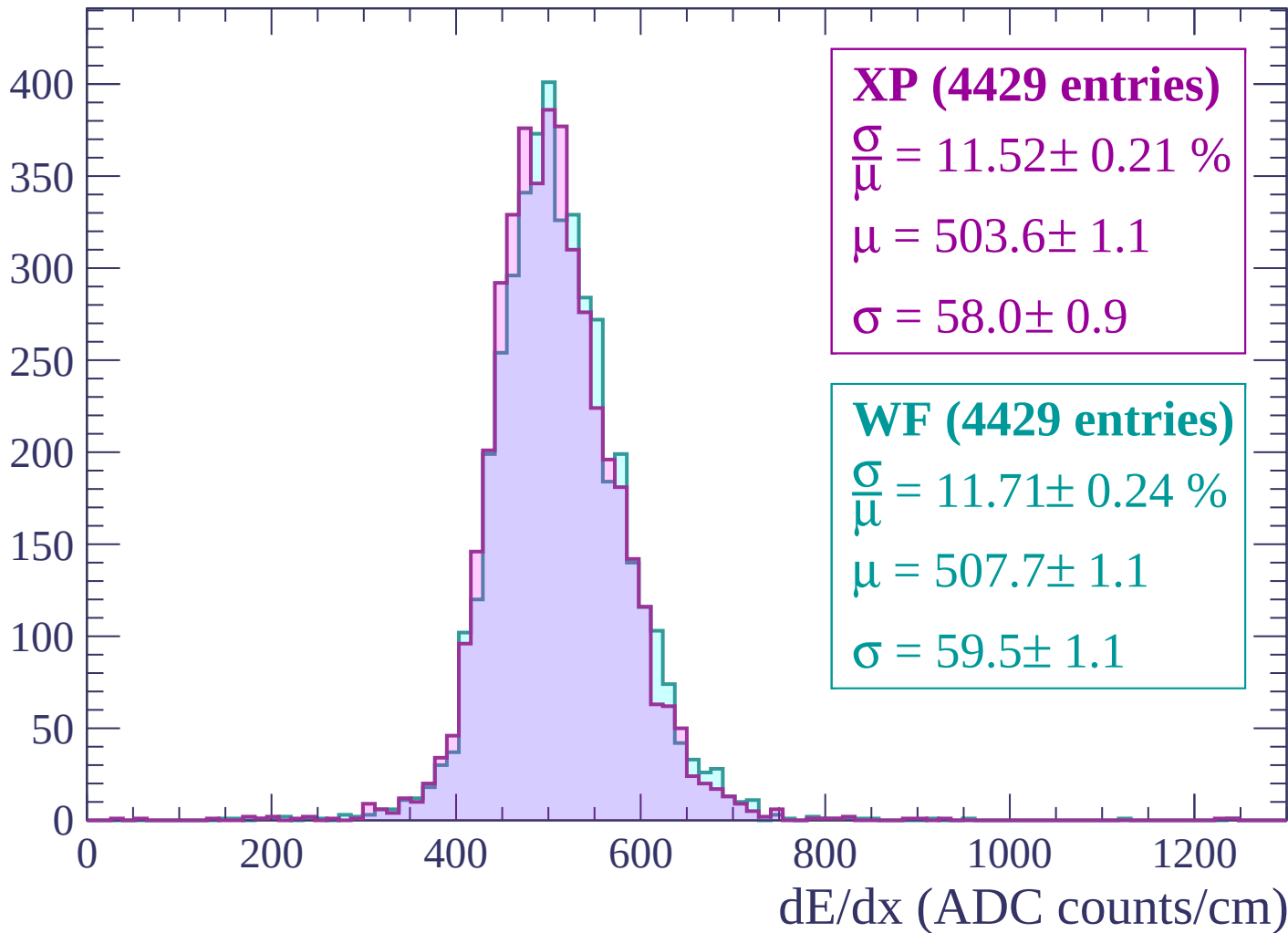
Energy loss | -2460 < p < -2400

Count



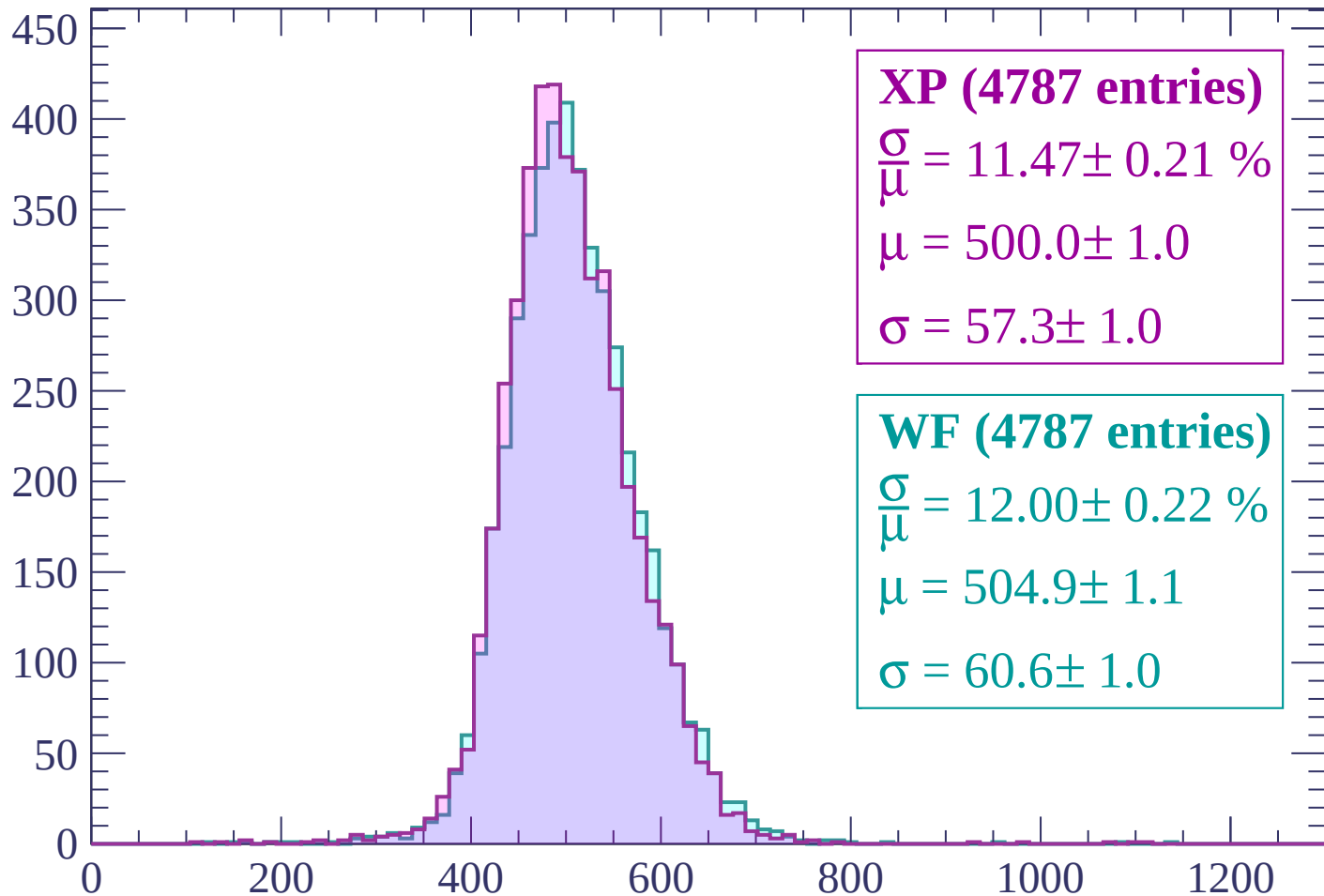
Energy loss | -2400 < p < -2340

Count



Energy loss | $-2340 < p < -2280$

Count



XP (4787 entries)

$\frac{\sigma}{\mu} = 11.47 \pm 0.21 \%$

$\mu = 500.0 \pm 1.0$

$\sigma = 57.3 \pm 1.0$

WF (4787 entries)

$\frac{\sigma}{\mu} = 12.00 \pm 0.22 \%$

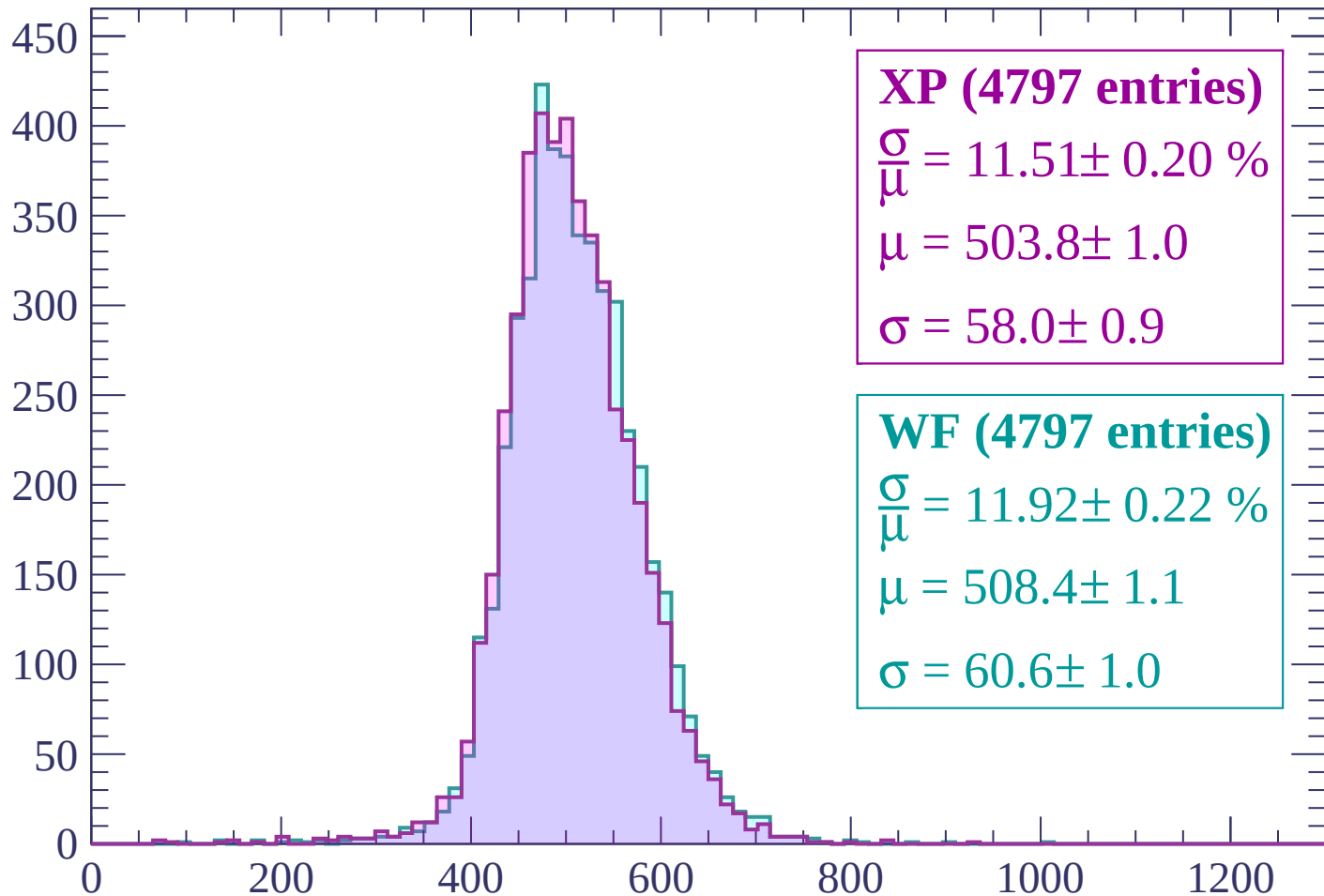
$\mu = 504.9 \pm 1.1$

$\sigma = 60.6 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -2280 < p < -2220

Count



XP (4797 entries)

$\frac{\sigma}{\mu} = 11.51 \pm 0.20 \%$

$\mu = 503.8 \pm 1.0$

$\sigma = 58.0 \pm 0.9$

WF (4797 entries)

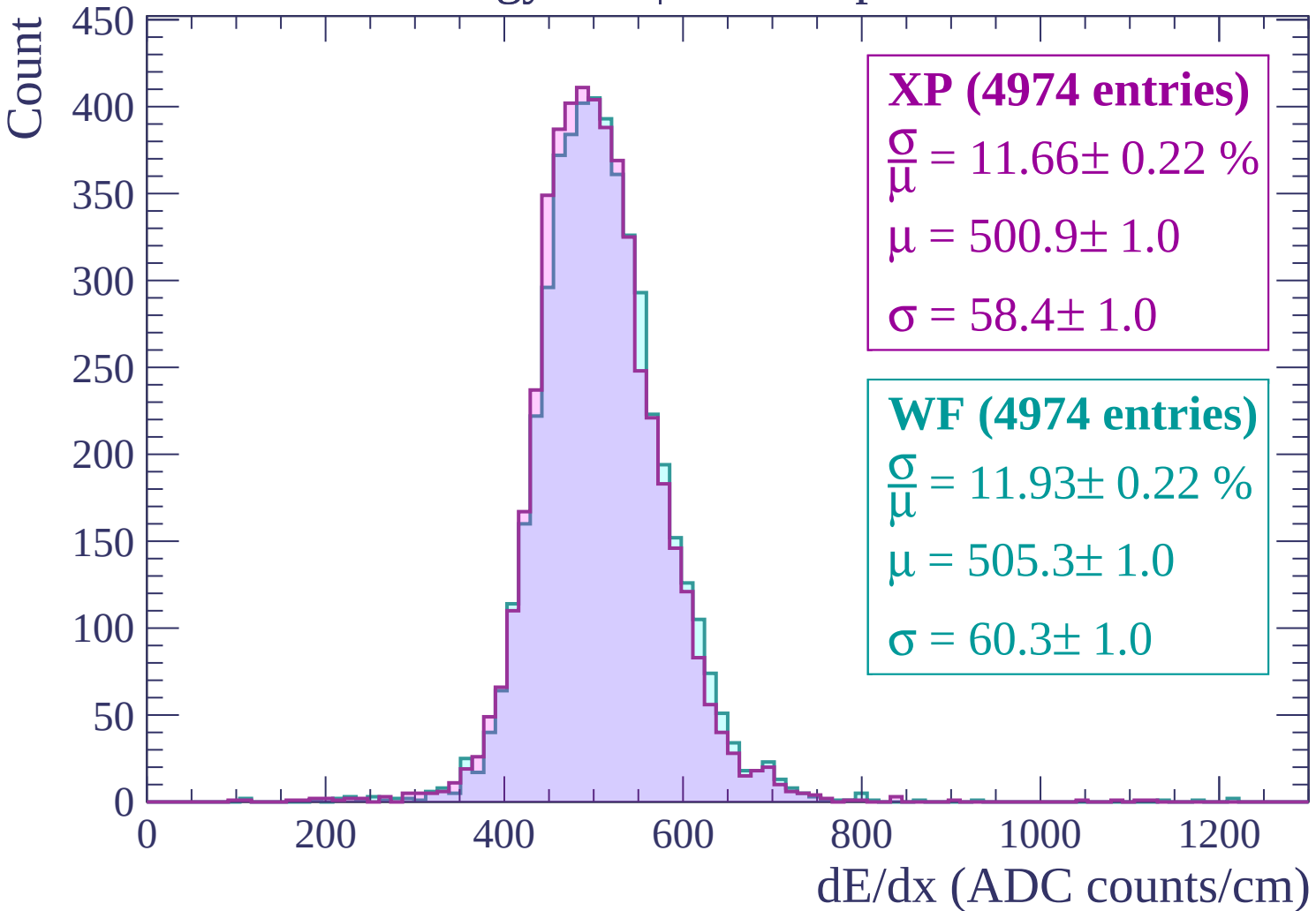
$\frac{\sigma}{\mu} = 11.92 \pm 0.22 \%$

$\mu = 508.4 \pm 1.1$

$\sigma = 60.6 \pm 1.0$

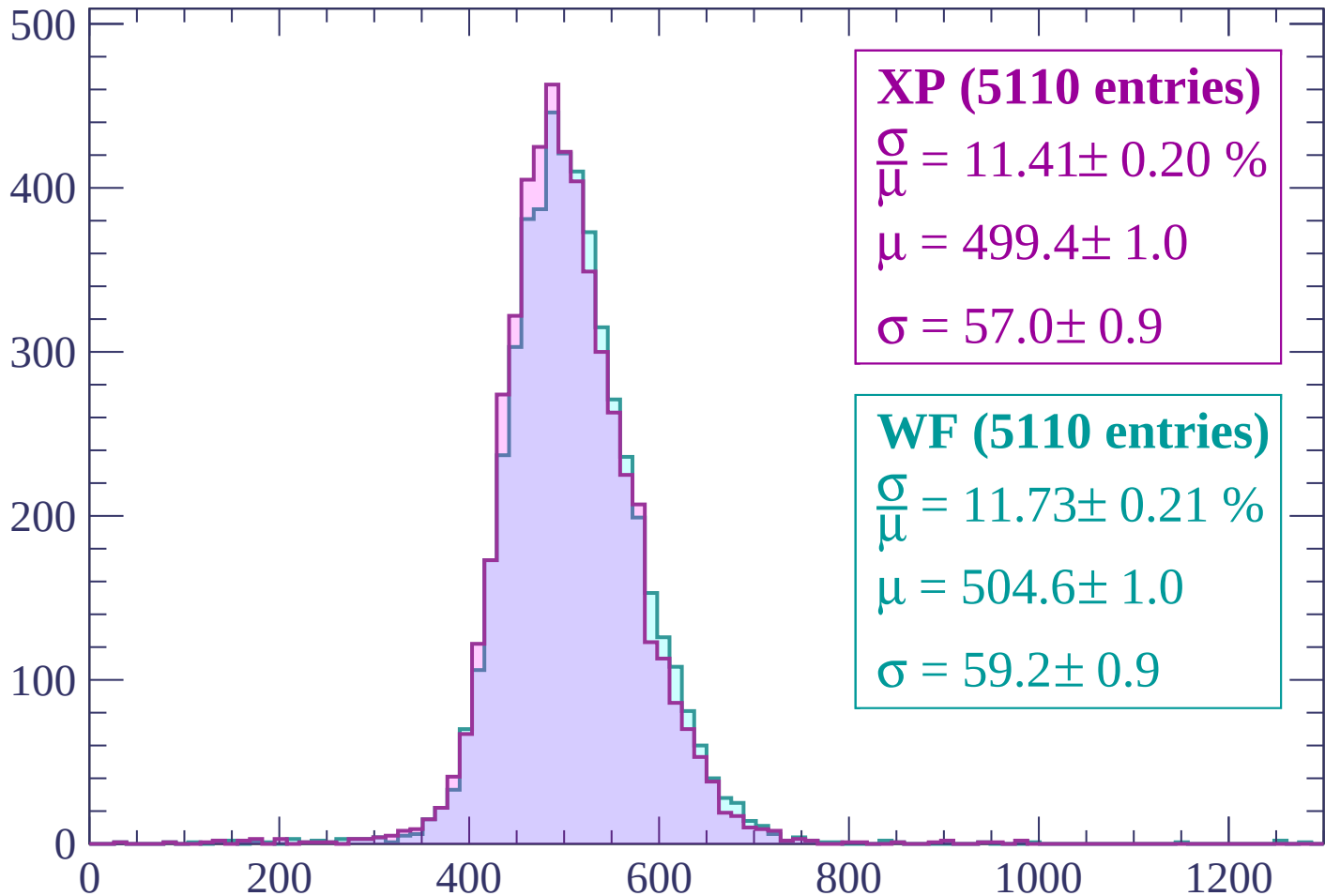
dE/dx (ADC counts/cm)

Energy loss | -2220 < p < -2160



Energy loss | -2160 < p < -2100

Count



XP (5110 entries)

$\frac{\sigma}{\mu} = 11.41 \pm 0.20 \%$

$\mu = 499.4 \pm 1.0$

$\sigma = 57.0 \pm 0.9$

WF (5110 entries)

$\frac{\sigma}{\mu} = 11.73 \pm 0.21 \%$

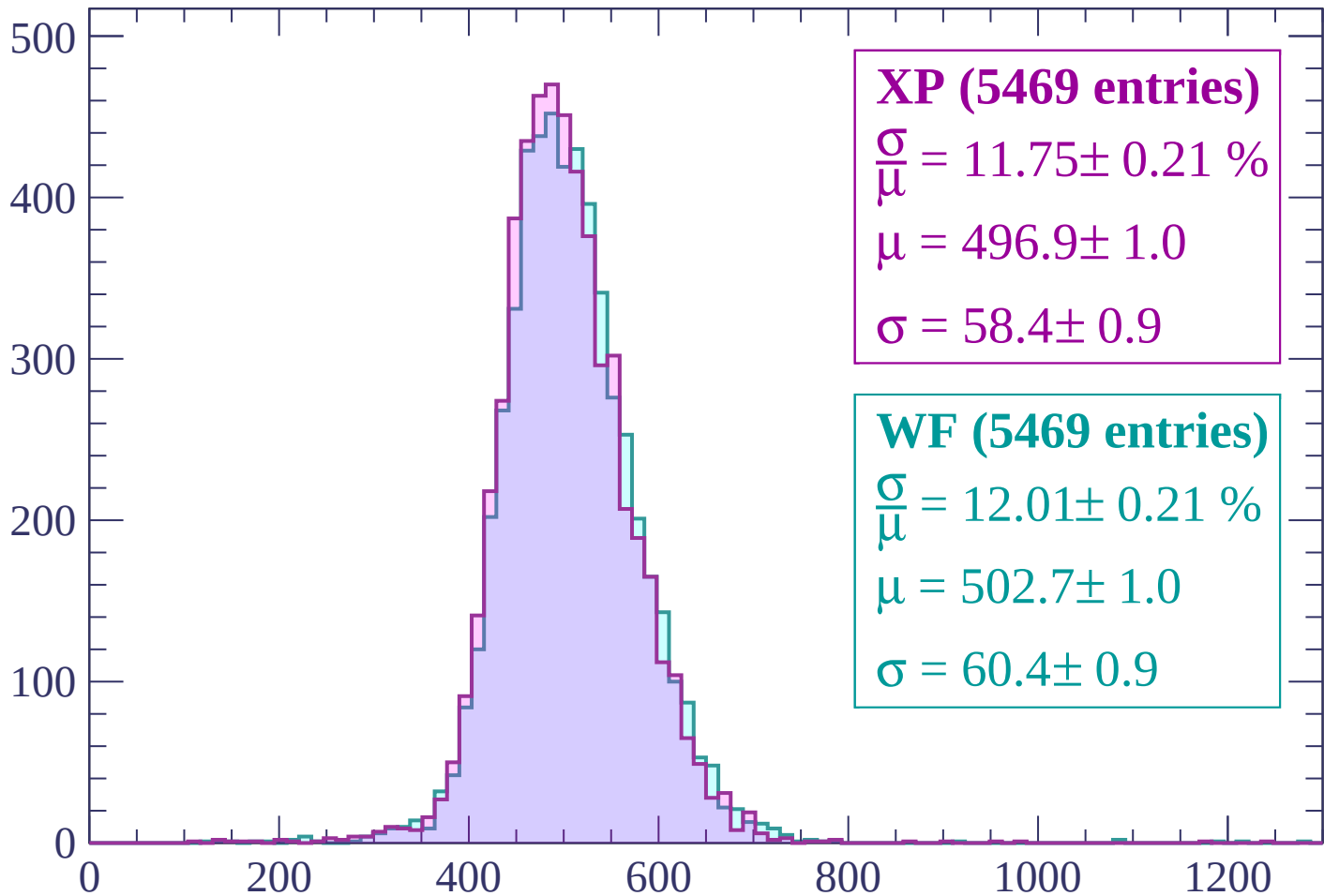
$\mu = 504.6 \pm 1.0$

$\sigma = 59.2 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | -2100 < p < -2040

Count



XP (5469 entries)

$$\frac{\sigma}{\mu} = 11.75 \pm 0.21 \%$$

$$\mu = 496.9 \pm 1.0$$

$$\sigma = 58.4 \pm 0.9$$

WF (5469 entries)

$$\frac{\sigma}{\mu} = 12.01 \pm 0.21 \%$$

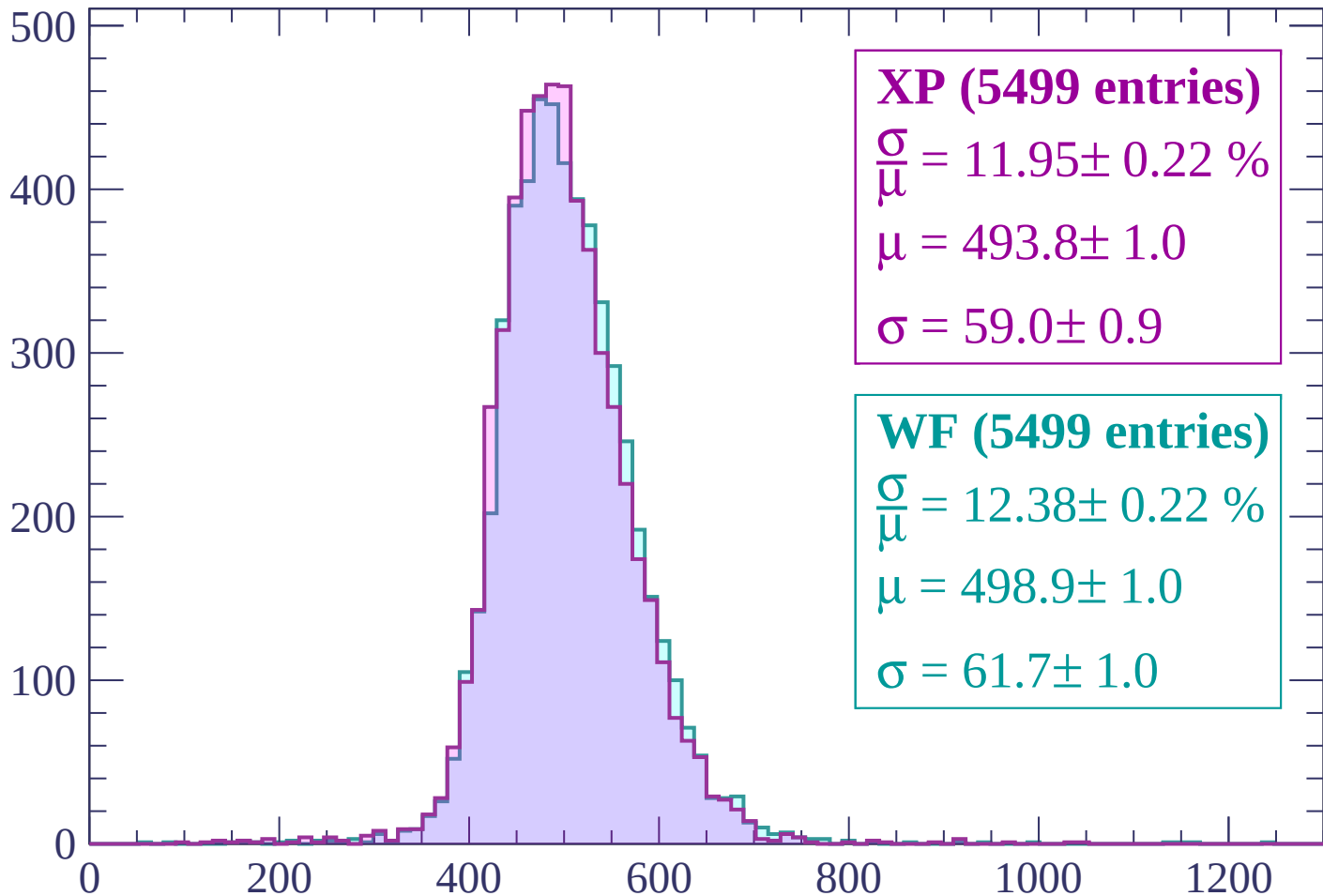
$$\mu = 502.7 \pm 1.0$$

$$\sigma = 60.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -2040 < p < -1980

Count



XP (5499 entries)

$\frac{\sigma}{\mu} = 11.95 \pm 0.22 \%$

$\mu = 493.8 \pm 1.0$

$\sigma = 59.0 \pm 0.9$

WF (5499 entries)

$\frac{\sigma}{\mu} = 12.38 \pm 0.22 \%$

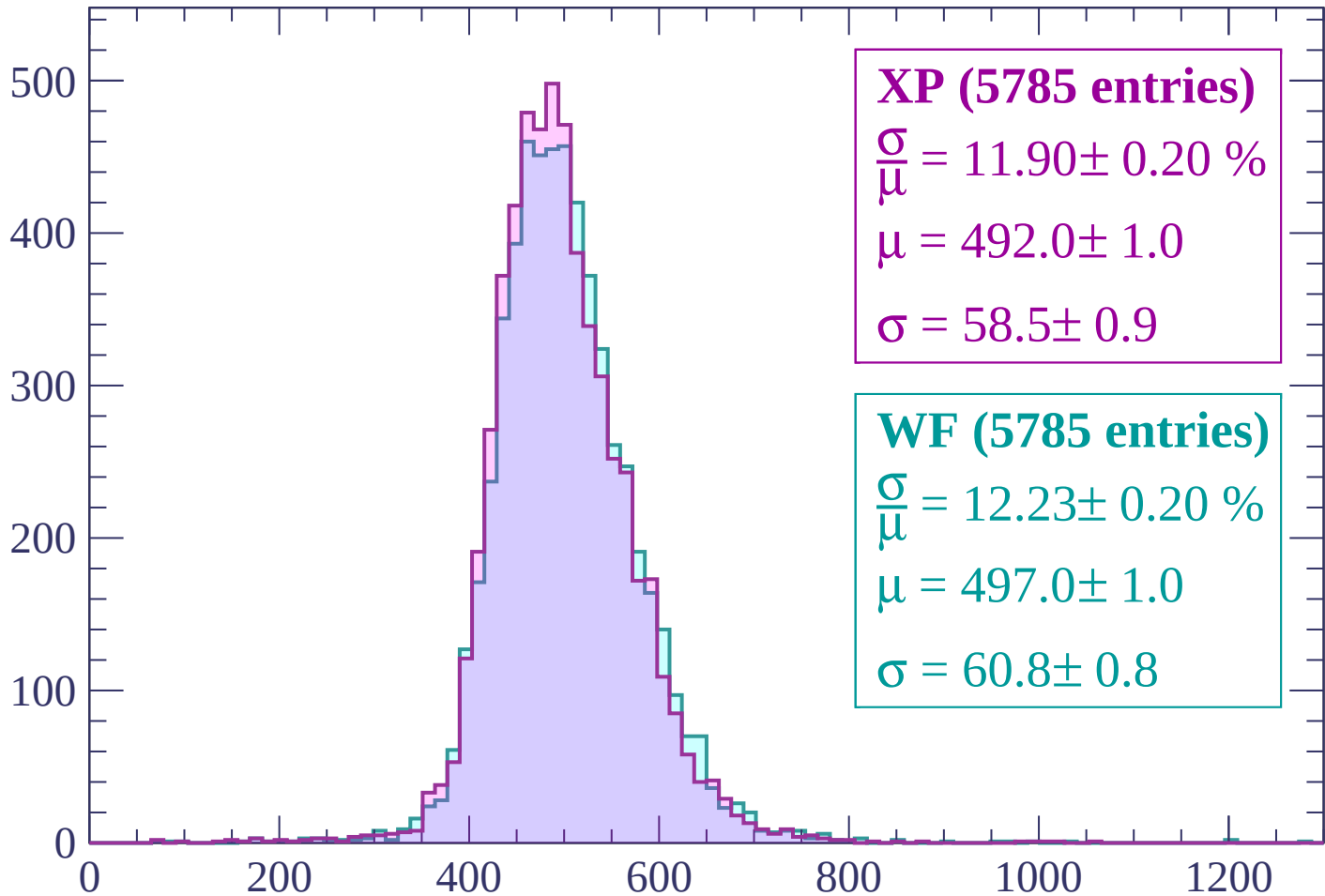
$\mu = 498.9 \pm 1.0$

$\sigma = 61.7 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $-1980 < p < -1920$

Count



XP (5785 entries)

$$\frac{\sigma}{\mu} = 11.90 \pm 0.20 \%$$

$$\mu = 492.0 \pm 1.0$$

$$\sigma = 58.5 \pm 0.9$$

WF (5785 entries)

$$\frac{\sigma}{\mu} = 12.23 \pm 0.20 \%$$

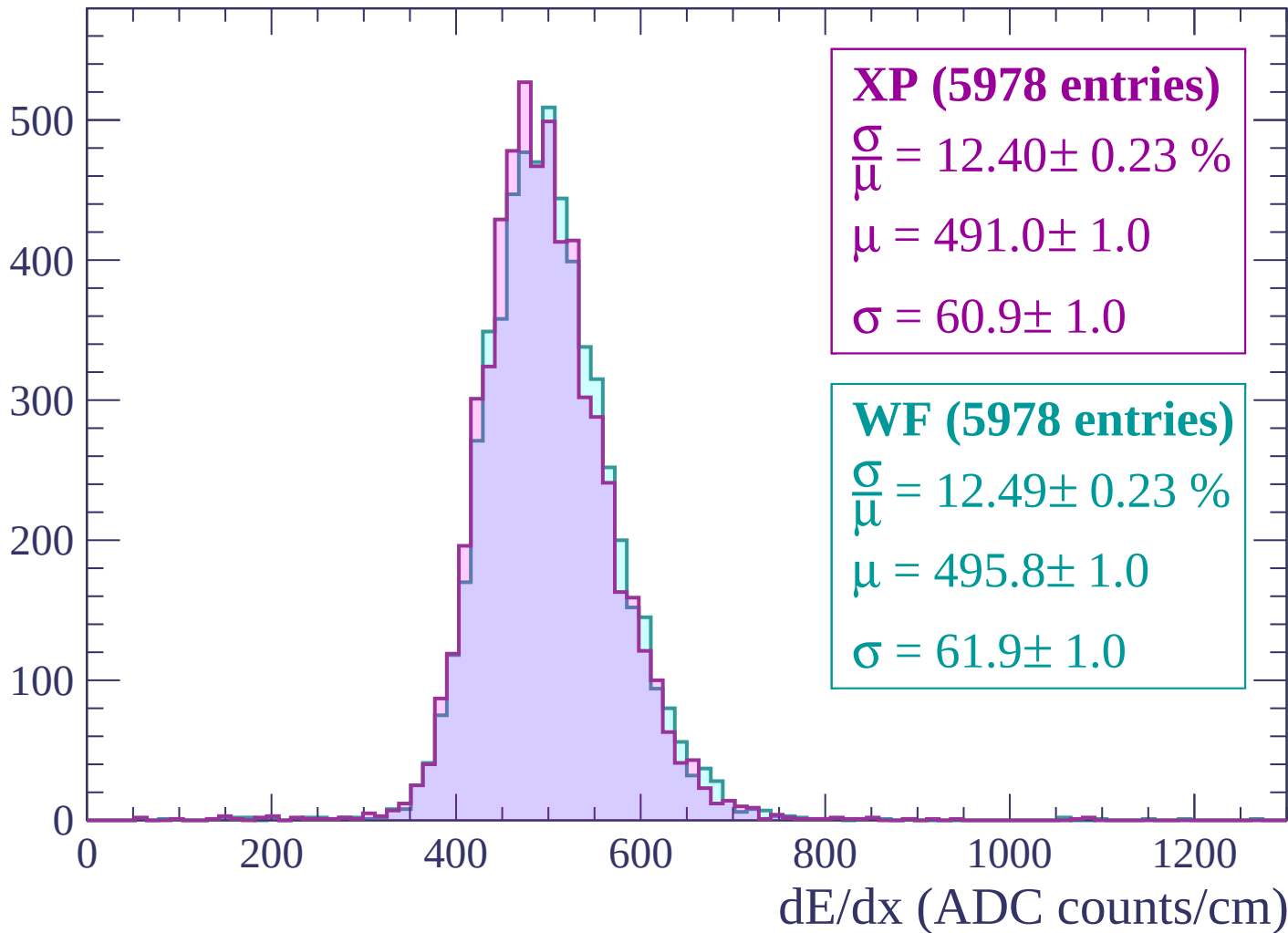
$$\mu = 497.0 \pm 1.0$$

$$\sigma = 60.8 \pm 0.8$$

dE/dx (ADC counts/cm)

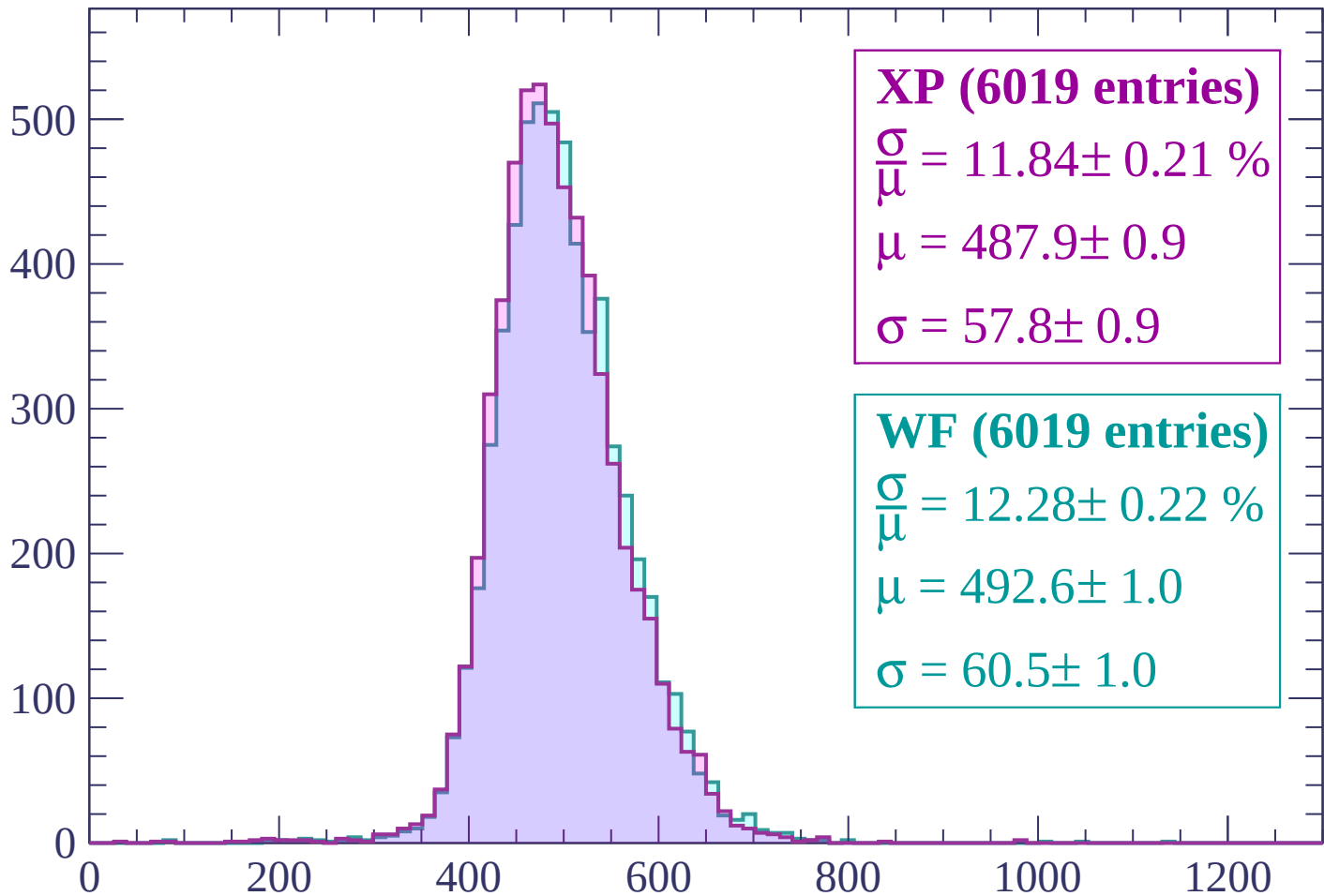
Energy loss | -1920 < p < -1860

Count



Energy loss | -1860 < p < -1800

Count



XP (6019 entries)

$$\frac{\sigma}{\mu} = 11.84 \pm 0.21 \%$$

$$\mu = 487.9 \pm 0.9$$

$$\sigma = 57.8 \pm 0.9$$

WF (6019 entries)

$$\frac{\sigma}{\mu} = 12.28 \pm 0.22 \%$$

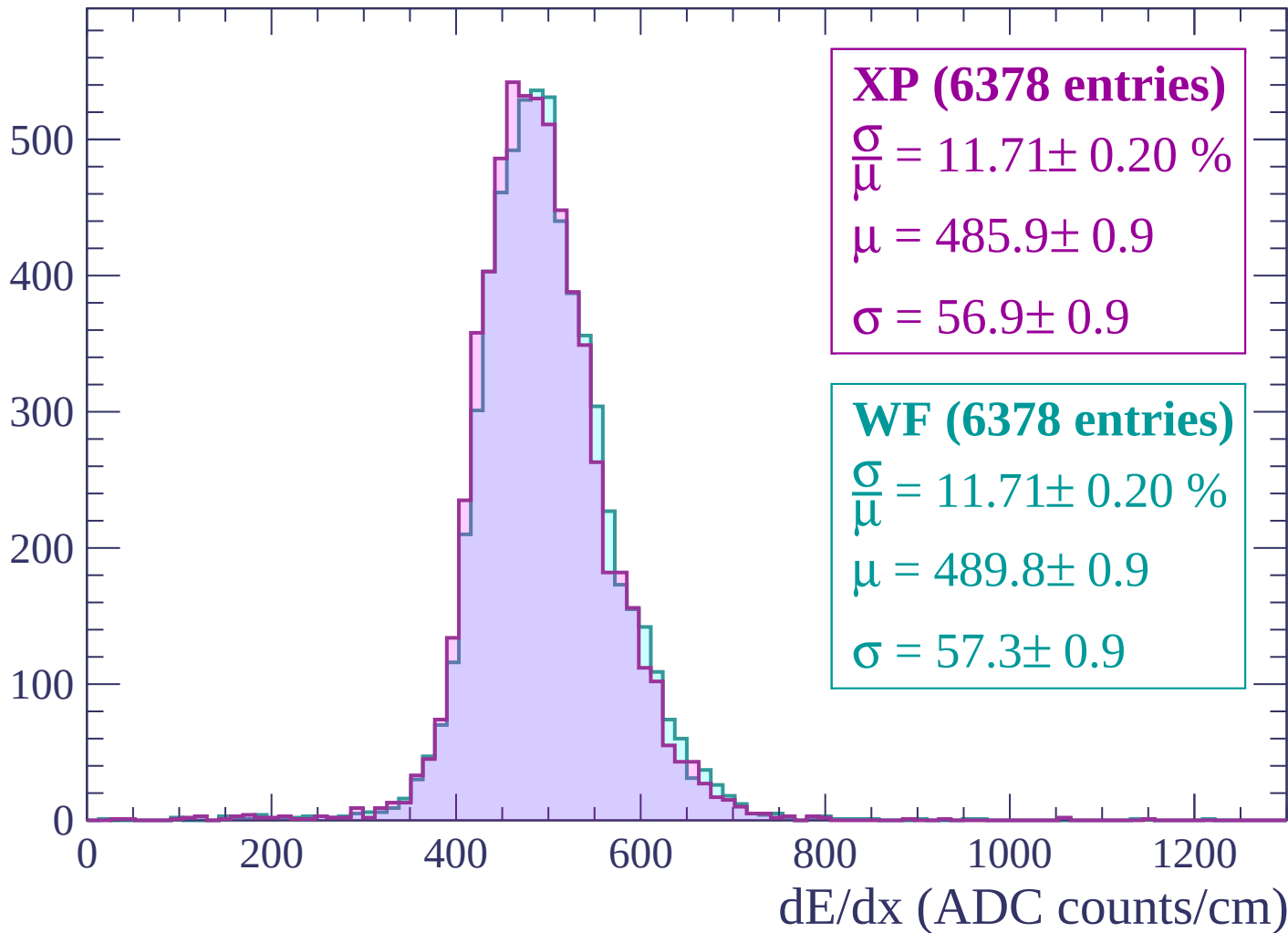
$$\mu = 492.6 \pm 1.0$$

$$\sigma = 60.5 \pm 1.0$$

dE/dx (ADC counts/cm)

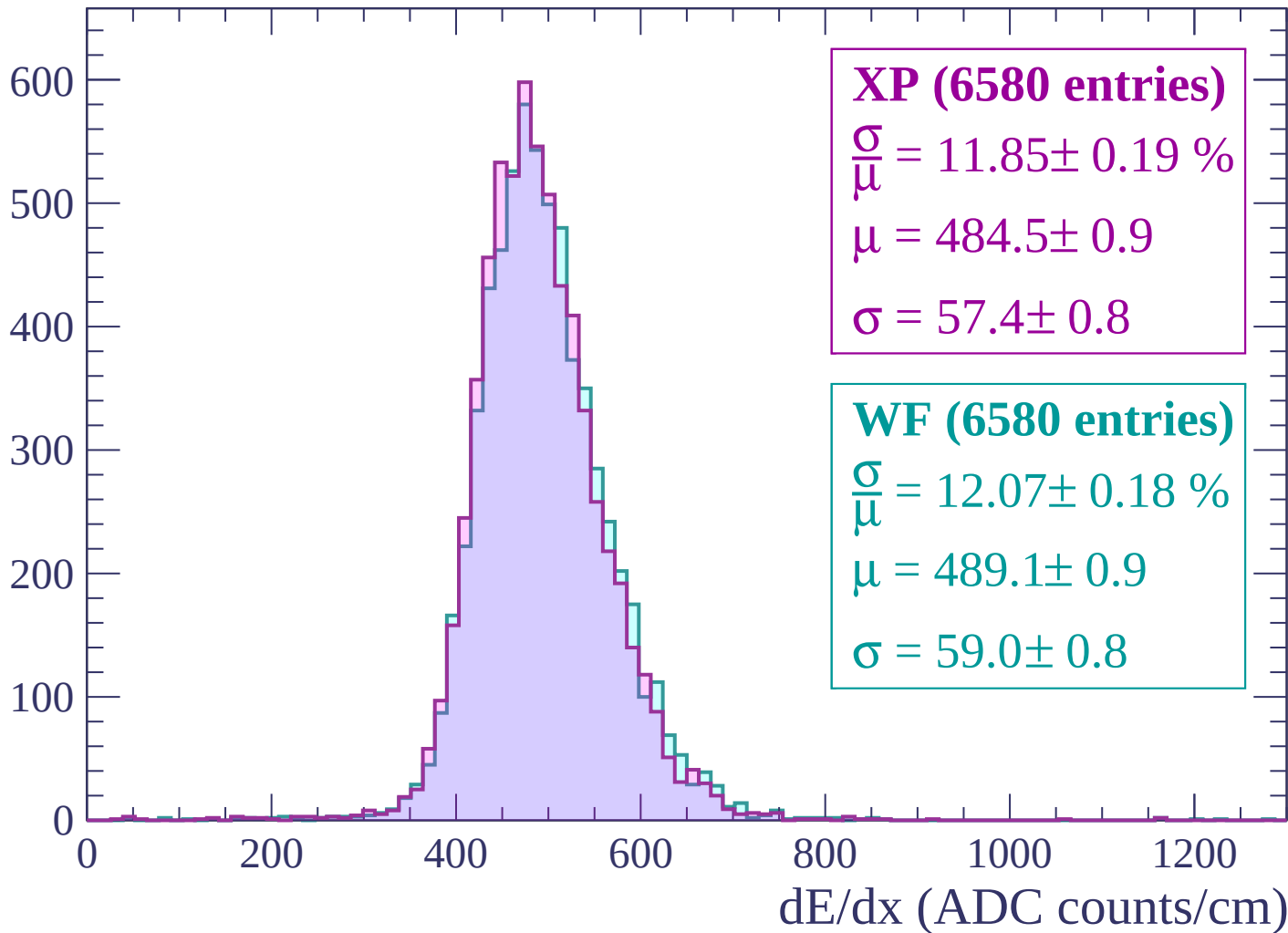
Energy loss | -1800 < p < -1740

Count



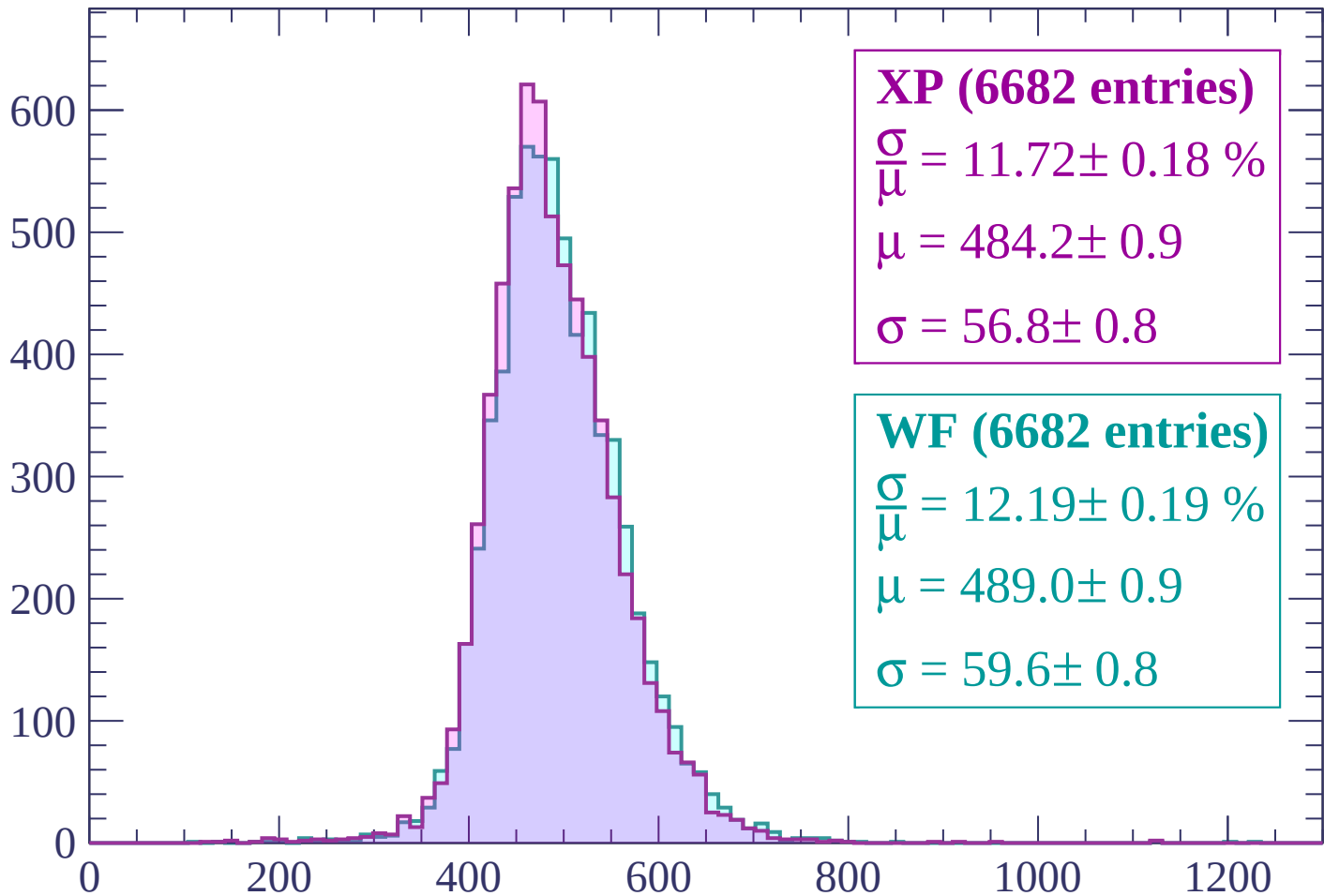
Energy loss | -1740 < p < -1680

Count



Energy loss | $-1680 < p < -1620$

Count



XP (6682 entries)

$$\frac{\sigma}{\mu} = 11.72 \pm 0.18 \%$$

$$\mu = 484.2 \pm 0.9$$

$$\sigma = 56.8 \pm 0.8$$

WF (6682 entries)

$$\frac{\sigma}{\mu} = 12.19 \pm 0.19 \%$$

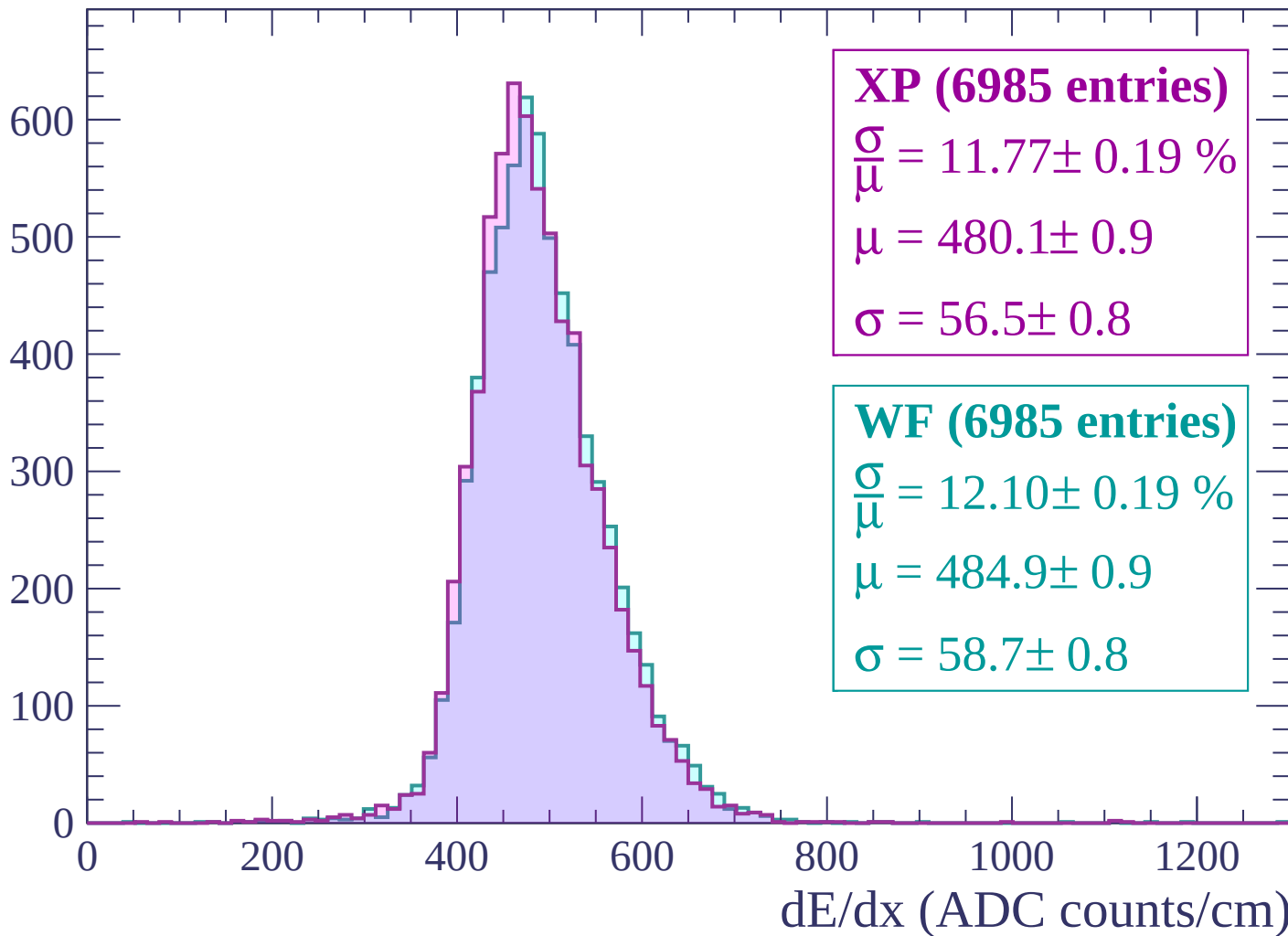
$$\mu = 489.0 \pm 0.9$$

$$\sigma = 59.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1620 < p < -1560$

Count



Energy loss | -1560 < p < -1500

Count

700
600
500
400
300
200
100
0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (7235 entries)

$$\frac{\sigma}{\mu} = 11.86 \pm 0.19 \%$$

$$\mu = 477.3 \pm 0.9$$

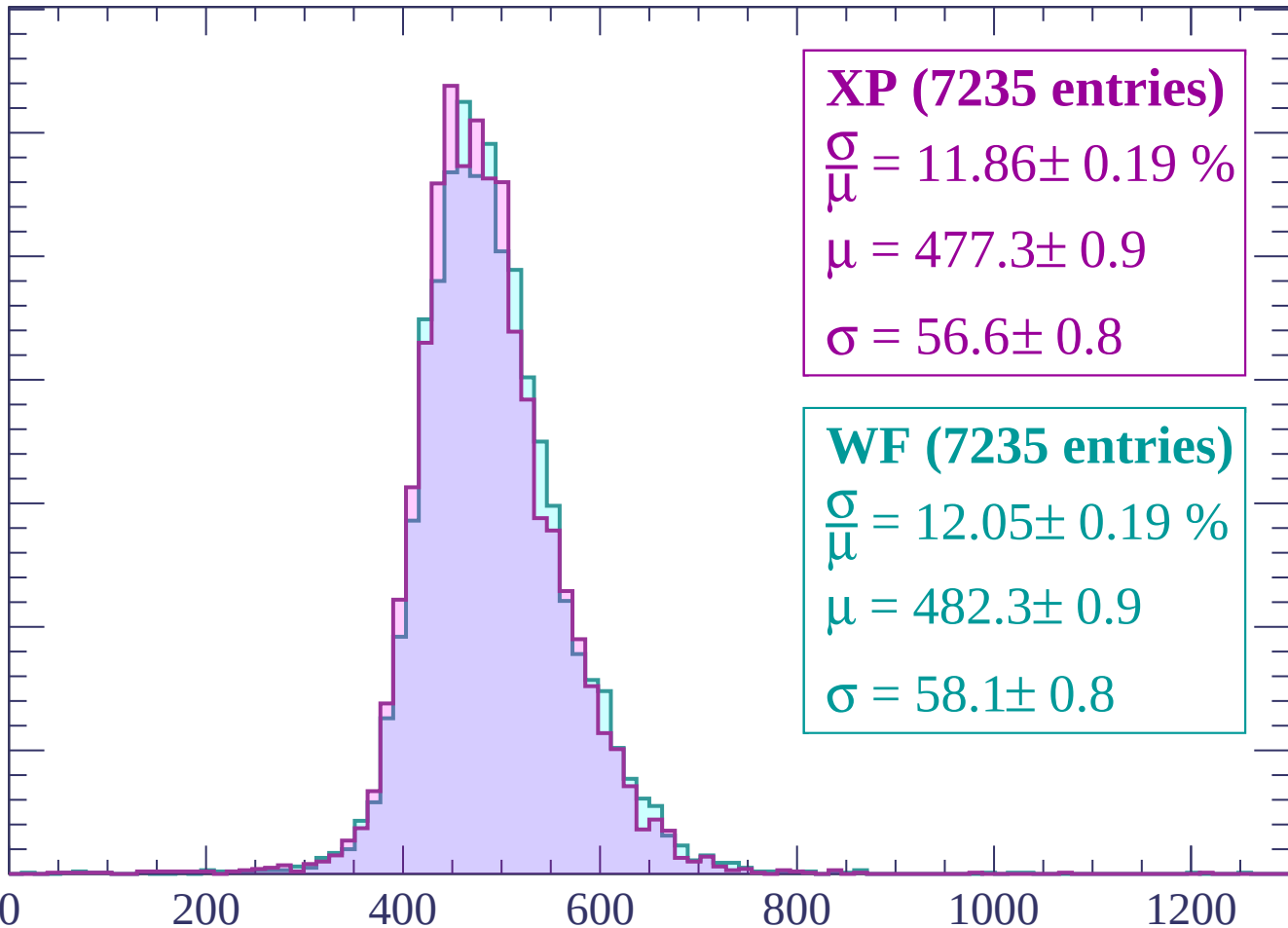
$$\sigma = 56.6 \pm 0.8$$

WF (7235 entries)

$$\frac{\sigma}{\mu} = 12.05 \pm 0.19 \%$$

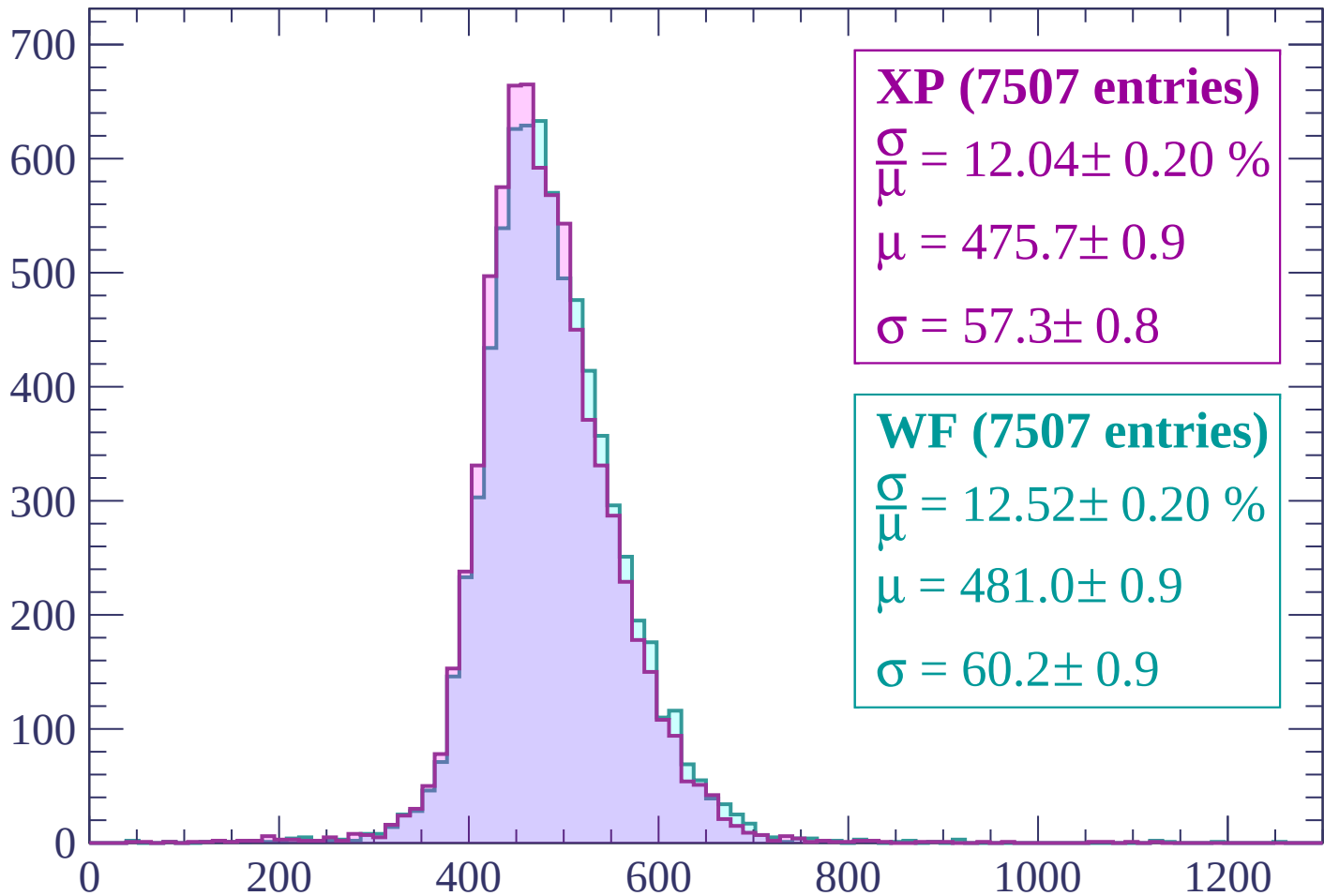
$$\mu = 482.3 \pm 0.9$$

$$\sigma = 58.1 \pm 0.8$$



Energy loss | $-1500 < p < -1440$

Count



XP (7507 entries)

$$\frac{\sigma}{\mu} = 12.04 \pm 0.20 \%$$

$$\mu = 475.7 \pm 0.9$$

$$\sigma = 57.3 \pm 0.8$$

WF (7507 entries)

$$\frac{\sigma}{\mu} = 12.52 \pm 0.20 \%$$

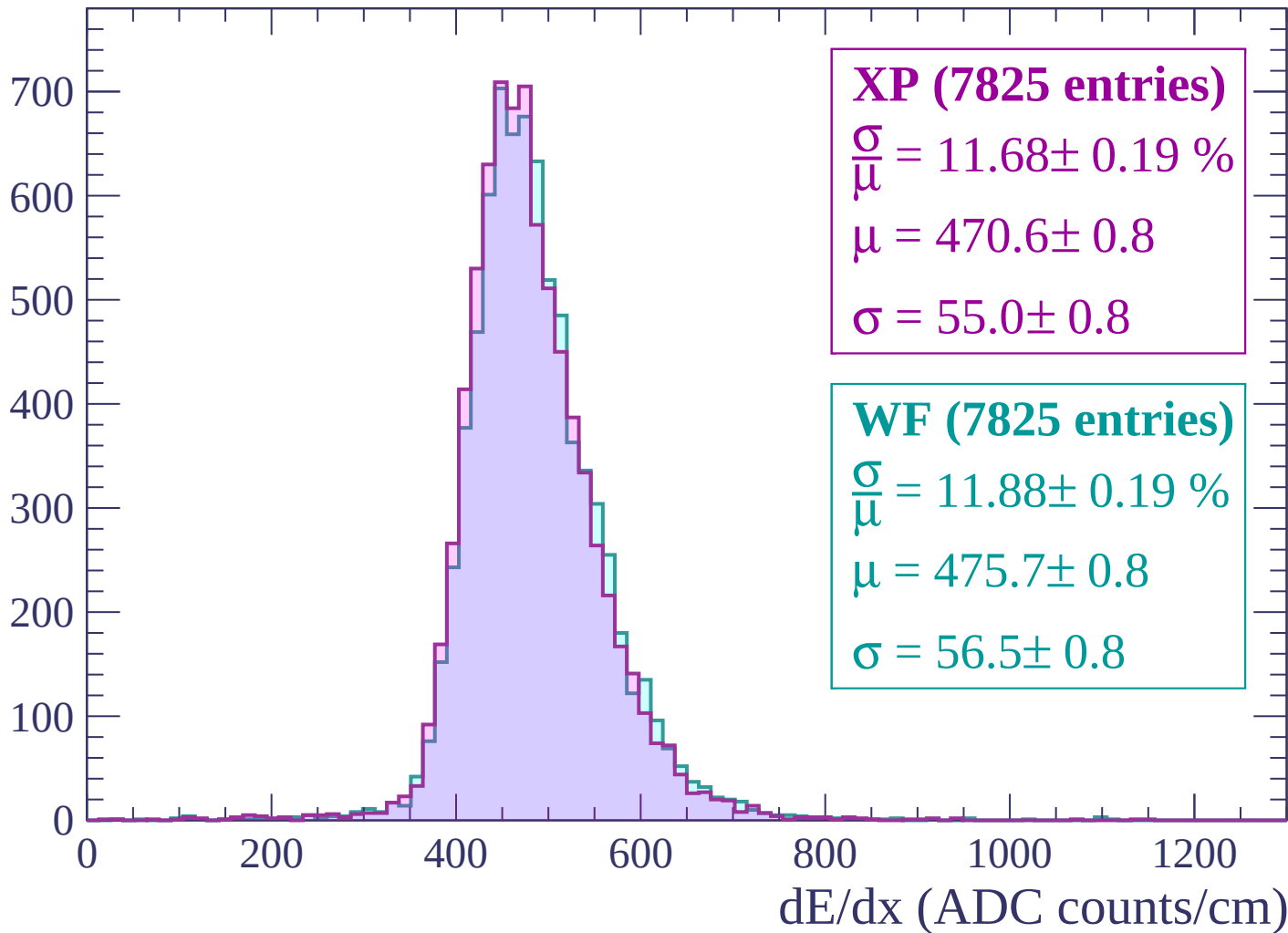
$$\mu = 481.0 \pm 0.9$$

$$\sigma = 60.2 \pm 0.9$$

dE/dx (ADC counts/cm)

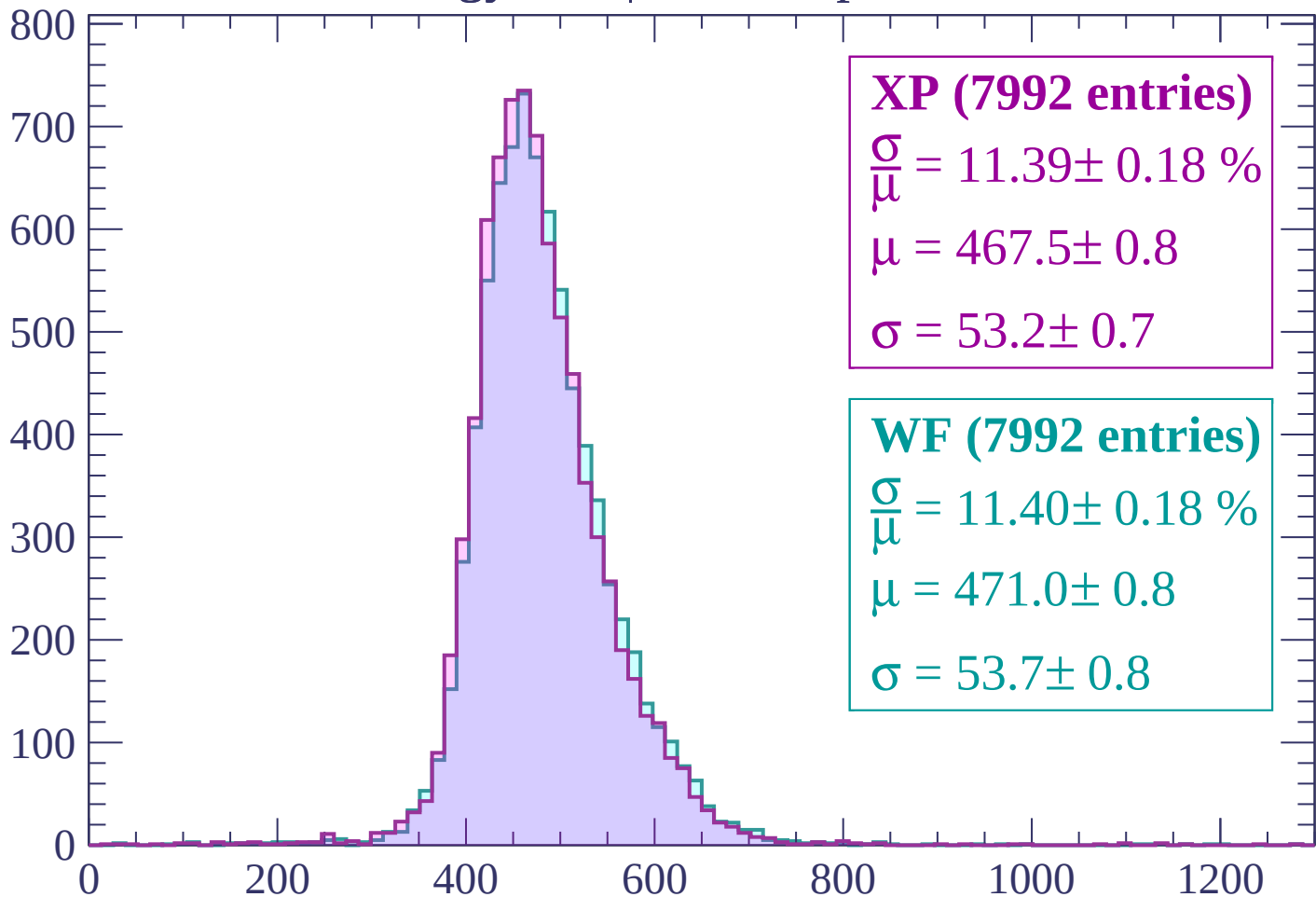
Energy loss | $-1440 < p < -1380$

Count



Energy loss | -1380 < p < -1320

Count



XP (7992 entries)

$$\frac{\sigma}{\mu} = 11.39 \pm 0.18 \%$$

$$\mu = 467.5 \pm 0.8$$

$$\sigma = 53.2 \pm 0.7$$

WF (7992 entries)

$$\frac{\sigma}{\mu} = 11.40 \pm 0.18 \%$$

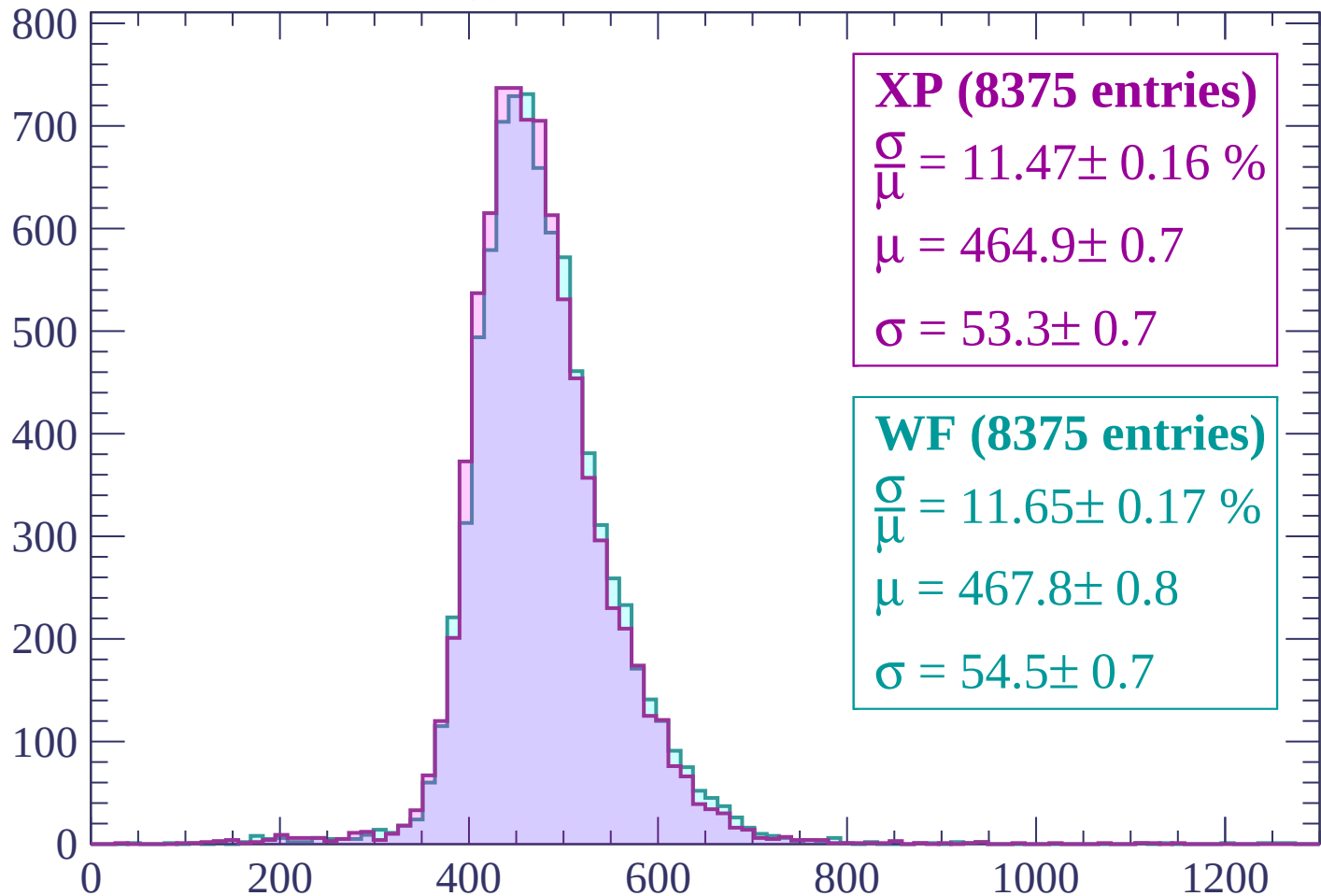
$$\mu = 471.0 \pm 0.8$$

$$\sigma = 53.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1260

Count



XP (8375 entries)

$$\frac{\sigma}{\mu} = 11.47 \pm 0.16 \%$$

$$\mu = 464.9 \pm 0.7$$

$$\sigma = 53.3 \pm 0.7$$

WF (8375 entries)

$$\frac{\sigma}{\mu} = 11.65 \pm 0.17 \%$$

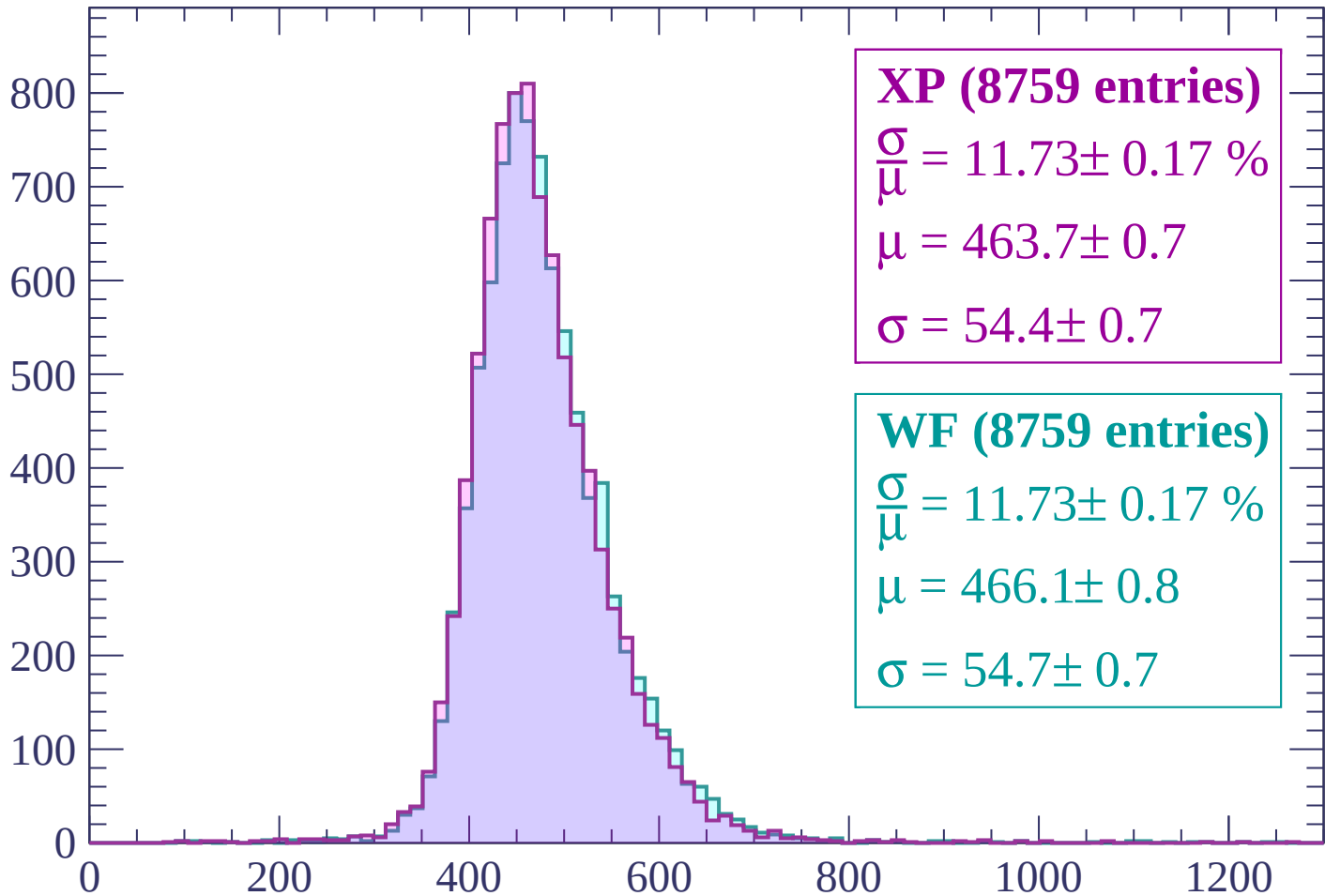
$$\mu = 467.8 \pm 0.8$$

$$\sigma = 54.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1260 < p < -1200

Count



XP (8759 entries)

$$\frac{\sigma}{\mu} = 11.73 \pm 0.17 \%$$

$$\mu = 463.7 \pm 0.7$$

$$\sigma = 54.4 \pm 0.7$$

WF (8759 entries)

$$\frac{\sigma}{\mu} = 11.73 \pm 0.17 \%$$

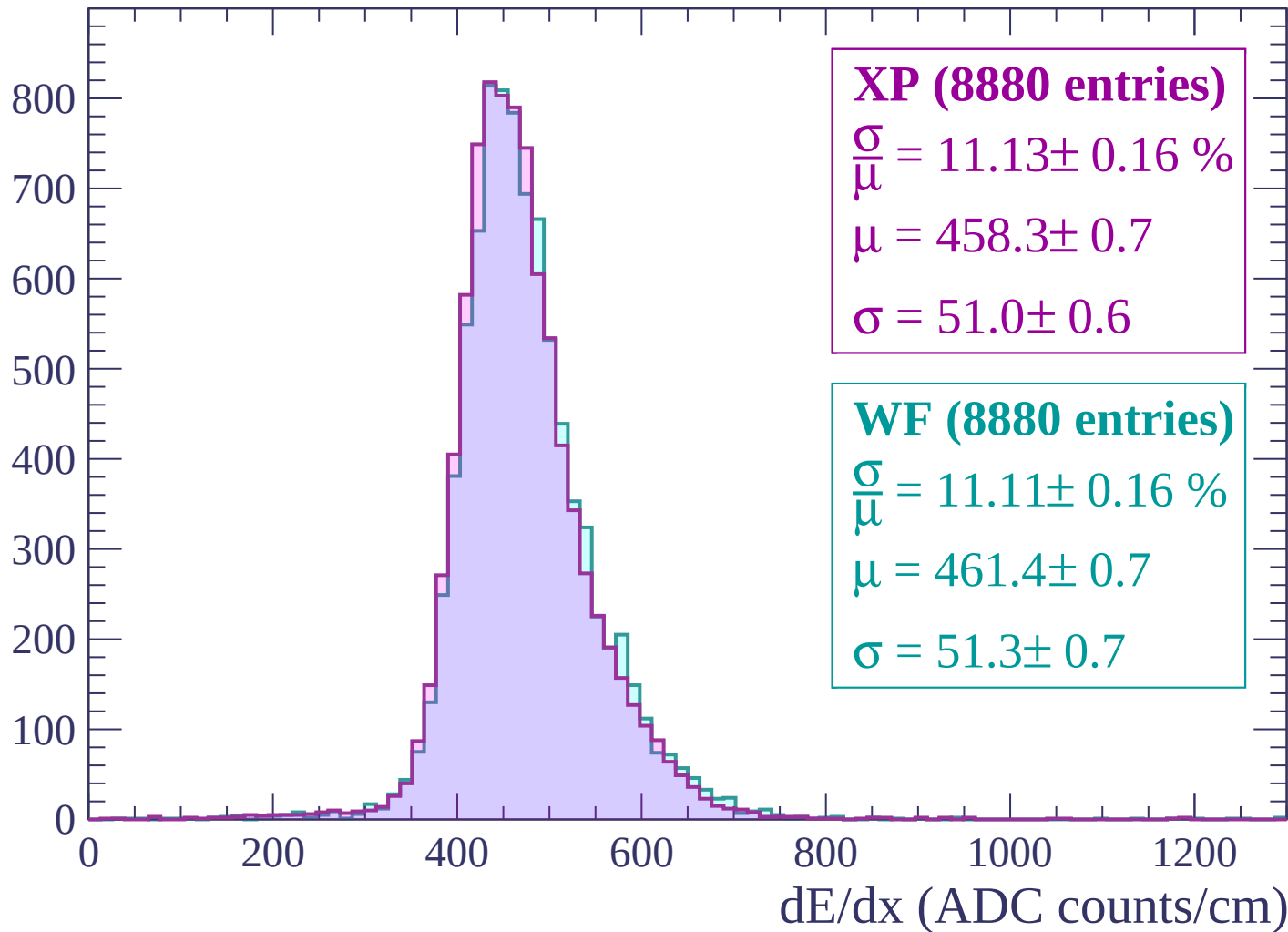
$$\mu = 466.1 \pm 0.8$$

$$\sigma = 54.7 \pm 0.7$$

dE/dx (ADC counts/cm)

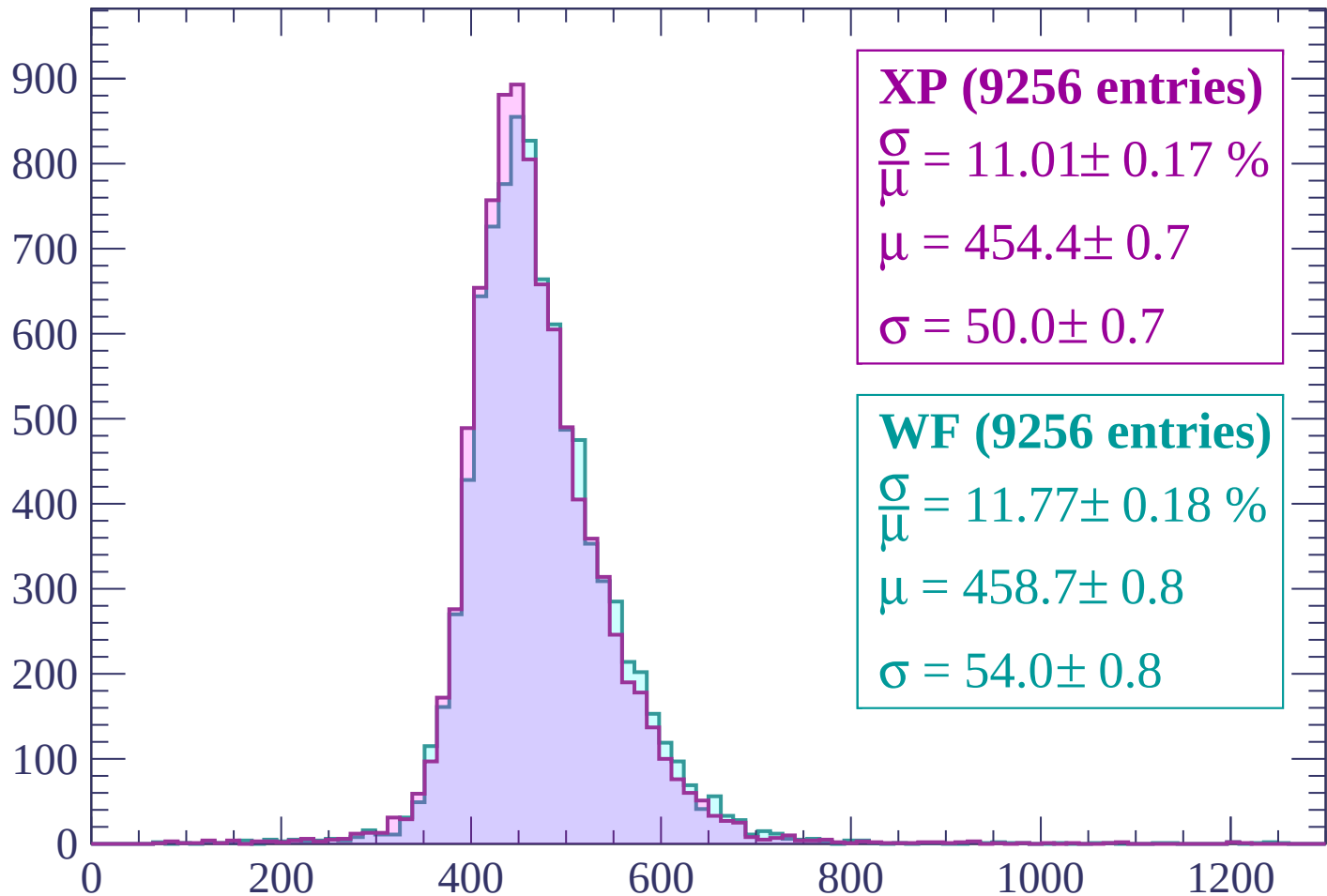
Energy loss | -1200 < p < -1140

Count



Energy loss | -1140 < p < -1080

Count



XP (9256 entries)

$$\frac{\sigma}{\mu} = 11.01 \pm 0.17 \%$$

$$\mu = 454.4 \pm 0.7$$

$$\sigma = 50.0 \pm 0.7$$

WF (9256 entries)

$$\frac{\sigma}{\mu} = 11.77 \pm 0.18 \%$$

$$\mu = 458.7 \pm 0.8$$

$$\sigma = 54.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1080 < p < -1020

Count

1000

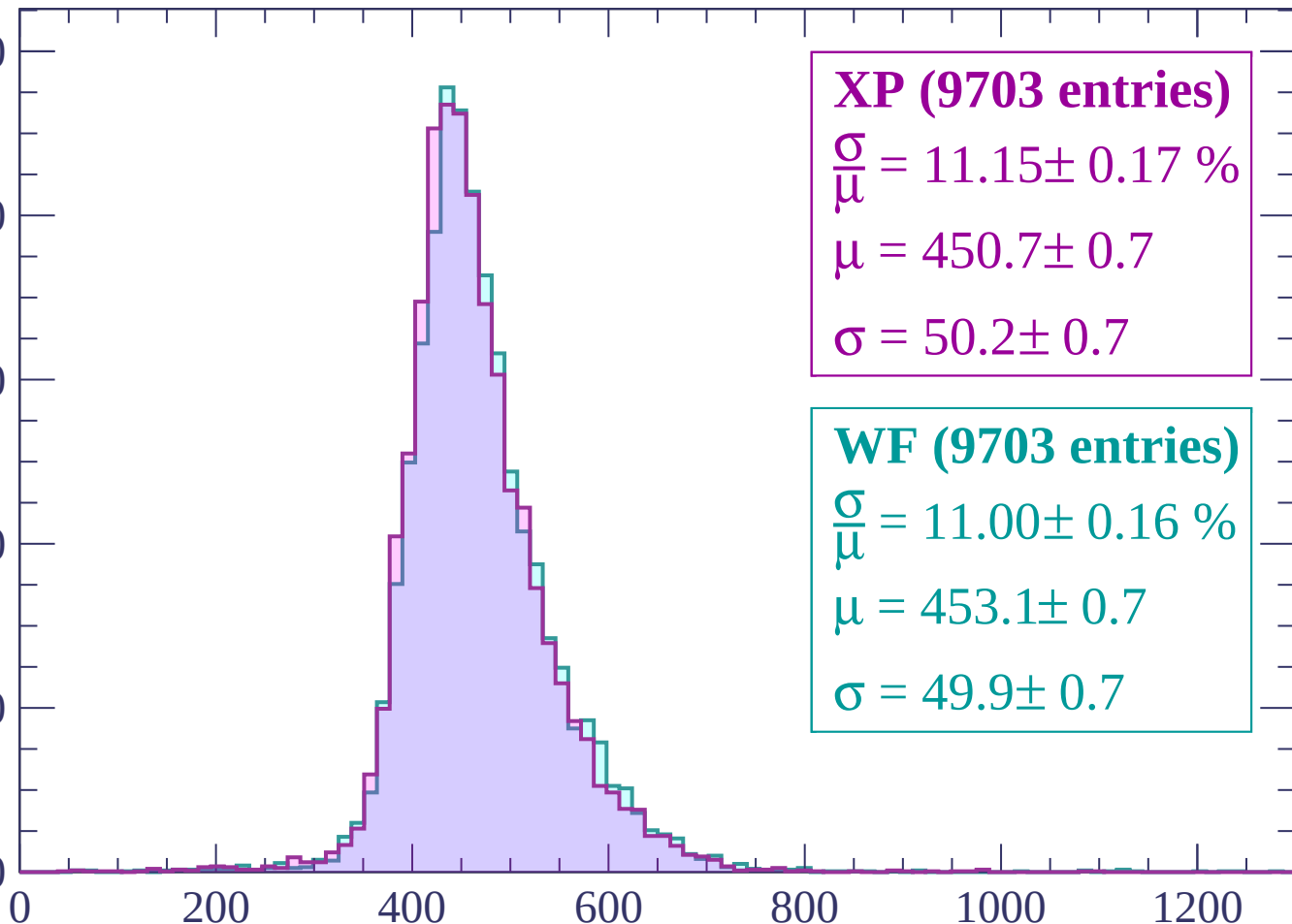
800

600

400

200

0



XP (9703 entries)

$\frac{\sigma}{\mu} = 11.15 \pm 0.17 \%$

$\mu = 450.7 \pm 0.7$

$\sigma = 50.2 \pm 0.7$

WF (9703 entries)

$\frac{\sigma}{\mu} = 11.00 \pm 0.16 \%$

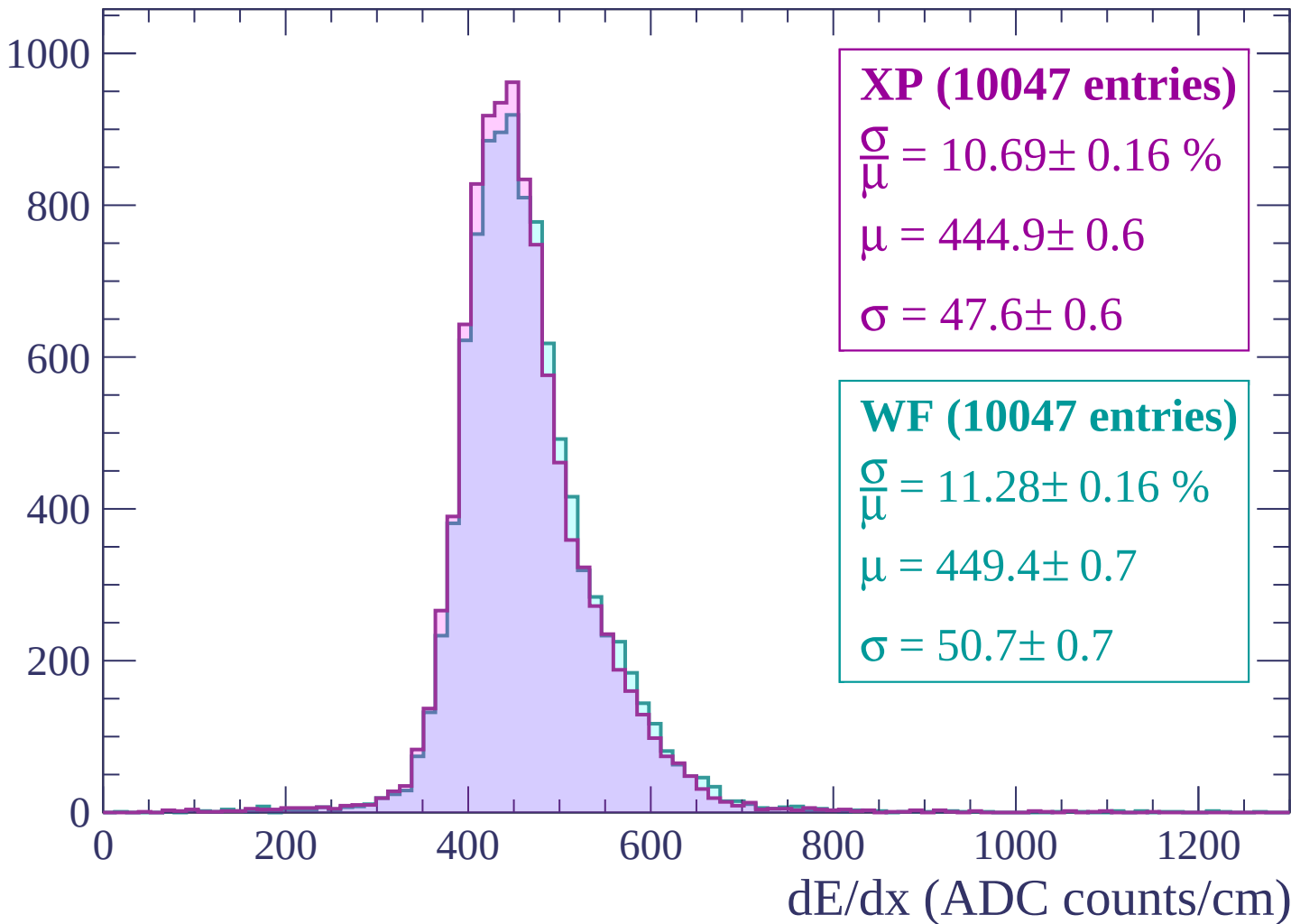
$\mu = 453.1 \pm 0.7$

$\sigma = 49.9 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count



Energy loss | $-960 < p < -900$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (10205 entries)

$\frac{\sigma}{\mu} = 10.79 \pm 0.14 \%$

$\mu = 442.0 \pm 0.6$

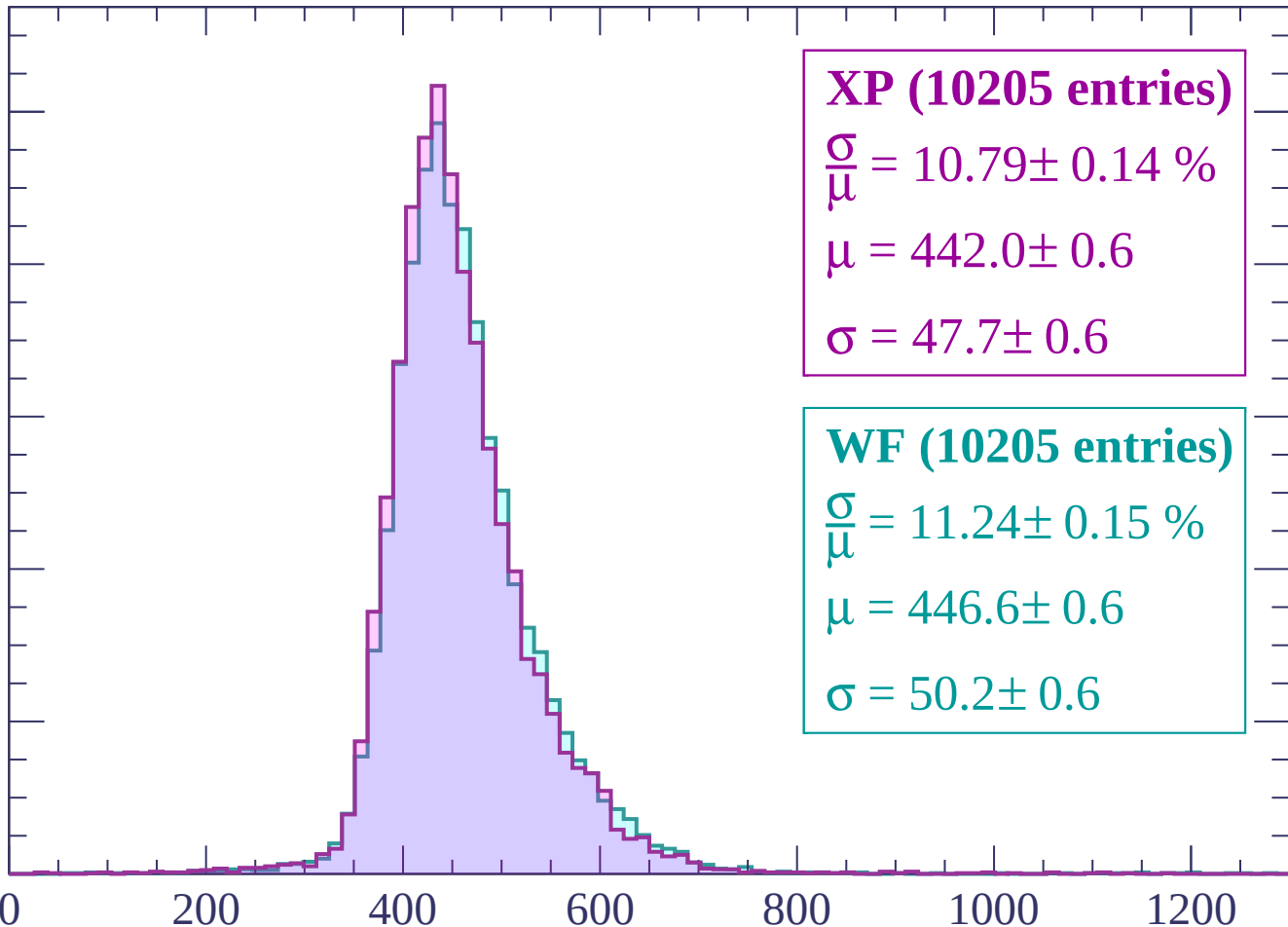
$\sigma = 47.7 \pm 0.6$

WF (10205 entries)

$\frac{\sigma}{\mu} = 11.24 \pm 0.15 \%$

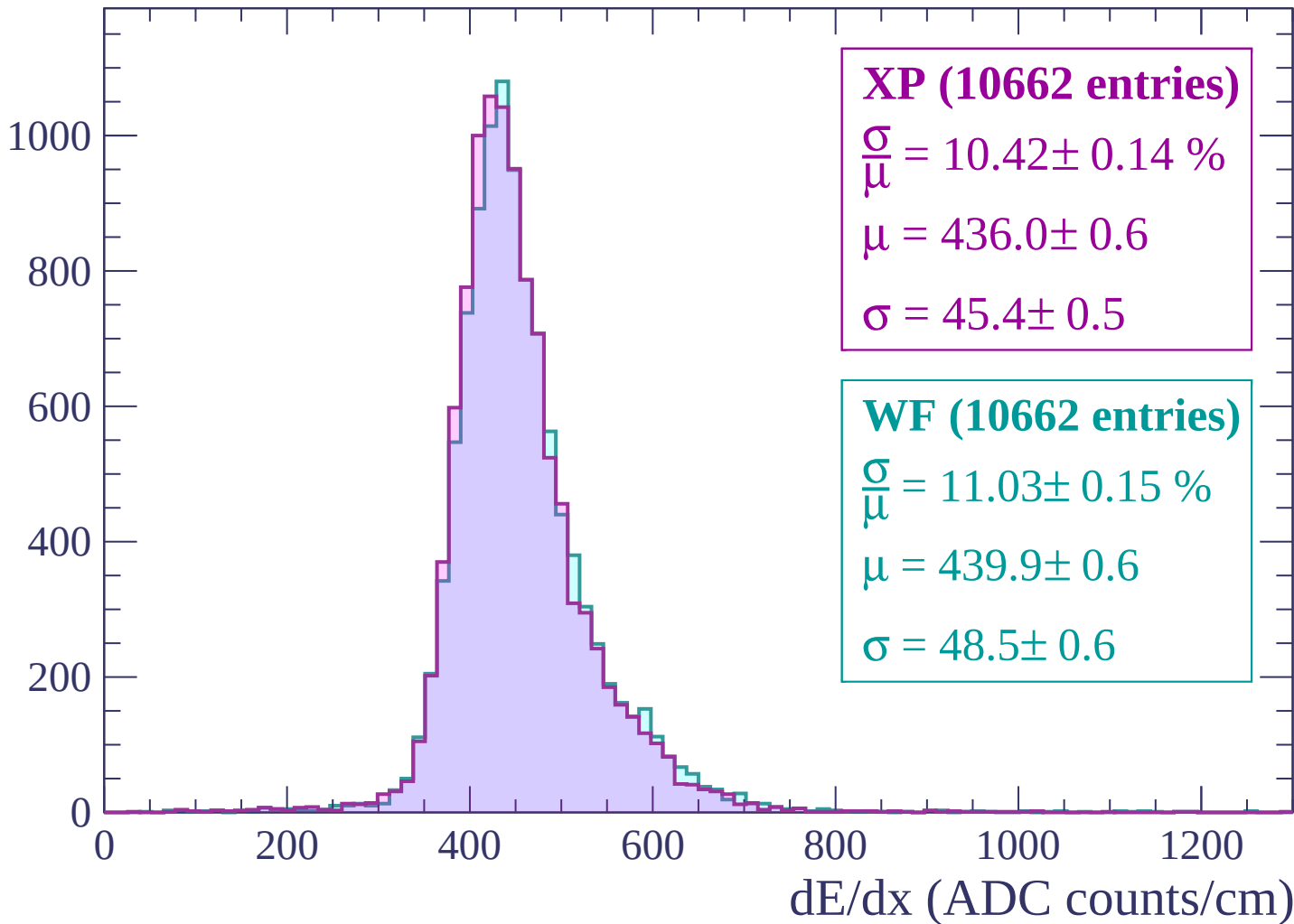
$\mu = 446.6 \pm 0.6$

$\sigma = 50.2 \pm 0.6$

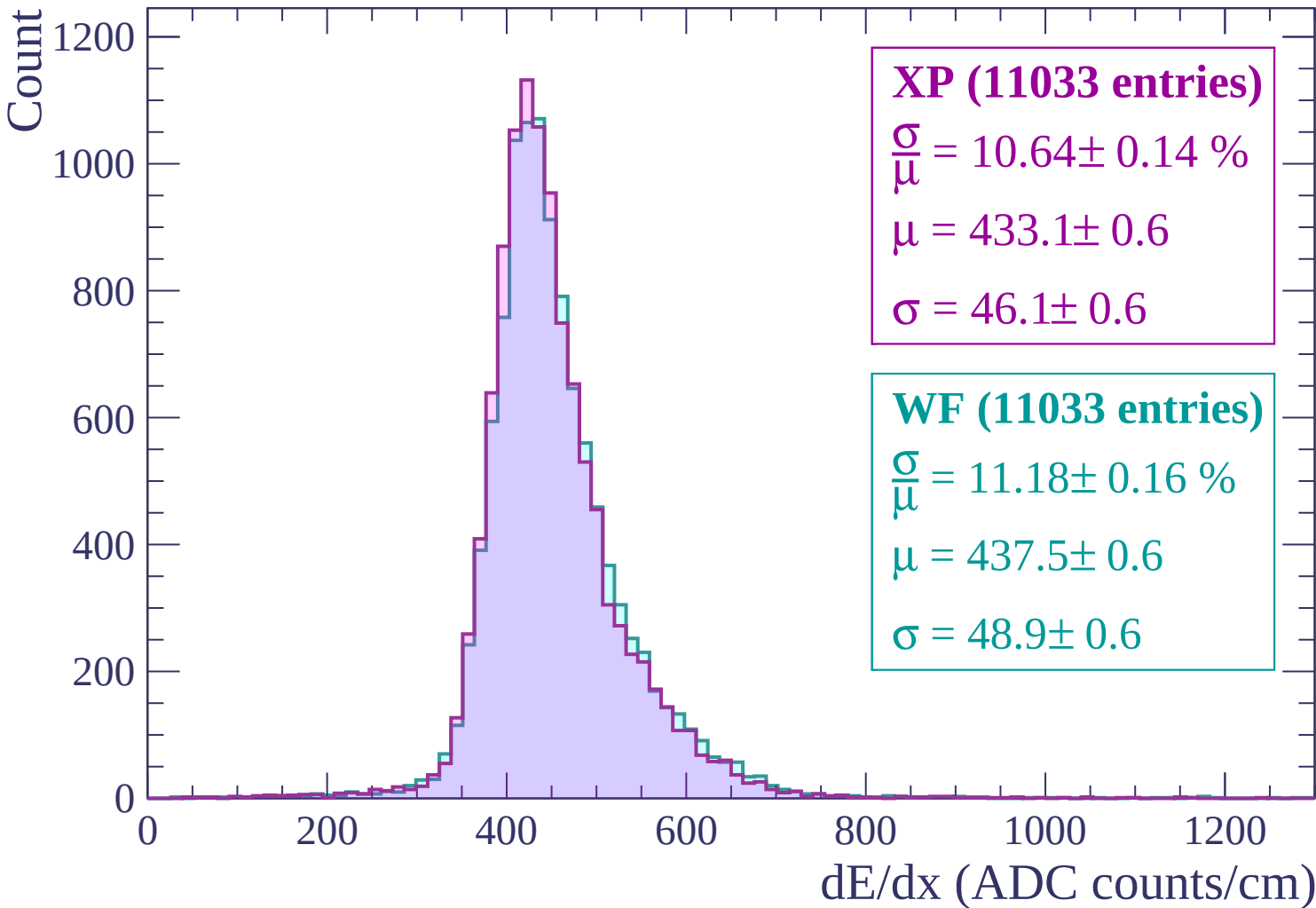


Energy loss | $-900 < p < -840$

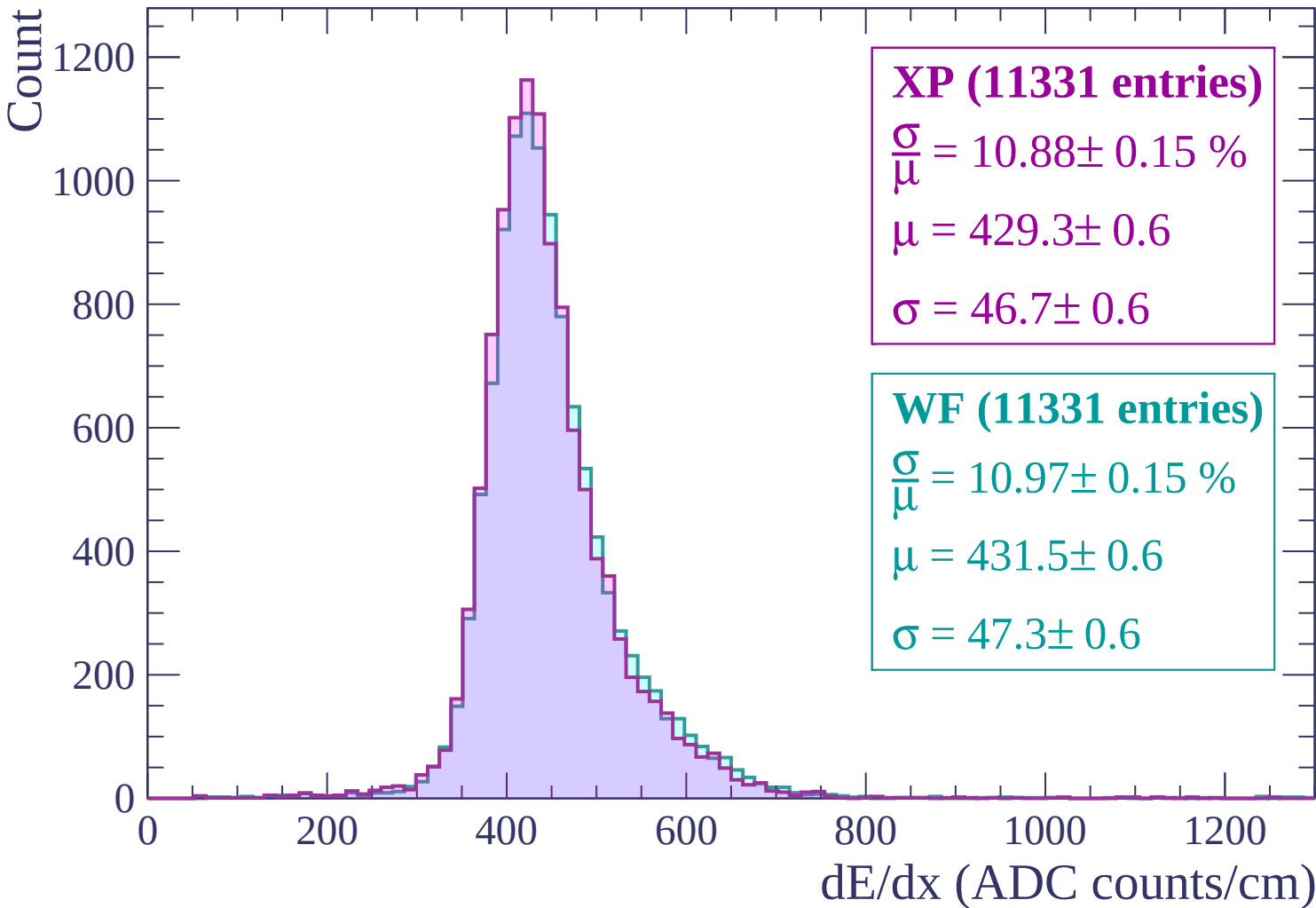
Count



Energy loss | $-840 < p < -780$



Energy loss | $-780 < p < -720$



Energy loss | $-720 < p < -660$

Count

1200

1000

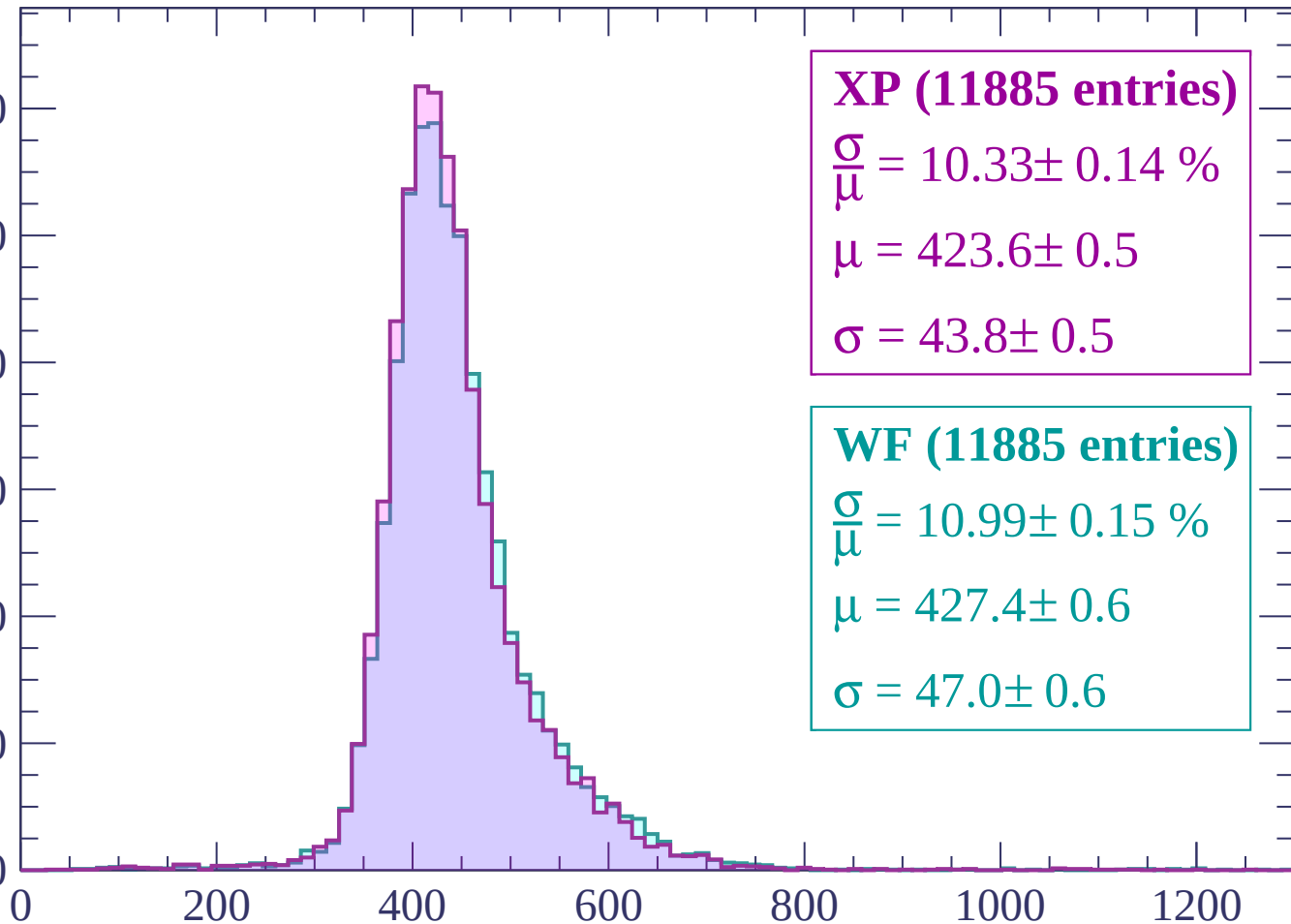
800

600

400

200

0



XP (11885 entries)

$\frac{\sigma}{\mu} = 10.33 \pm 0.14 \%$

$\mu = 423.6 \pm 0.5$

$\sigma = 43.8 \pm 0.5$

WF (11885 entries)

$\frac{\sigma}{\mu} = 10.99 \pm 0.15 \%$

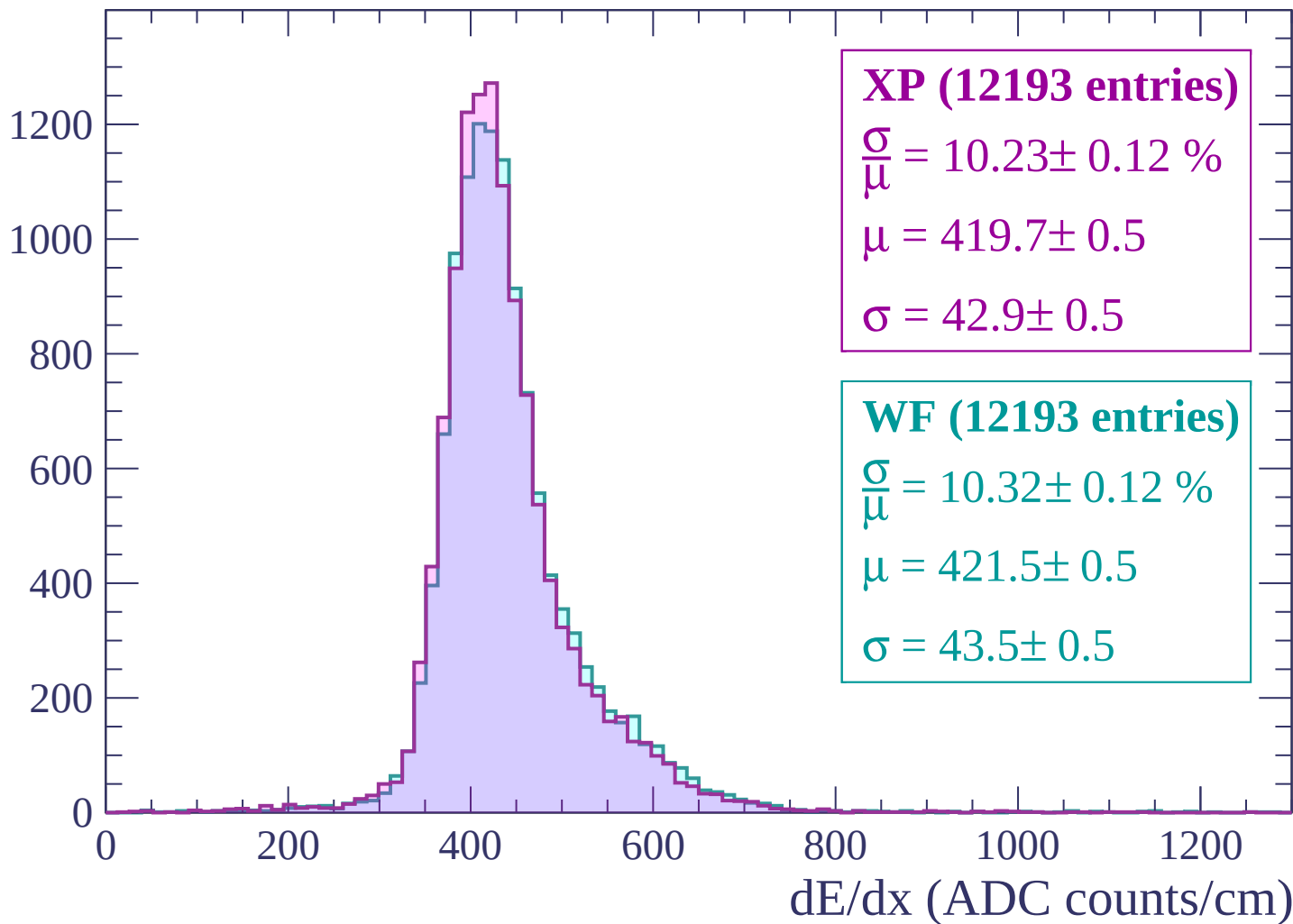
$\mu = 427.4 \pm 0.6$

$\sigma = 47.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

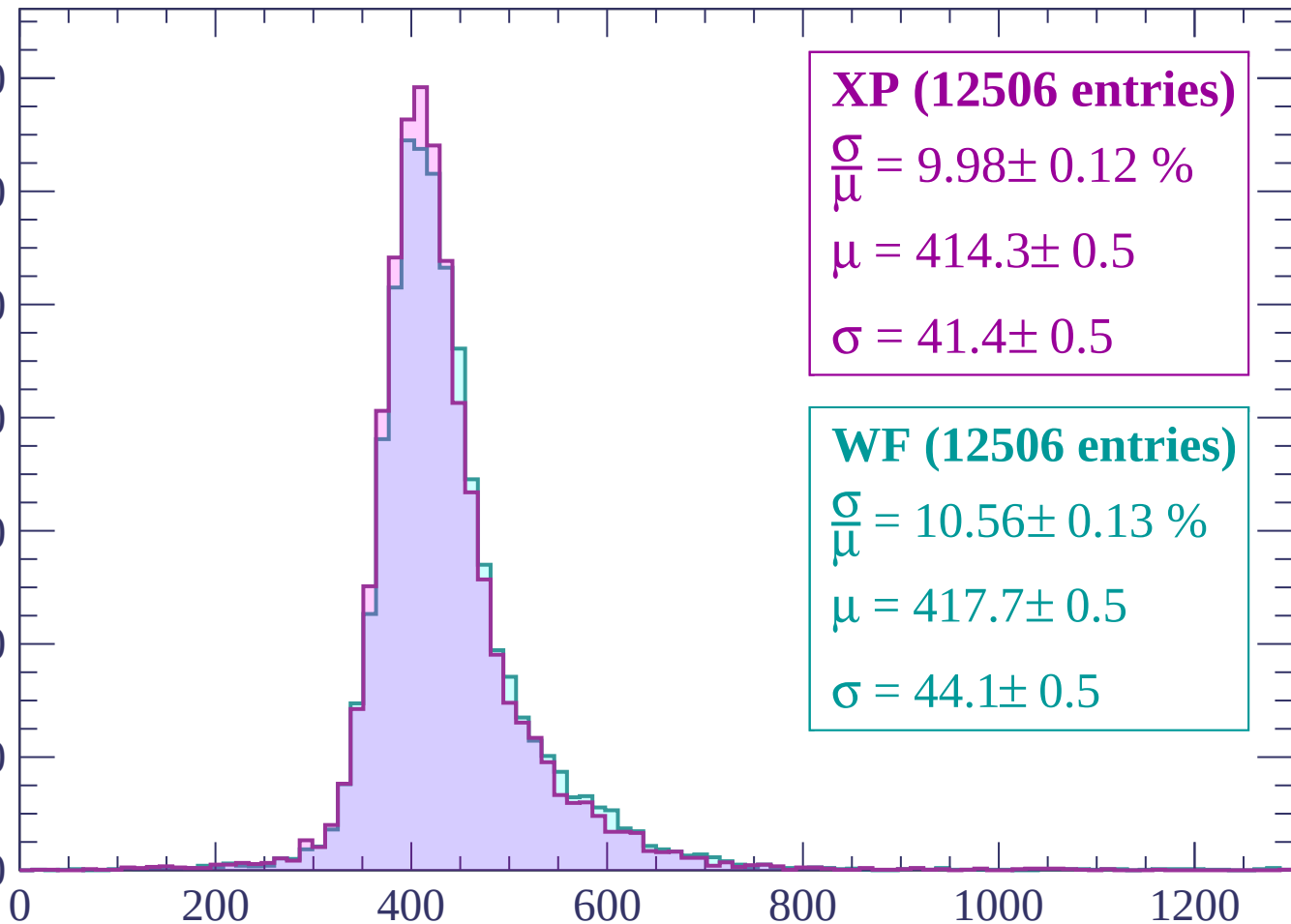
Count



Energy loss | $-600 < p < -540$

Count

1400
1200
1000
800
600
400
200
0



XP (12506 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.12 \%$$

$$\mu = 414.3 \pm 0.5$$

$$\sigma = 41.4 \pm 0.5$$

WF (12506 entries)

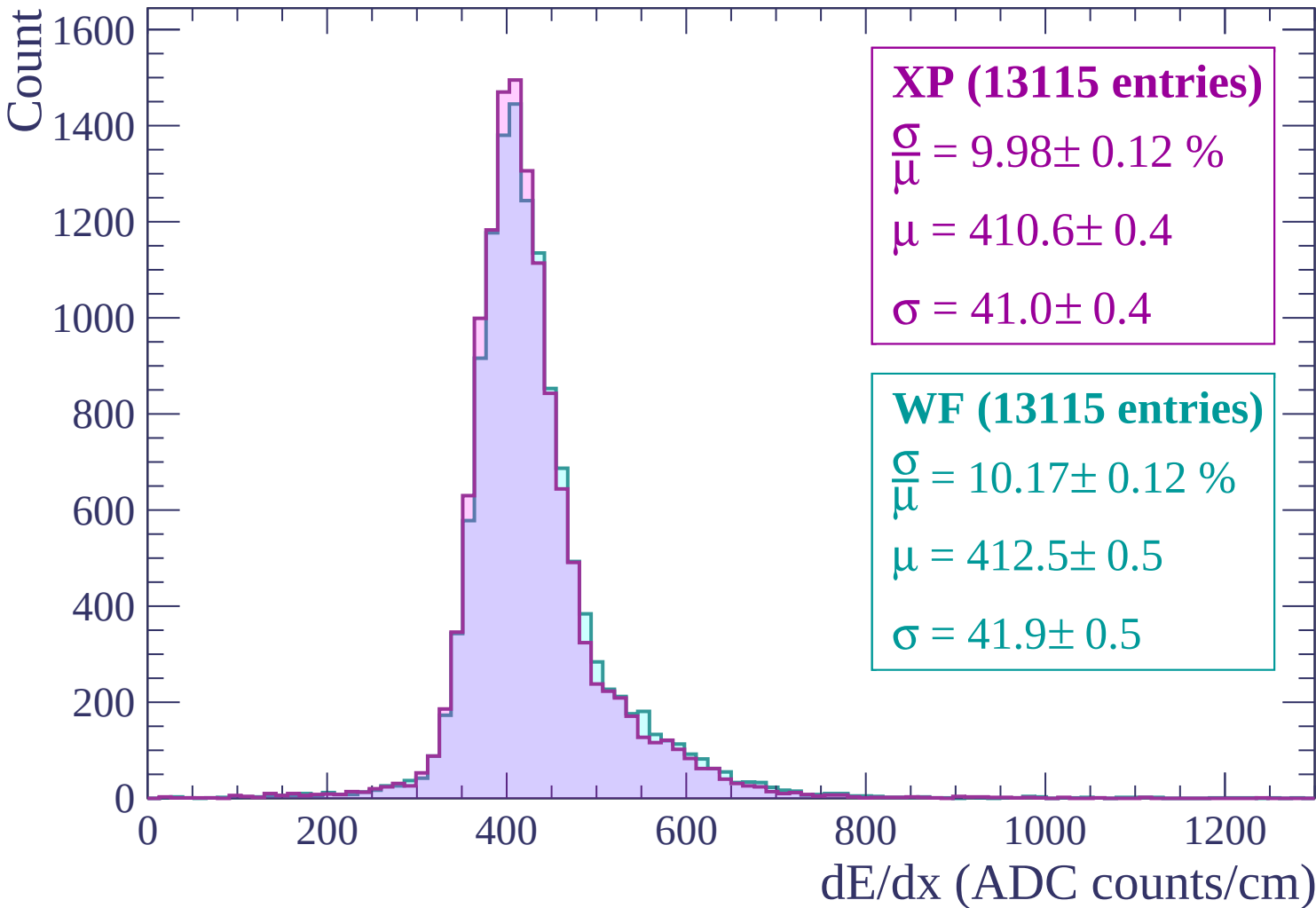
$$\frac{\sigma}{\mu} = 10.56 \pm 0.13 \%$$

$$\mu = 417.7 \pm 0.5$$

$$\sigma = 44.1 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$



Energy loss | $-480 < p < -420$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13540 entries)

$$\frac{\sigma}{\mu} = 9.58 \pm 0.11 \%$$

$$\mu = 405.7 \pm 0.4$$

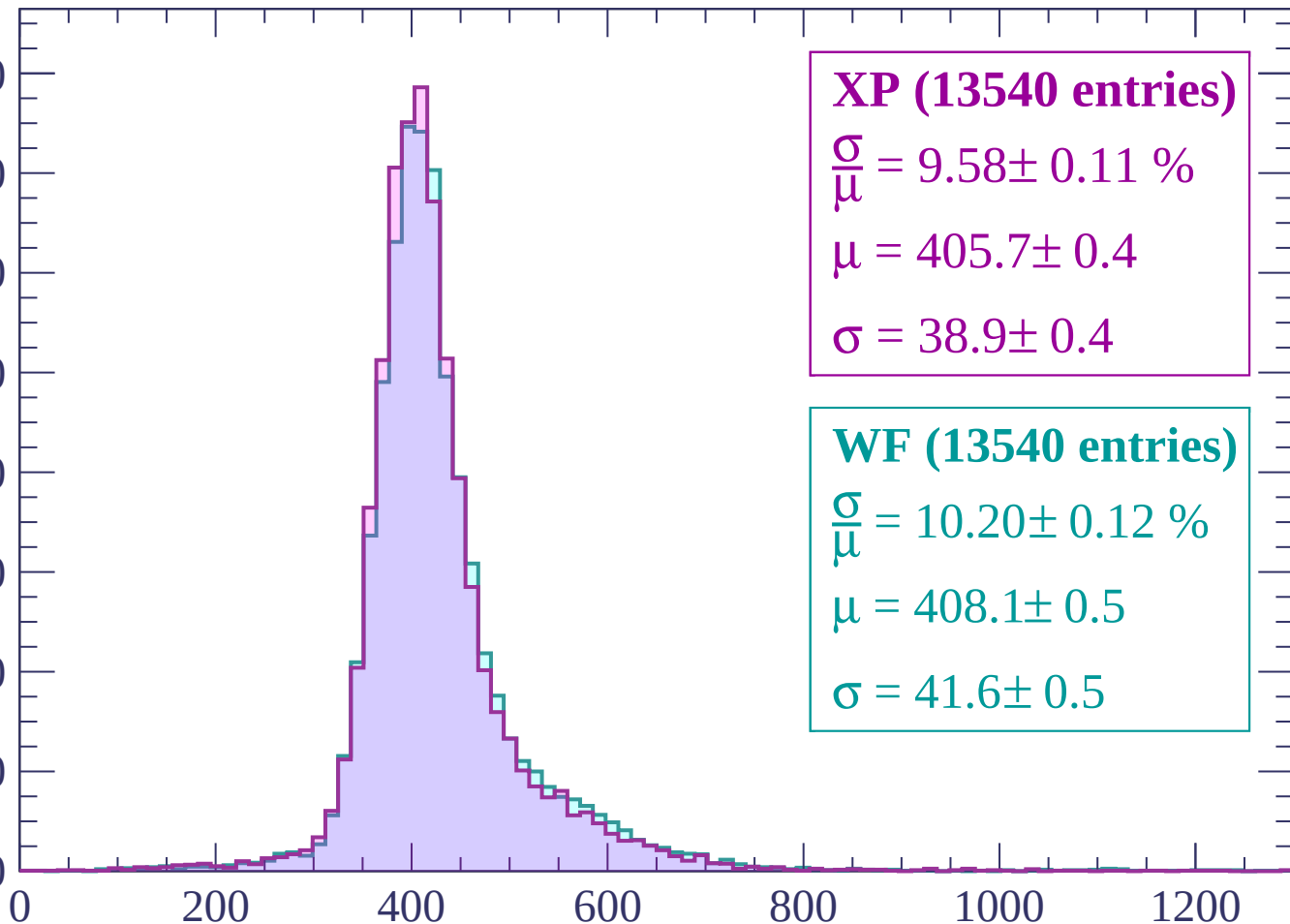
$$\sigma = 38.9 \pm 0.4$$

WF (13540 entries)

$$\frac{\sigma}{\mu} = 10.20 \pm 0.12 \%$$

$$\mu = 408.1 \pm 0.5$$

$$\sigma = 41.6 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13901 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.11 \%$$

$$\mu = 402.7 \pm 0.4$$

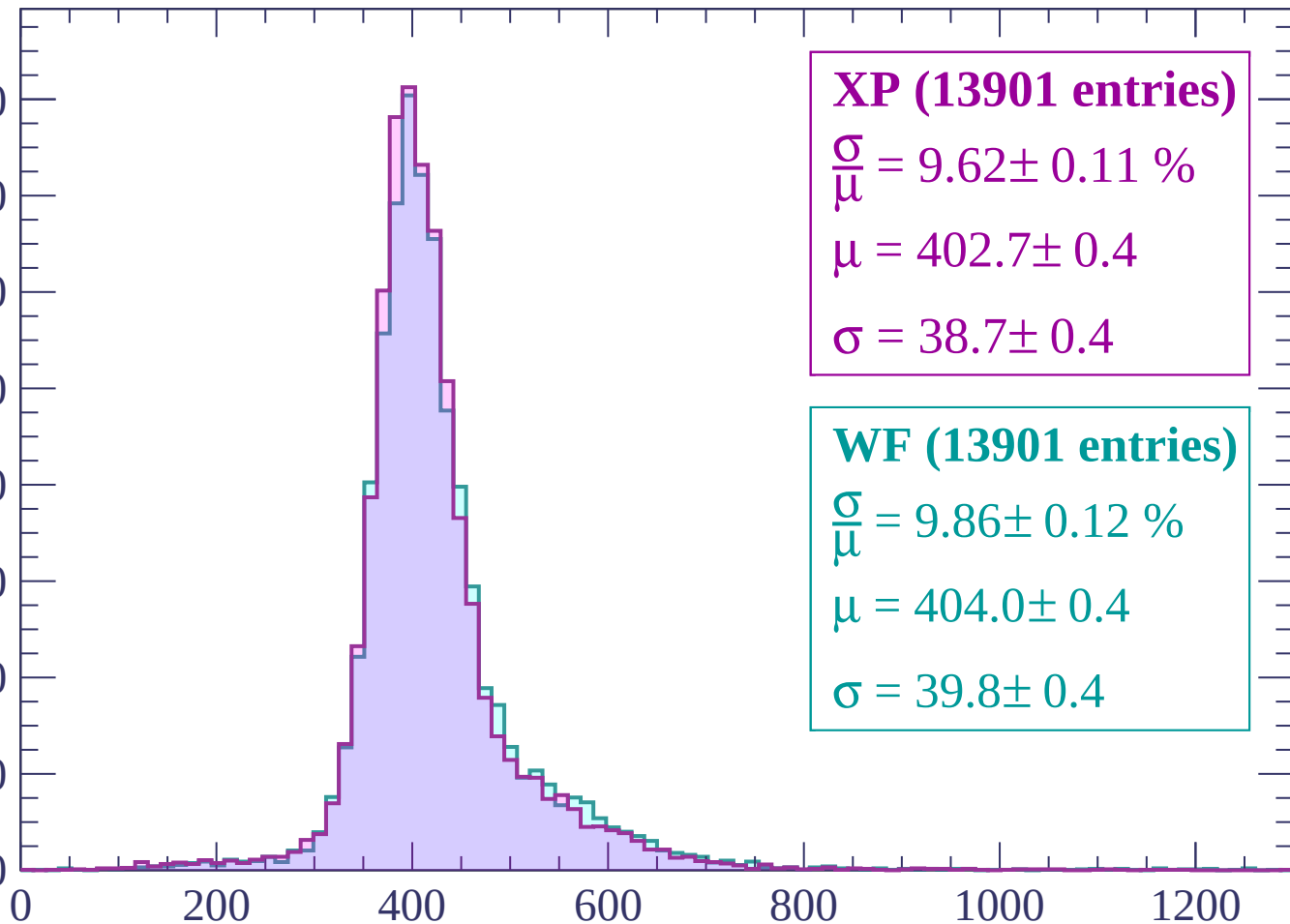
$$\sigma = 38.7 \pm 0.4$$

WF (13901 entries)

$$\frac{\sigma}{\mu} = 9.86 \pm 0.12 \%$$

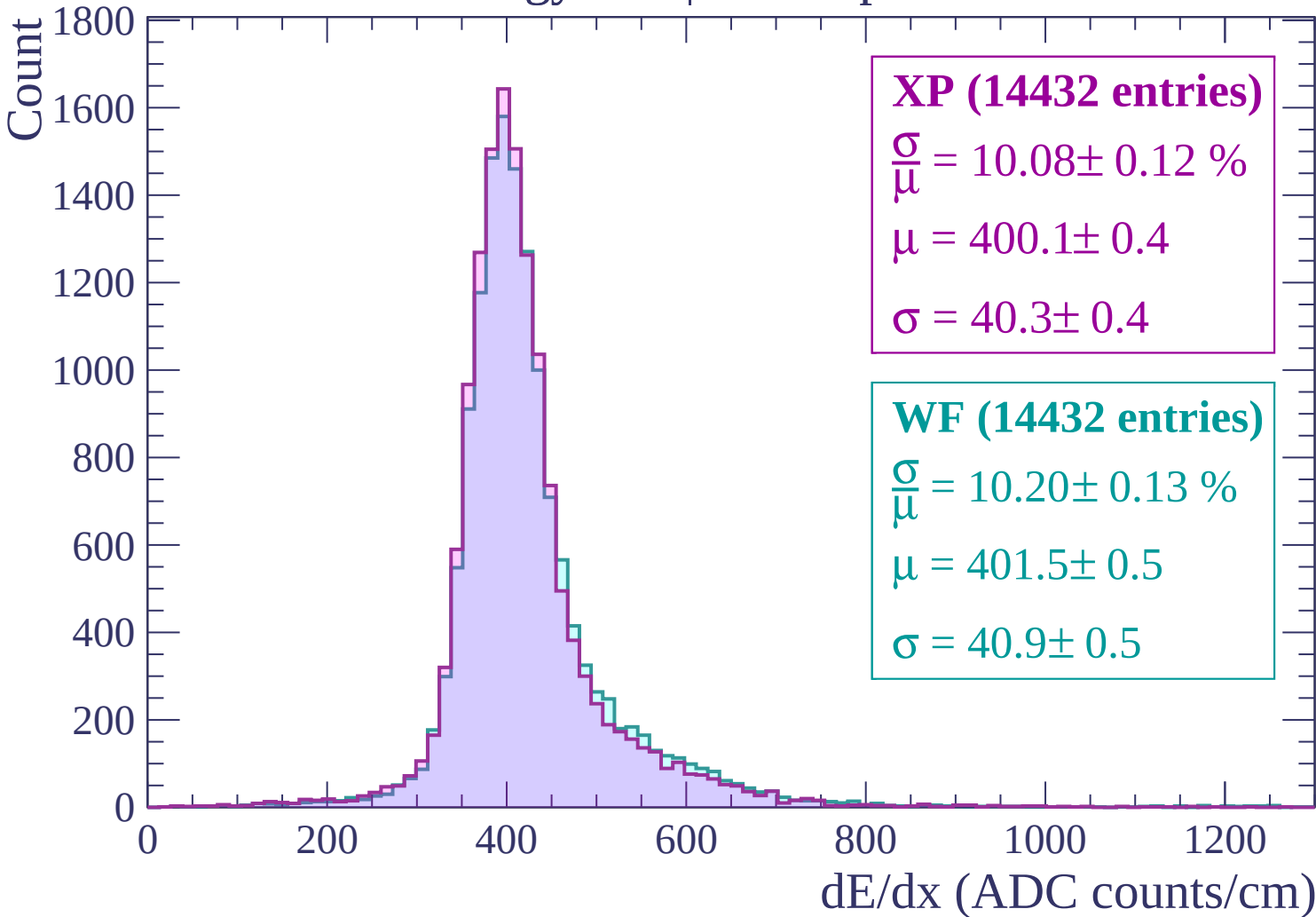
$$\mu = 404.0 \pm 0.4$$

$$\sigma = 39.8 \pm 0.4$$

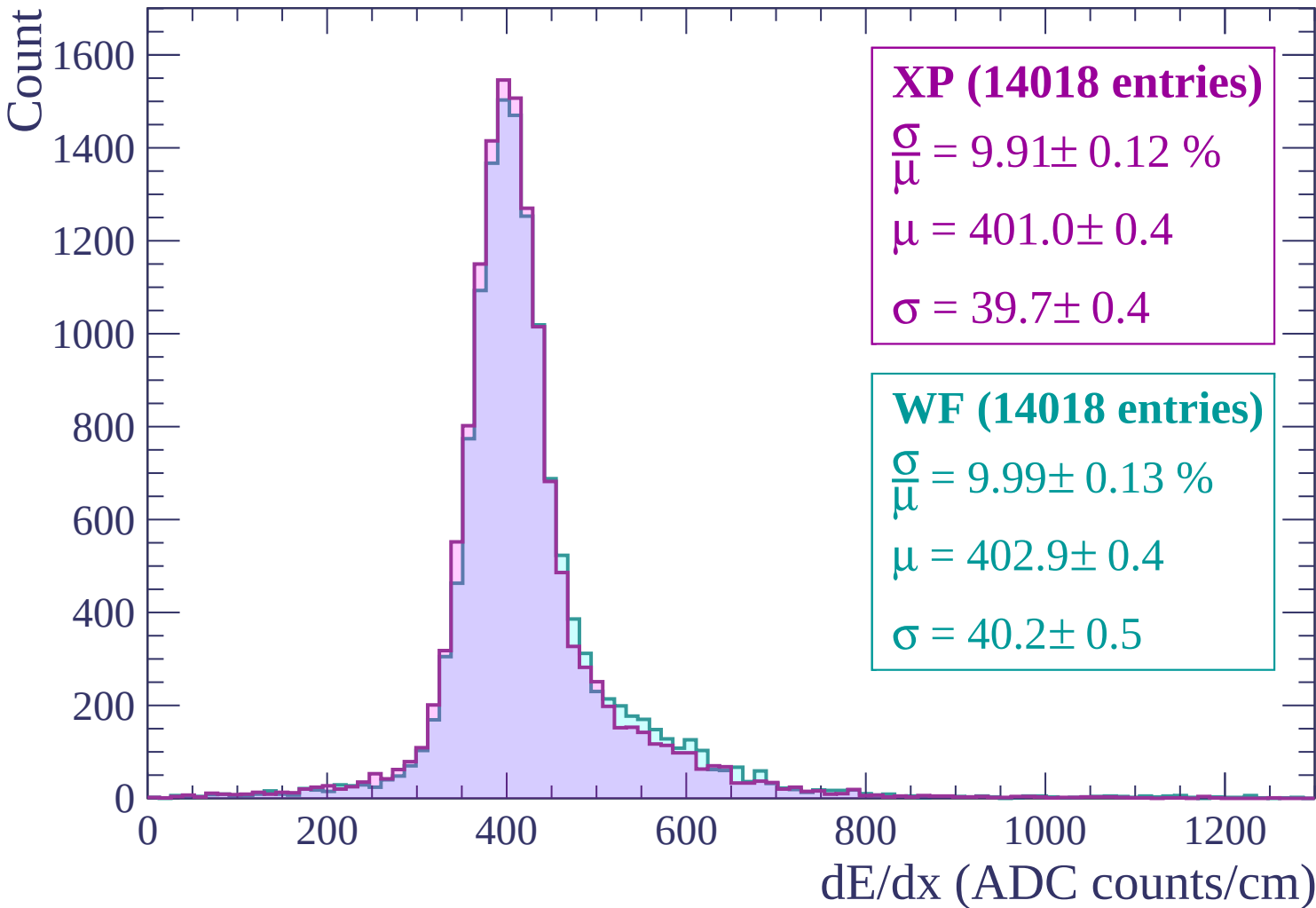


dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$



Energy loss | $-300 < p < -240$



Energy loss | $-240 < p < -180$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (12643 entries)

$\frac{\sigma}{\mu} = 11.39 \pm 0.16 \%$

$\mu = 412.5 \pm 0.6$

$\sigma = 47.0 \pm 0.6$

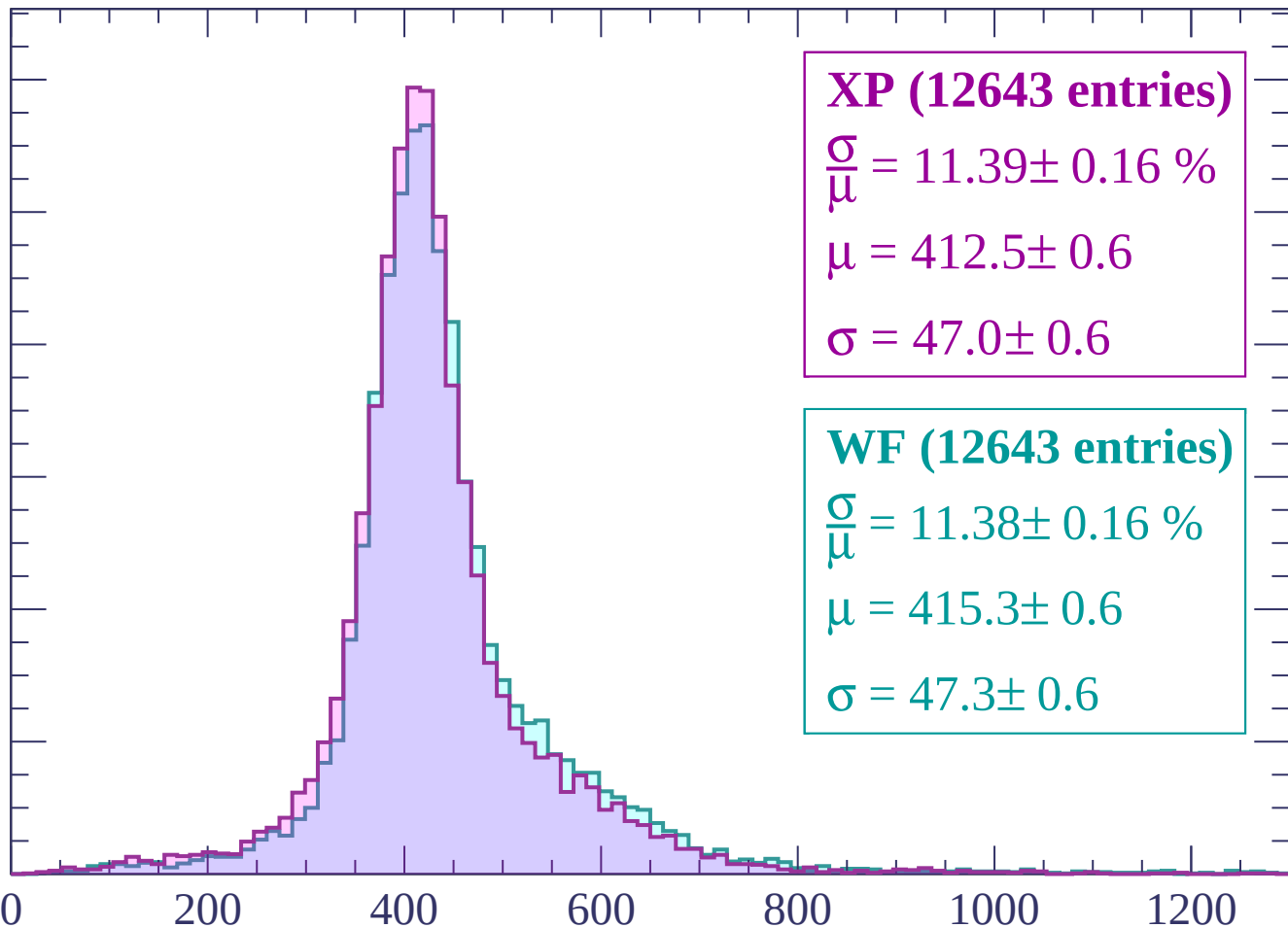
WF (12643 entries)

$\frac{\sigma}{\mu} = 11.38 \pm 0.16 \%$

$\mu = 415.3 \pm 0.6$

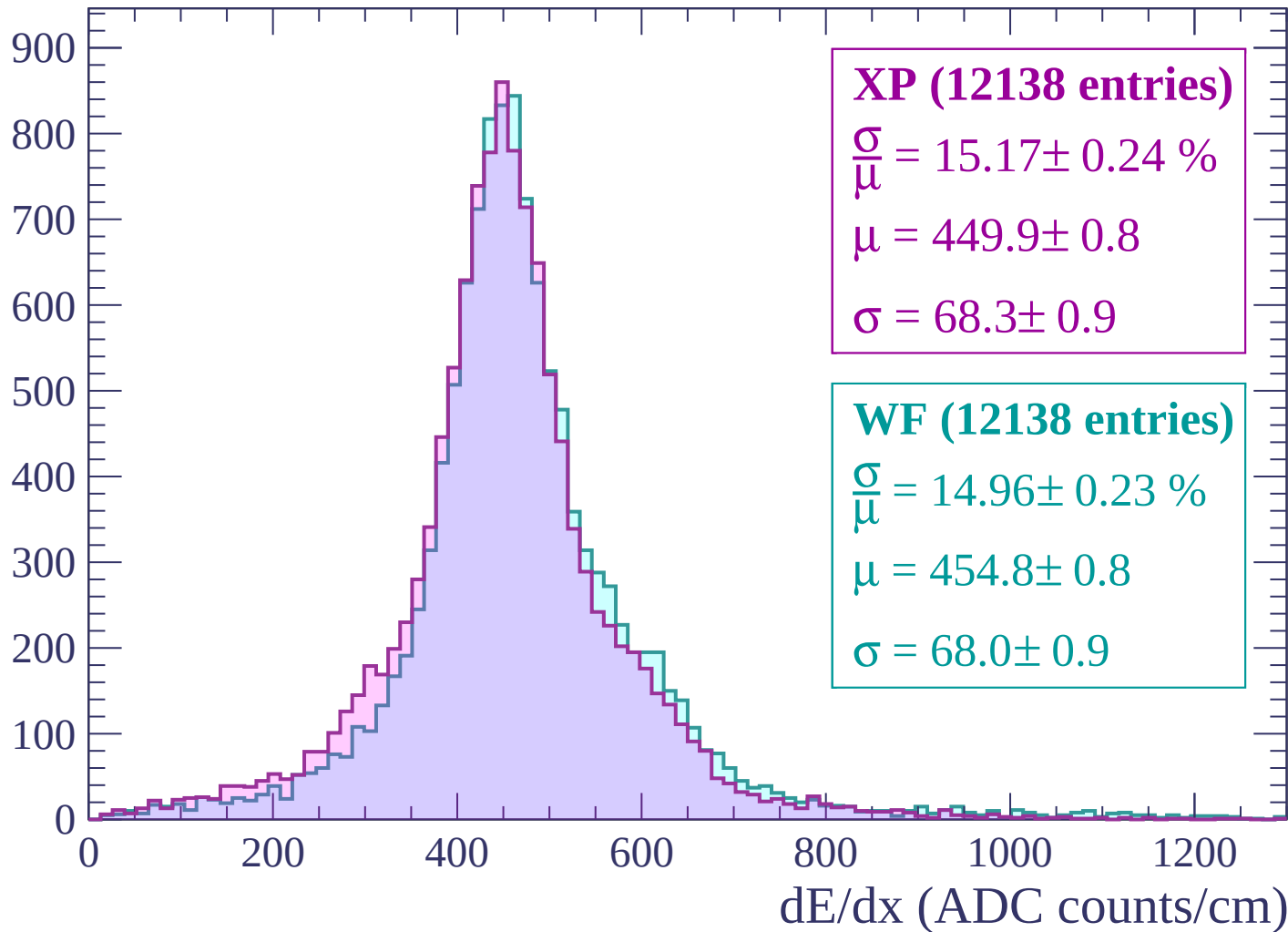
$\sigma = 47.3 \pm 0.6$

dE/dx (ADC counts/cm)



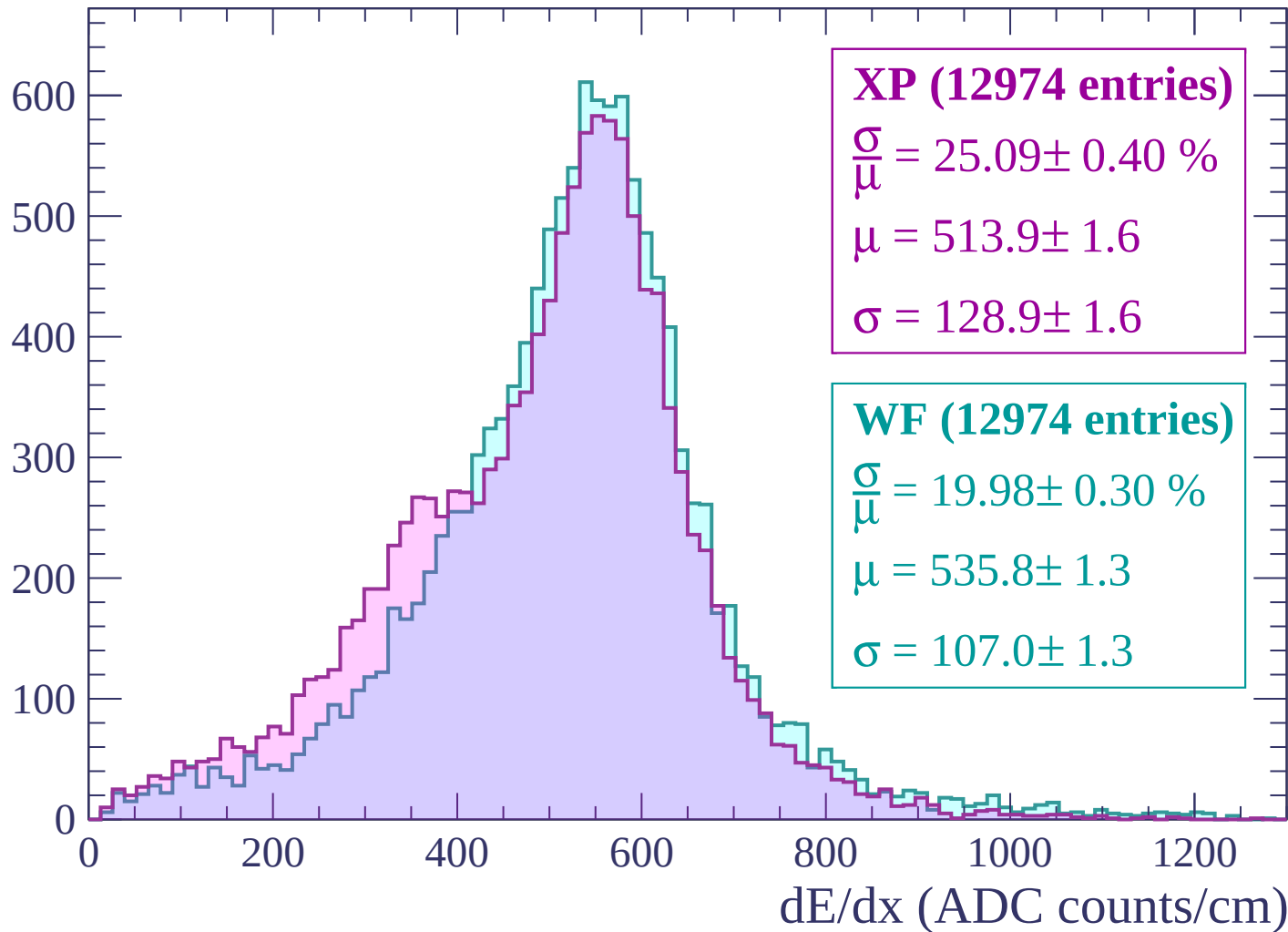
Energy loss | $-180 < p < -120$

Count



Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (26261 entries)

$$\frac{\sigma}{\mu} = 11.55 \pm 0.12 \%$$

$$\mu = 549.2 \pm 0.6$$

$$\sigma = 63.4 \pm 0.6$$

WF (26261 entries)

$$\frac{\sigma}{\mu} = 12.34 \pm 0.12 \%$$

$$\mu = 548.0 \pm 0.6$$

$$\sigma = 67.6 \pm 0.6$$

0

200

400

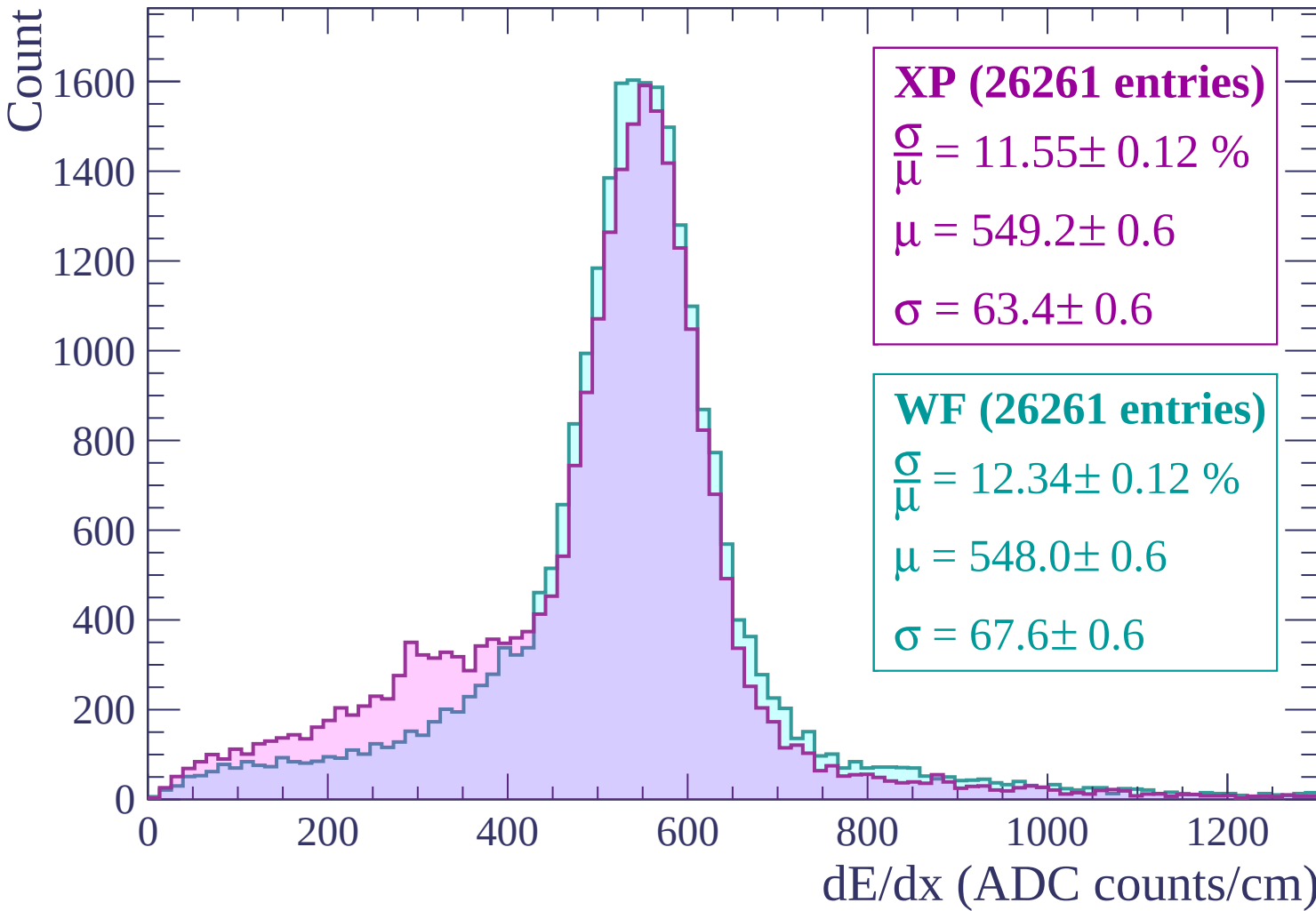
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $0 < p < 60$

Count

3500
3000
2500
2000
1500
1000
500
0

XP (54448 entries)

$\frac{\sigma}{\mu} = 15.38 \pm 0.11 \%$

$\mu = 496.0 \pm 0.4$

$\sigma = 76.3 \pm 0.5$

WF (54448 entries)

$\frac{\sigma}{\mu} = 14.78 \pm 0.10 \%$

$\mu = 501.4 \pm 0.4$

$\sigma = 74.1 \pm 0.4$

0

200

400

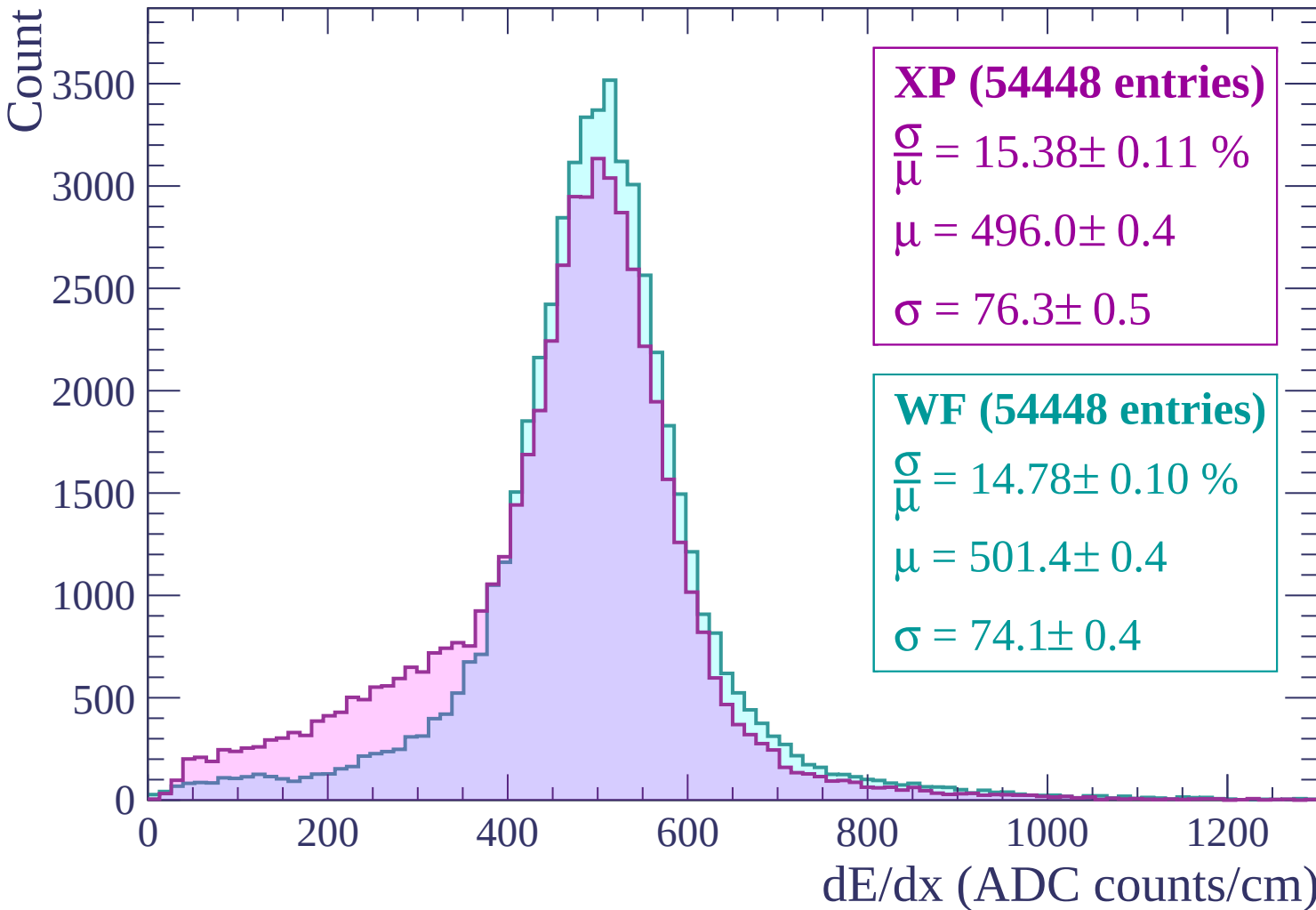
600

800

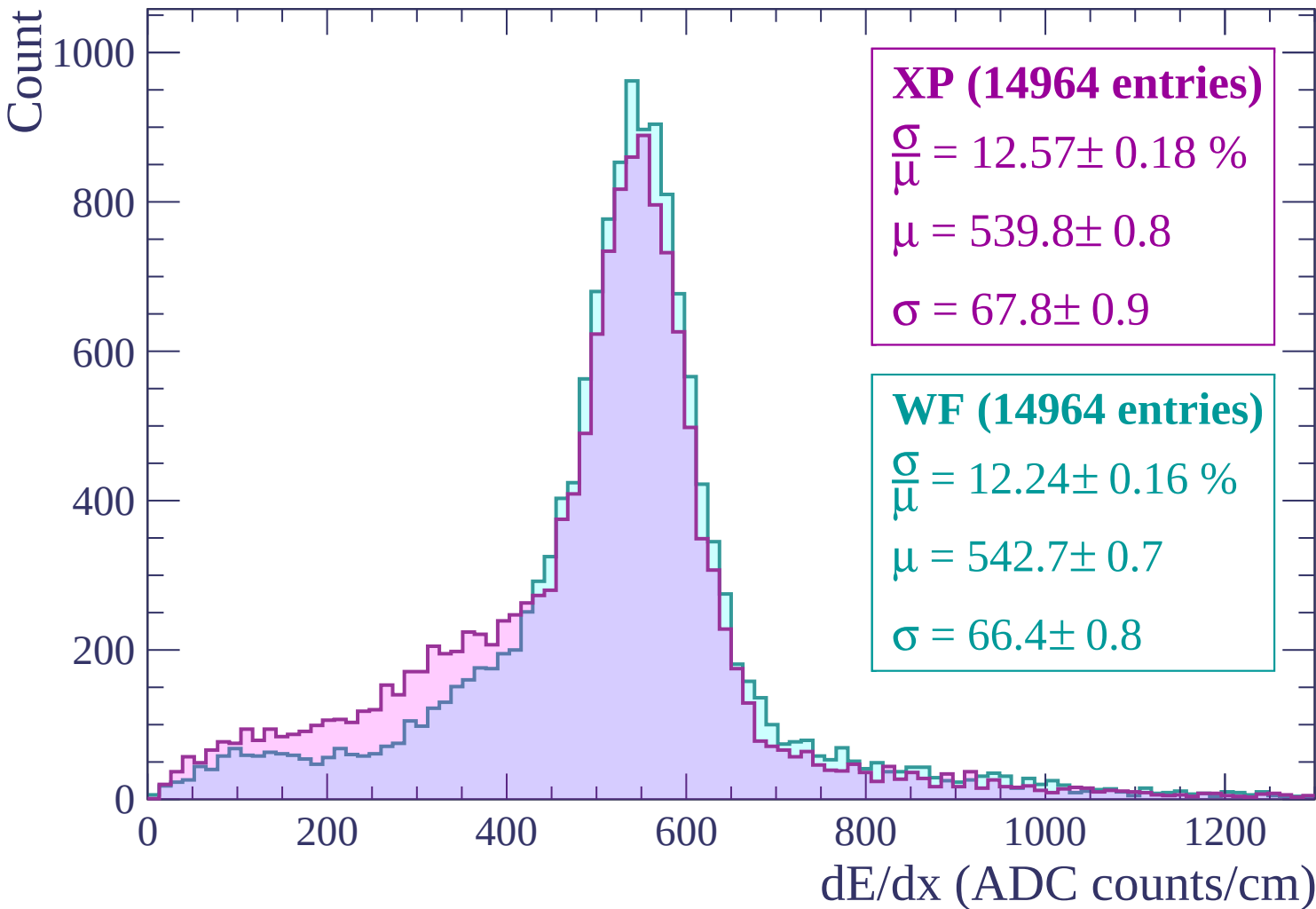
1000

1200

dE/dx (ADC counts/cm)

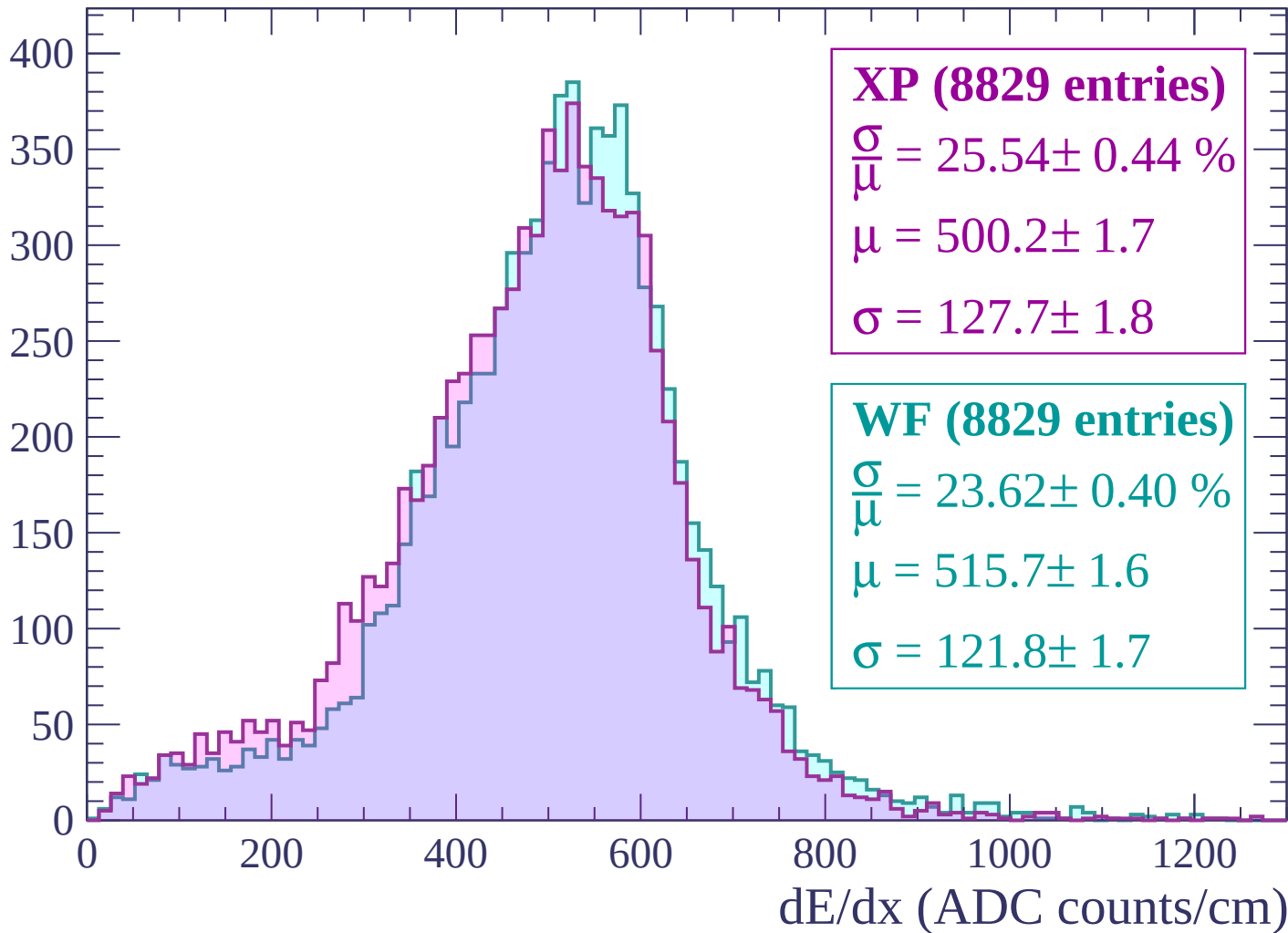


Energy loss | $60 < p < 120$



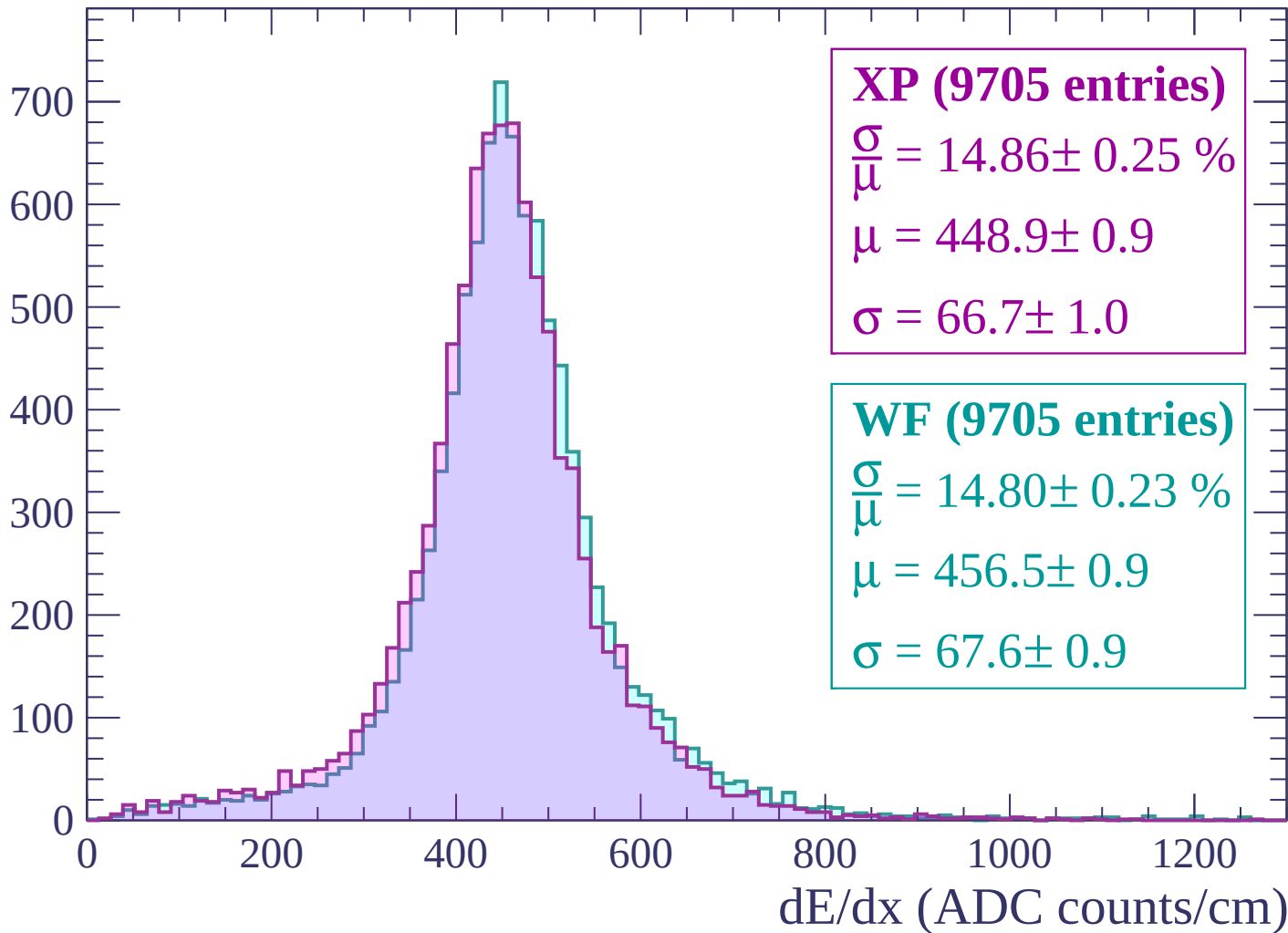
Energy loss | $120 < p < 180$

Count



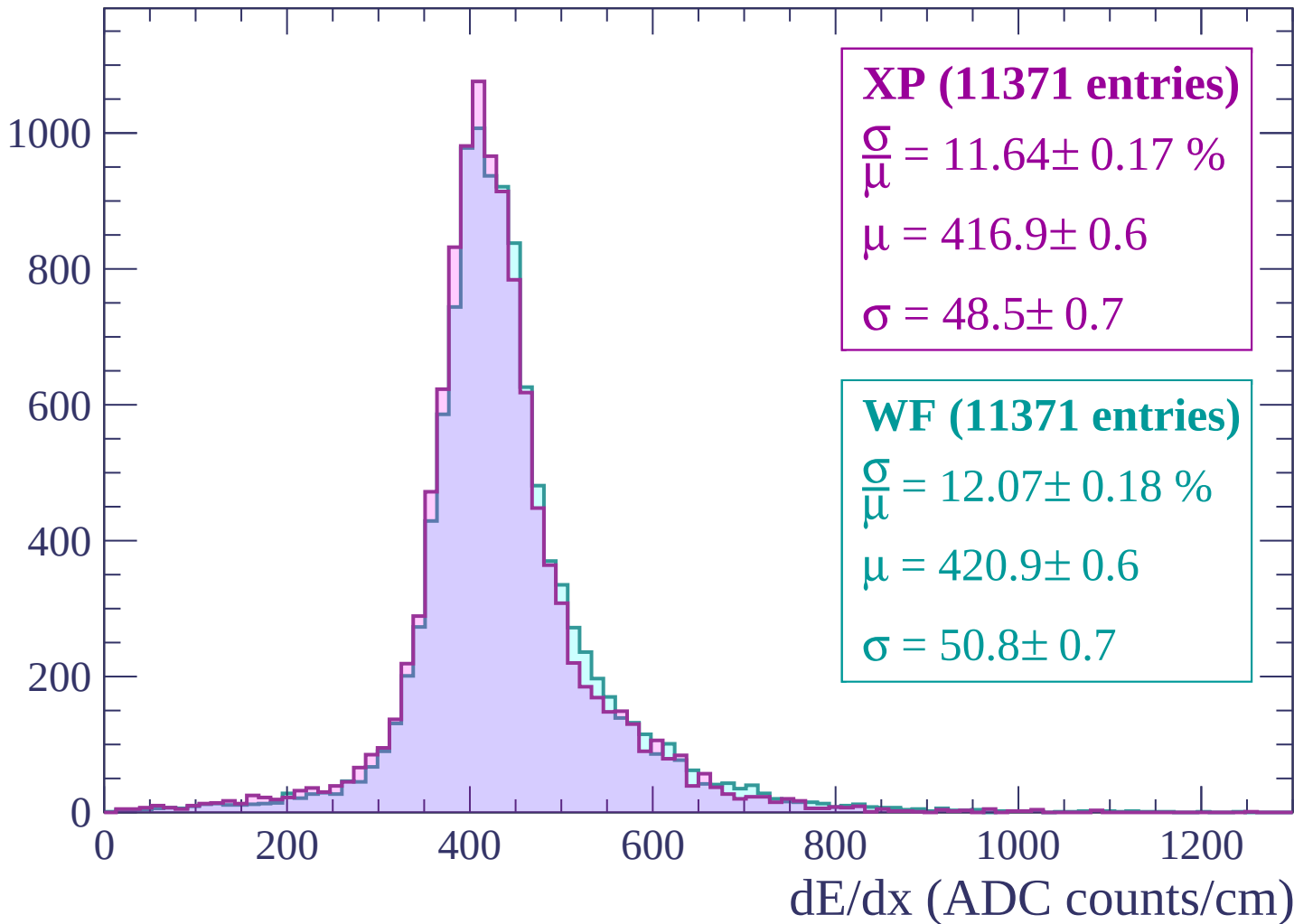
Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count



Energy loss | $300 < p < 360$

Count

1400
1200
1000
800
600
400
200
0

XP (13744 entries)

$\frac{\sigma}{\mu} = 10.25 \pm 0.13 \%$

$\mu = 404.2 \pm 0.5$

$\sigma = 41.4 \pm 0.5$

WF (13744 entries)

$\frac{\sigma}{\mu} = 10.59 \pm 0.13 \%$

$\mu = 407.5 \pm 0.5$

$\sigma = 43.1 \pm 0.5$

0

200

400

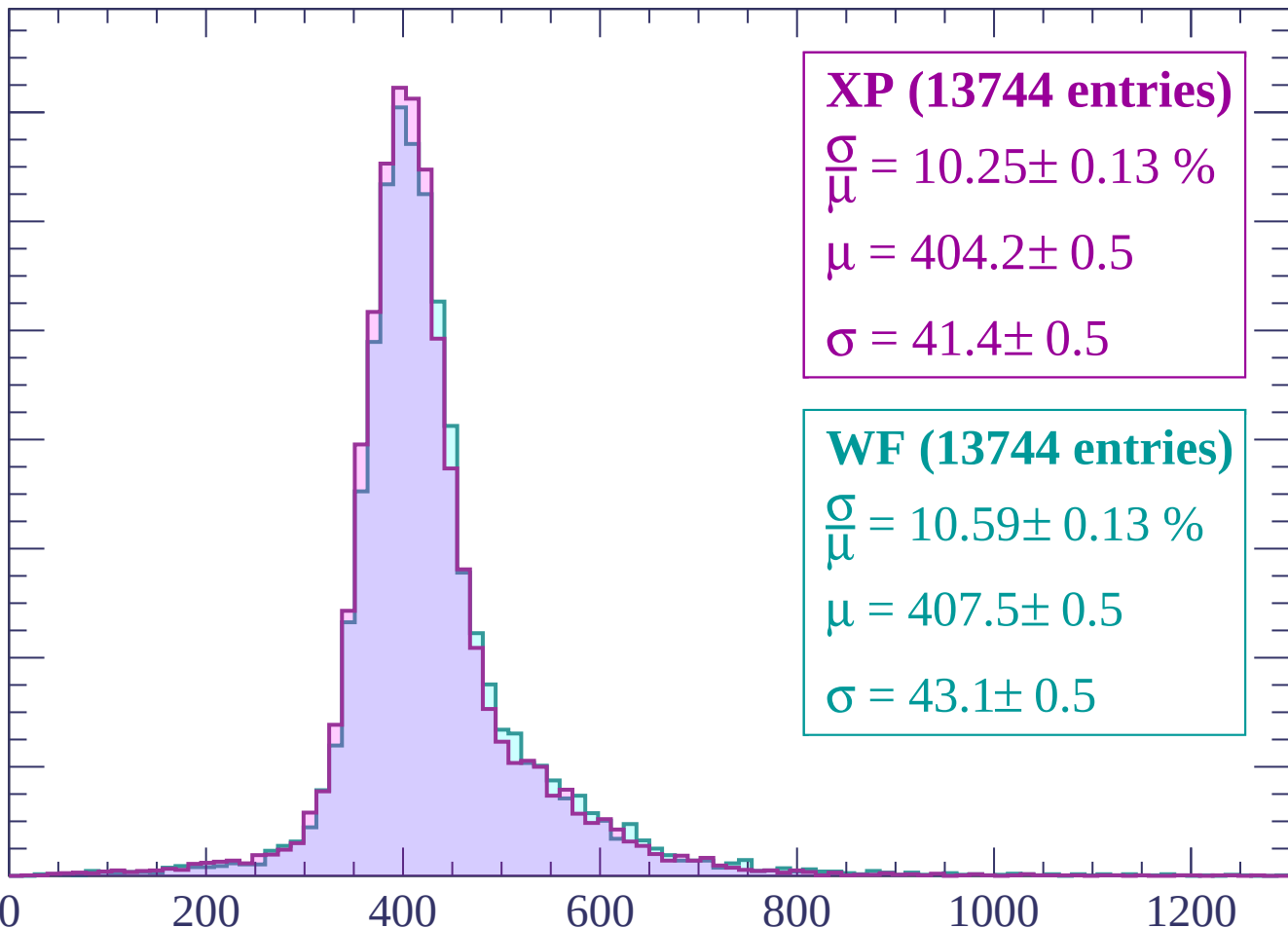
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $360 < p < 420$

Count

1600

1400

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (14259 entries)

$\frac{\sigma}{\mu} = 10.08 \pm 0.13 \%$

$\mu = 402.3 \pm 0.4$

$\sigma = 40.5 \pm 0.5$

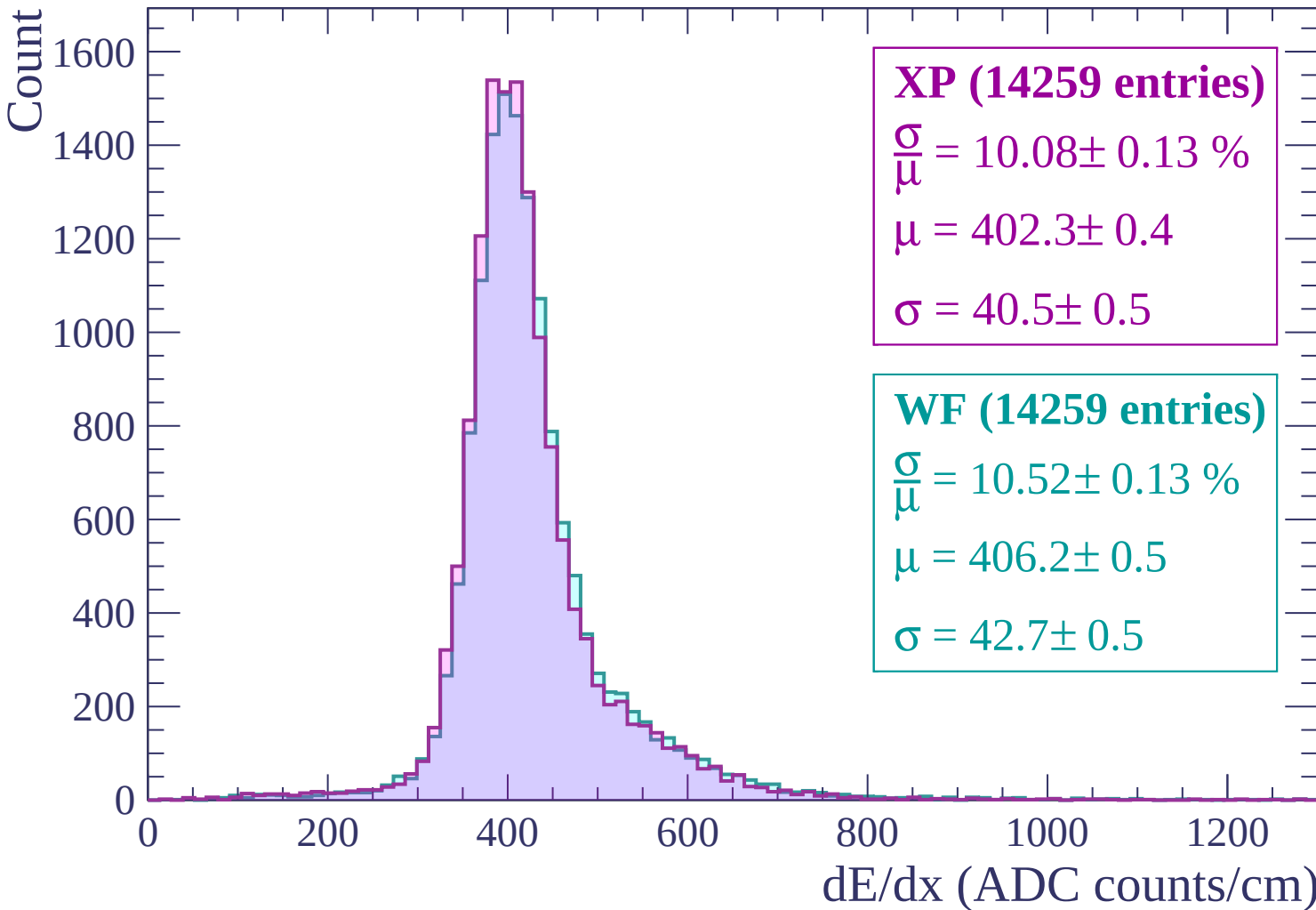
WF (14259 entries)

$\frac{\sigma}{\mu} = 10.52 \pm 0.13 \%$

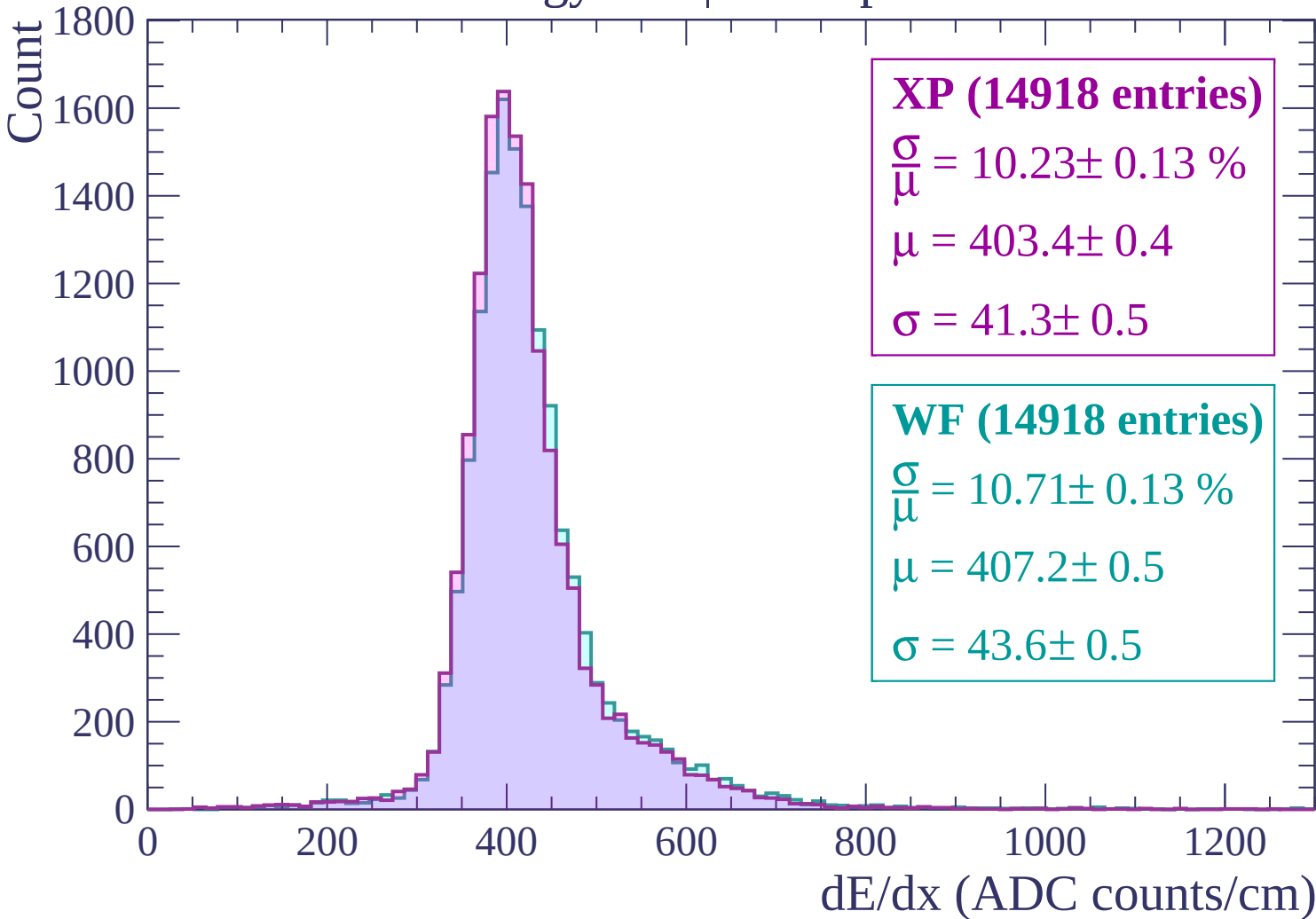
$\mu = 406.2 \pm 0.5$

$\sigma = 42.7 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $420 < p < 480$



Energy loss | $480 < p < 540$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14549 entries)

$\frac{\sigma}{\mu} = 10.18 \pm 0.12 \%$

$\mu = 408.4 \pm 0.4$

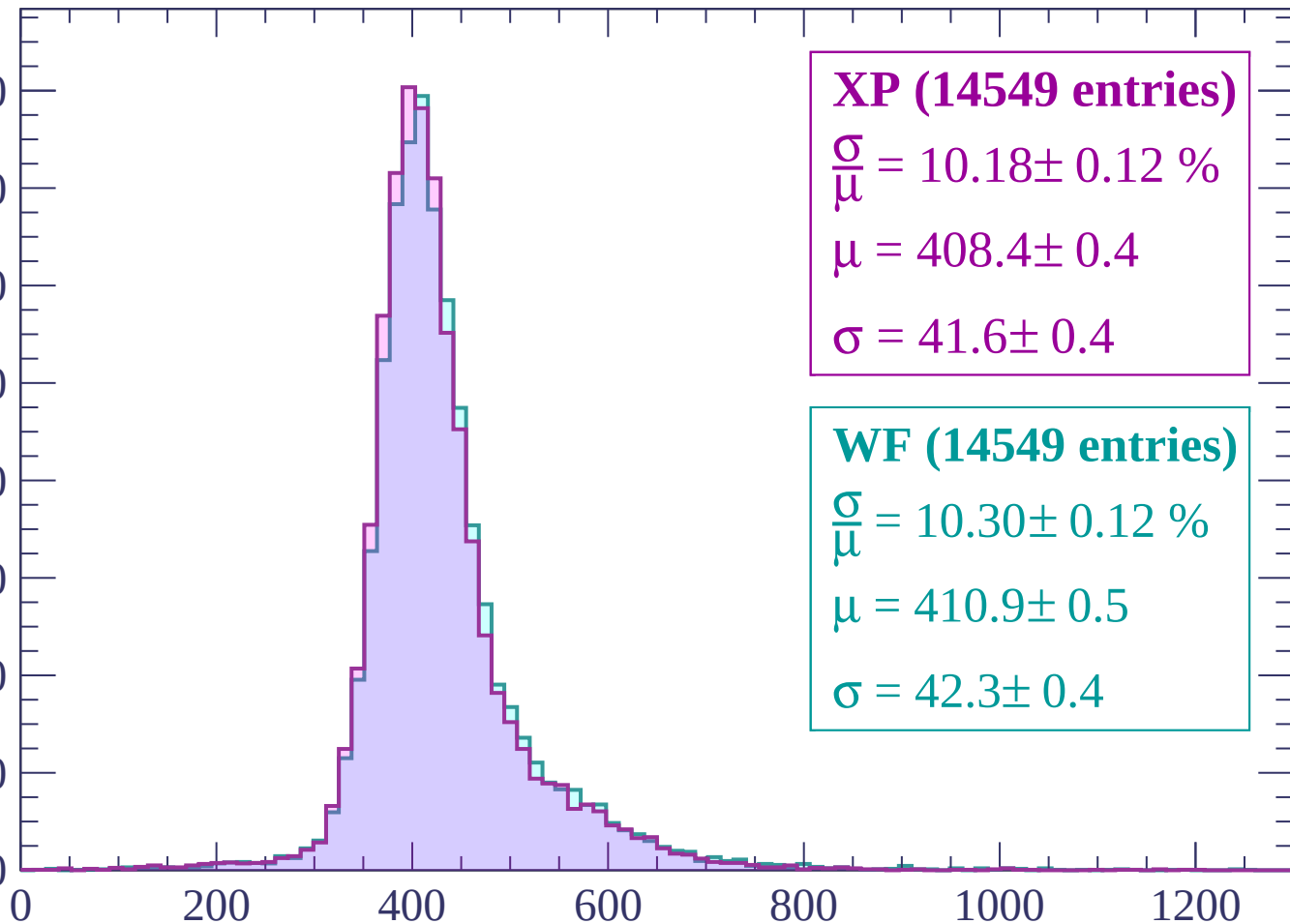
$\sigma = 41.6 \pm 0.4$

WF (14549 entries)

$\frac{\sigma}{\mu} = 10.30 \pm 0.12 \%$

$\mu = 410.9 \pm 0.5$

$\sigma = 42.3 \pm 0.4$



dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14358 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.11 \%$$

$$\mu = 411.4 \pm 0.4$$

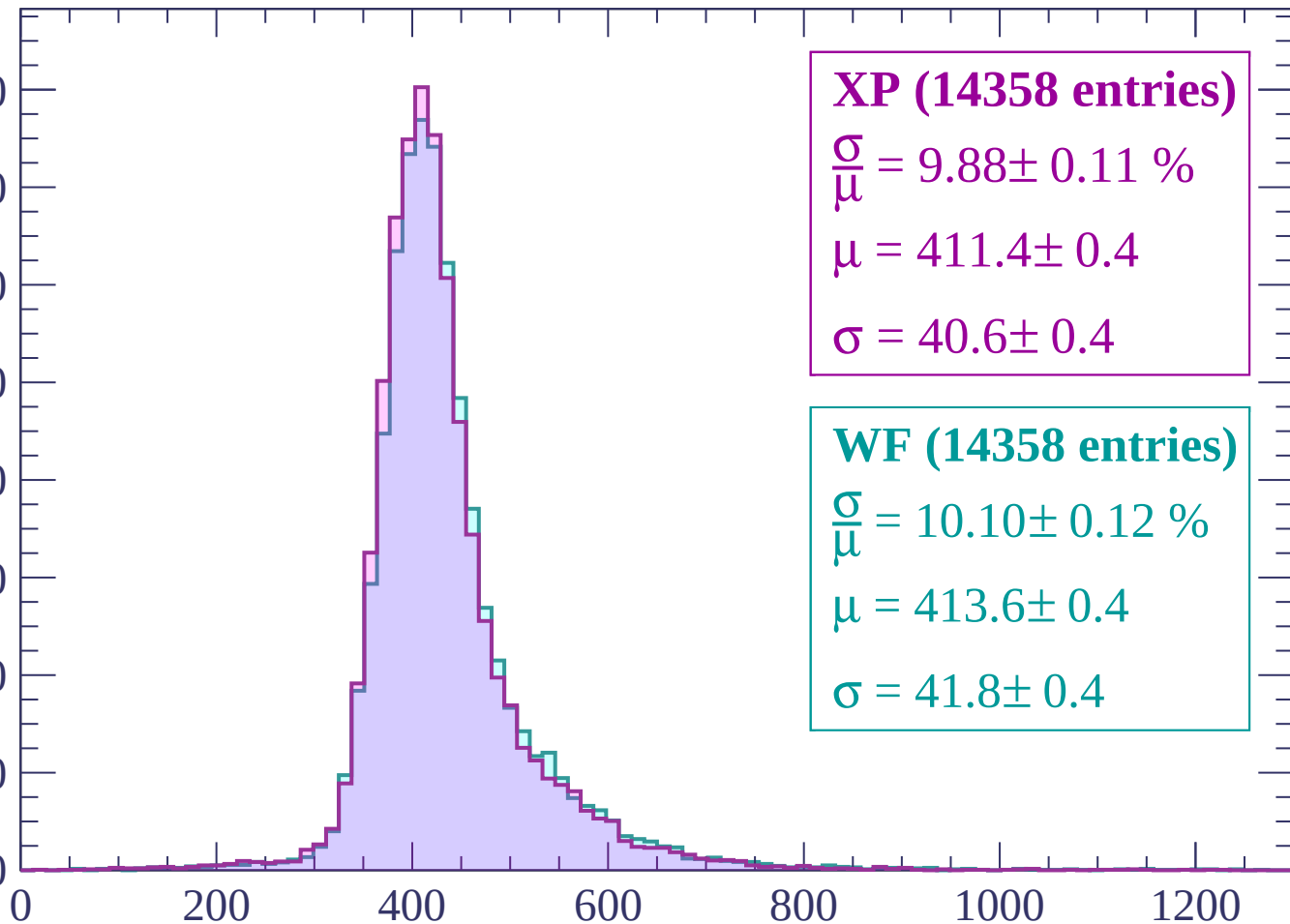
$$\sigma = 40.6 \pm 0.4$$

WF (14358 entries)

$$\frac{\sigma}{\mu} = 10.10 \pm 0.12 \%$$

$$\mu = 413.6 \pm 0.4$$

$$\sigma = 41.8 \pm 0.4$$



dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13897 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.13 \%$$

$$\mu = 415.3 \pm 0.5$$

$$\sigma = 41.0 \pm 0.5$$

WF (13897 entries)

$$\frac{\sigma}{\mu} = 10.36 \pm 0.13 \%$$

$$\mu = 418.7 \pm 0.5$$

$$\sigma = 43.4 \pm 0.5$$

0

200

400

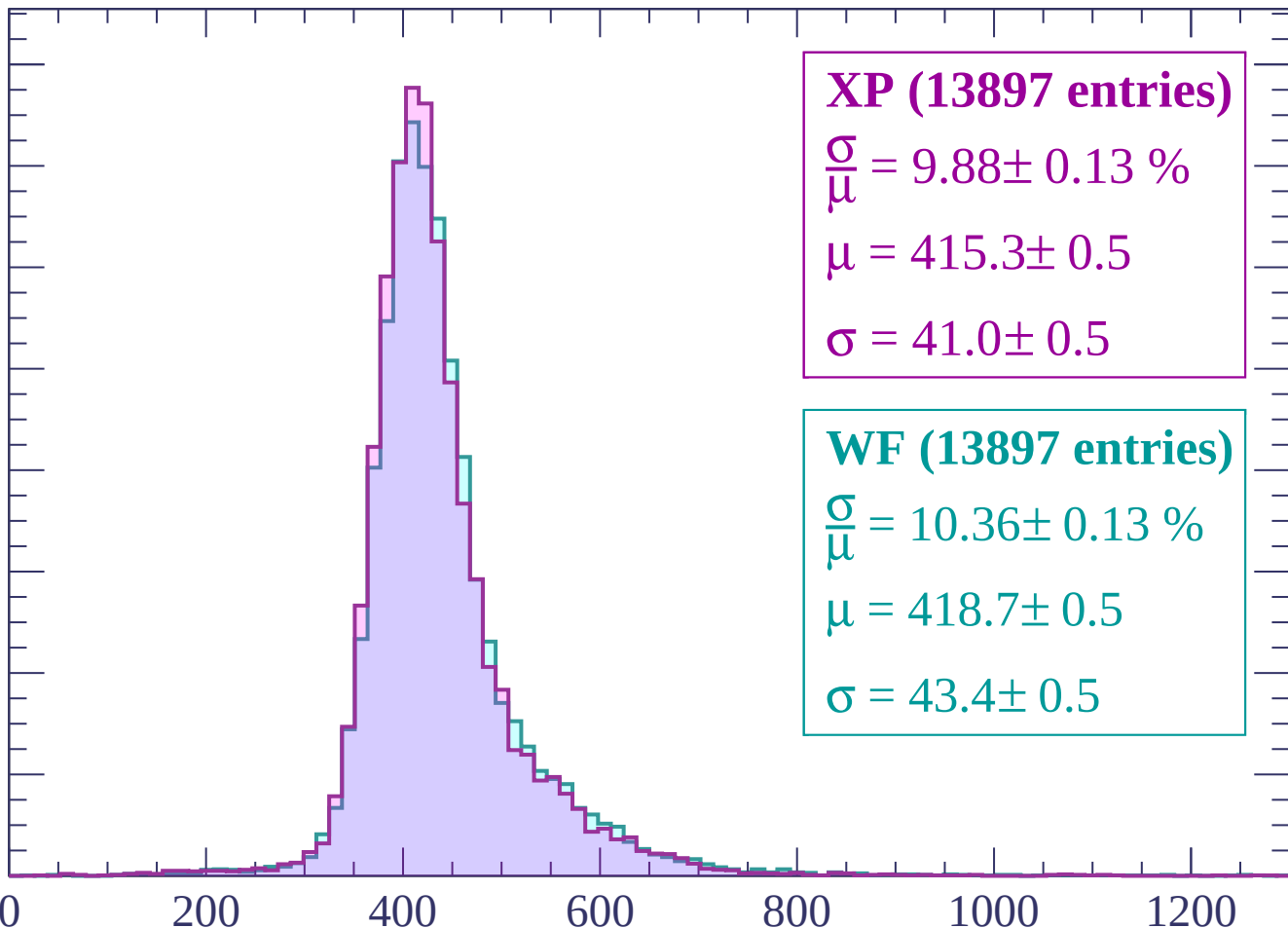
600

800

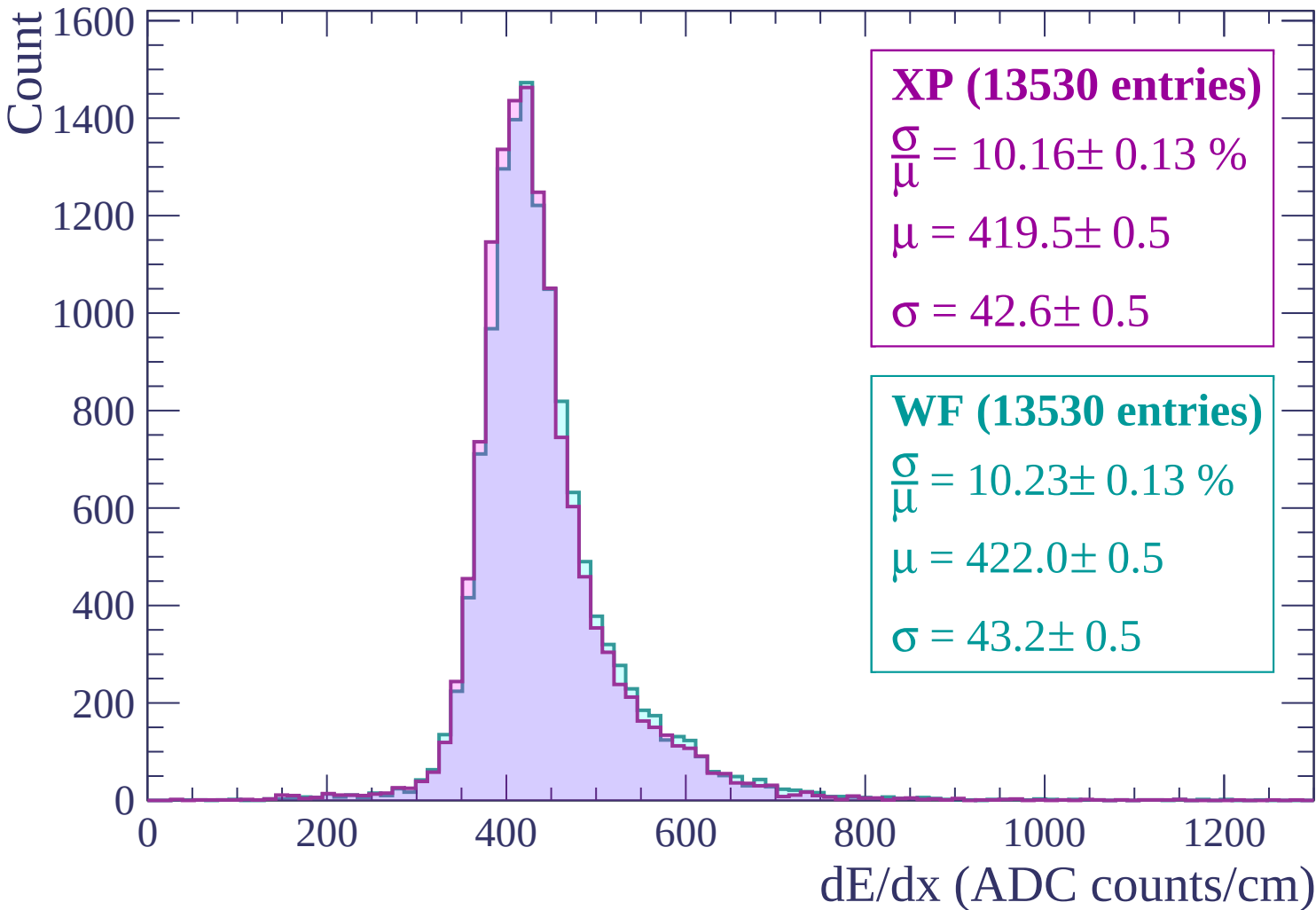
1000

1200

dE/dx (ADC counts/cm)



Energy loss | $660 < p < 720$



Energy loss | $720 < p < 780$

Count

1400
1200
1000
800
600
400
200
0

XP (13330 entries)

$$\frac{\sigma}{\mu} = 10.14 \pm 0.13 \%$$

$$\mu = 423.8 \pm 0.5$$

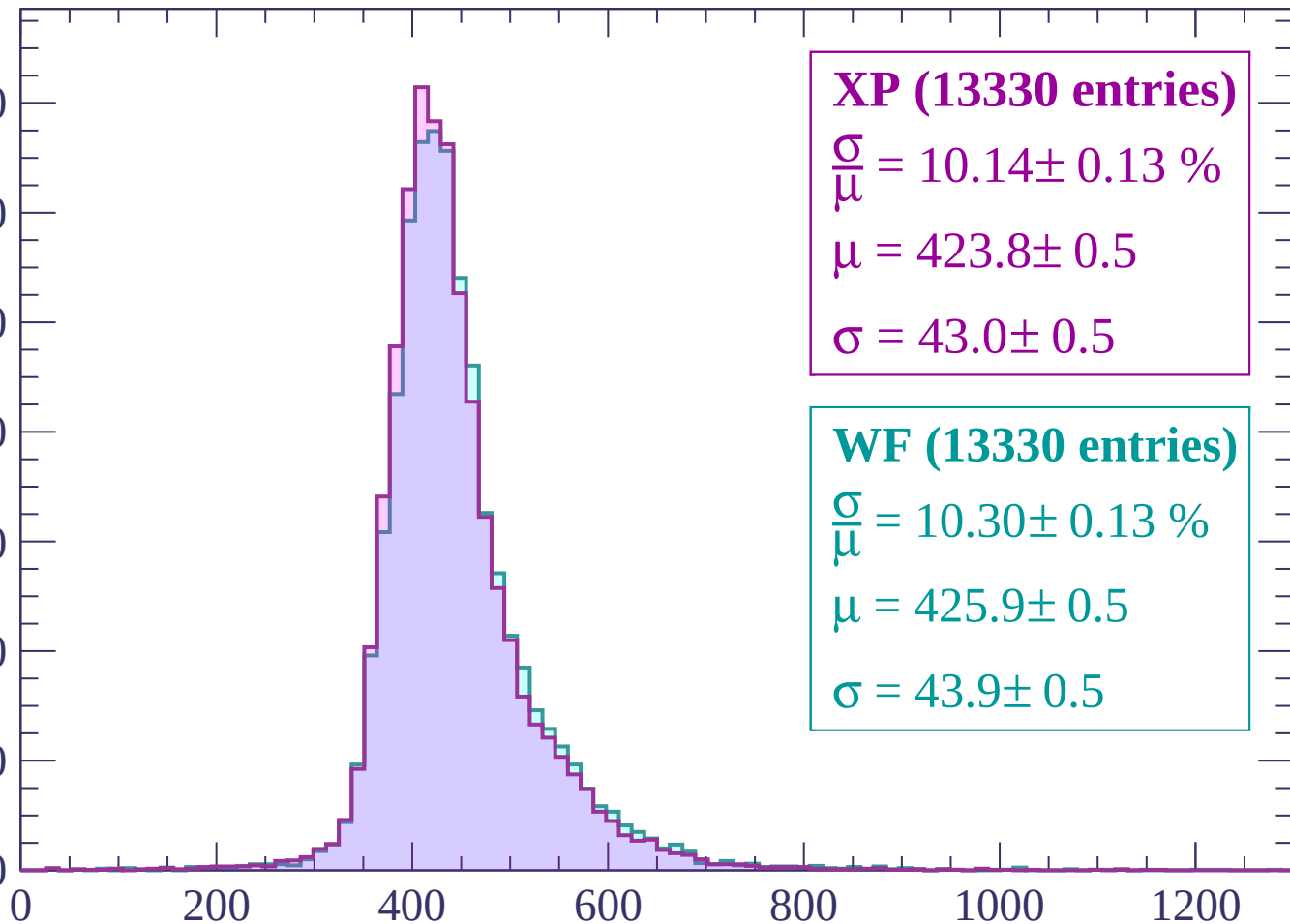
$$\sigma = 43.0 \pm 0.5$$

WF (13330 entries)

$$\frac{\sigma}{\mu} = 10.30 \pm 0.13 \%$$

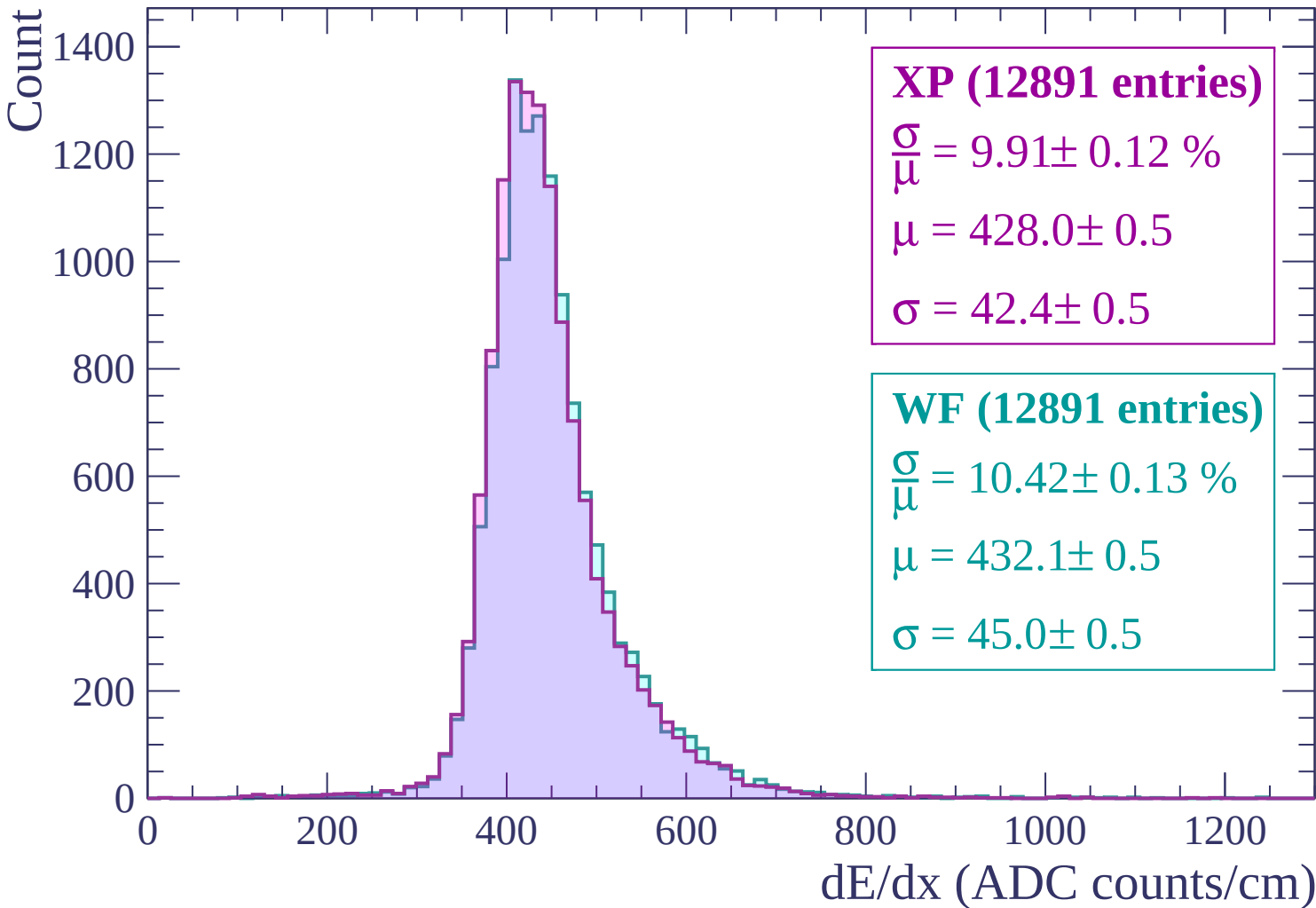
$$\mu = 425.9 \pm 0.5$$

$$\sigma = 43.9 \pm 0.5$$



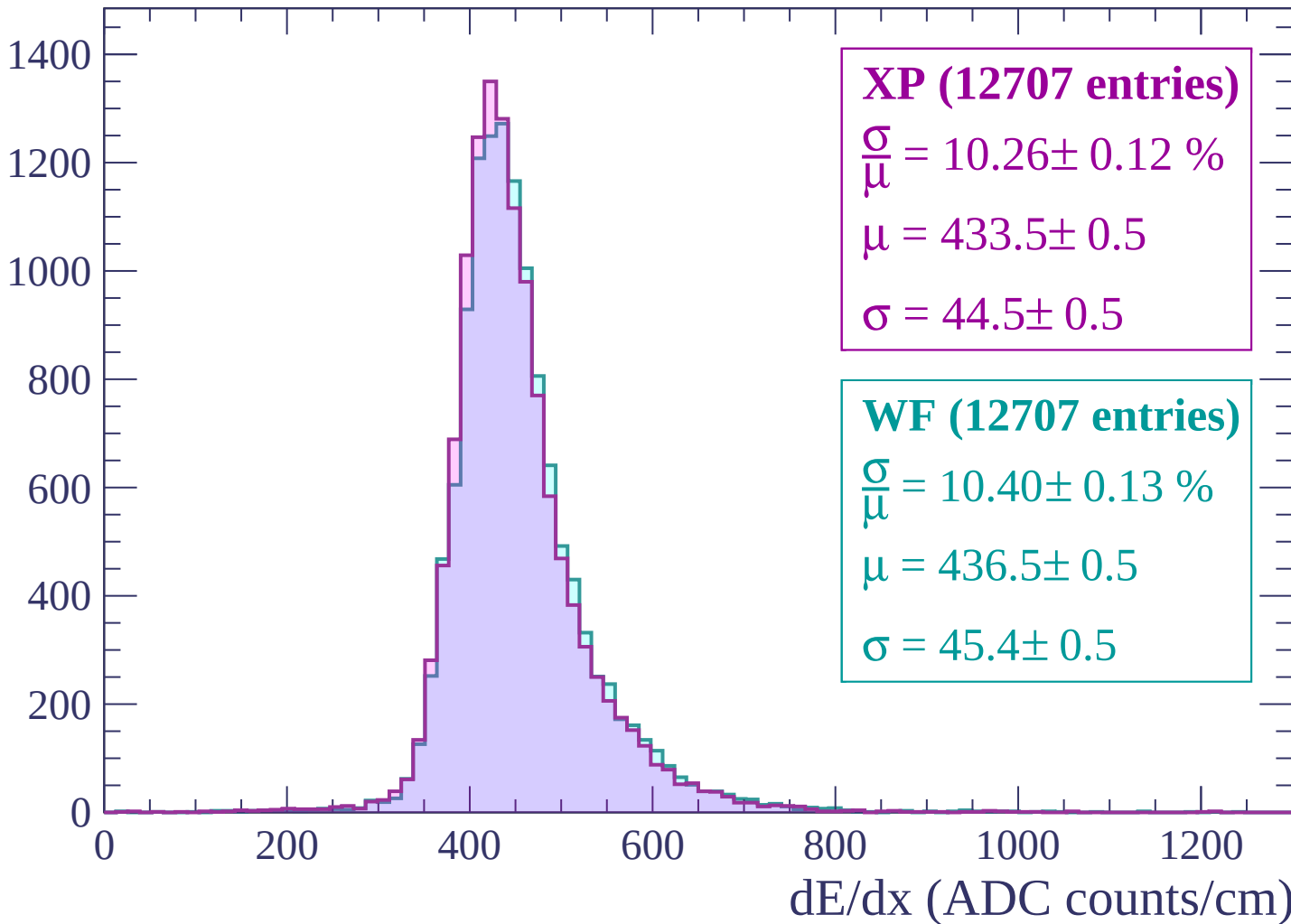
dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

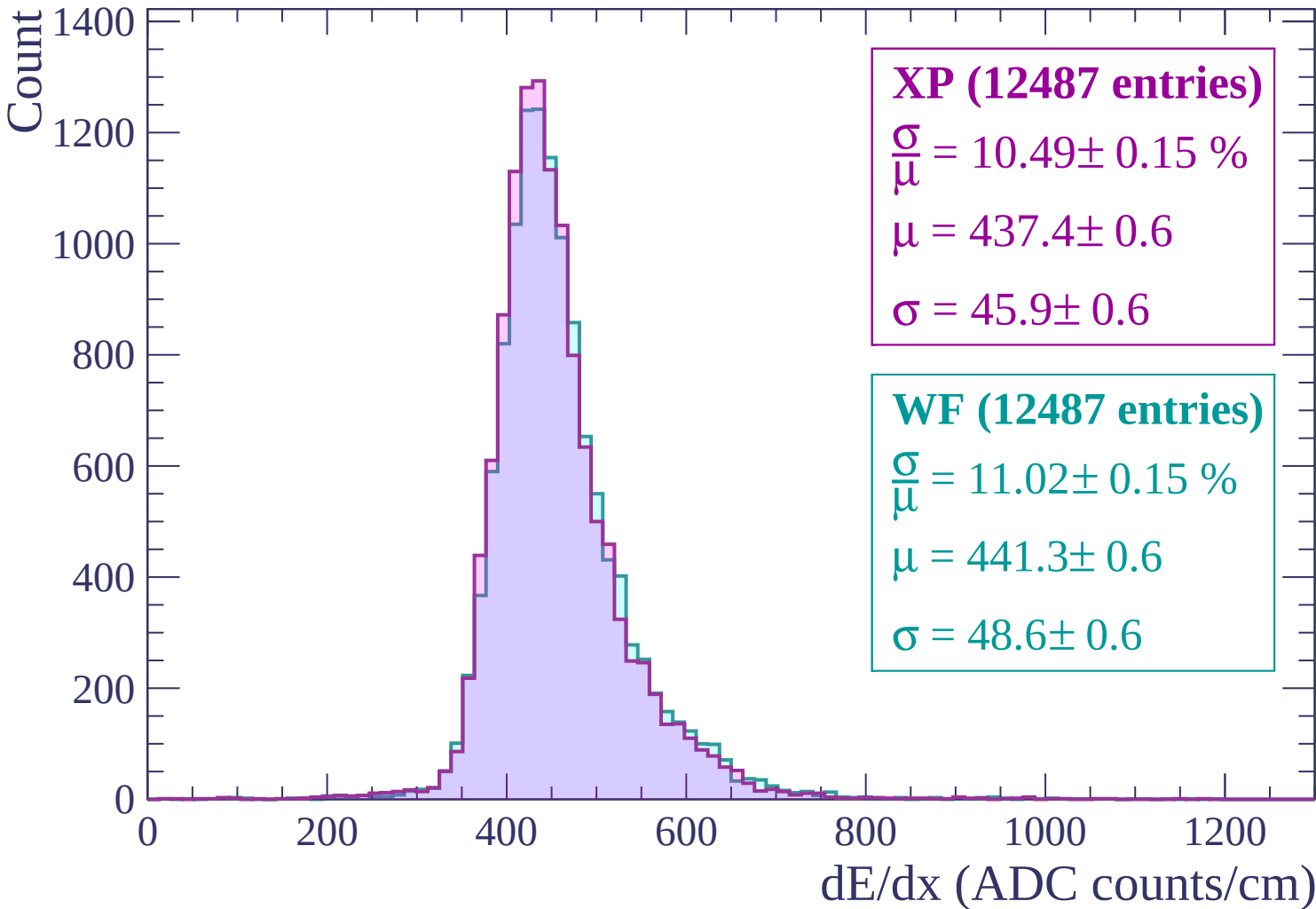


Energy loss | $840 < p < 900$

Count



Energy loss | $900 < p < 960$



Energy loss | $960 < p < 1020$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (11767 entries)

$\frac{\sigma}{\mu} = 10.54 \pm 0.15 \%$

$\mu = 442.1 \pm 0.6$

$\sigma = 46.6 \pm 0.6$

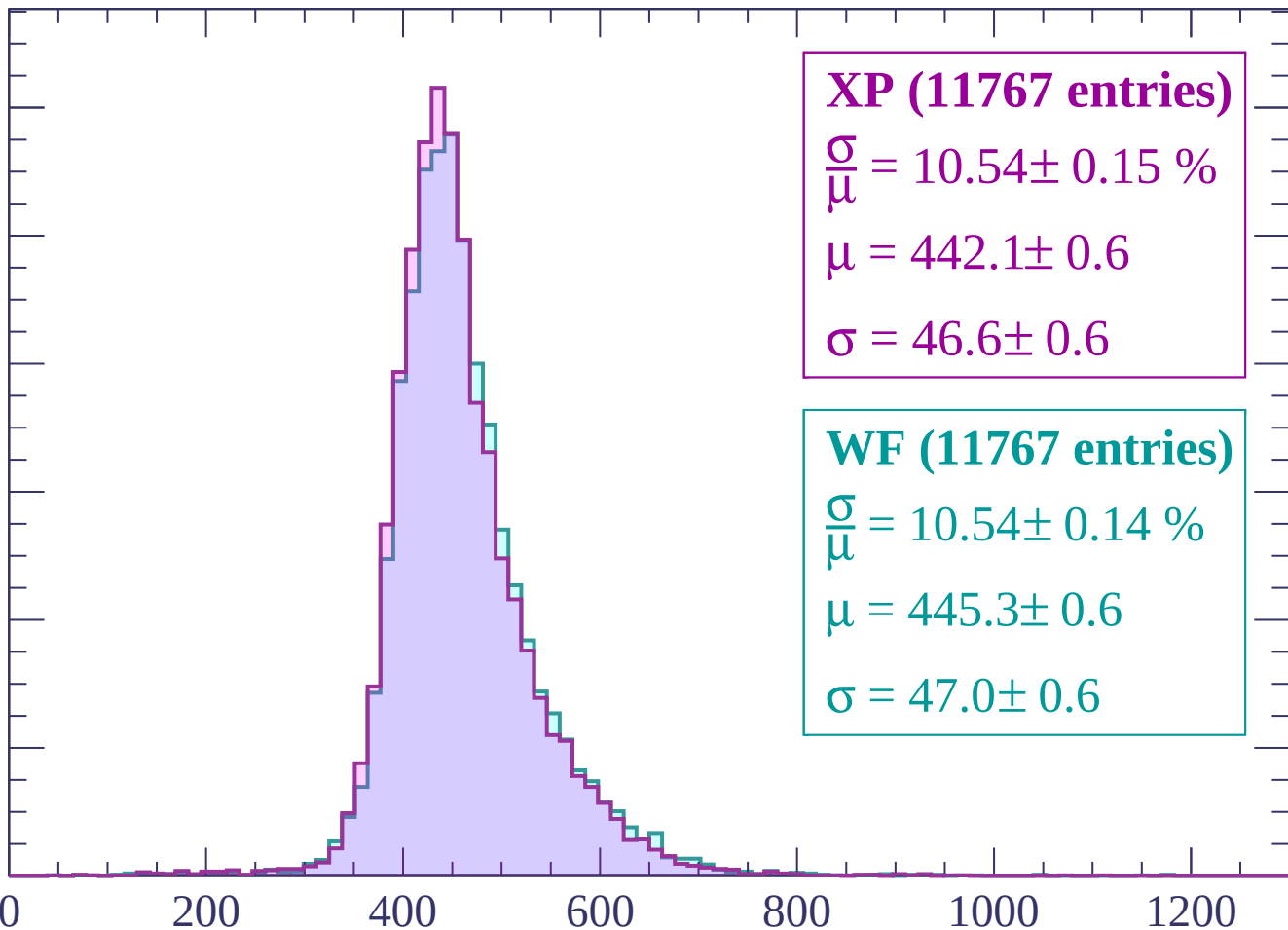
WF (11767 entries)

$\frac{\sigma}{\mu} = 10.54 \pm 0.14 \%$

$\mu = 445.3 \pm 0.6$

$\sigma = 47.0 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $1020 < p < 1080$

Count

1200

1000

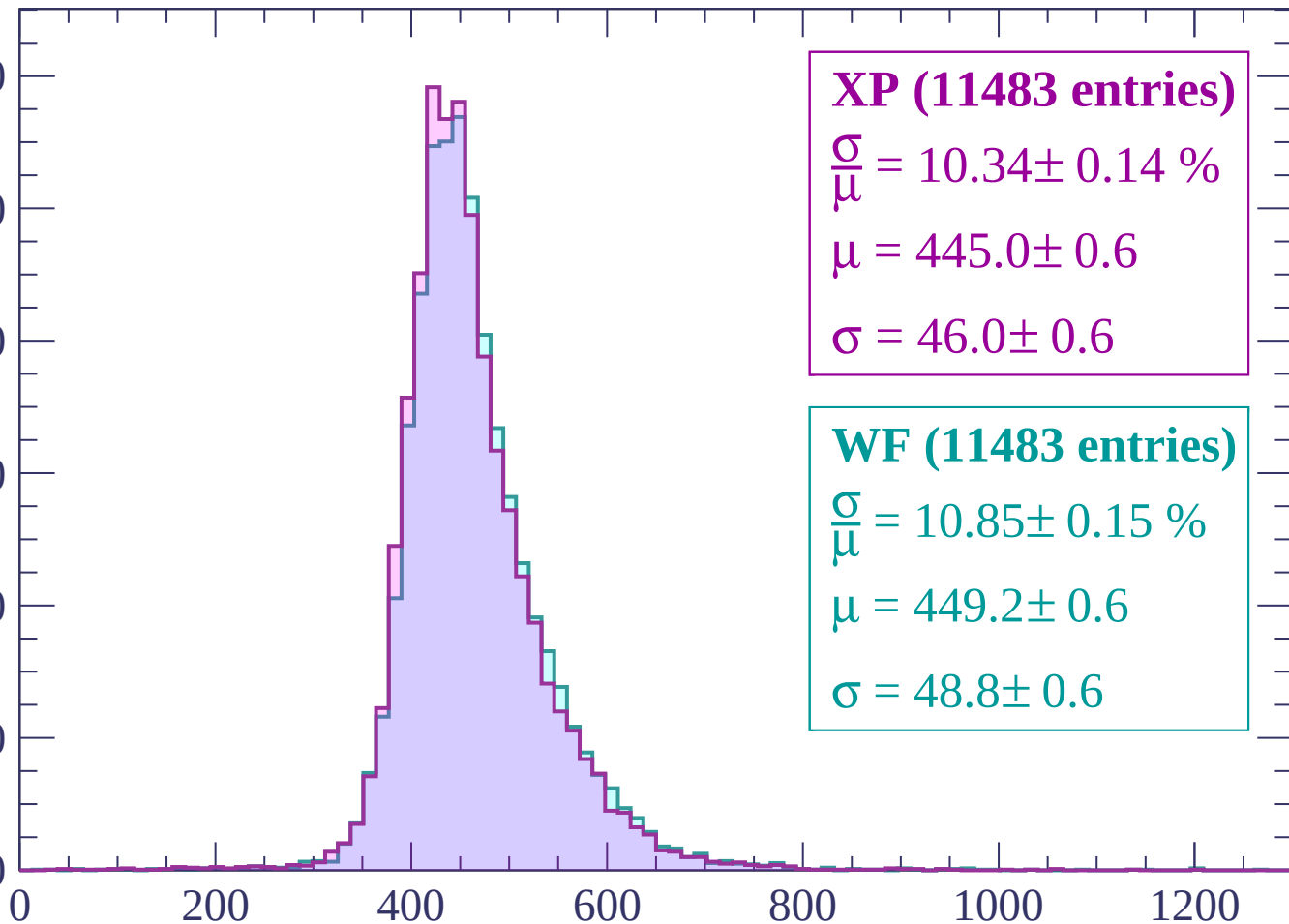
800

600

400

200

0



XP (11483 entries)

$\frac{\sigma}{\mu} = 10.34 \pm 0.14 \%$

$\mu = 445.0 \pm 0.6$

$\sigma = 46.0 \pm 0.6$

WF (11483 entries)

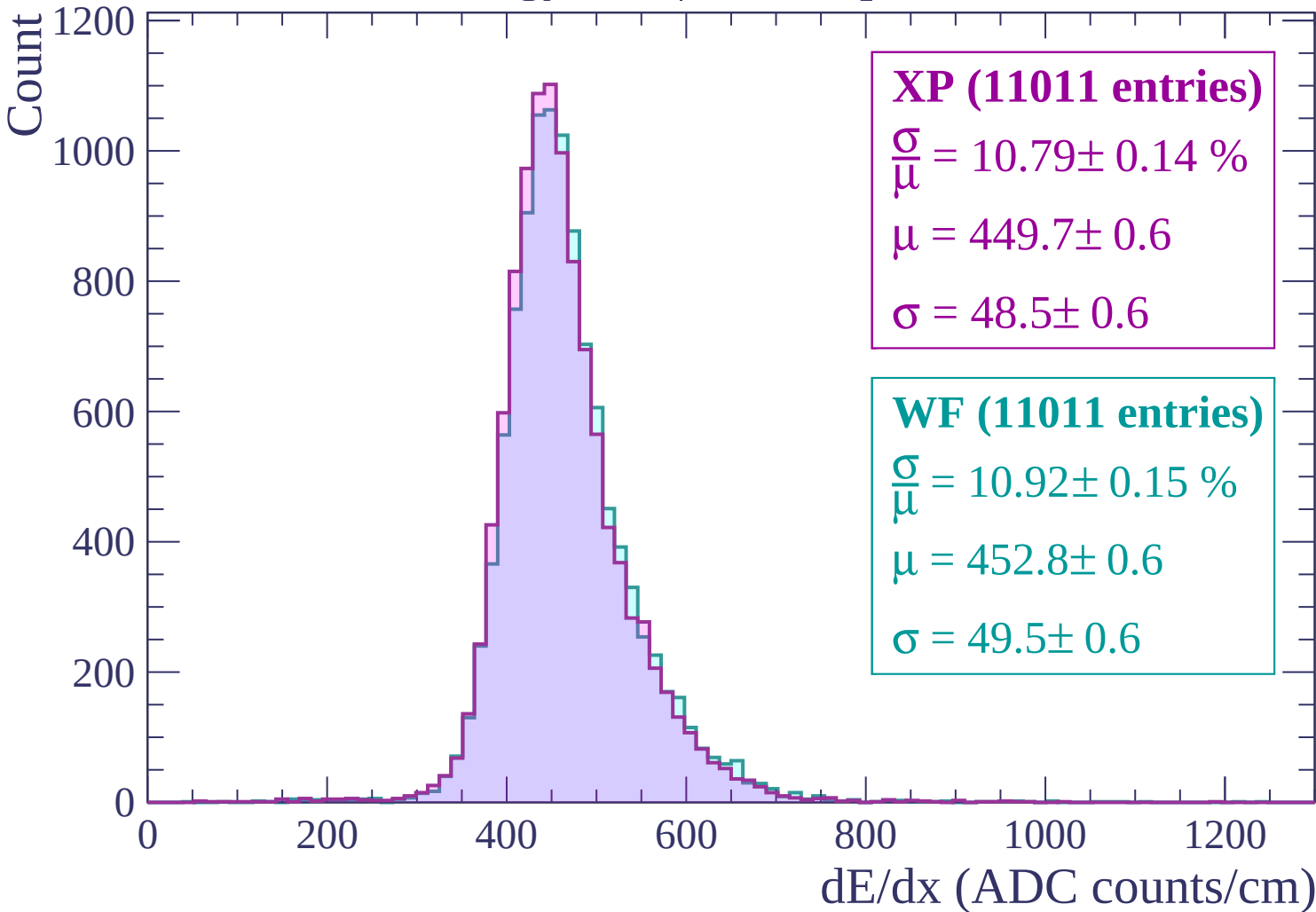
$\frac{\sigma}{\mu} = 10.85 \pm 0.15 \%$

$\mu = 449.2 \pm 0.6$

$\sigma = 48.8 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$



Energy loss | $1140 < p < 1200$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (10611 entries)

$\frac{\sigma}{\mu} = 10.54 \pm 0.13 \%$

$\mu = 452.1 \pm 0.6$

$\sigma = 47.6 \pm 0.5$

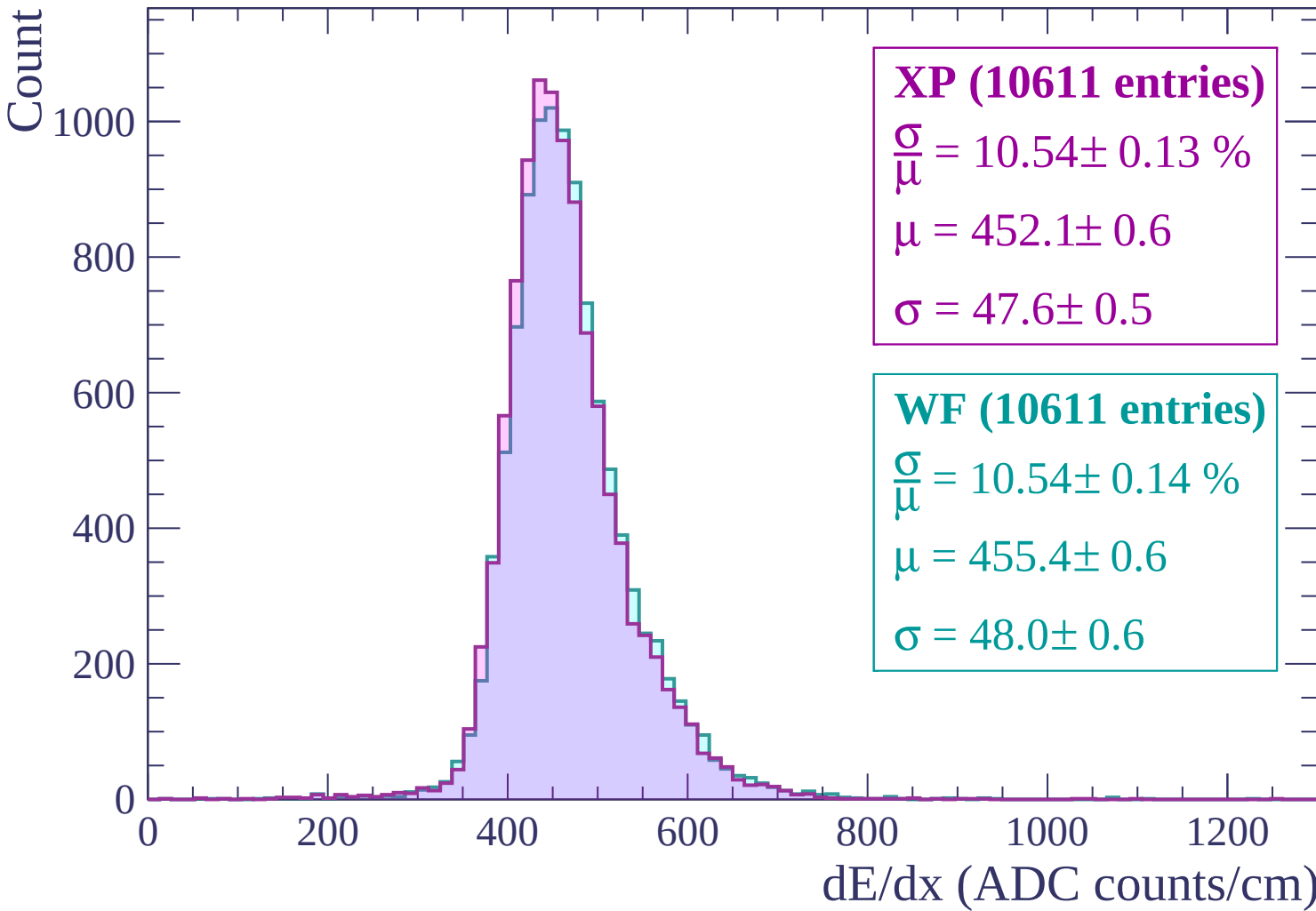
WF (10611 entries)

$\frac{\sigma}{\mu} = 10.54 \pm 0.14 \%$

$\mu = 455.4 \pm 0.6$

$\sigma = 48.0 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $1200 < p < 1260$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (10408 entries)

$\frac{\sigma}{\mu} = 11.08 \pm 0.15 \%$

$\mu = 457.3 \pm 0.6$

$\sigma = 50.7 \pm 0.6$

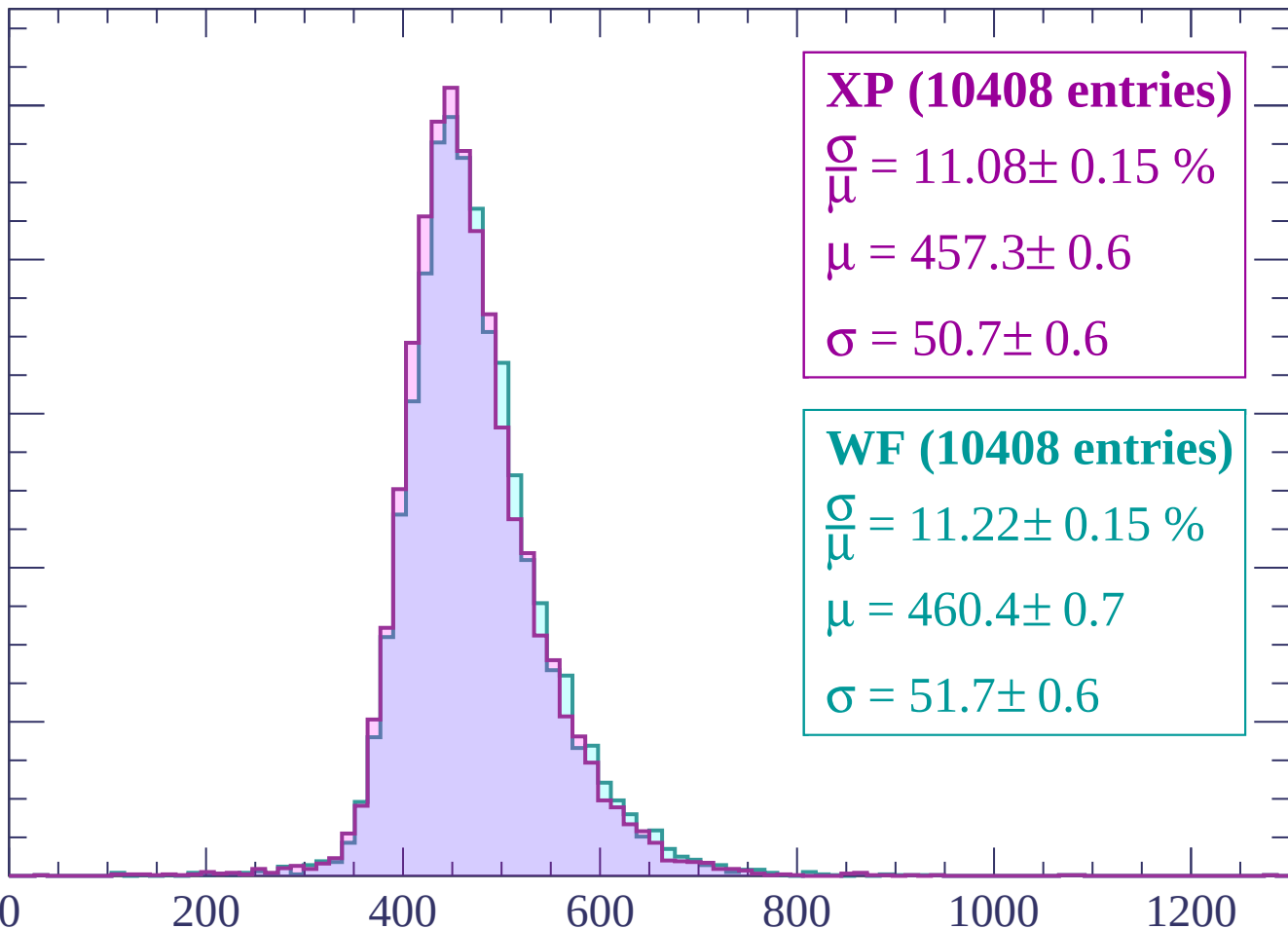
WF (10408 entries)

$\frac{\sigma}{\mu} = 11.22 \pm 0.15 \%$

$\mu = 460.4 \pm 0.7$

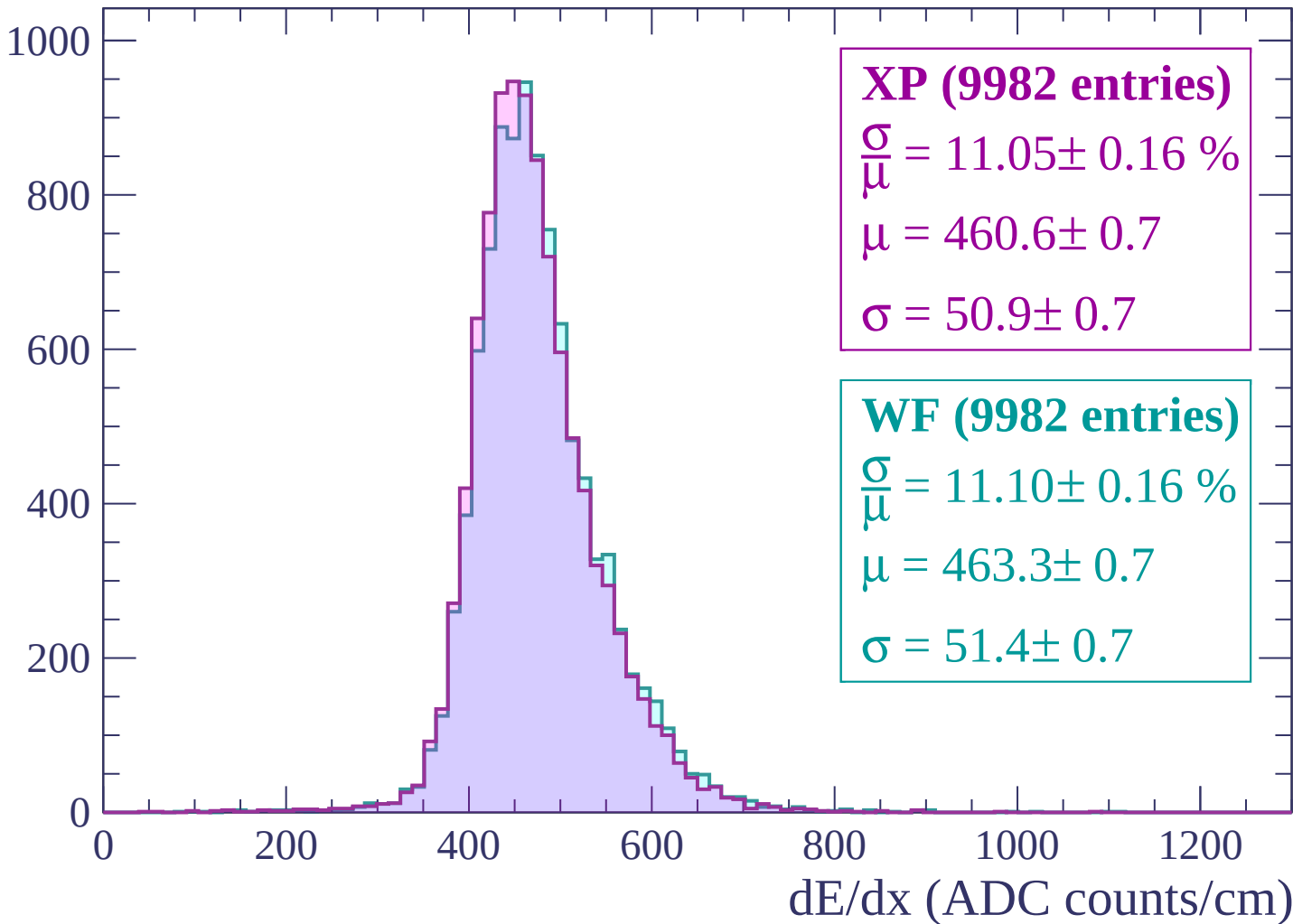
$\sigma = 51.7 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $1260 < p < 1320$

Count



Energy loss | $1320 < p < 1380$

Count

1000

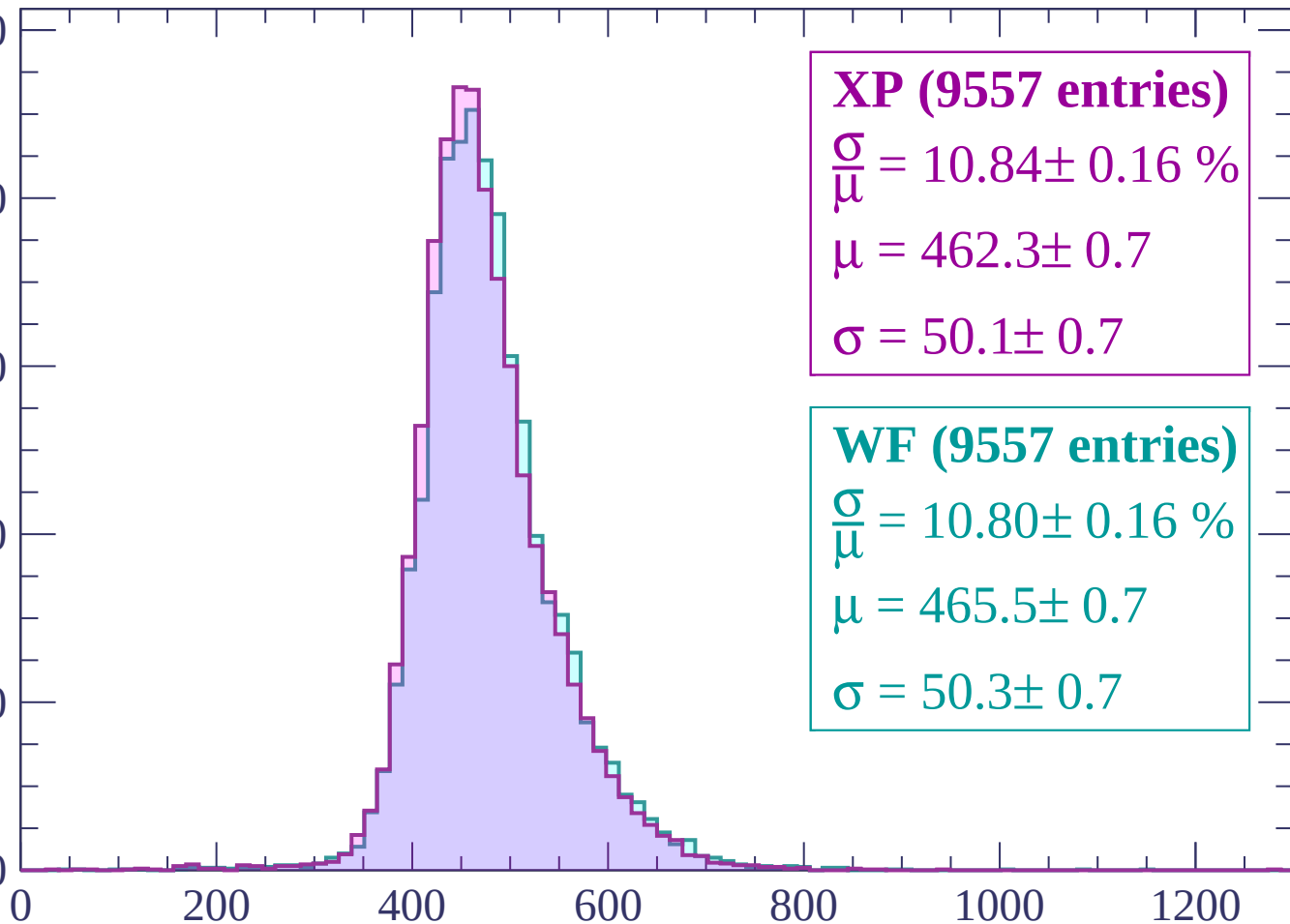
800

600

400

200

0



XP (9557 entries)

$\frac{\sigma}{\mu} = 10.84 \pm 0.16 \%$

$\mu = 462.3 \pm 0.7$

$\sigma = 50.1 \pm 0.7$

WF (9557 entries)

$\frac{\sigma}{\mu} = 10.80 \pm 0.16 \%$

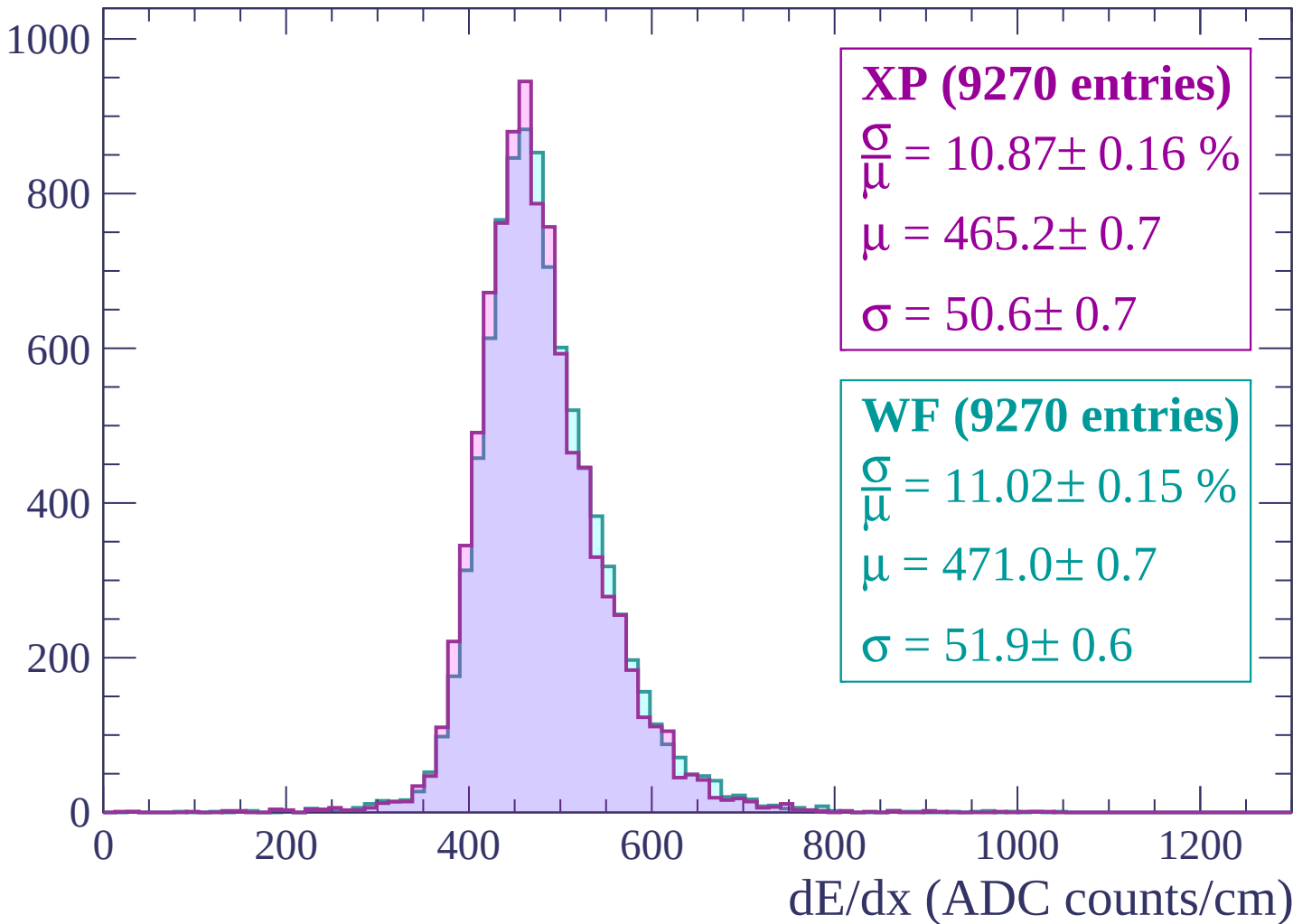
$\mu = 465.5 \pm 0.7$

$\sigma = 50.3 \pm 0.7$

dE/dx (ADC counts/cm)

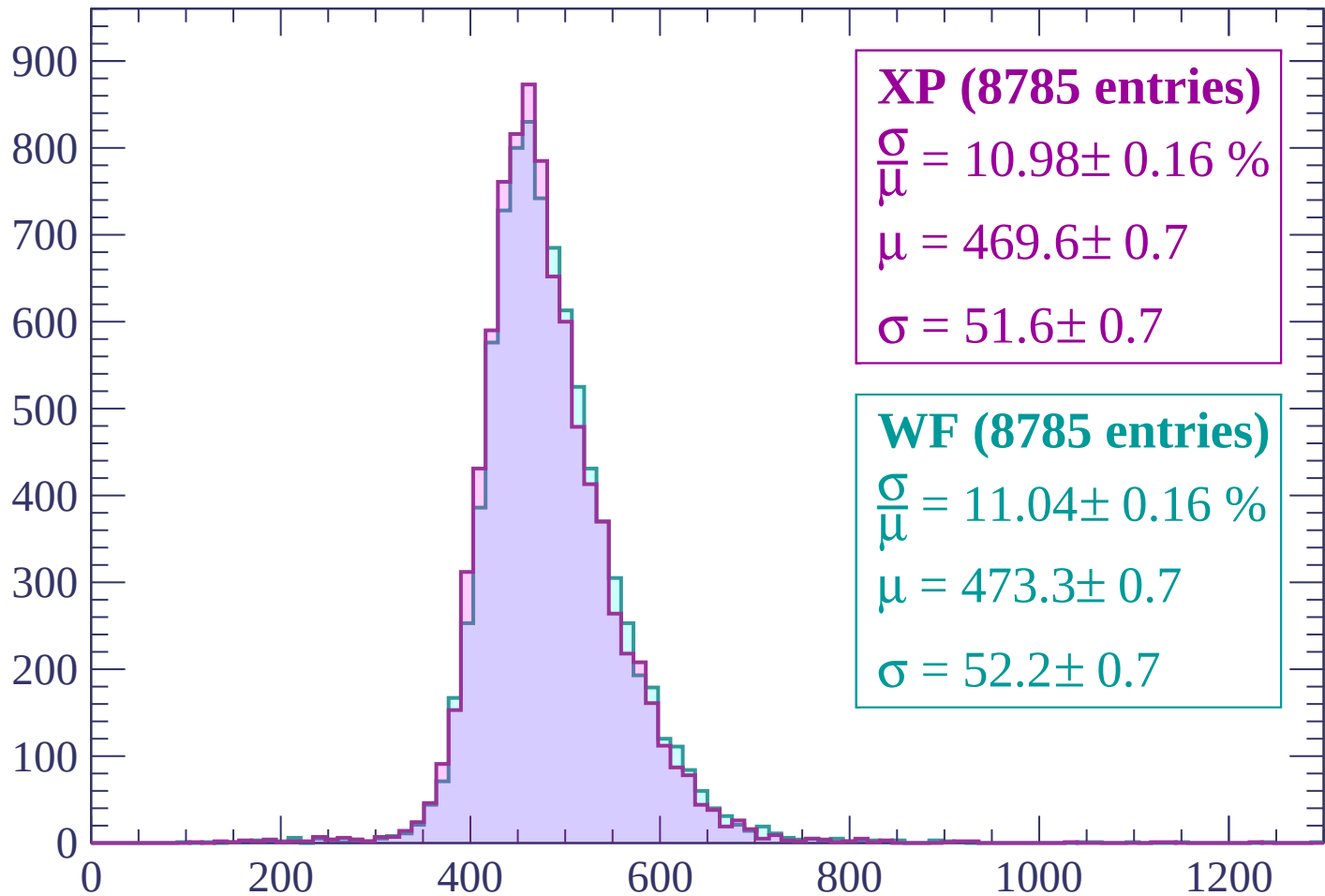
Energy loss | $1380 < p < 1440$

Count



Energy loss | $1440 < p < 1500$

Count



XP (8785 entries)

$\frac{\sigma}{\mu} = 10.98 \pm 0.16 \%$

$\mu = 469.6 \pm 0.7$

$\sigma = 51.6 \pm 0.7$

WF (8785 entries)

$\frac{\sigma}{\mu} = 11.04 \pm 0.16 \%$

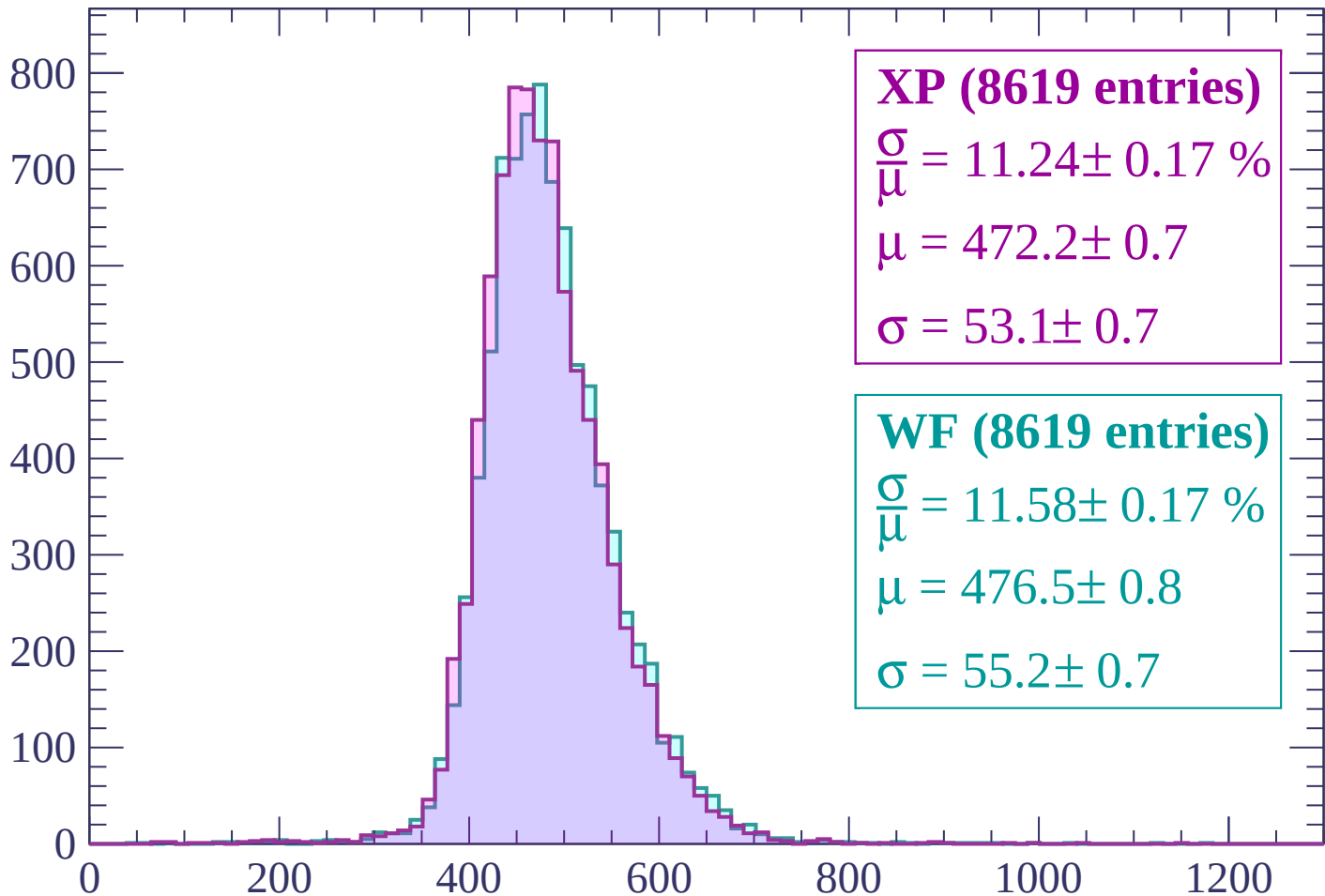
$\mu = 473.3 \pm 0.7$

$\sigma = 52.2 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP (8619 entries)

$$\frac{\sigma}{\mu} = 11.24 \pm 0.17 \%$$

$$\mu = 472.2 \pm 0.7$$

$$\sigma = 53.1 \pm 0.7$$

WF (8619 entries)

$$\frac{\sigma}{\mu} = 11.58 \pm 0.17 \%$$

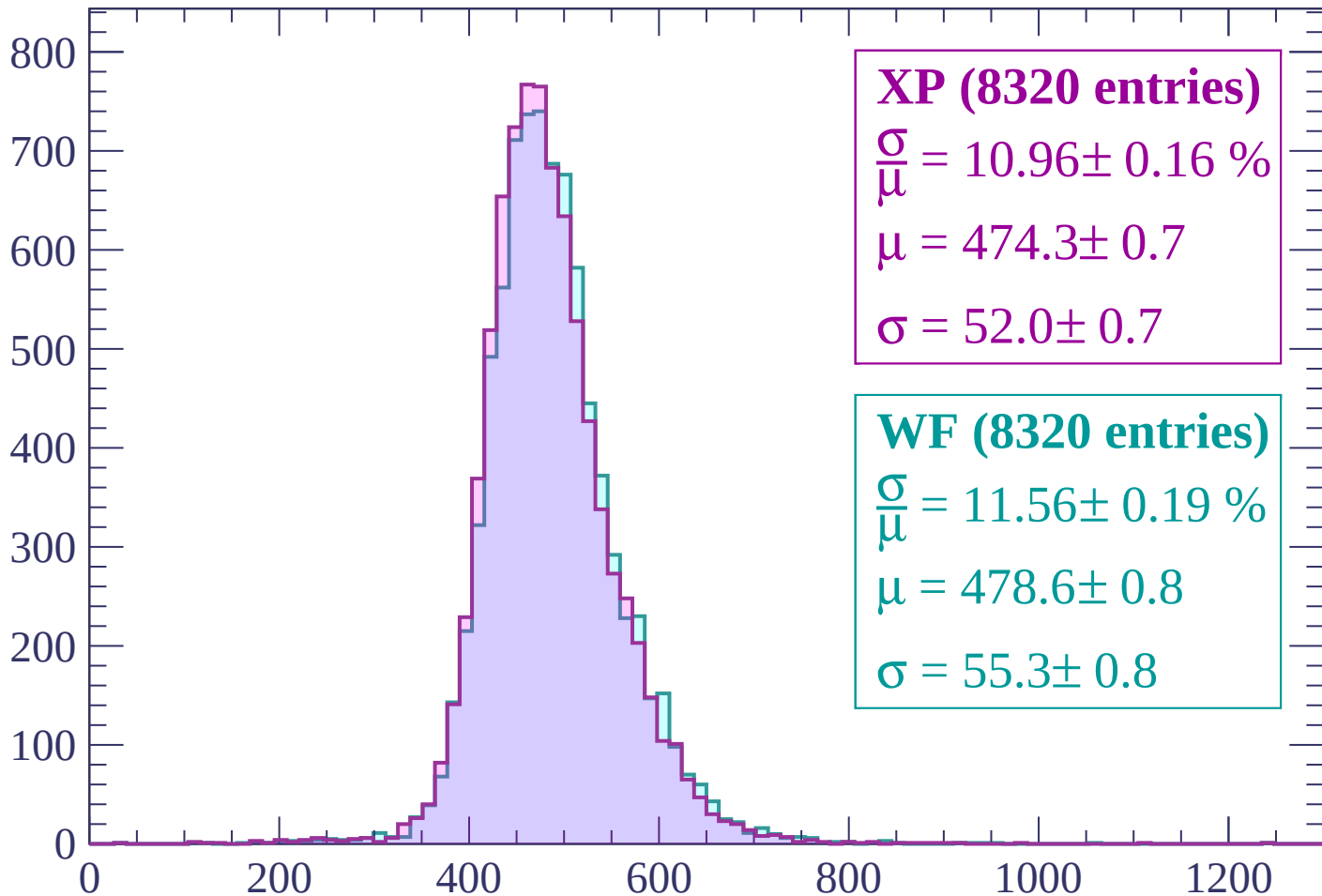
$$\mu = 476.5 \pm 0.8$$

$$\sigma = 55.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP (8320 entries)

$$\frac{\sigma}{\mu} = 10.96 \pm 0.16 \%$$

$$\mu = 474.3 \pm 0.7$$

$$\sigma = 52.0 \pm 0.7$$

WF (8320 entries)

$$\frac{\sigma}{\mu} = 11.56 \pm 0.19 \%$$

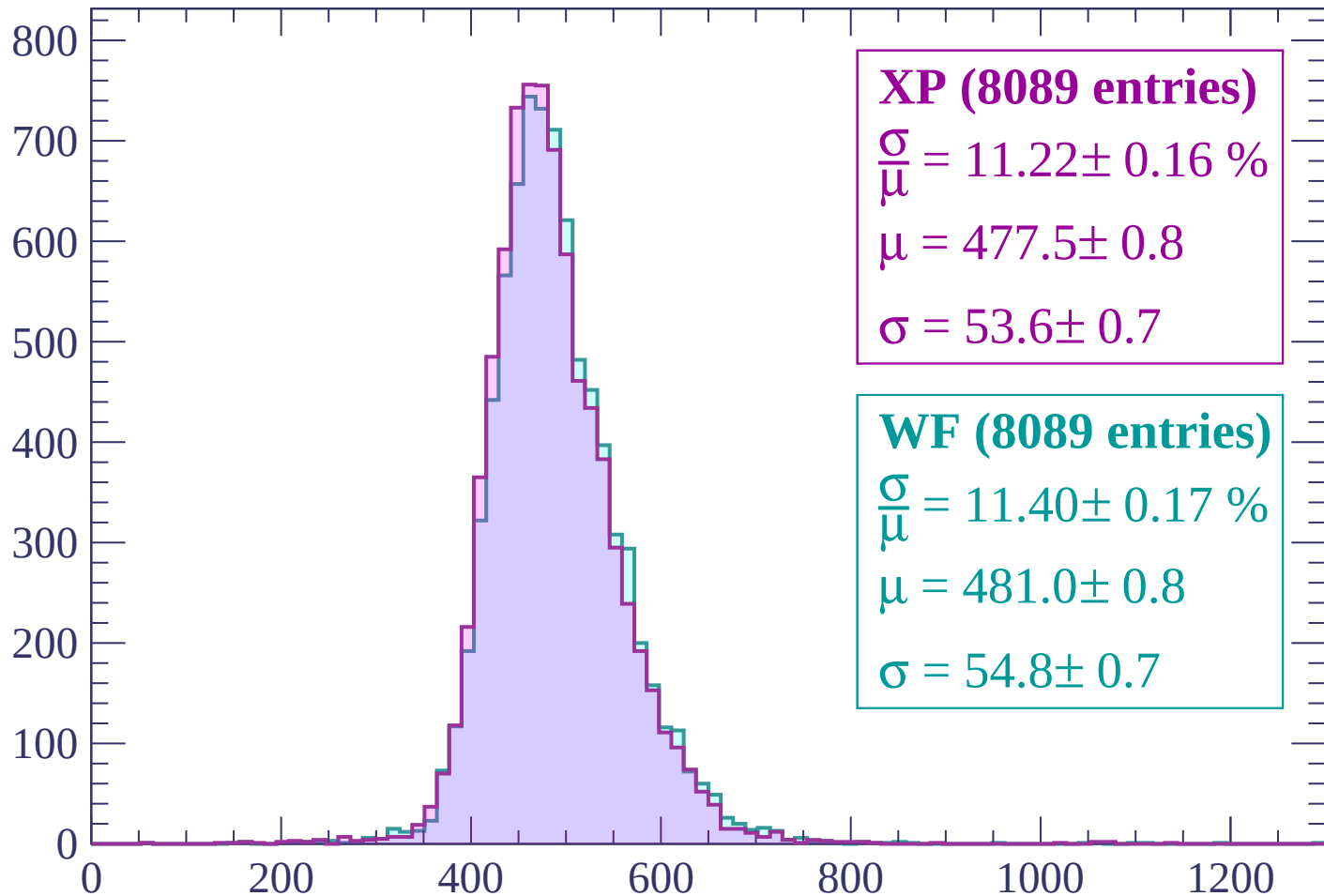
$$\mu = 478.6 \pm 0.8$$

$$\sigma = 55.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (8089 entries)

$\frac{\sigma}{\mu} = 11.22 \pm 0.16 \%$

$\mu = 477.5 \pm 0.8$

$\sigma = 53.6 \pm 0.7$

WF (8089 entries)

$\frac{\sigma}{\mu} = 11.40 \pm 0.17 \%$

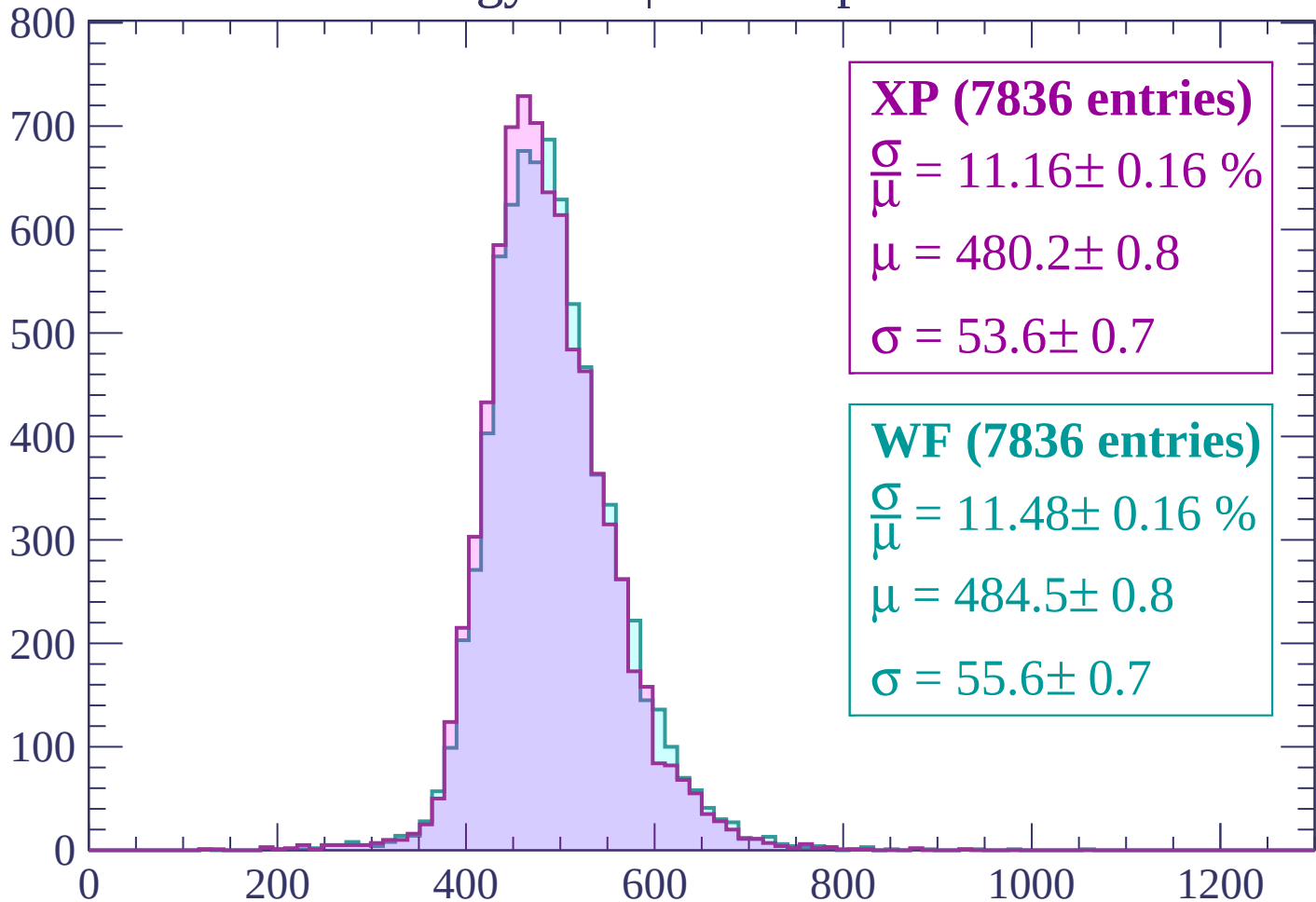
$\mu = 481.0 \pm 0.8$

$\sigma = 54.8 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (7836 entries)

$$\frac{\sigma}{\mu} = 11.16 \pm 0.16 \%$$

$$\mu = 480.2 \pm 0.8$$

$$\sigma = 53.6 \pm 0.7$$

WF (7836 entries)

$$\frac{\sigma}{\mu} = 11.48 \pm 0.16 \%$$

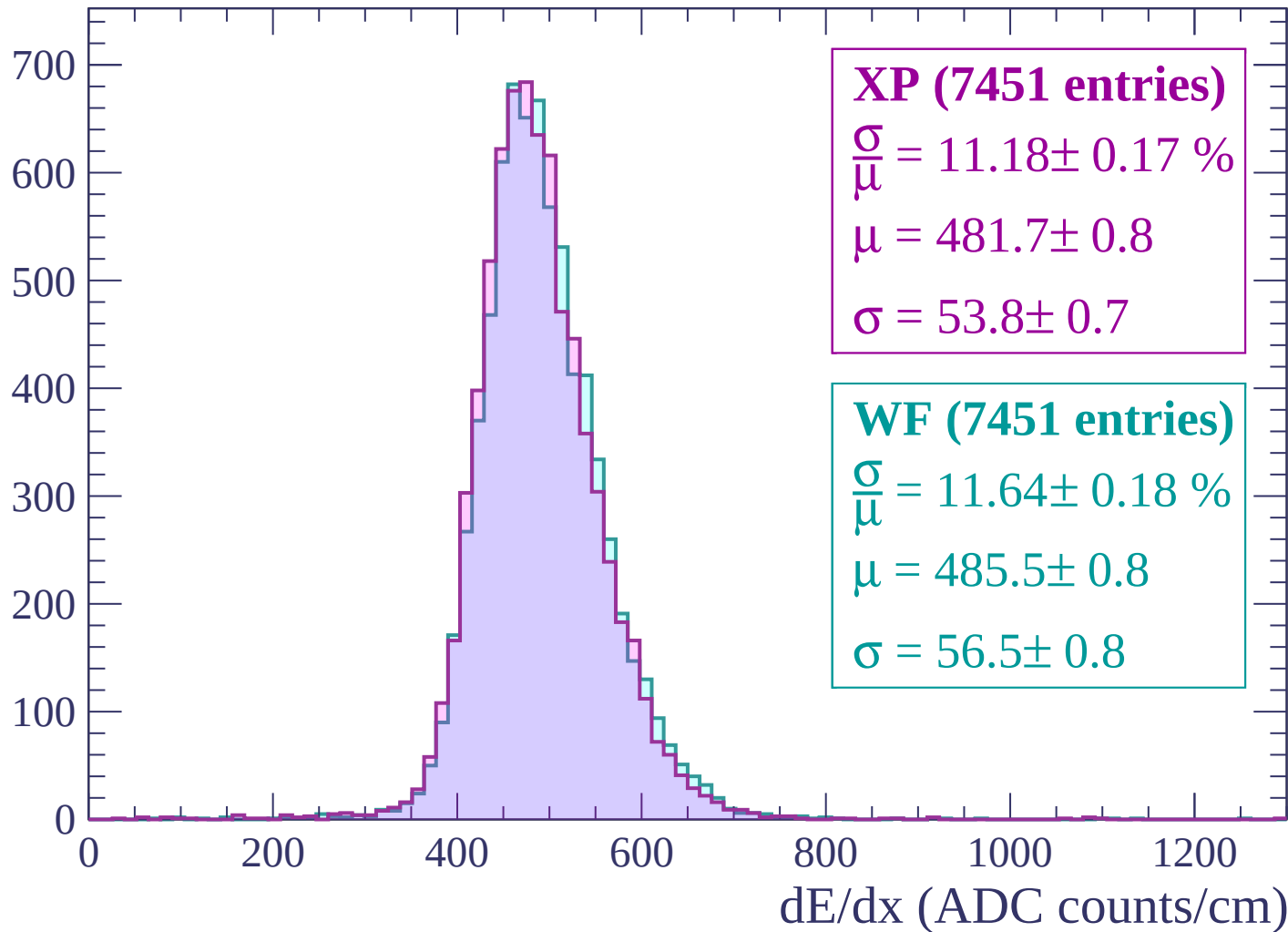
$$\mu = 484.5 \pm 0.8$$

$$\sigma = 55.6 \pm 0.7$$

dE/dx (ADC counts/cm)

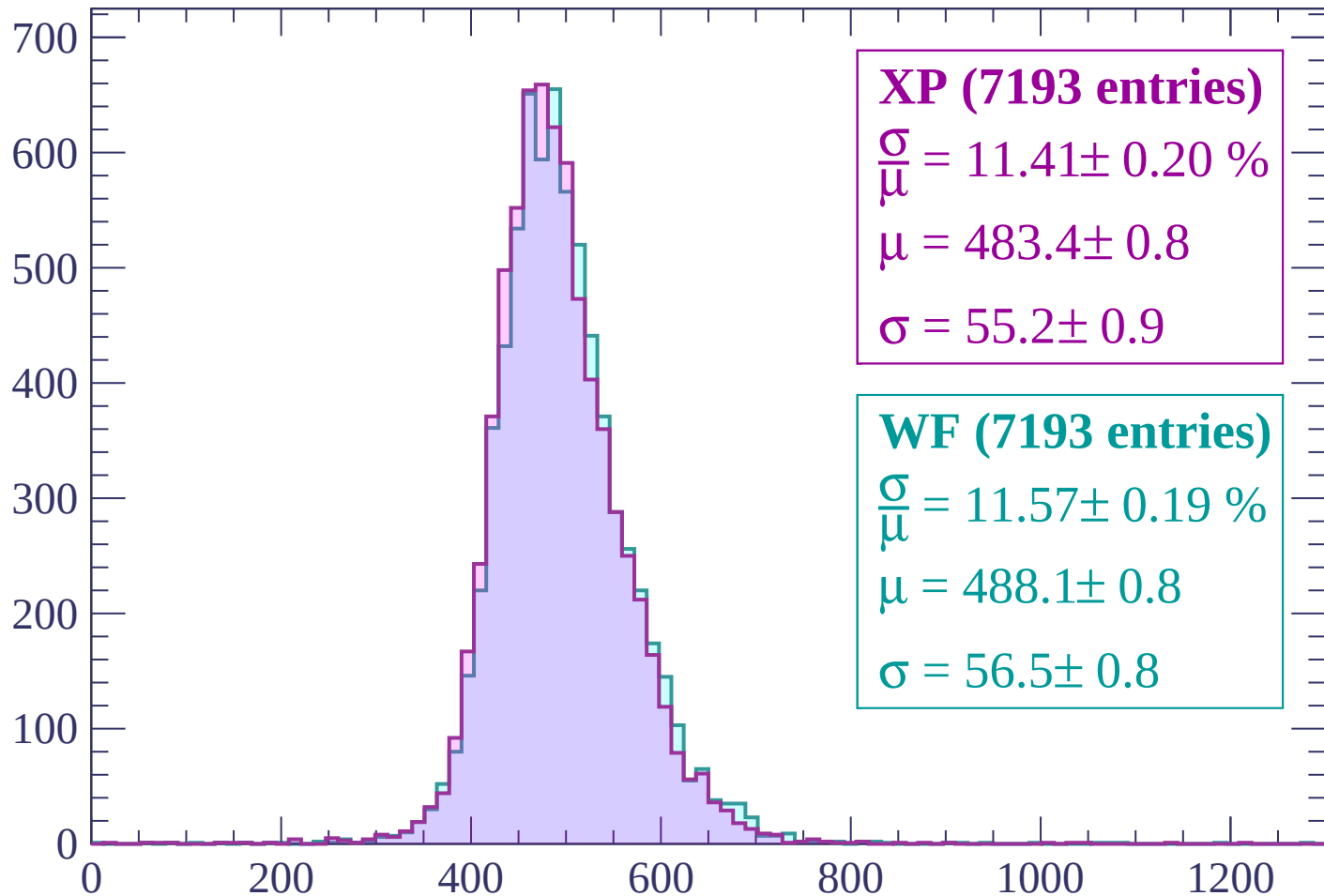
Energy loss | $1740 < p < 1800$

Count



Energy loss | $1800 < p < 1860$

Count



XP (7193 entries)

$$\frac{\sigma}{\mu} = 11.41 \pm 0.20 \%$$

$$\mu = 483.4 \pm 0.8$$

$$\sigma = 55.2 \pm 0.9$$

WF (7193 entries)

$$\frac{\sigma}{\mu} = 11.57 \pm 0.19 \%$$

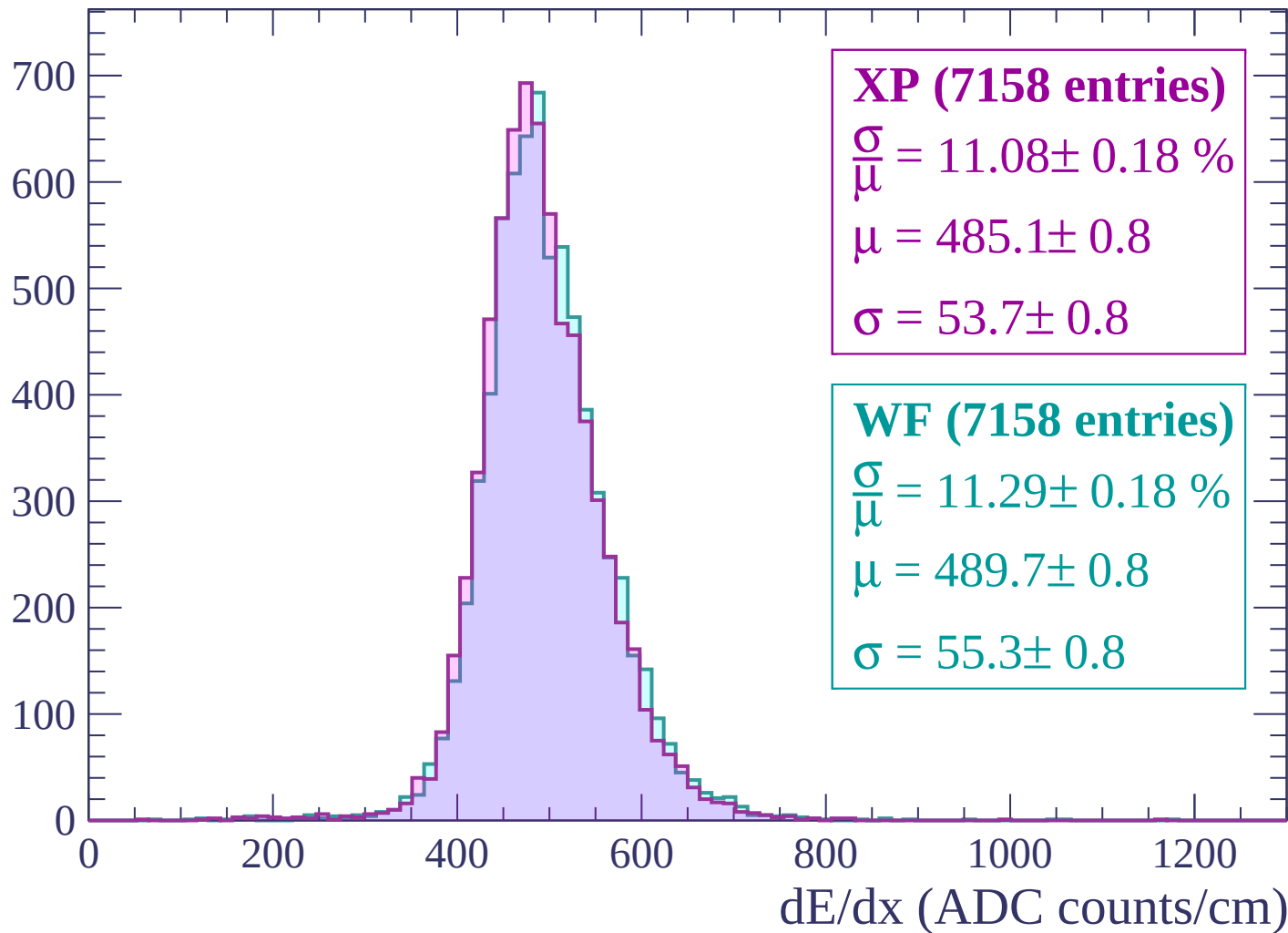
$$\mu = 488.1 \pm 0.8$$

$$\sigma = 56.5 \pm 0.8$$

dE/dx (ADC counts/cm)

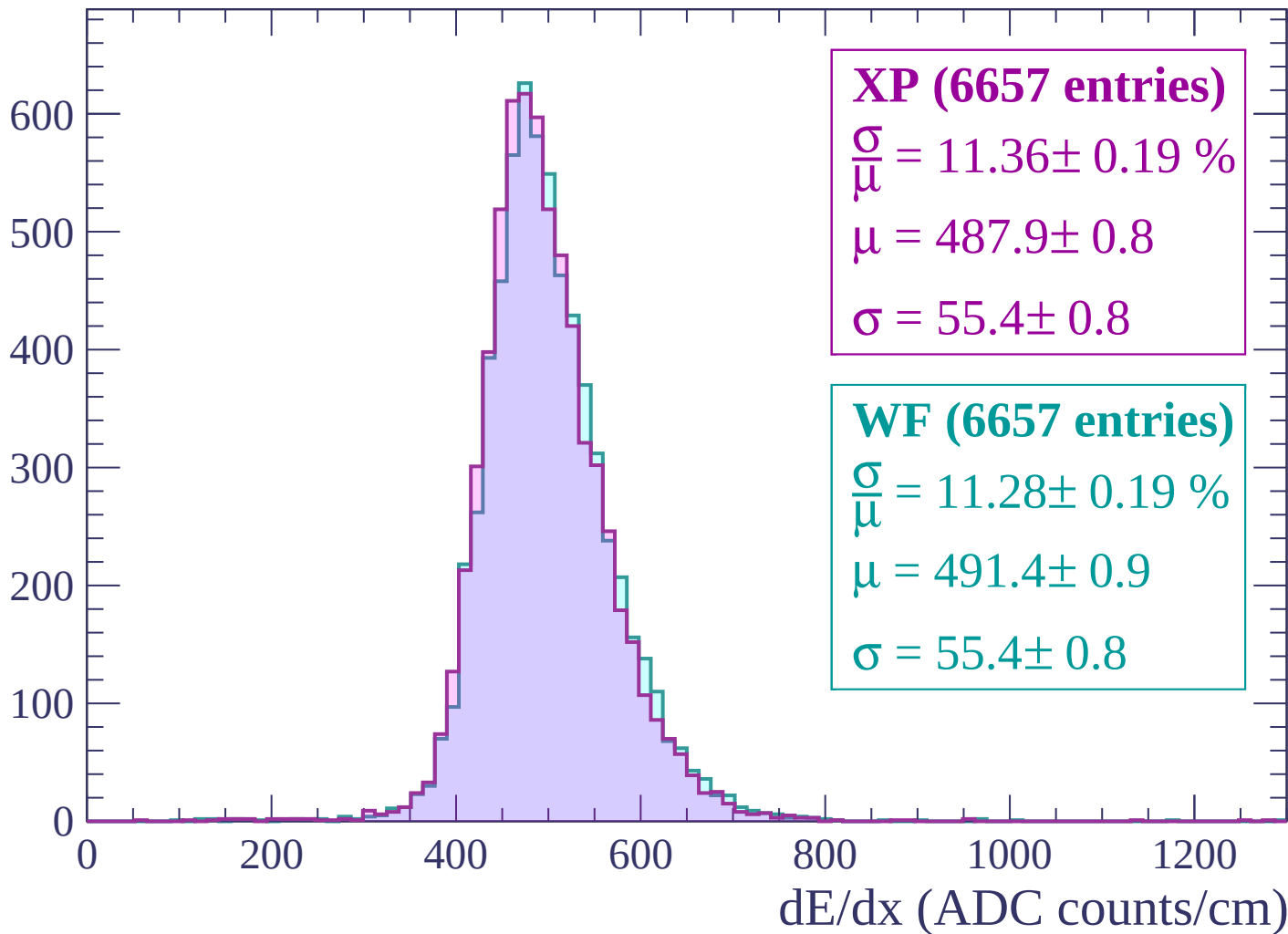
Energy loss | $1860 < p < 1920$

Count



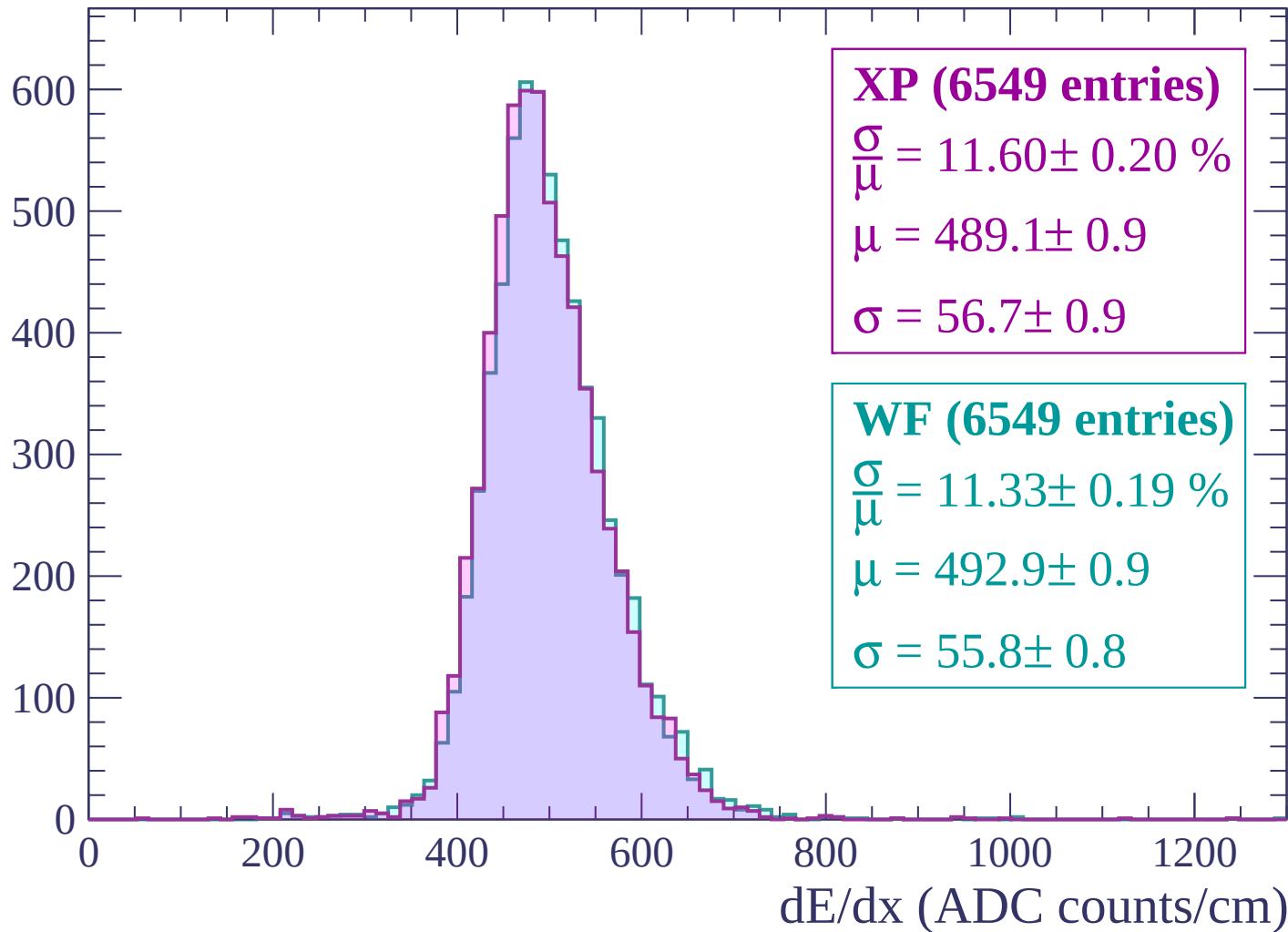
Energy loss | $1920 < p < 1980$

Count



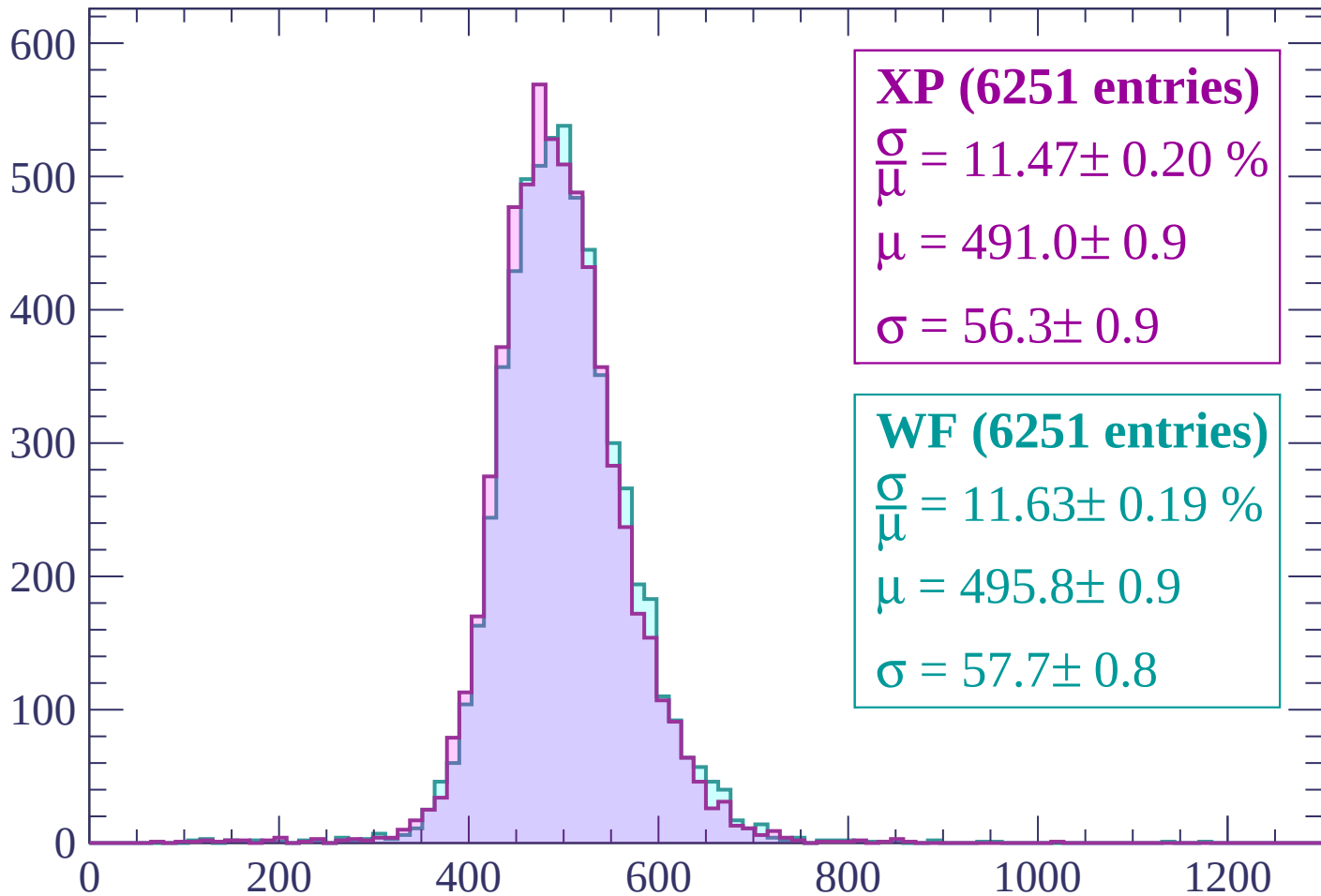
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count



XP (6251 entries)

$$\frac{\sigma}{\mu} = 11.47 \pm 0.20 \%$$

$$\mu = 491.0 \pm 0.9$$

$$\sigma = 56.3 \pm 0.9$$

WF (6251 entries)

$$\frac{\sigma}{\mu} = 11.63 \pm 0.19 \%$$

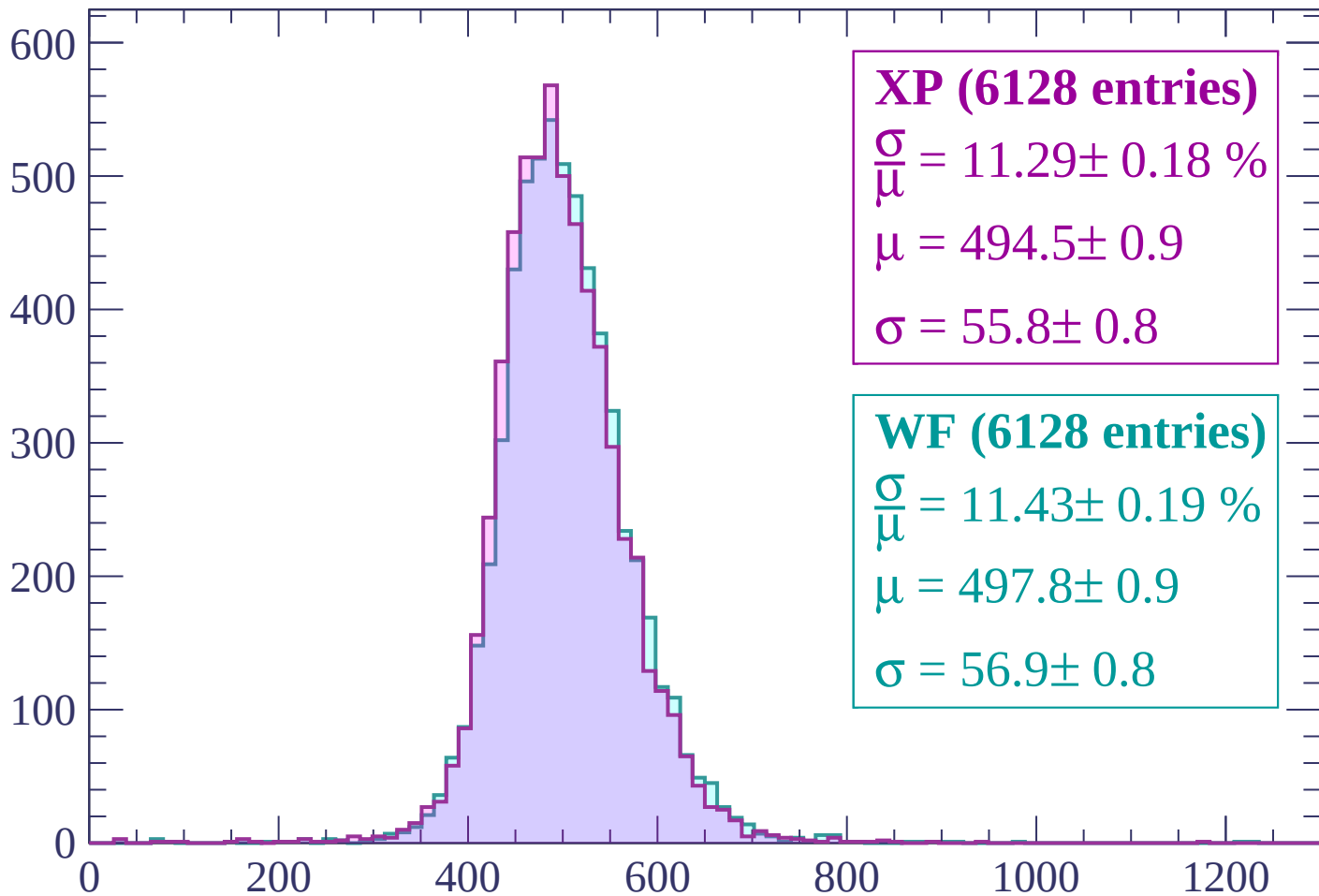
$$\mu = 495.8 \pm 0.9$$

$$\sigma = 57.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (6128 entries)

$$\frac{\sigma}{\mu} = 11.29 \pm 0.18 \%$$

$$\mu = 494.5 \pm 0.9$$

$$\sigma = 55.8 \pm 0.8$$

WF (6128 entries)

$$\frac{\sigma}{\mu} = 11.43 \pm 0.19 \%$$

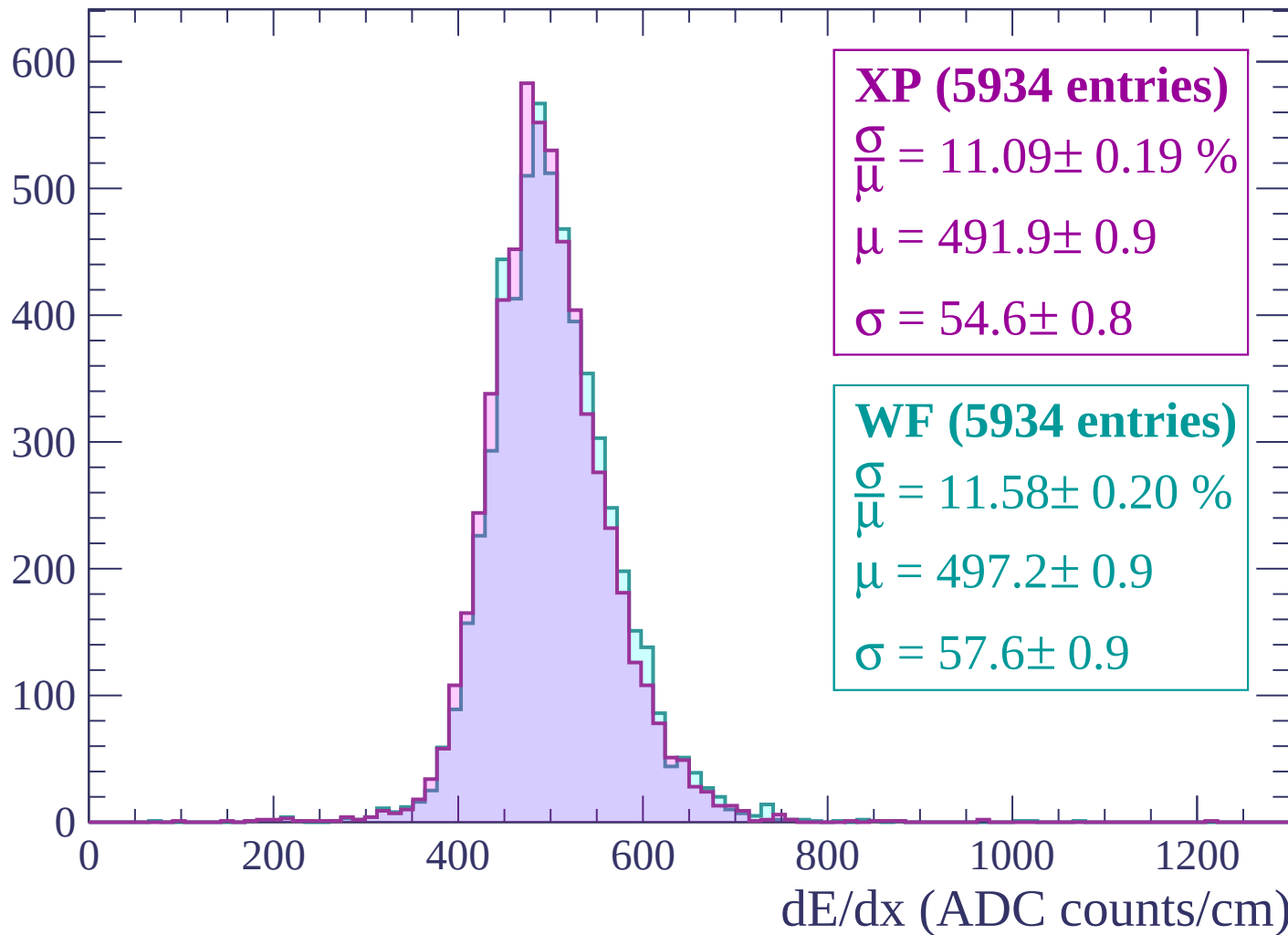
$$\mu = 497.8 \pm 0.9$$

$$\sigma = 56.9 \pm 0.8$$

dE/dx (ADC counts/cm)

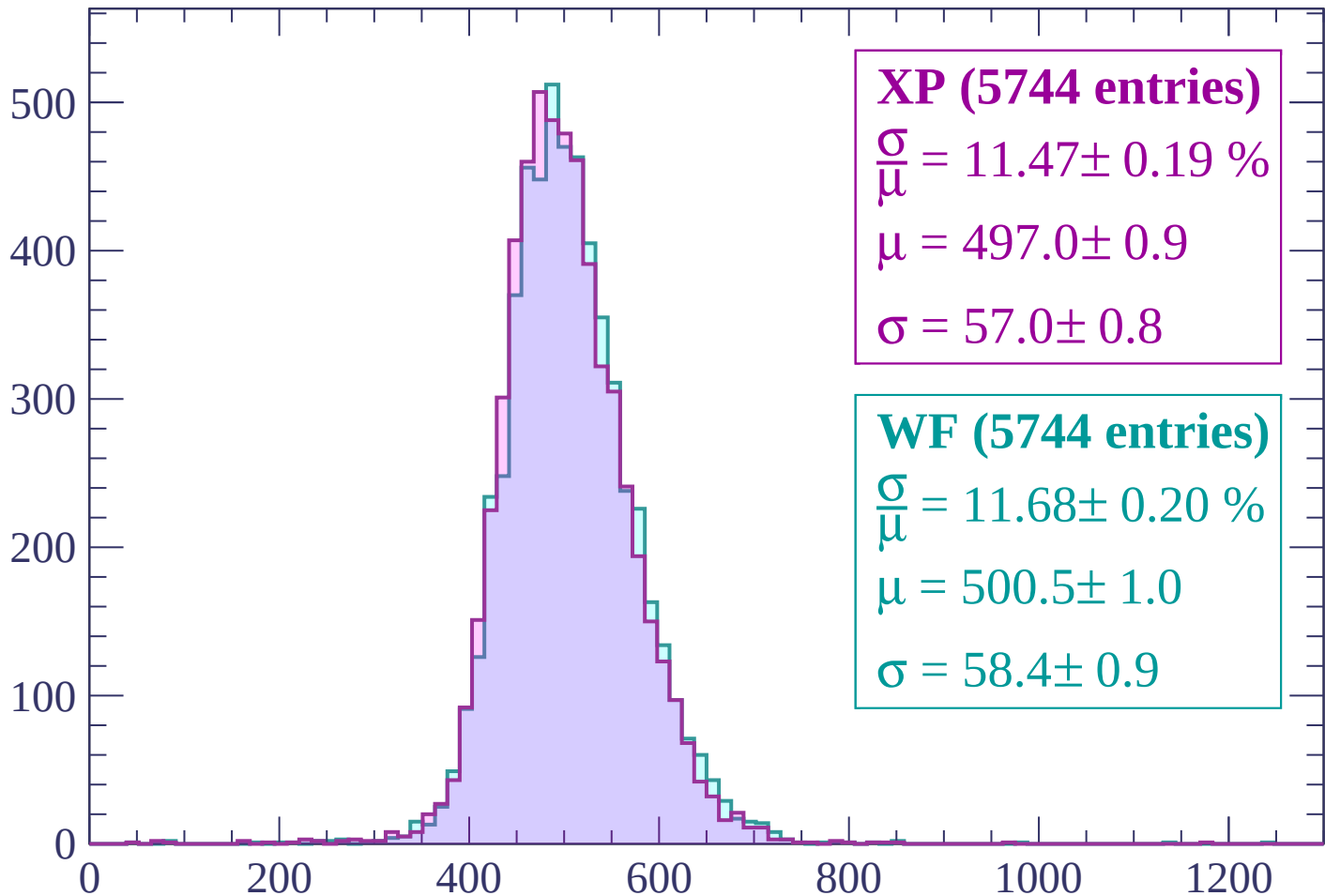
Energy loss | $2160 < p < 2220$

Count



Energy loss | $2220 < p < 2280$

Count



XP (5744 entries)

$\frac{\sigma}{\mu} = 11.47 \pm 0.19 \%$

$\mu = 497.0 \pm 0.9$

$\sigma = 57.0 \pm 0.8$

WF (5744 entries)

$\frac{\sigma}{\mu} = 11.68 \pm 0.20 \%$

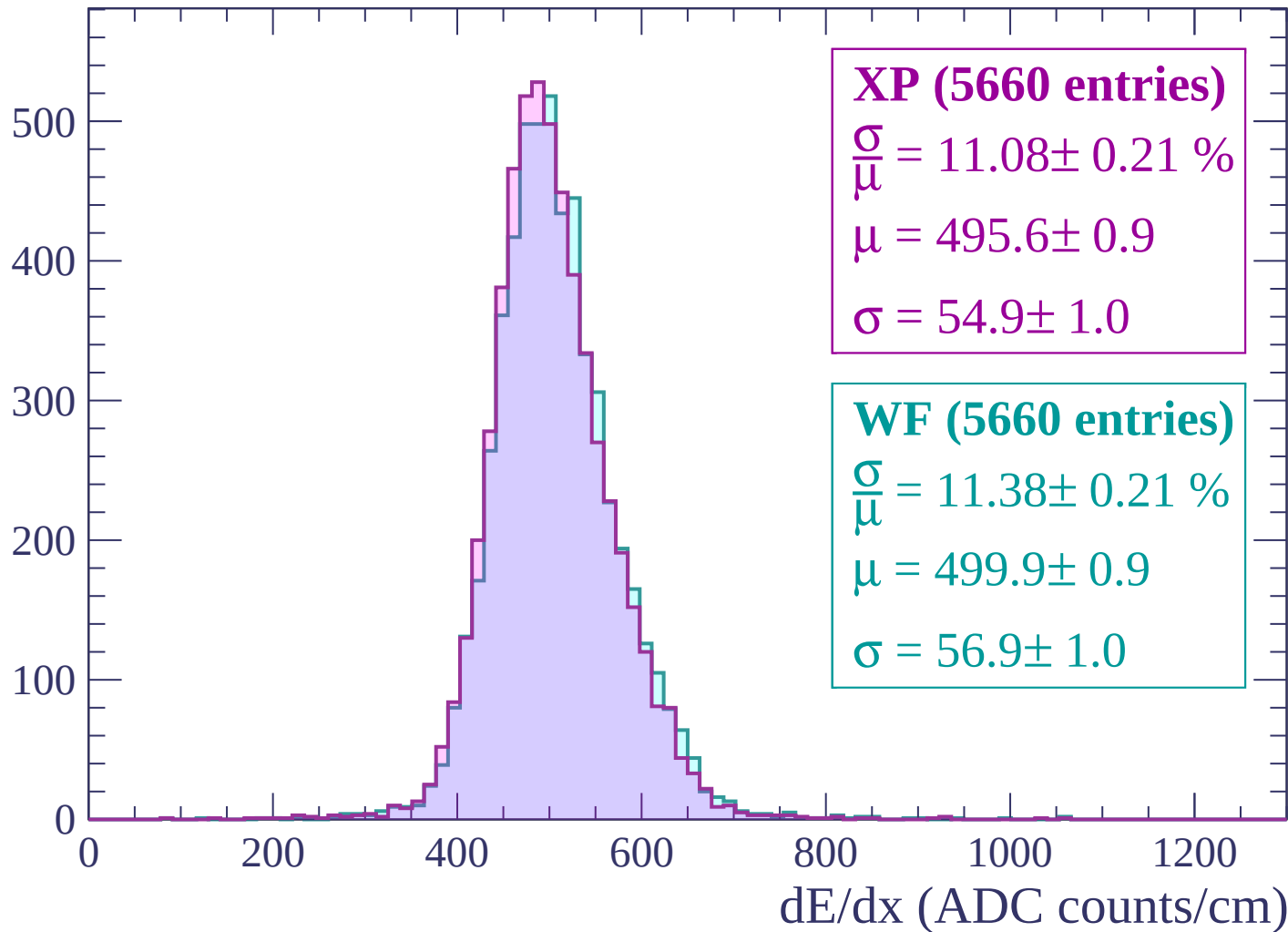
$\mu = 500.5 \pm 1.0$

$\sigma = 58.4 \pm 0.9$

dE/dx (ADC counts/cm)

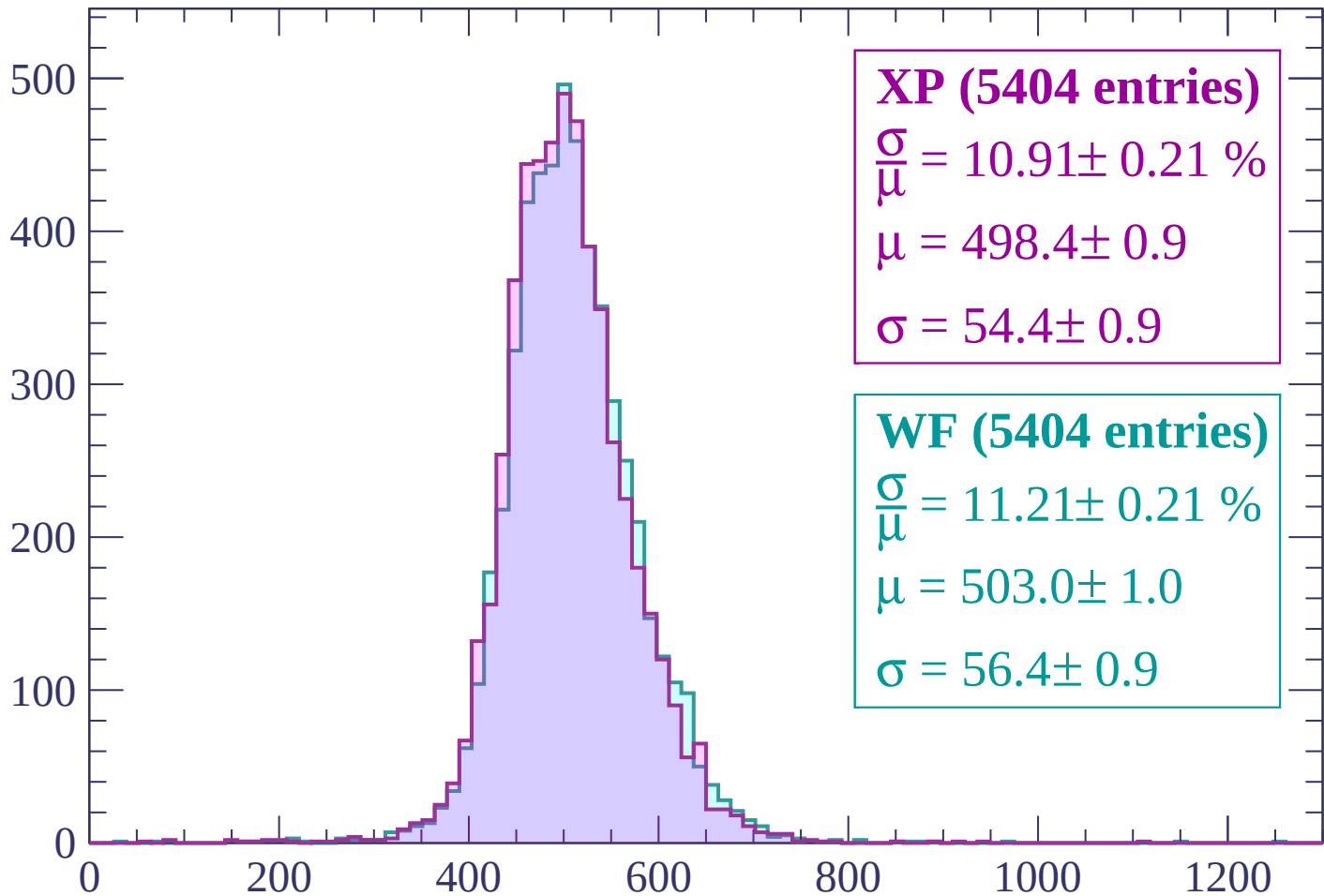
Energy loss | $2280 < p < 2340$

Count



Energy loss | $2340 < p < 2400$

Count



XP (5404 entries)

$\frac{\sigma}{\mu} = 10.91 \pm 0.21 \%$

$\mu = 498.4 \pm 0.9$

$\sigma = 54.4 \pm 0.9$

WF (5404 entries)

$\frac{\sigma}{\mu} = 11.21 \pm 0.21 \%$

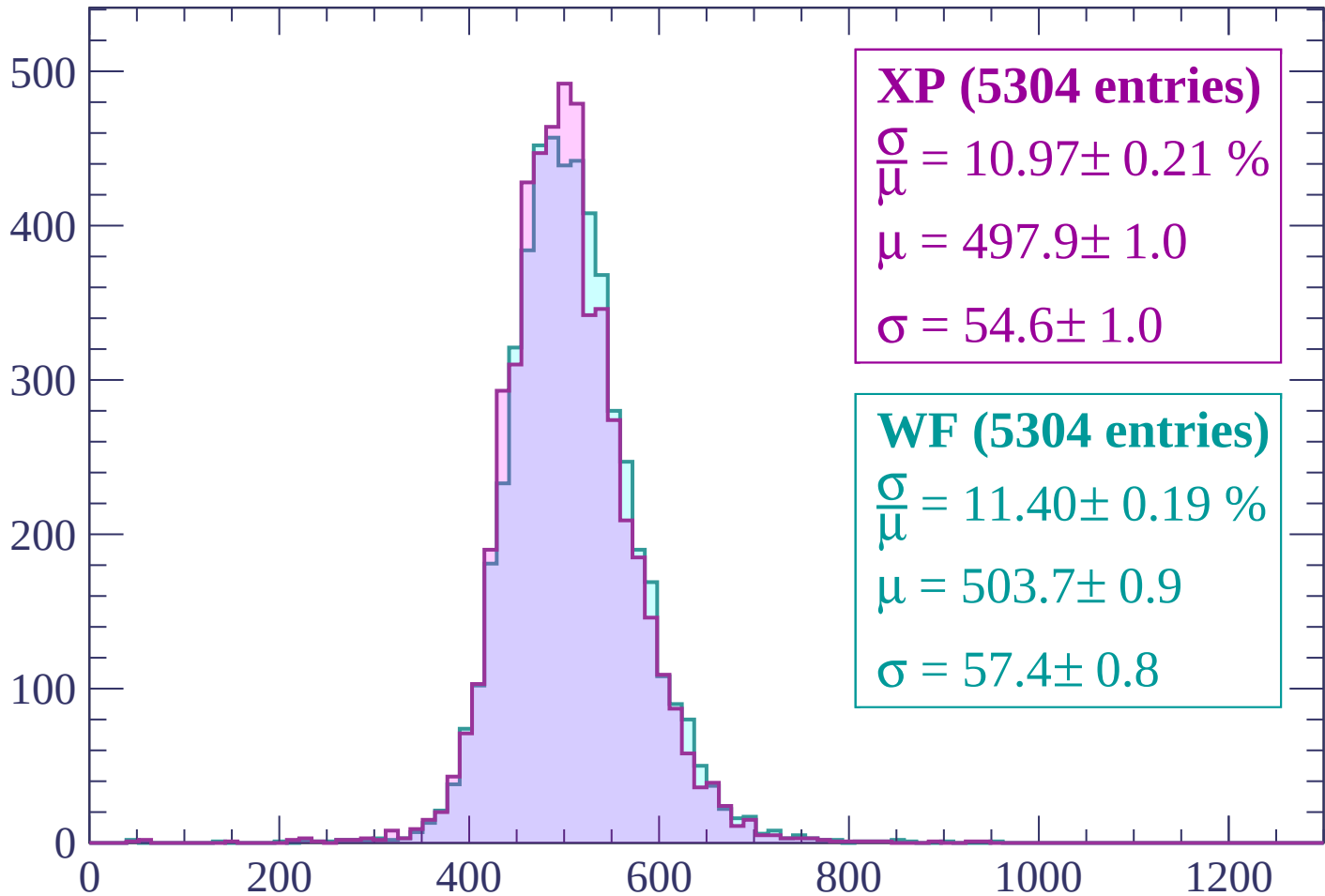
$\mu = 503.0 \pm 1.0$

$\sigma = 56.4 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $2400 < p < 2460$

Count



XP (5304 entries)

$$\frac{\sigma}{\mu} = 10.97 \pm 0.21 \%$$

$$\mu = 497.9 \pm 1.0$$

$$\sigma = 54.6 \pm 1.0$$

WF (5304 entries)

$$\frac{\sigma}{\mu} = 11.40 \pm 0.19 \%$$

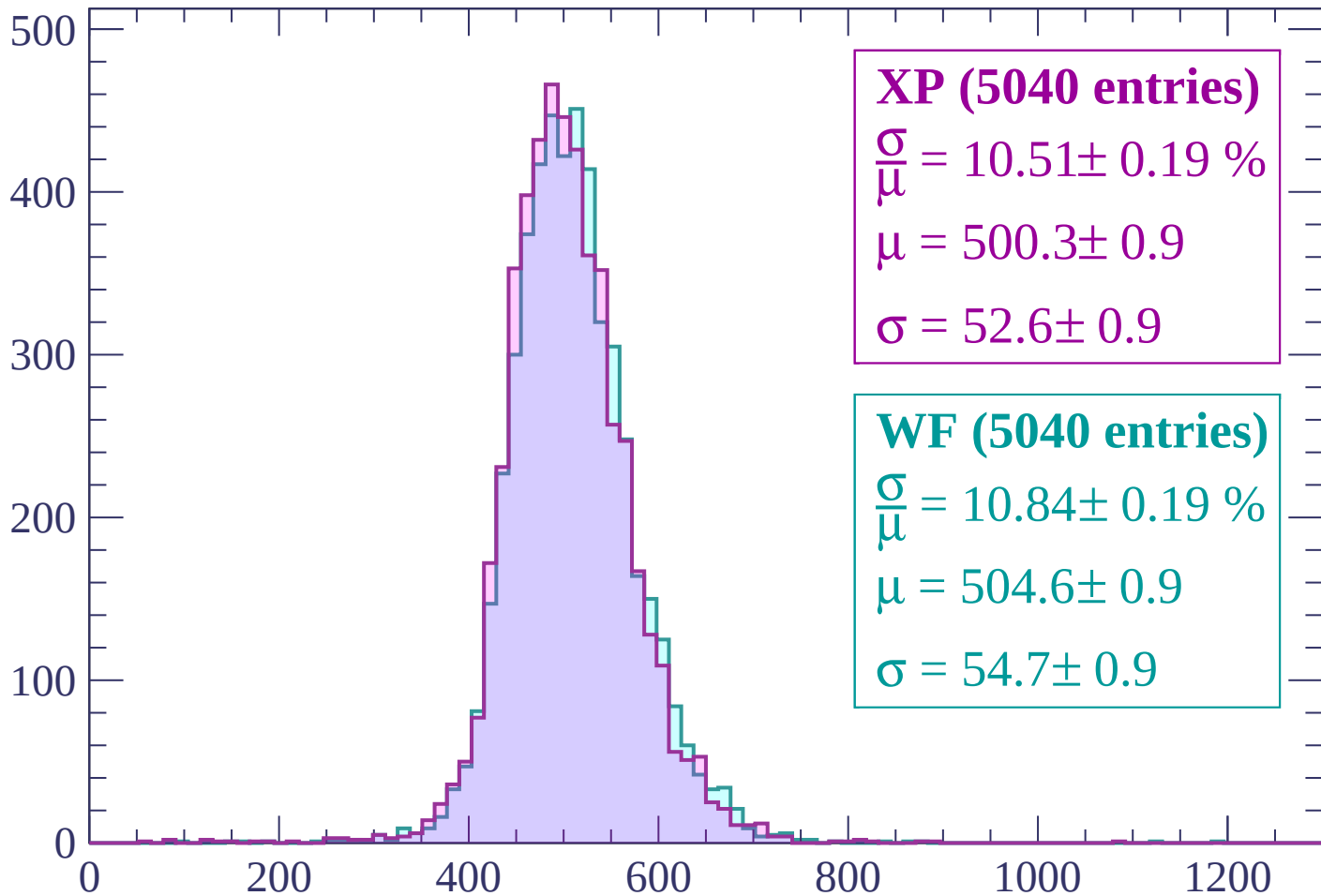
$$\mu = 503.7 \pm 0.9$$

$$\sigma = 57.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2460 < p < 2520$

Count



XP (5040 entries)

$\frac{\sigma}{\mu} = 10.51 \pm 0.19 \%$

$\mu = 500.3 \pm 0.9$

$\sigma = 52.6 \pm 0.9$

WF (5040 entries)

$\frac{\sigma}{\mu} = 10.84 \pm 0.19 \%$

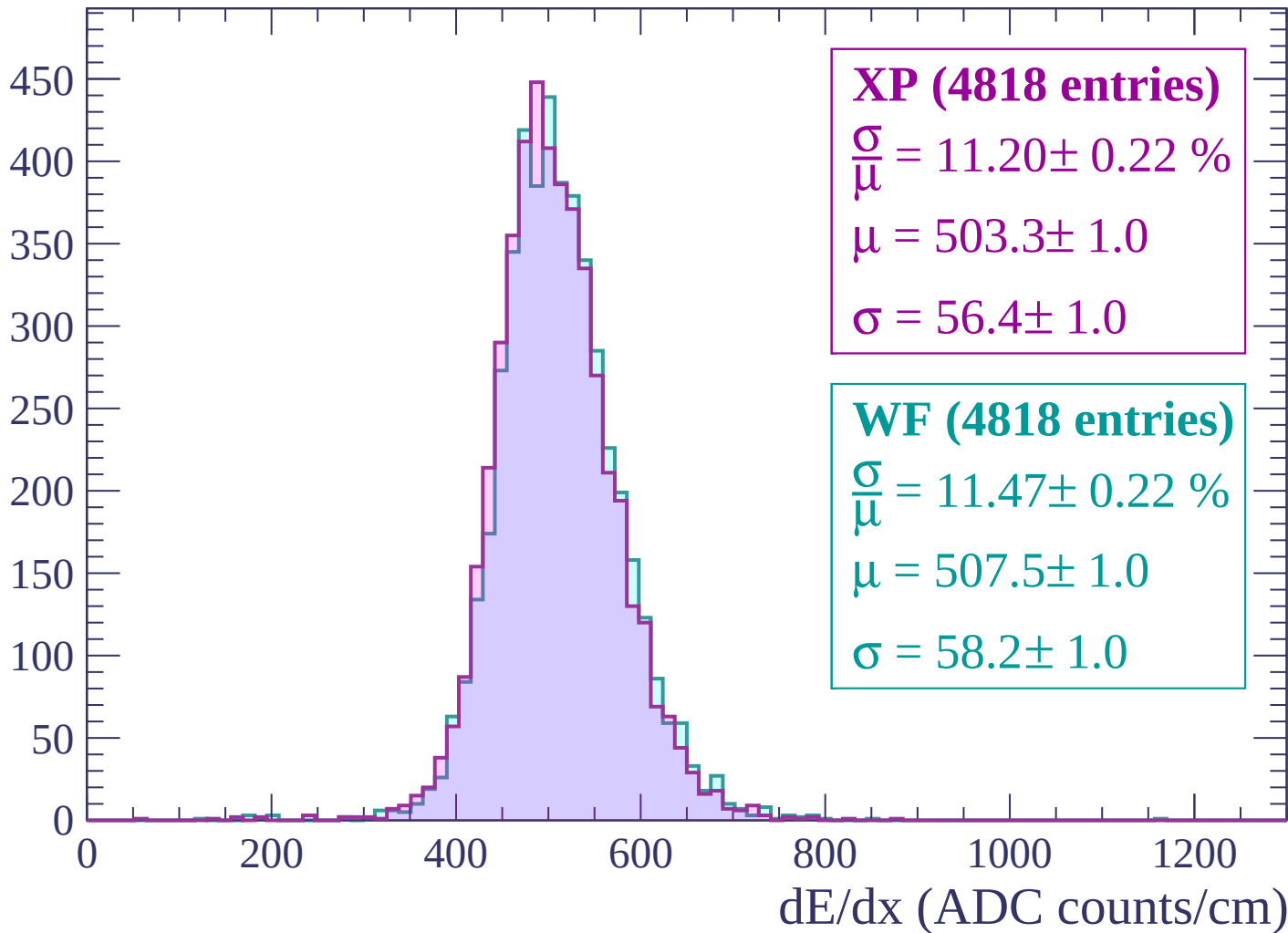
$\mu = 504.6 \pm 0.9$

$\sigma = 54.7 \pm 0.9$

dE/dx (ADC counts/cm)

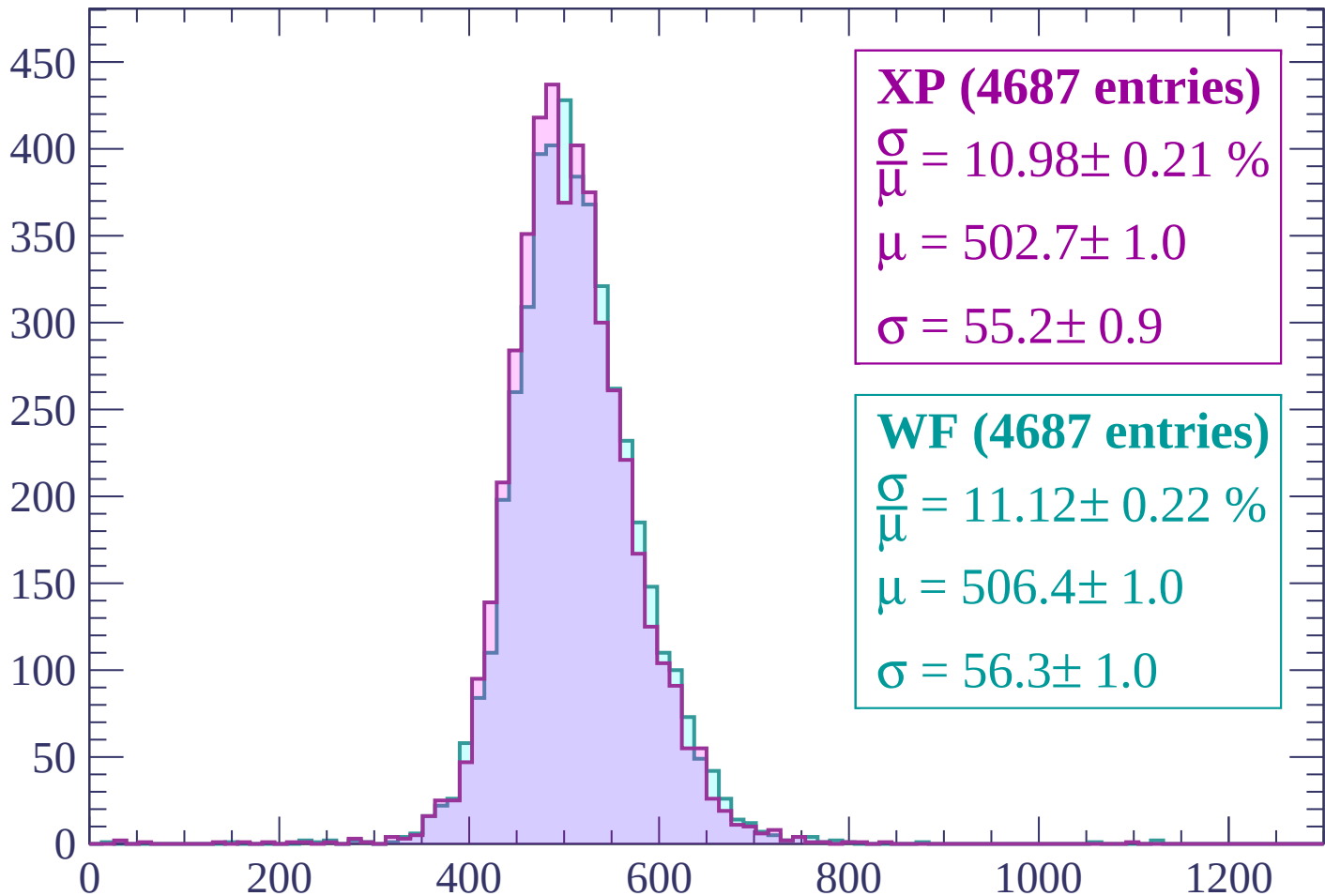
Energy loss | $2520 < p < 2580$

Count



Energy loss | $2580 < p < 2640$

Count



XP (4687 entries)

$\frac{\sigma}{\mu} = 10.98 \pm 0.21 \%$

$\mu = 502.7 \pm 1.0$

$\sigma = 55.2 \pm 0.9$

WF (4687 entries)

$\frac{\sigma}{\mu} = 11.12 \pm 0.22 \%$

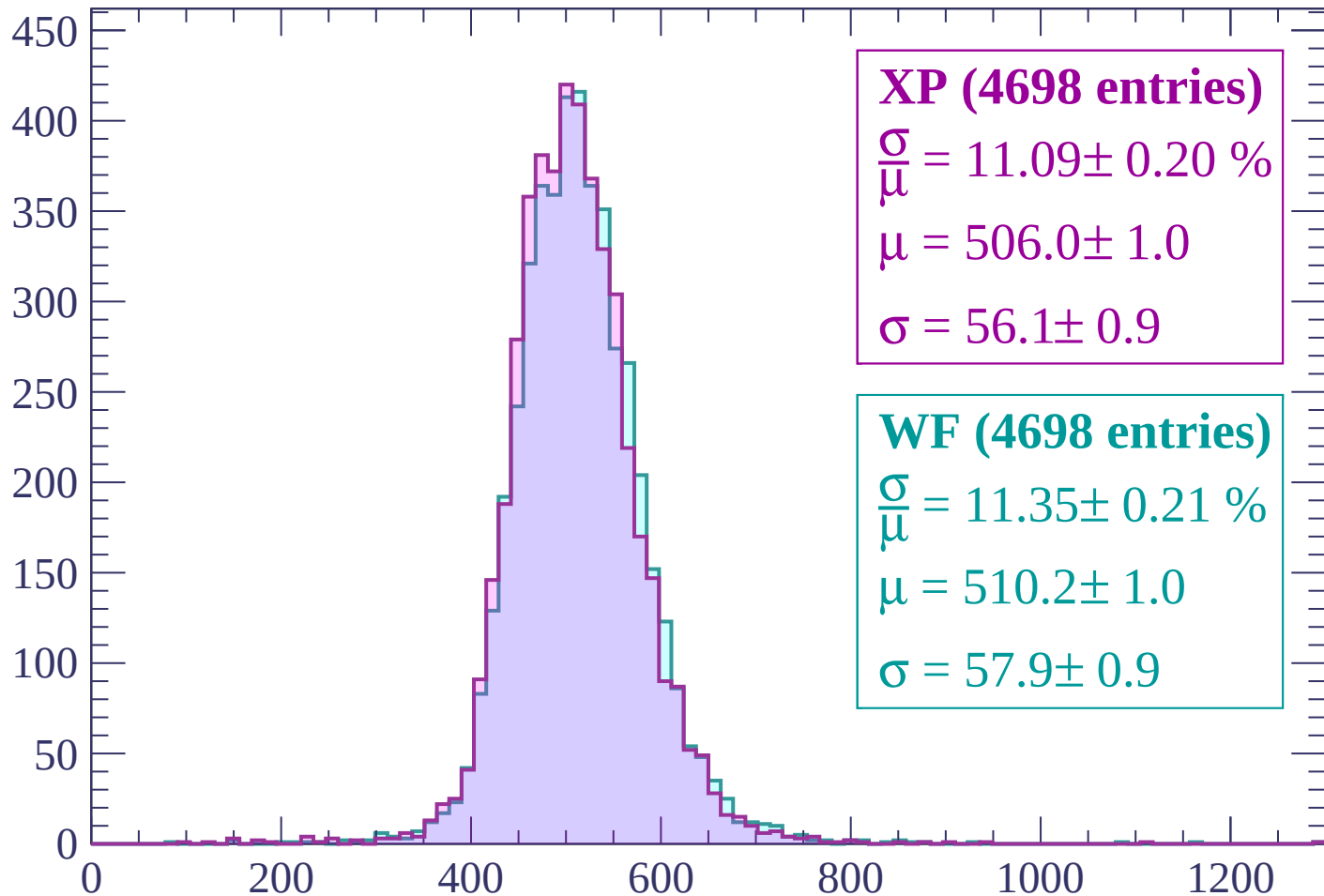
$\mu = 506.4 \pm 1.0$

$\sigma = 56.3 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2640 < p < 2700$

Count



XP (4698 entries)

$\frac{\sigma}{\mu} = 11.09 \pm 0.20 \%$

$\mu = 506.0 \pm 1.0$

$\sigma = 56.1 \pm 0.9$

WF (4698 entries)

$\frac{\sigma}{\mu} = 11.35 \pm 0.21 \%$

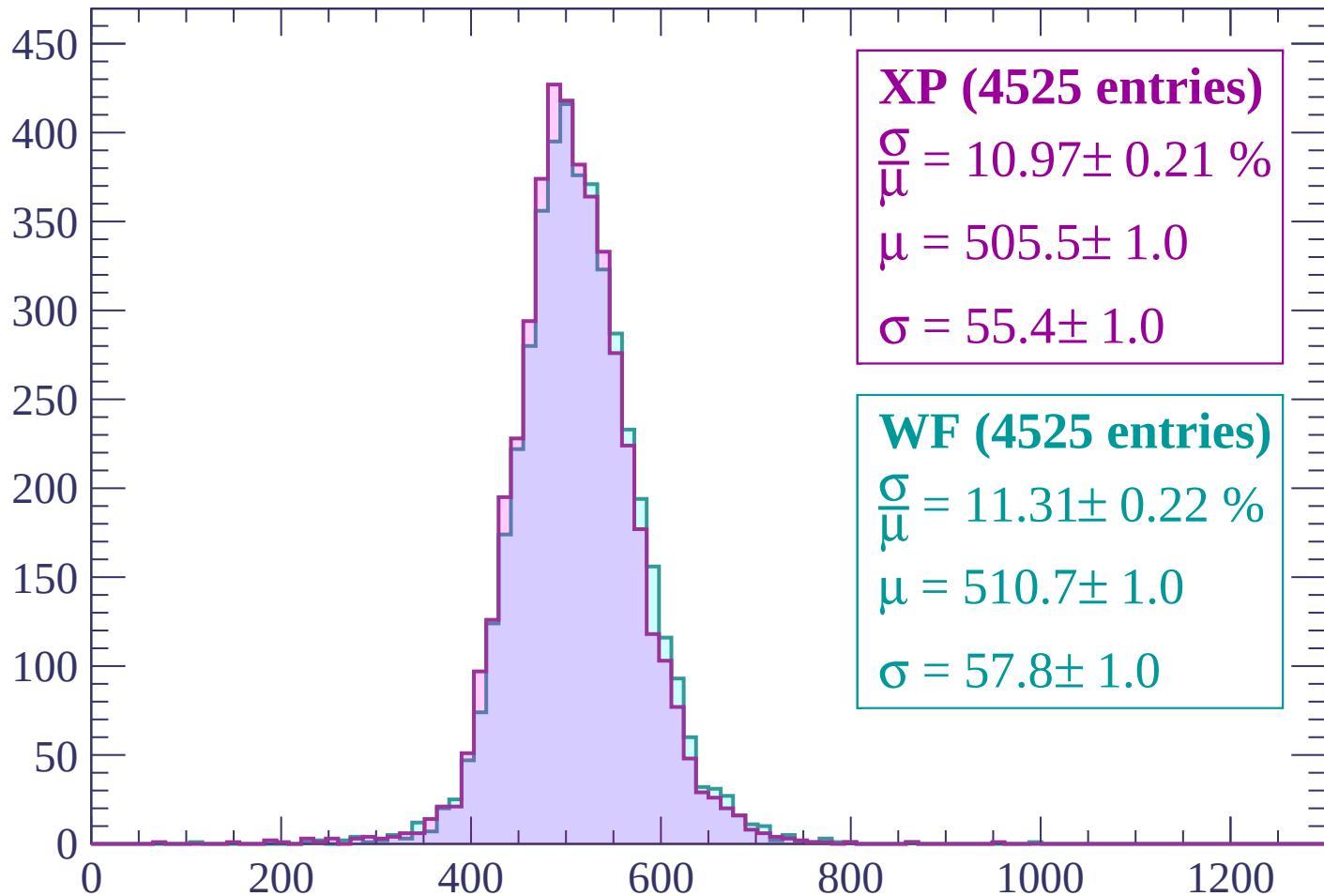
$\mu = 510.2 \pm 1.0$

$\sigma = 57.9 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP (4525 entries)

$\frac{\sigma}{\mu} = 10.97 \pm 0.21 \%$

$\mu = 505.5 \pm 1.0$

$\sigma = 55.4 \pm 1.0$

WF (4525 entries)

$\frac{\sigma}{\mu} = 11.31 \pm 0.22 \%$

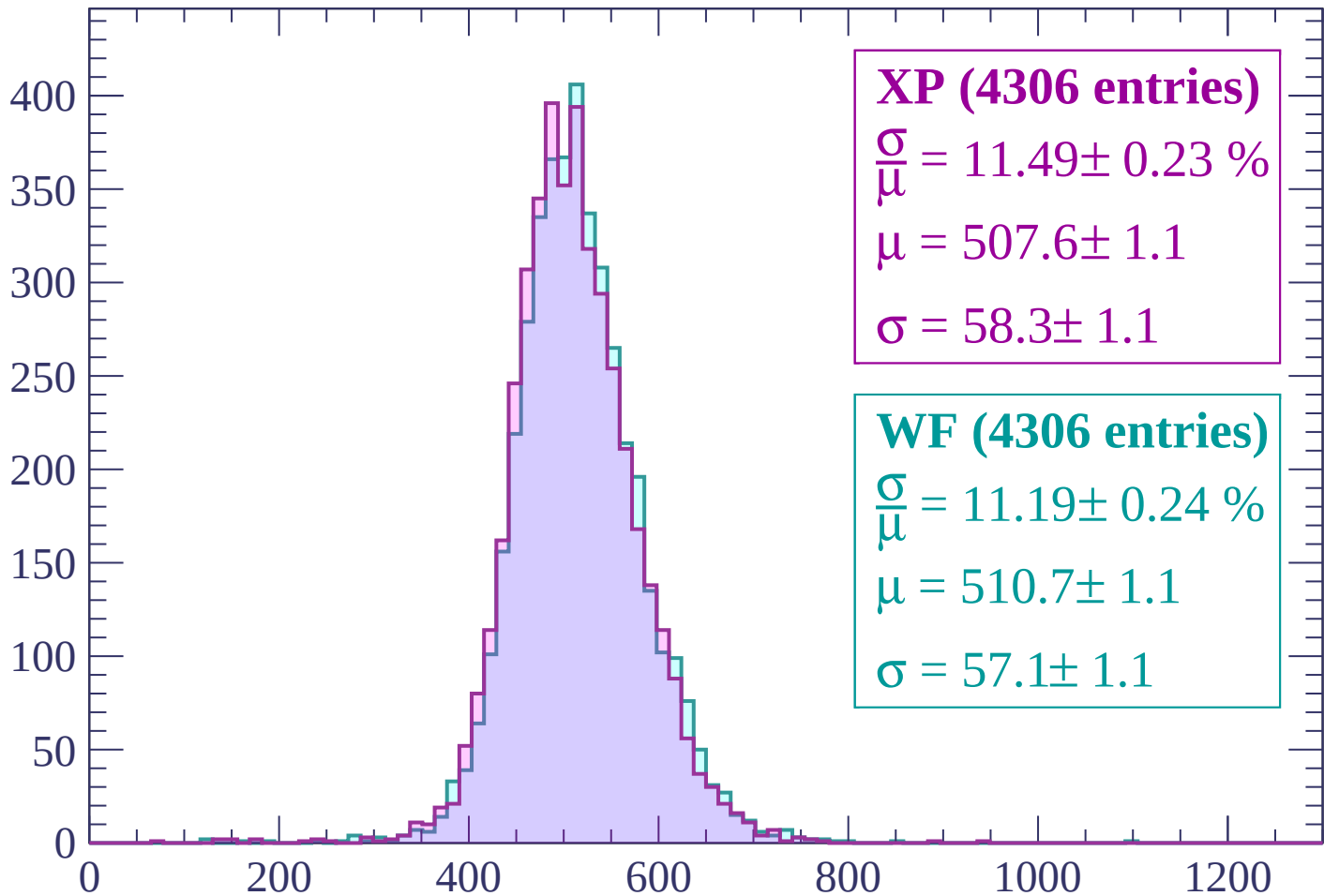
$\mu = 510.7 \pm 1.0$

$\sigma = 57.8 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2760 < p < 2820$

Count



XP (4306 entries)

$\frac{\sigma}{\mu} = 11.49 \pm 0.23 \%$

$\mu = 507.6 \pm 1.1$

$\sigma = 58.3 \pm 1.1$

WF (4306 entries)

$\frac{\sigma}{\mu} = 11.19 \pm 0.24 \%$

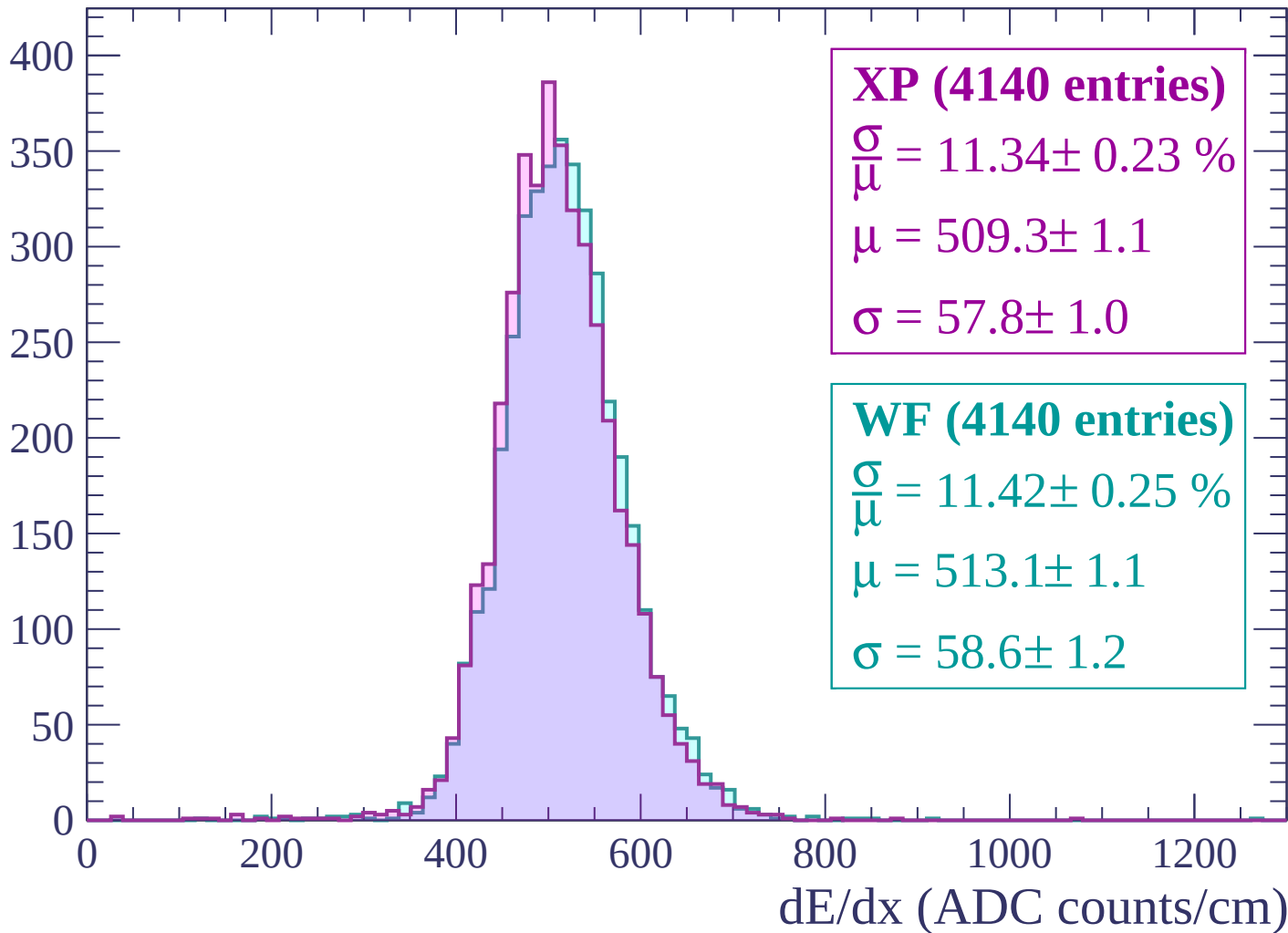
$\mu = 510.7 \pm 1.1$

$\sigma = 57.1 \pm 1.1$

dE/dx (ADC counts/cm)

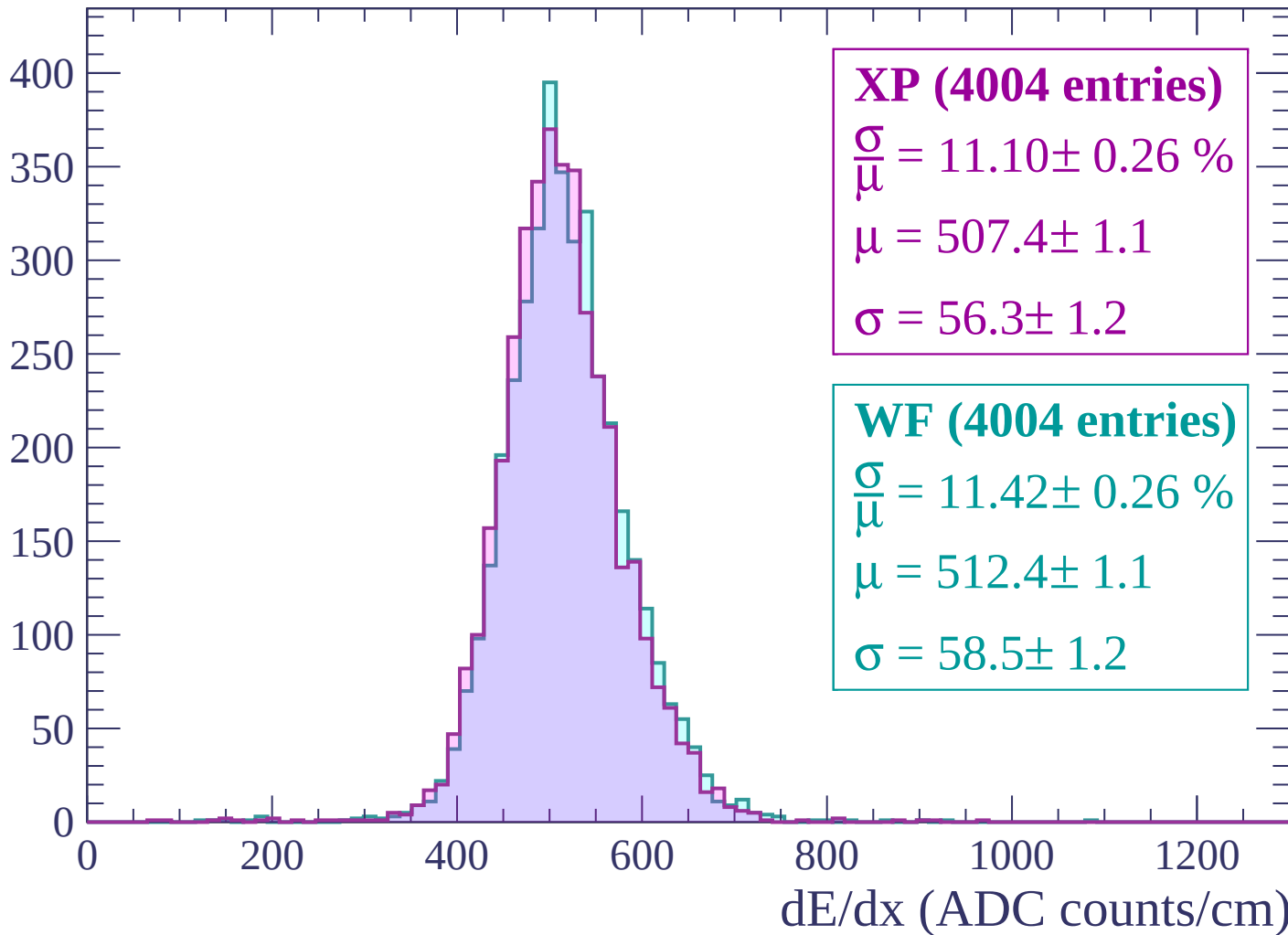
Energy loss | $2820 < p < 2880$

Count



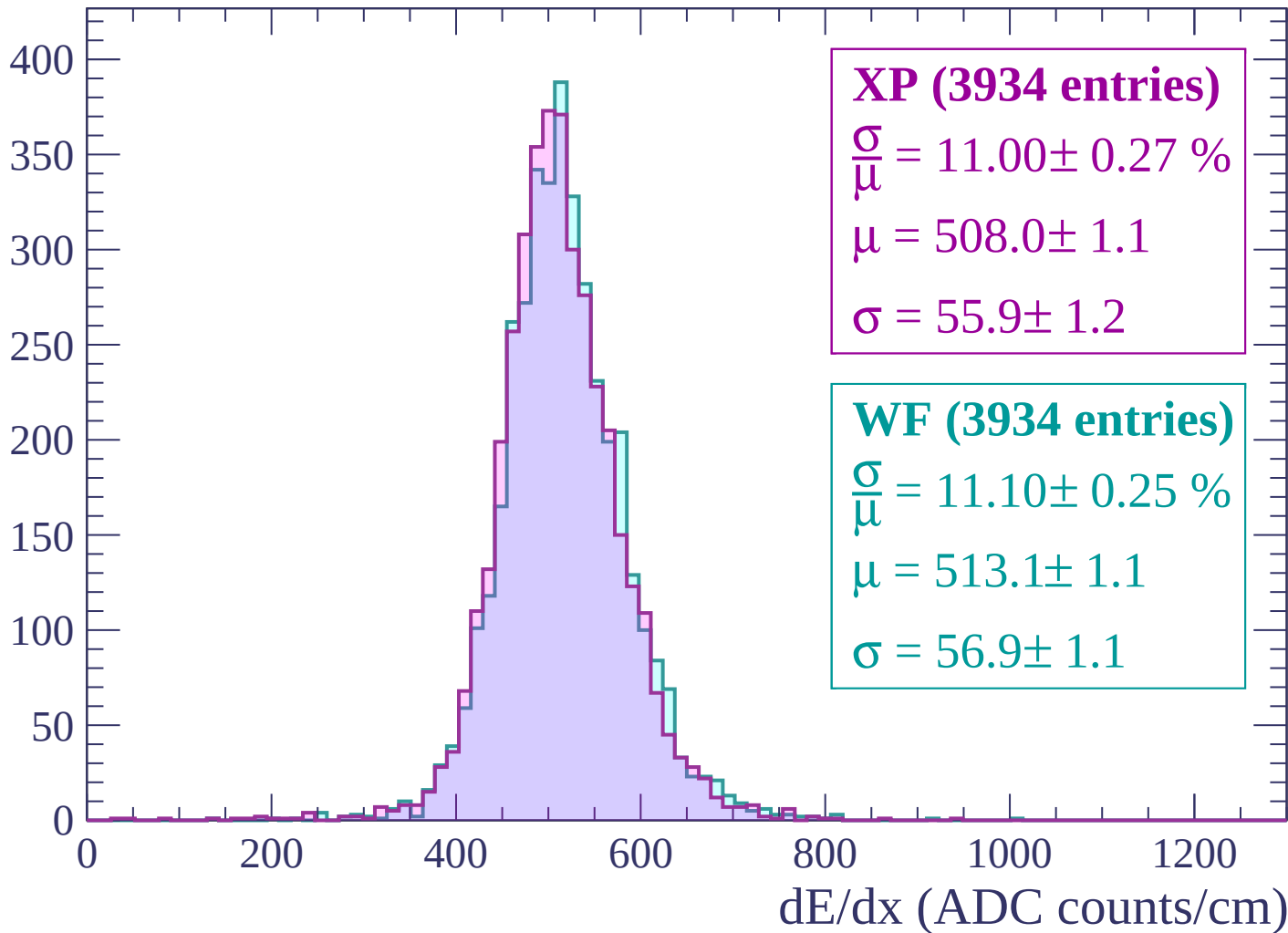
Energy loss | $2880 < p < 2940$

Count



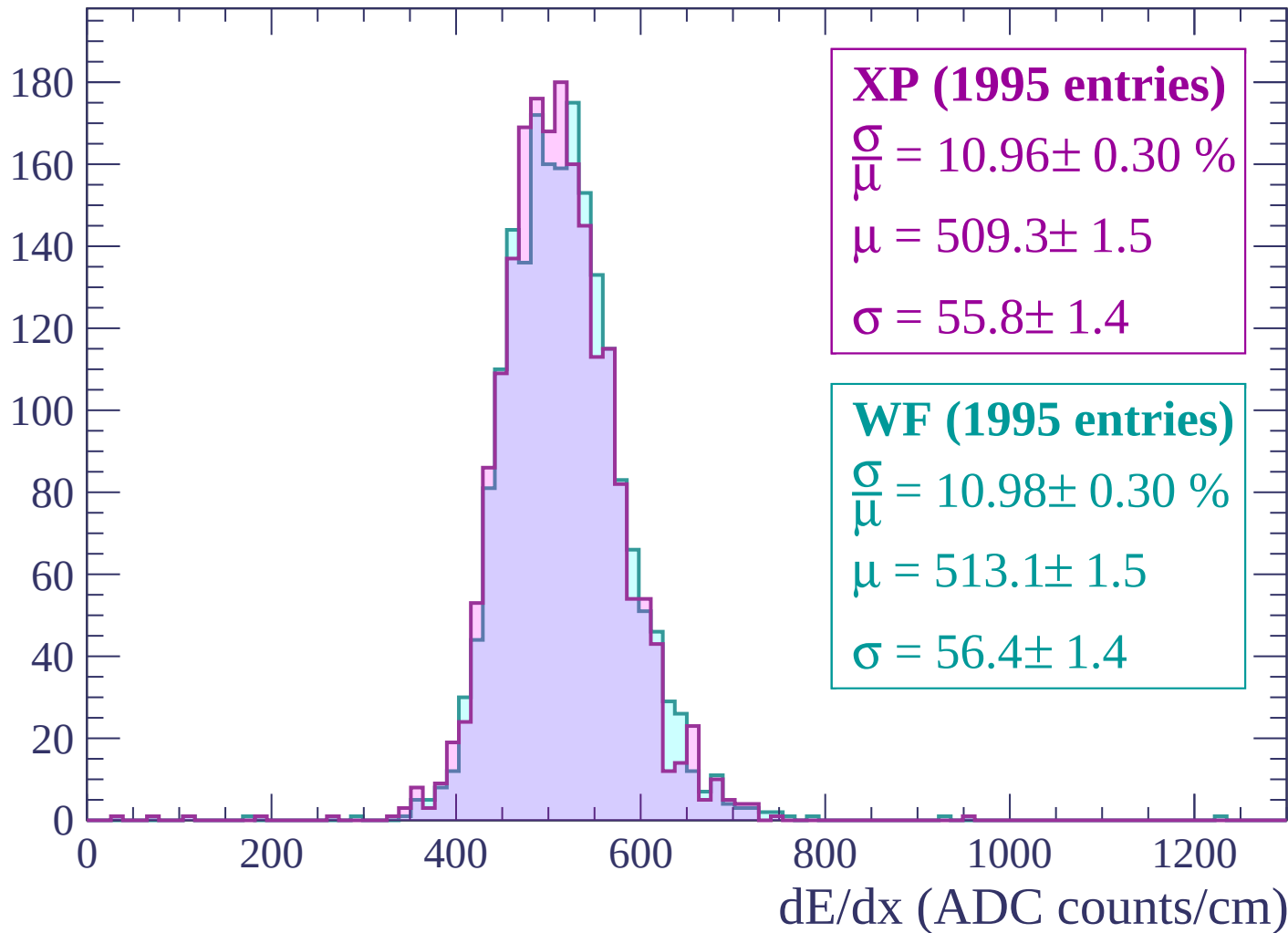
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

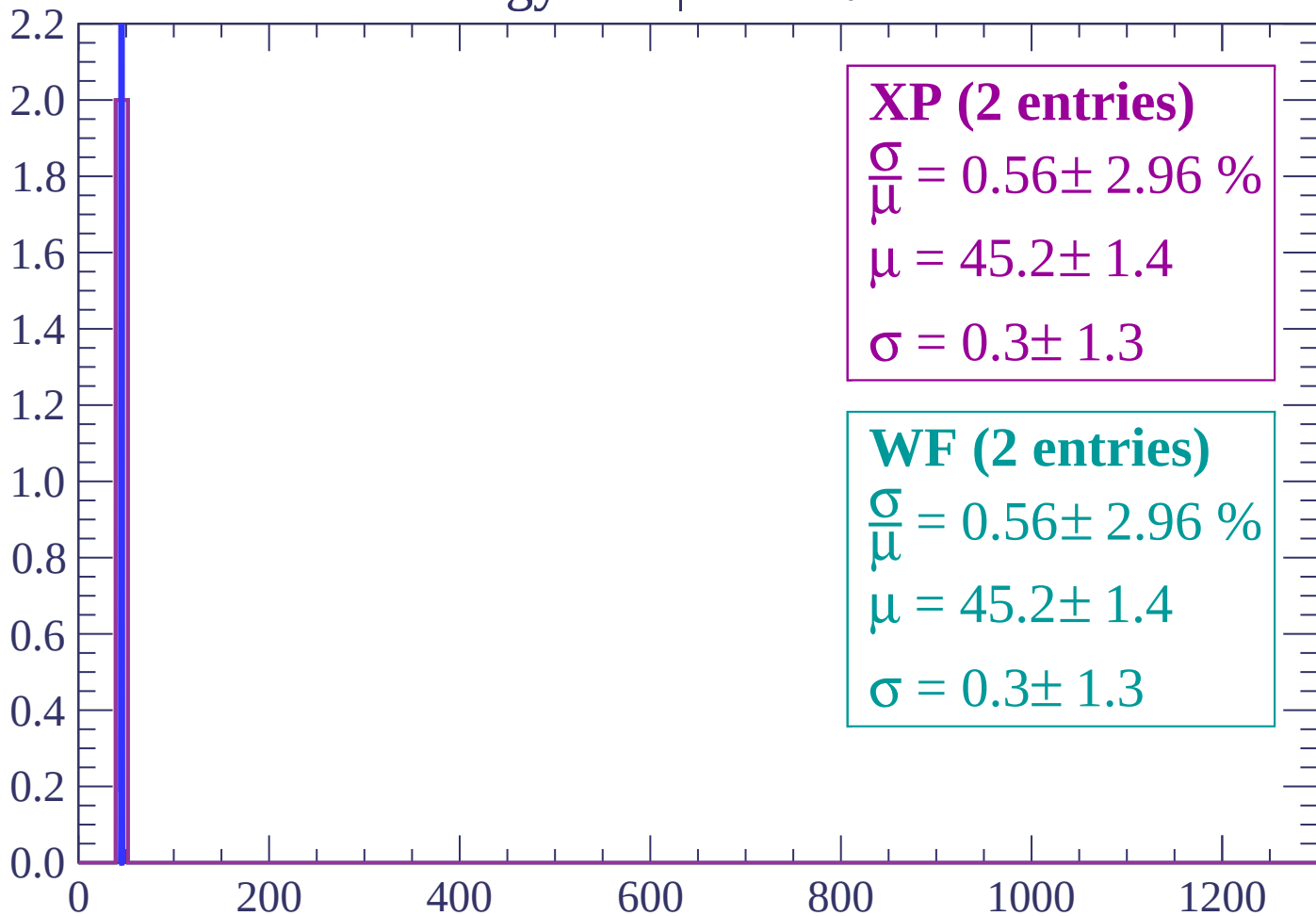
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count



XP (2 entries)

$\frac{\sigma}{\mu} = 0.56 \pm 2.96 \%$

$\mu = 45.2 \pm 1.4$

$\sigma = 0.3 \pm 1.3$

WF (2 entries)

$\frac{\sigma}{\mu} = 0.56 \pm 2.96 \%$

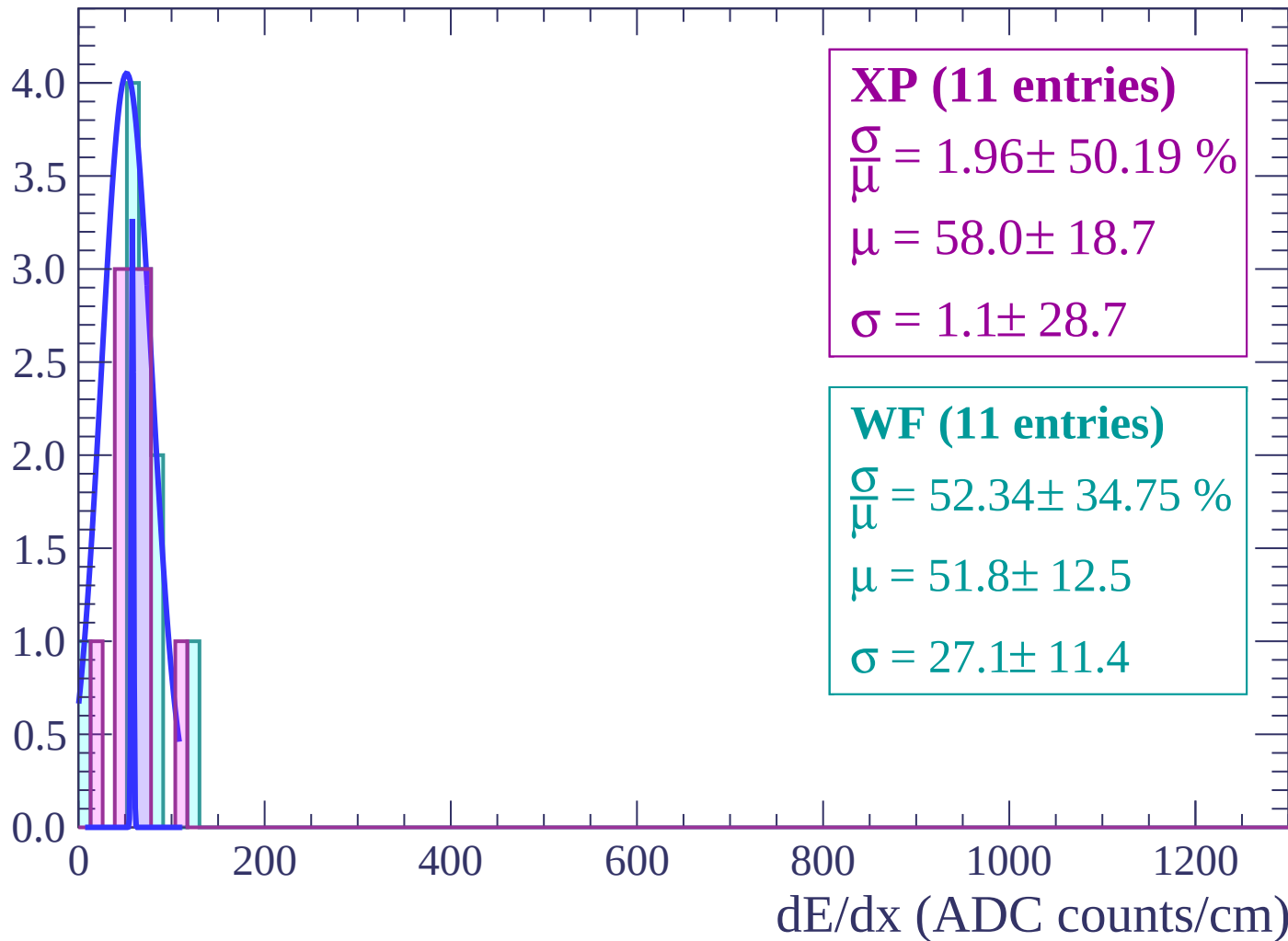
$\mu = 45.2 \pm 1.4$

$\sigma = 0.3 \pm 1.3$

dE/dx (ADC counts/cm)

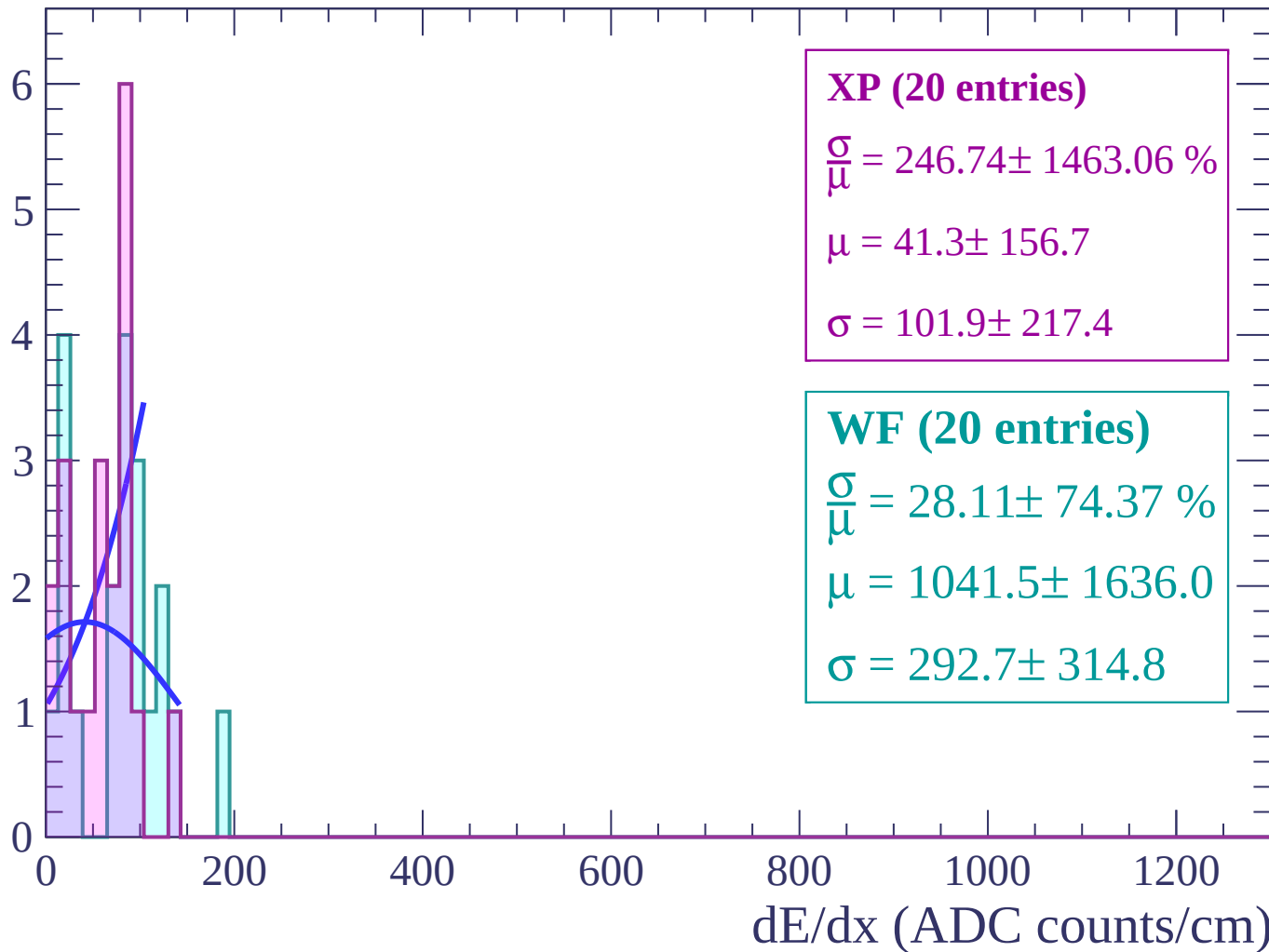
Energy loss | $-82 < \theta < -80$

Count



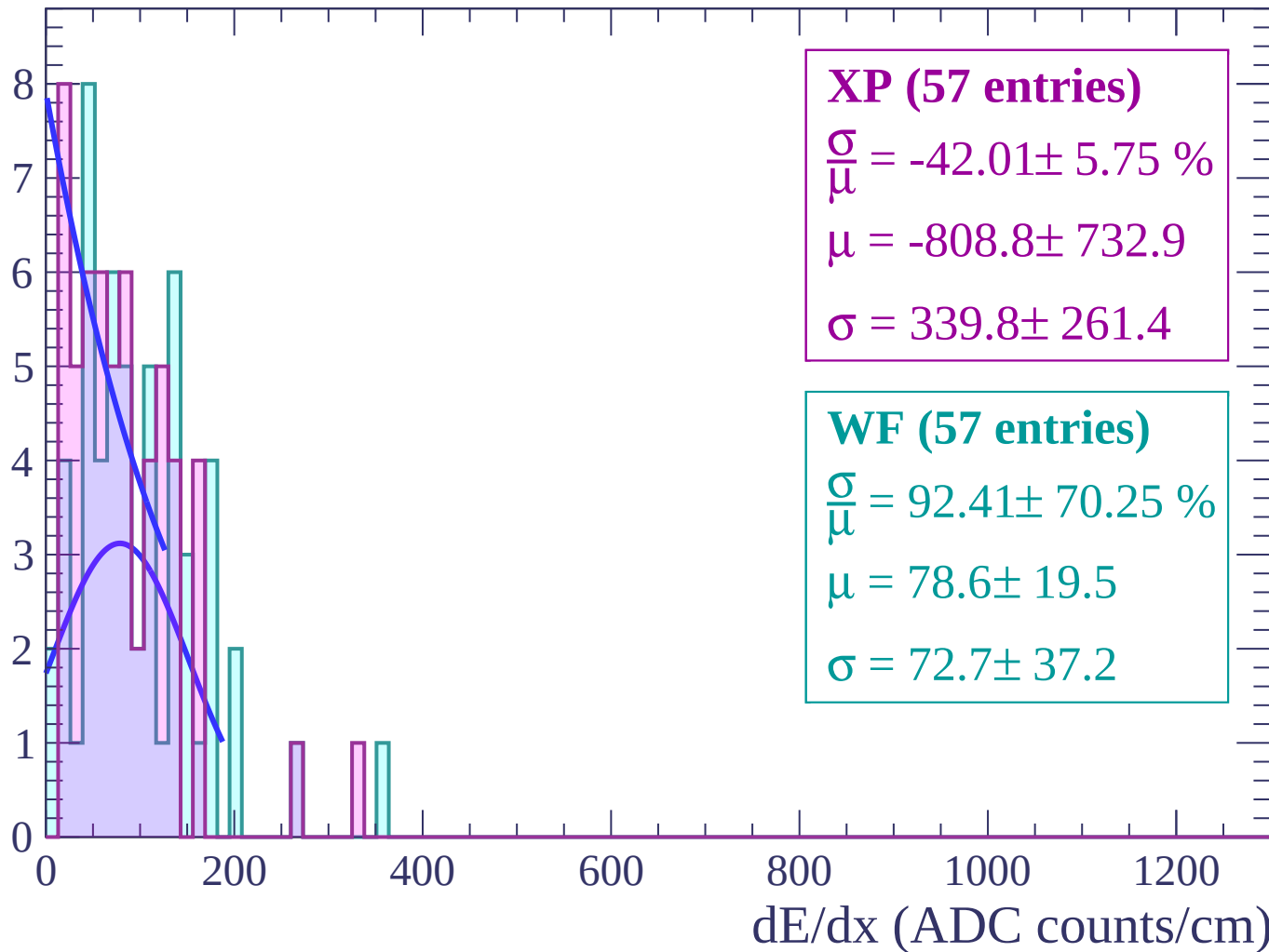
Energy loss | $-80 < \theta < -78$

Count



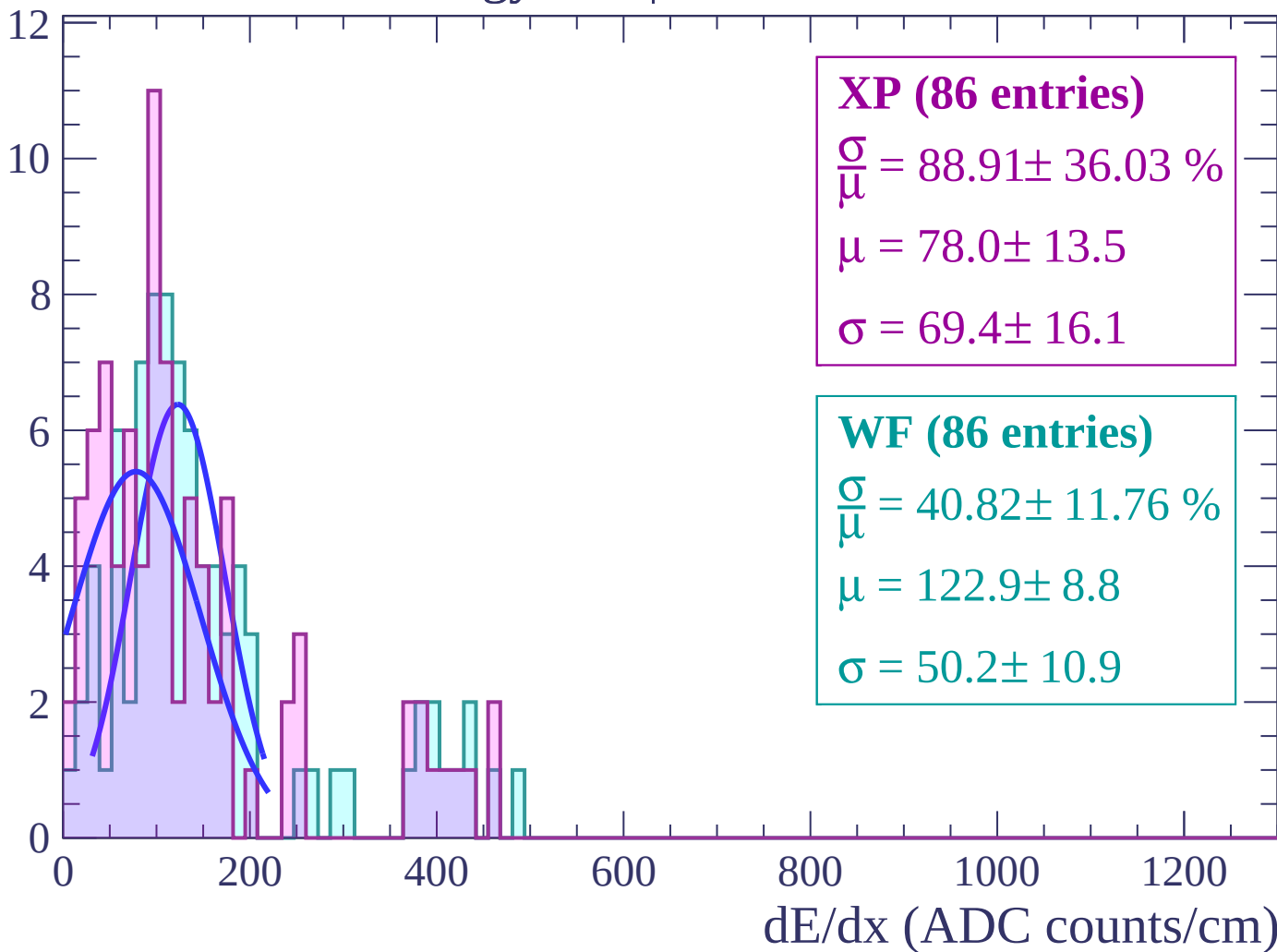
Energy loss | $-78 < \theta < -76$

Count



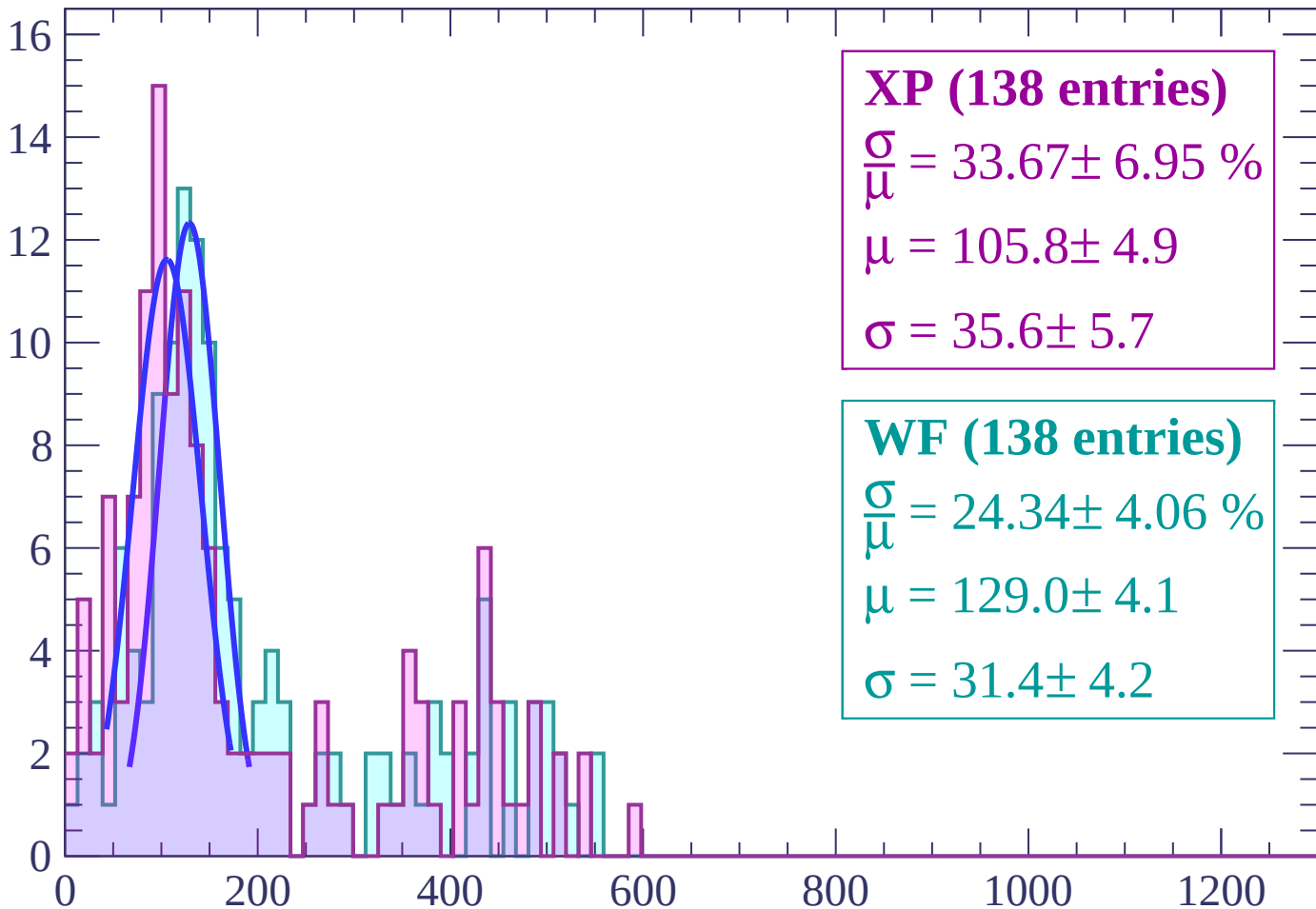
Energy loss | $-76 < \theta < -74$

Count



Energy loss | $-74 < \theta < -72$

Count



XP (138 entries)

$$\frac{\sigma}{\mu} = 33.67 \pm 6.95 \%$$

$$\mu = 105.8 \pm 4.9$$

$$\sigma = 35.6 \pm 5.7$$

WF (138 entries)

$$\frac{\sigma}{\mu} = 24.34 \pm 4.06 \%$$

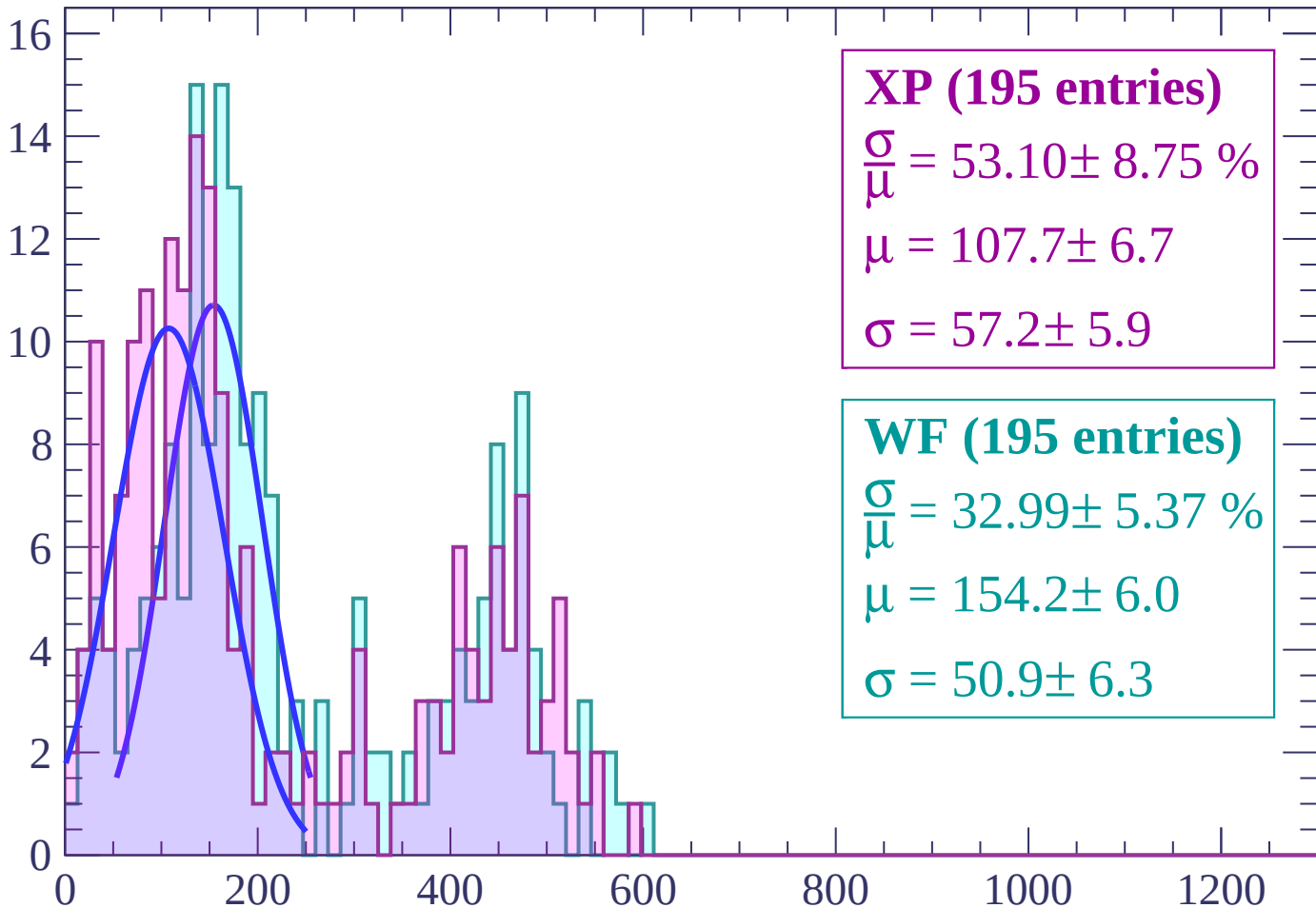
$$\mu = 129.0 \pm 4.1$$

$$\sigma = 31.4 \pm 4.2$$

dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

Count



XP (195 entries)

$$\frac{\sigma}{\mu} = 53.10 \pm 8.75 \%$$

$$\mu = 107.7 \pm 6.7$$

$$\sigma = 57.2 \pm 5.9$$

WF (195 entries)

$$\frac{\sigma}{\mu} = 32.99 \pm 5.37 \%$$

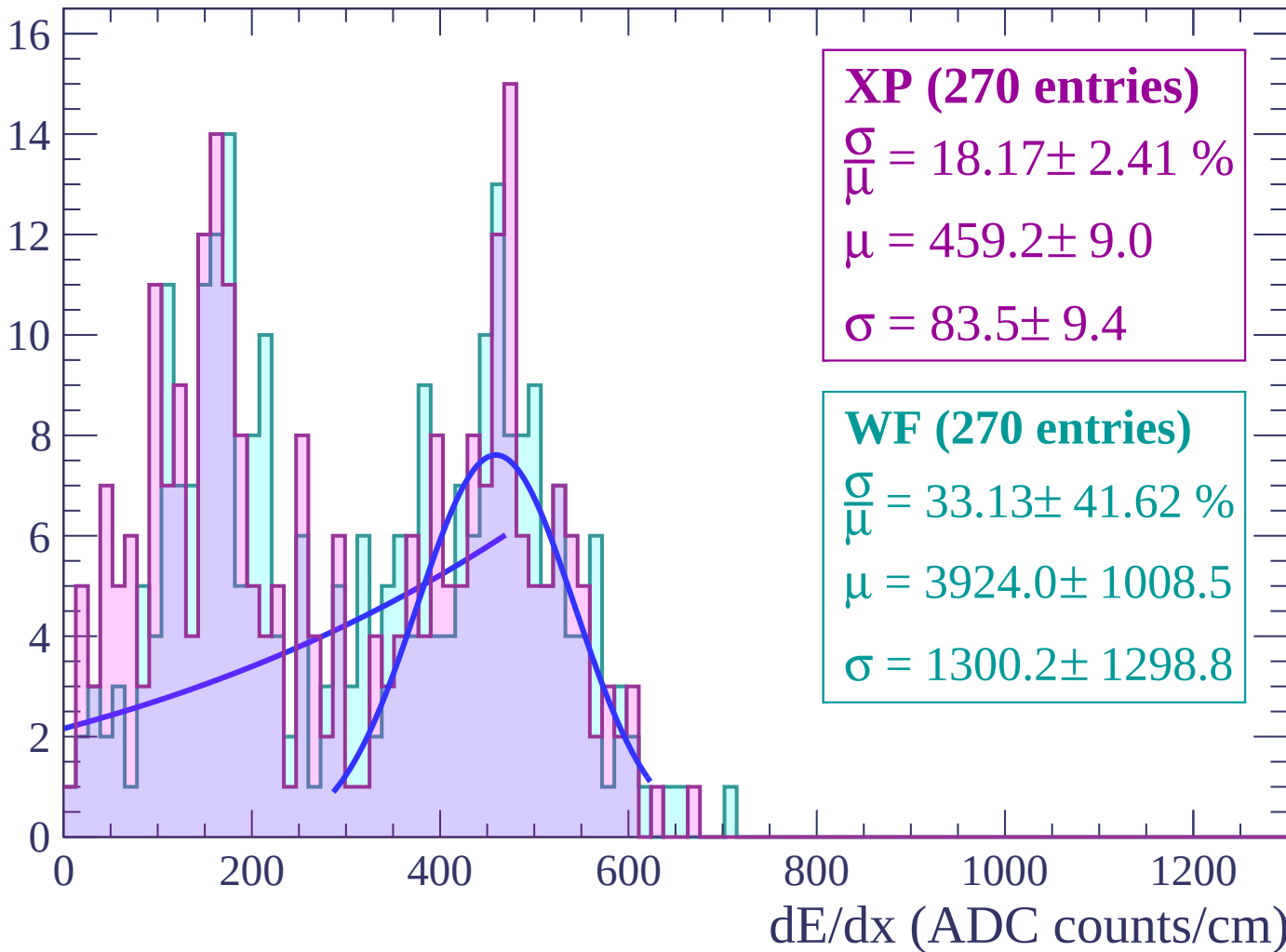
$$\mu = 154.2 \pm 6.0$$

$$\sigma = 50.9 \pm 6.3$$

dE/dx (ADC counts/cm)

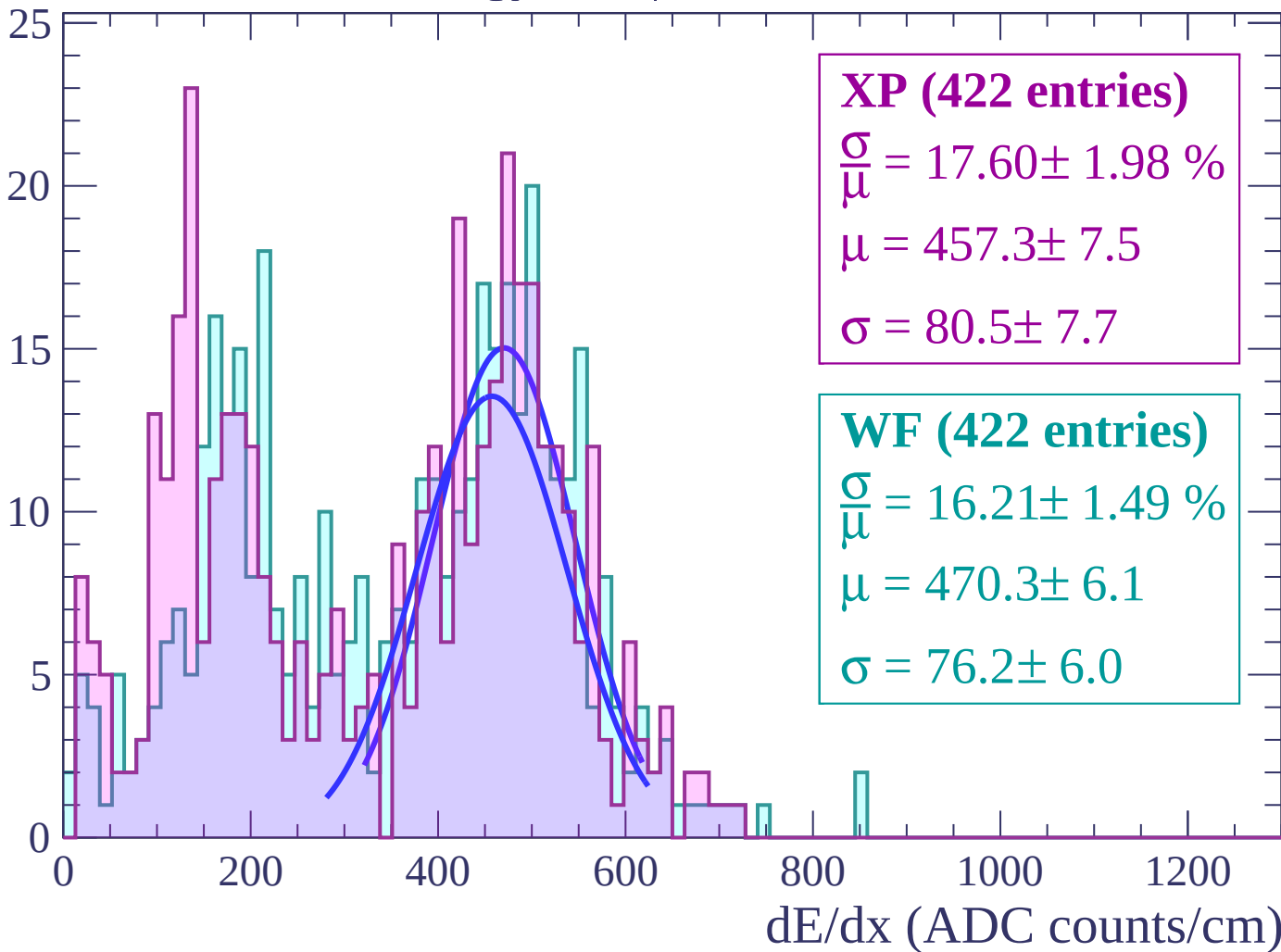
Energy loss | $-70 < \theta < -68$

Count



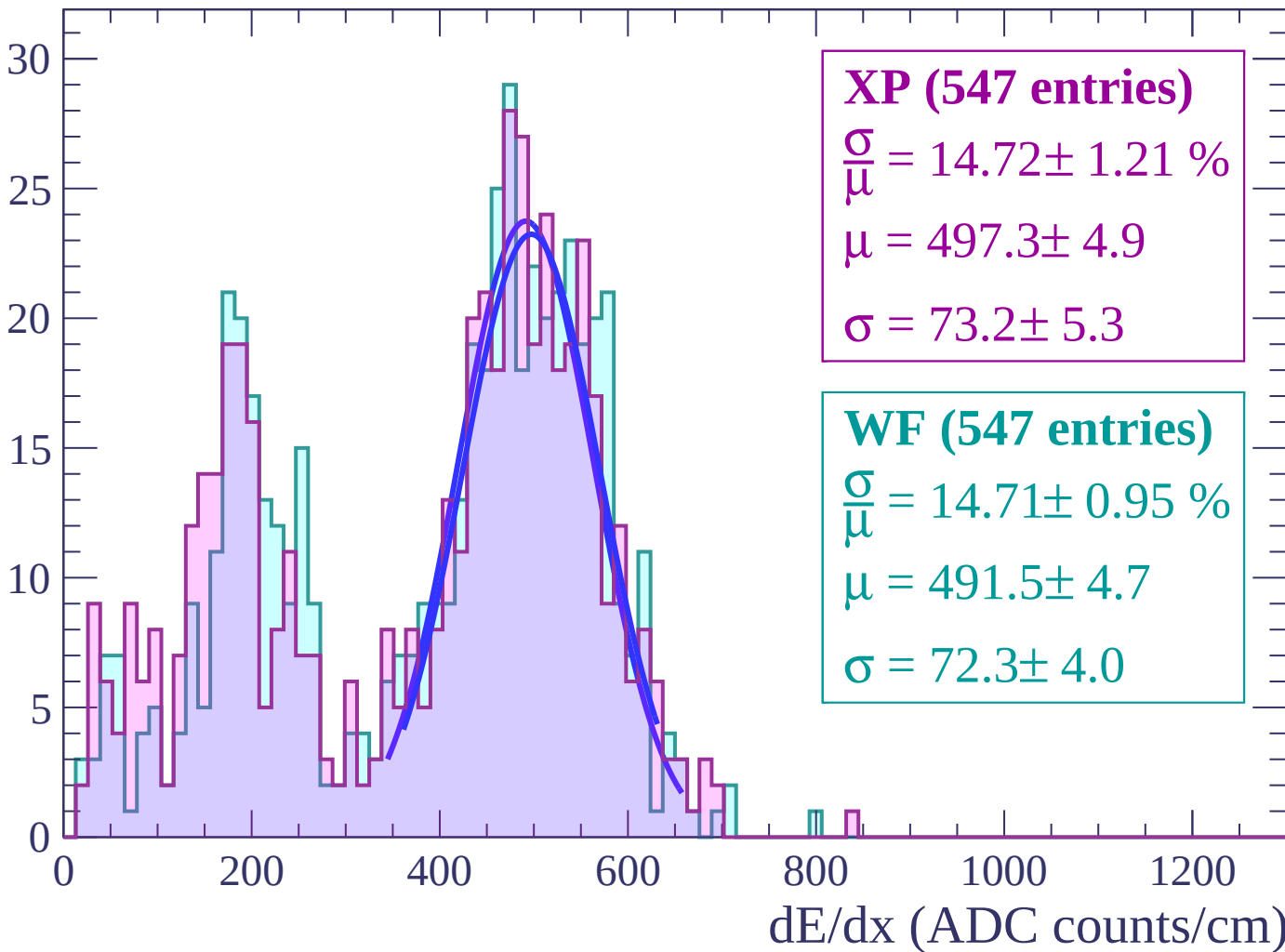
Energy loss | $-68 < \theta < -66$

Count



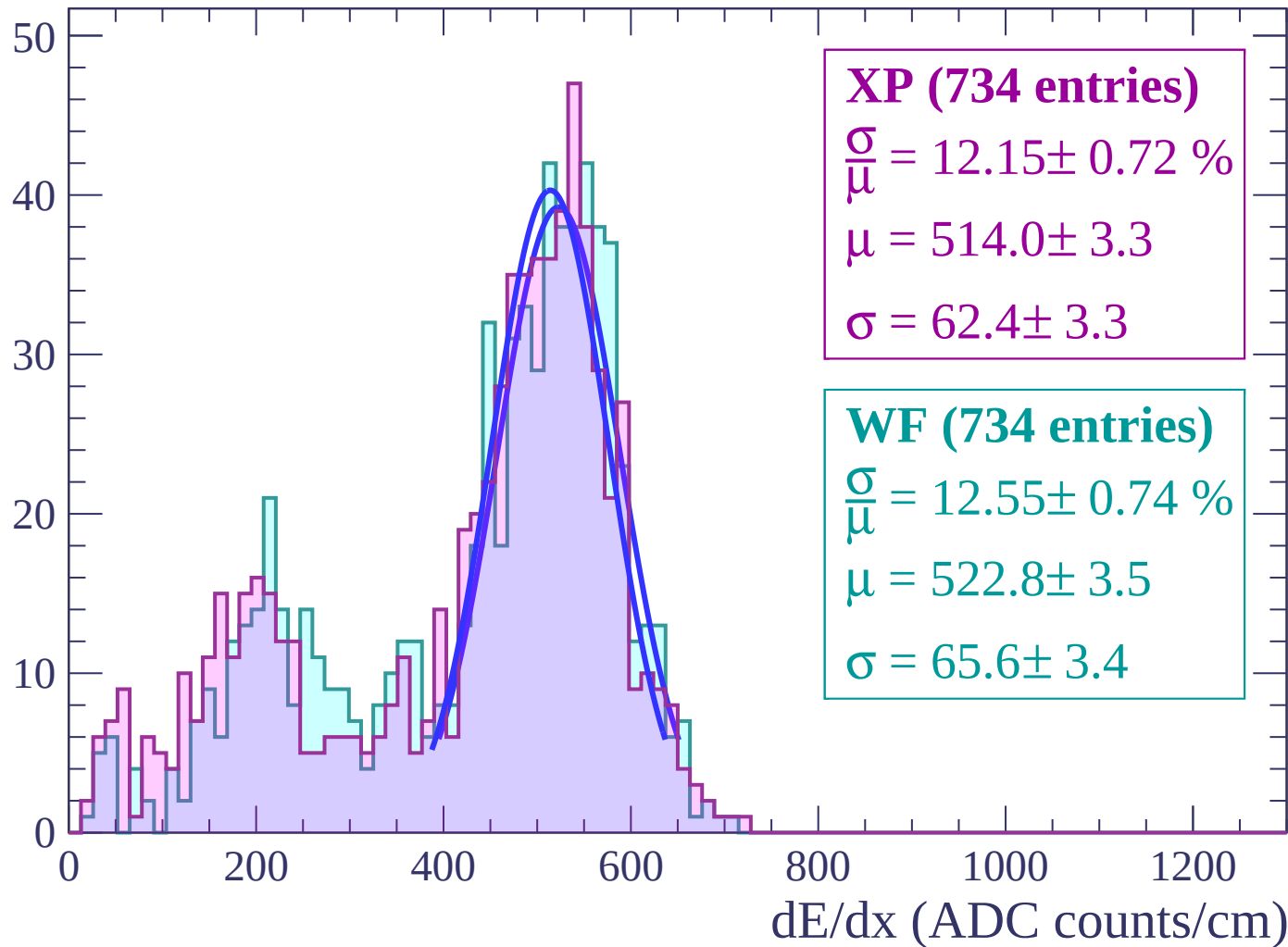
Energy loss | $-66 < \theta < -64$

Count



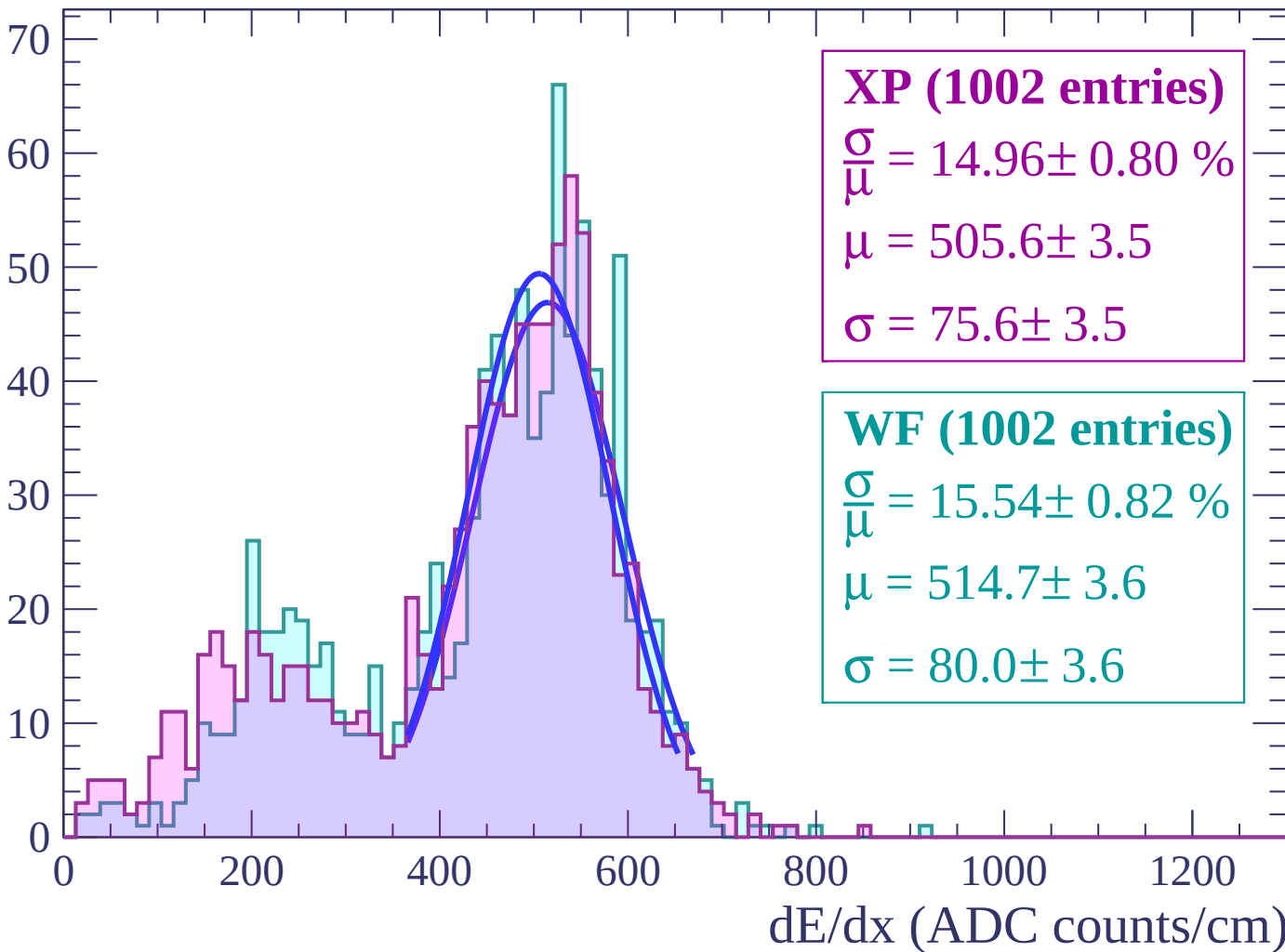
Energy loss | $-64 < \theta < -62$

Count



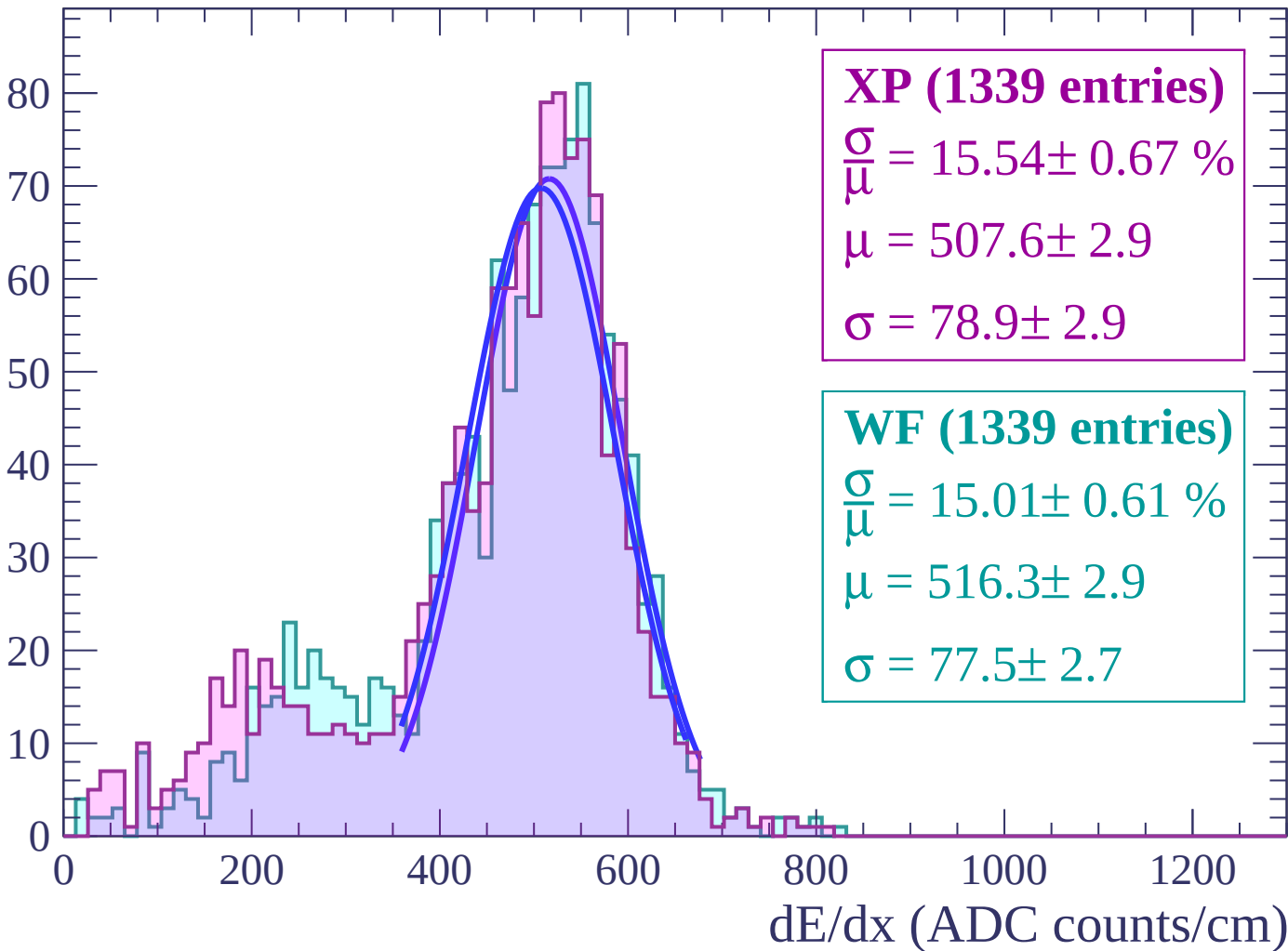
Energy loss | $-62 < \theta < -60$

Count



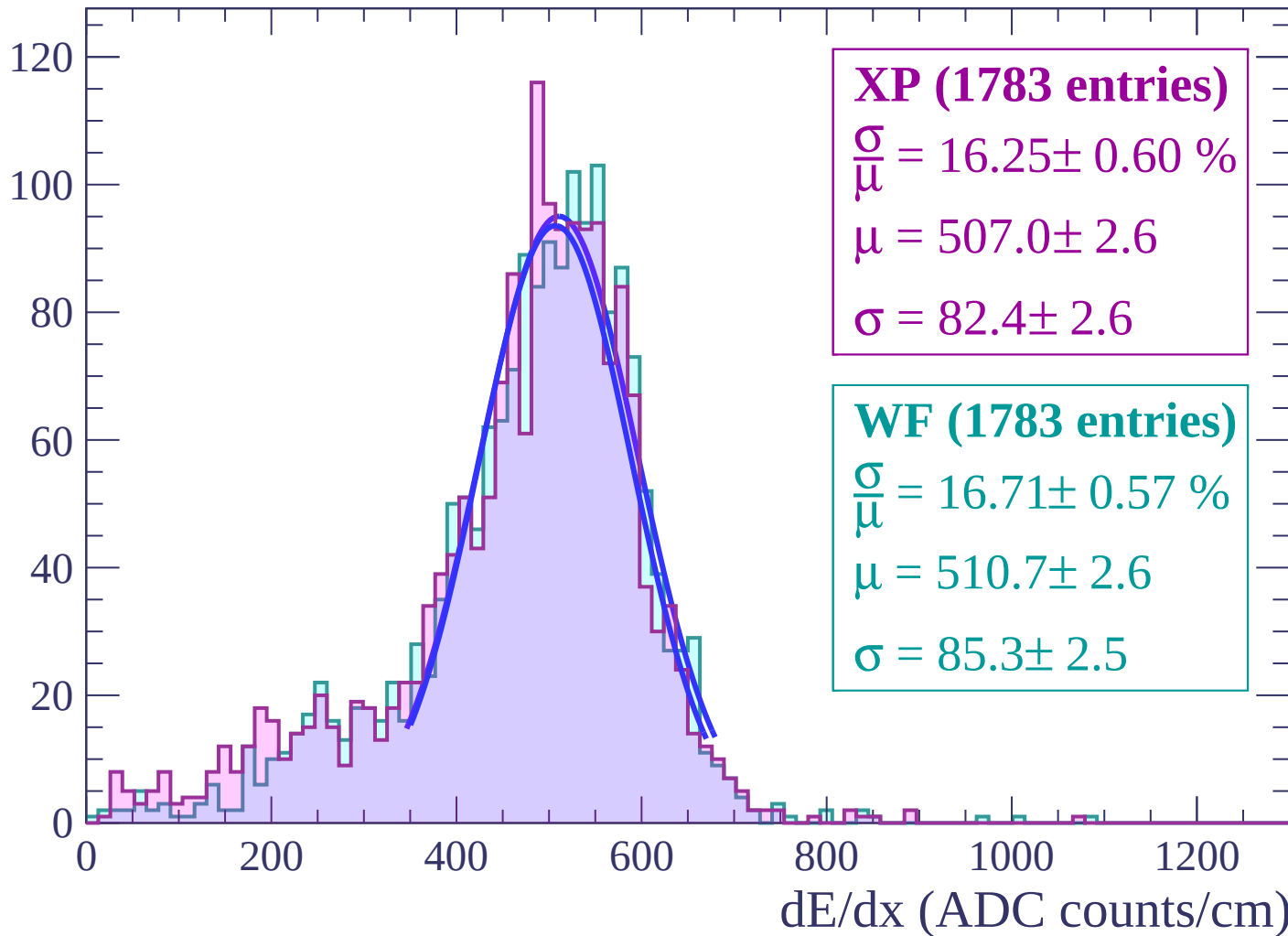
Energy loss | $-60 < \theta < -58$

Count



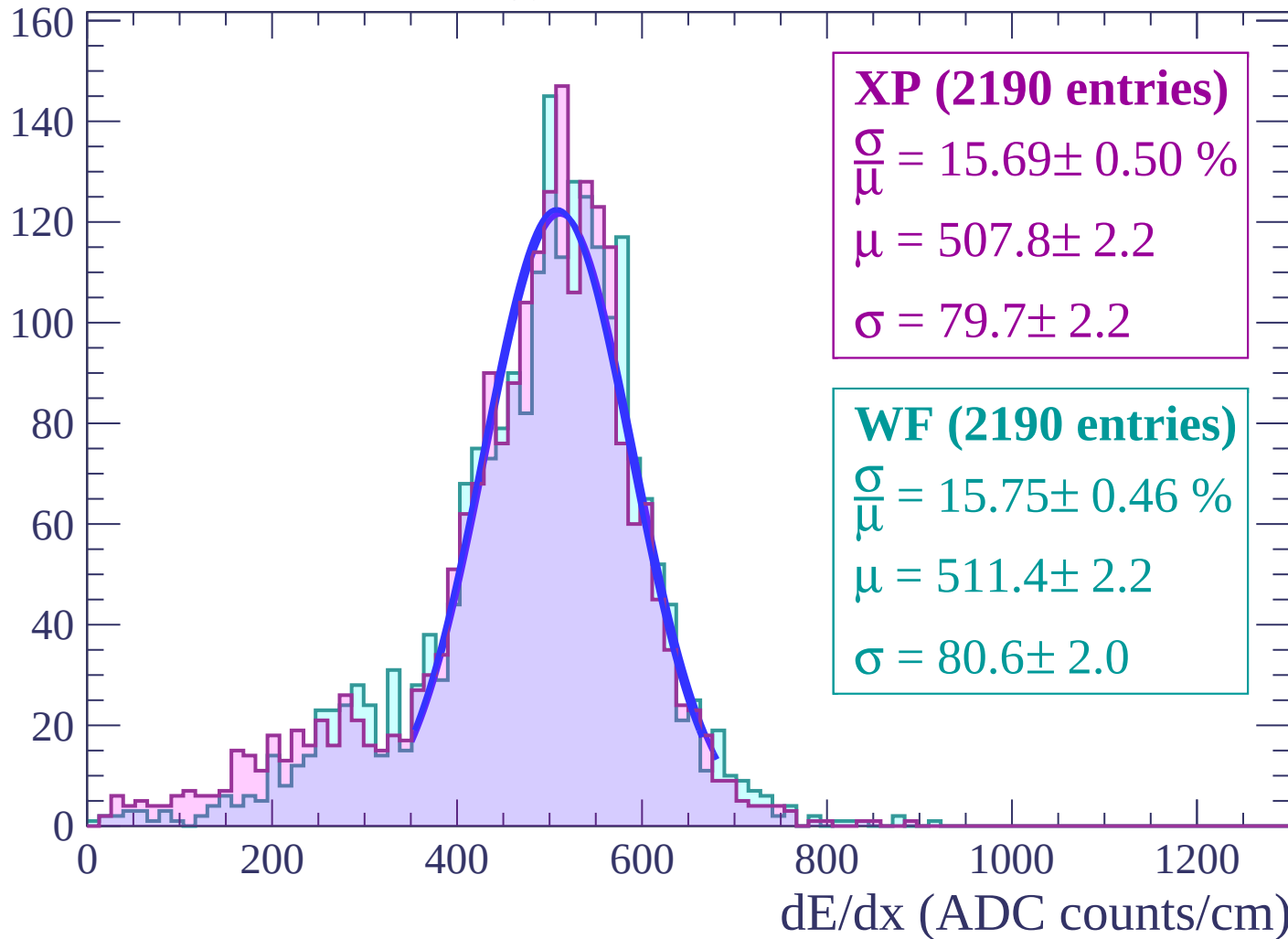
Energy loss | $-58 < \theta < -56$

Count



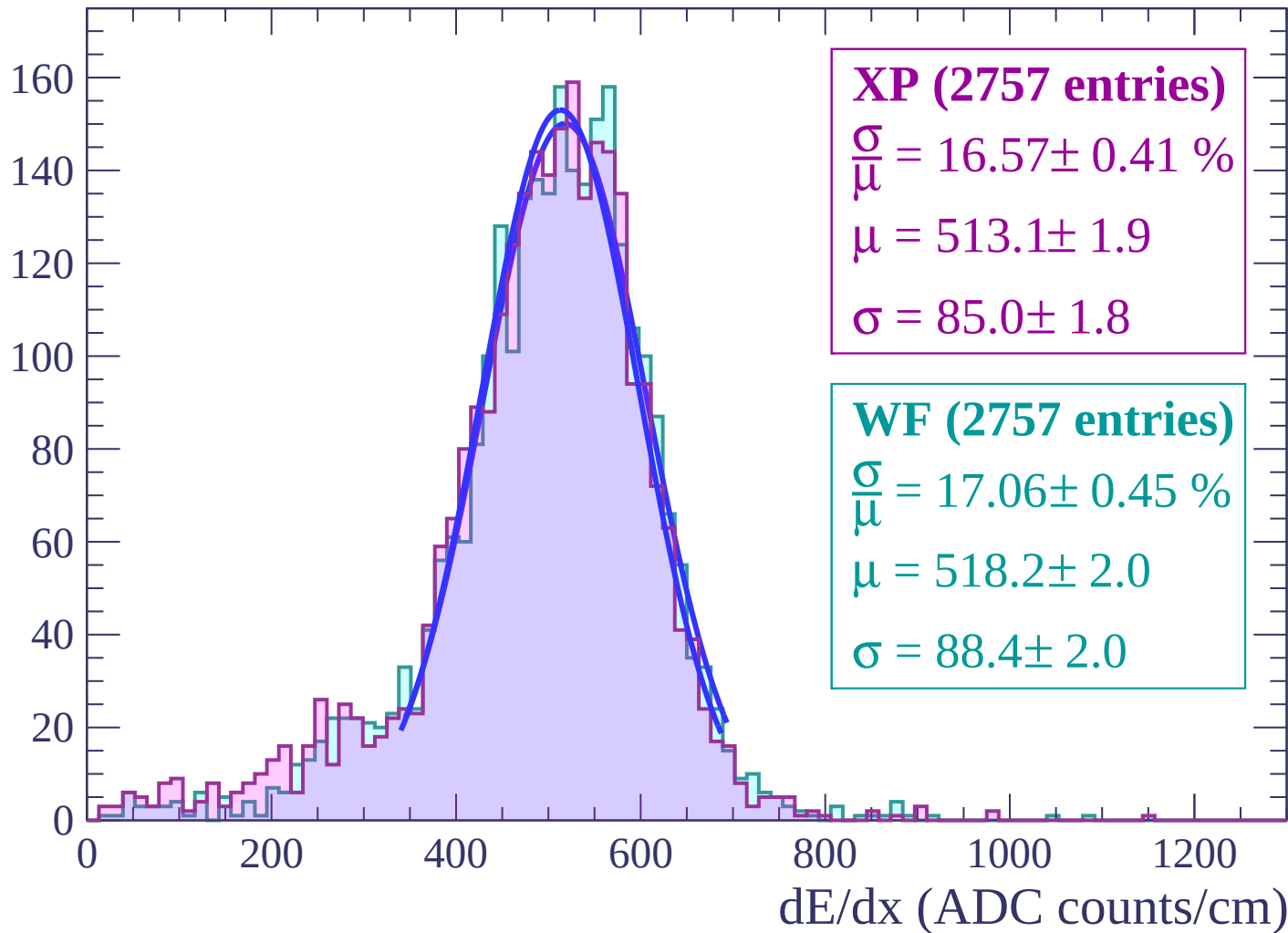
Energy loss | $-56 < \theta < -54$

Count



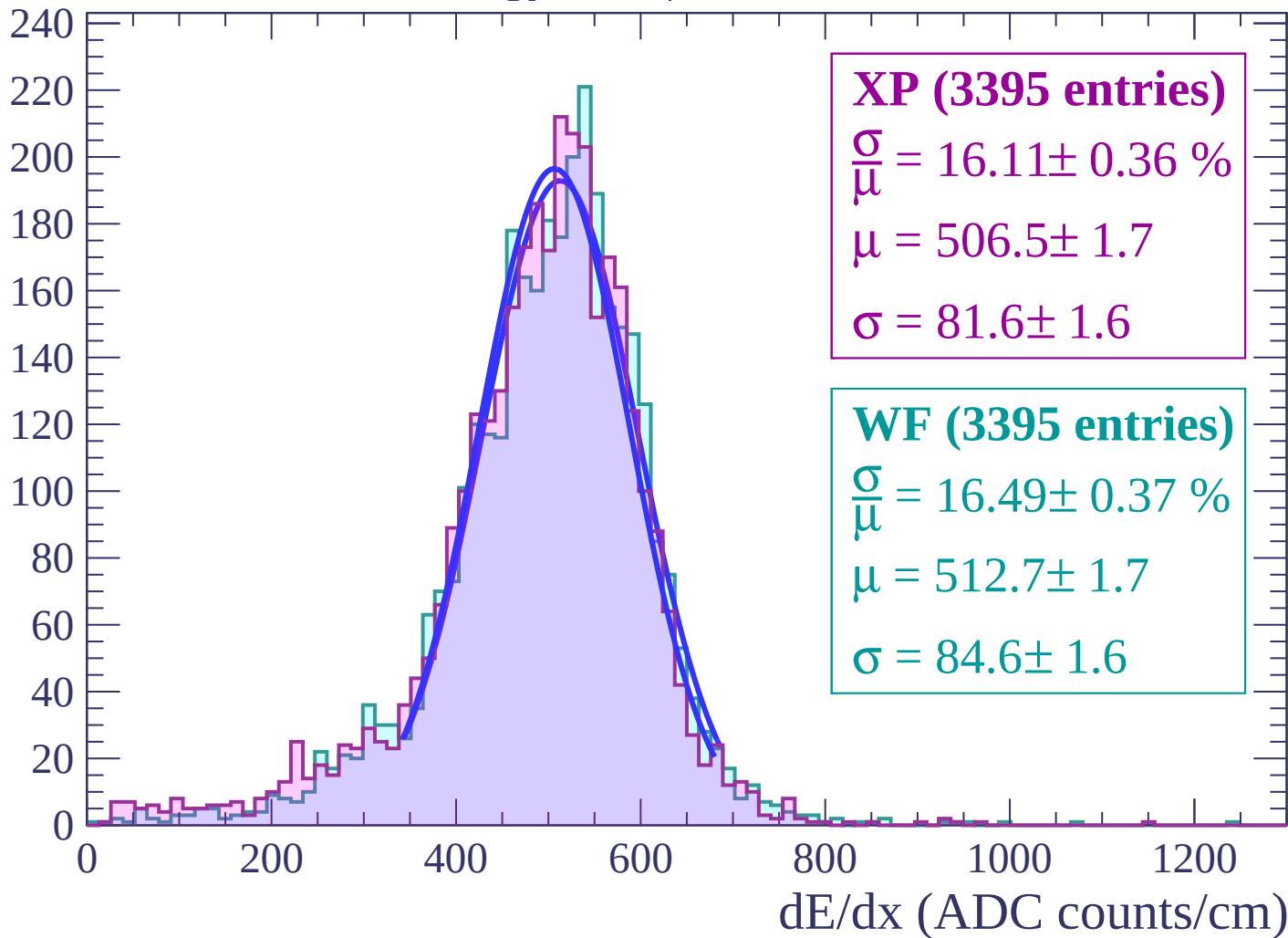
Energy loss | $-54 < \theta < -52$

Count



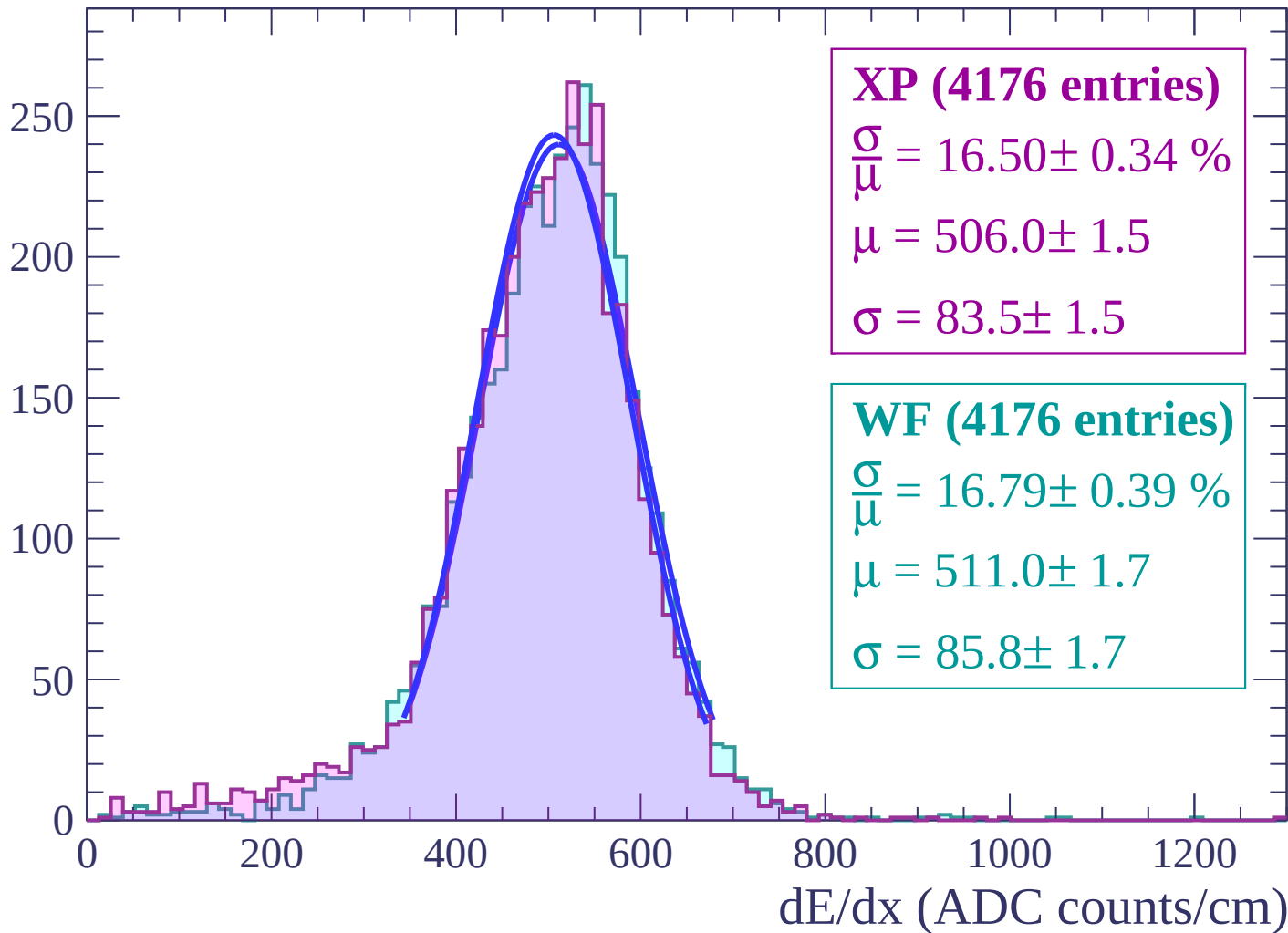
Energy loss | $-52 < \theta < -50$

Count



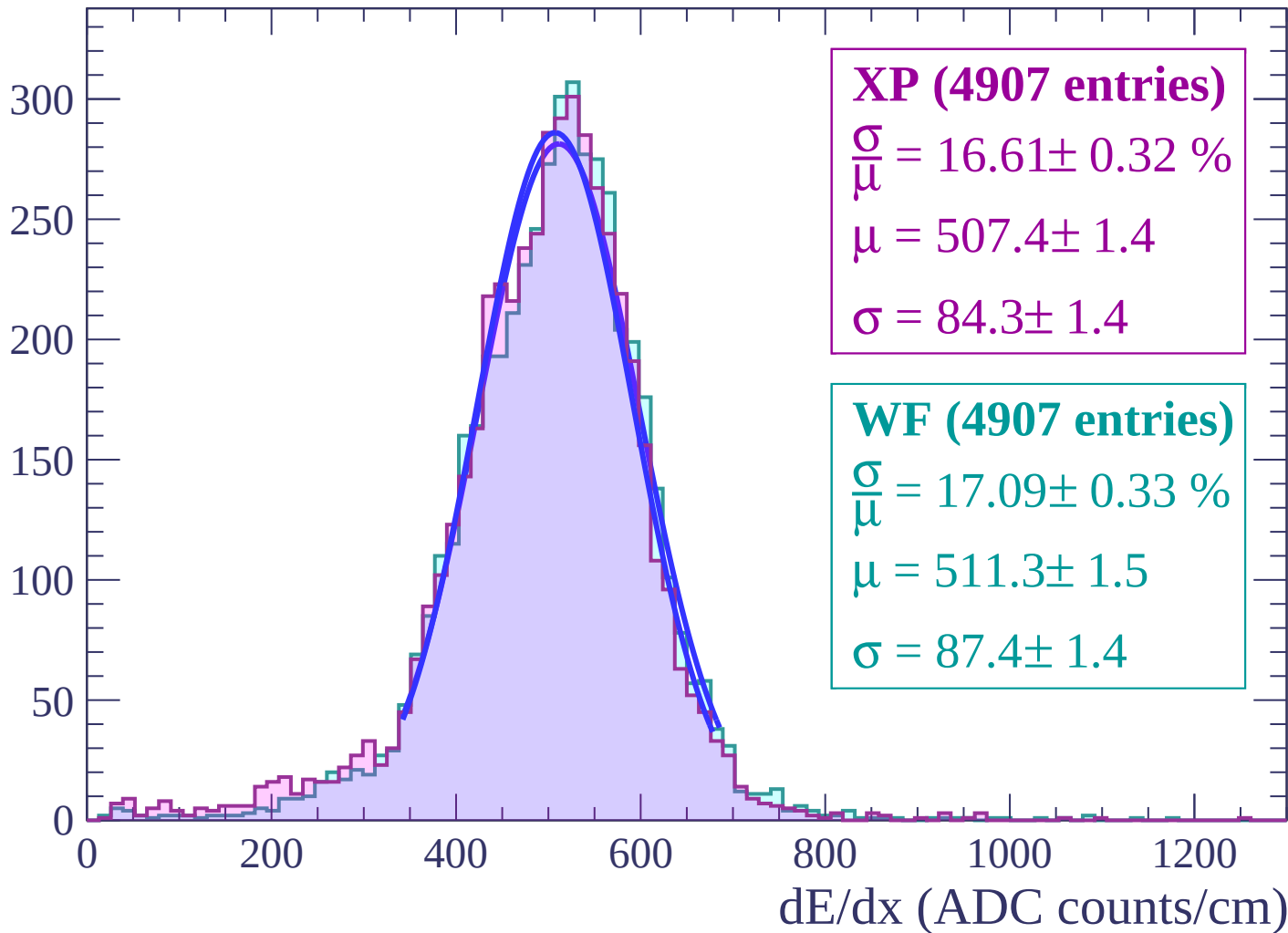
Energy loss | $-50 < \theta < -48$

Count



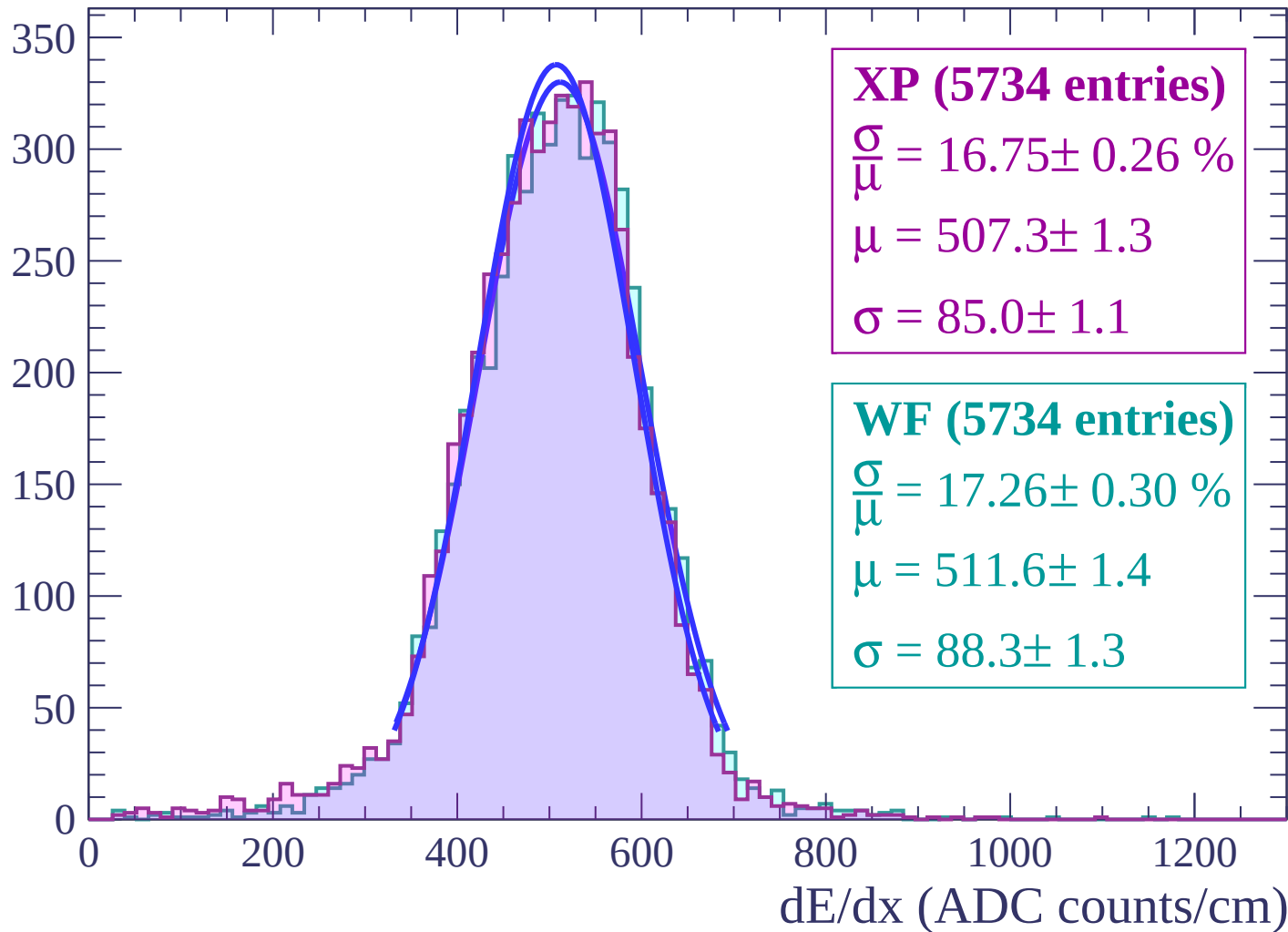
Energy loss | $-48 < \theta < -46$

Count



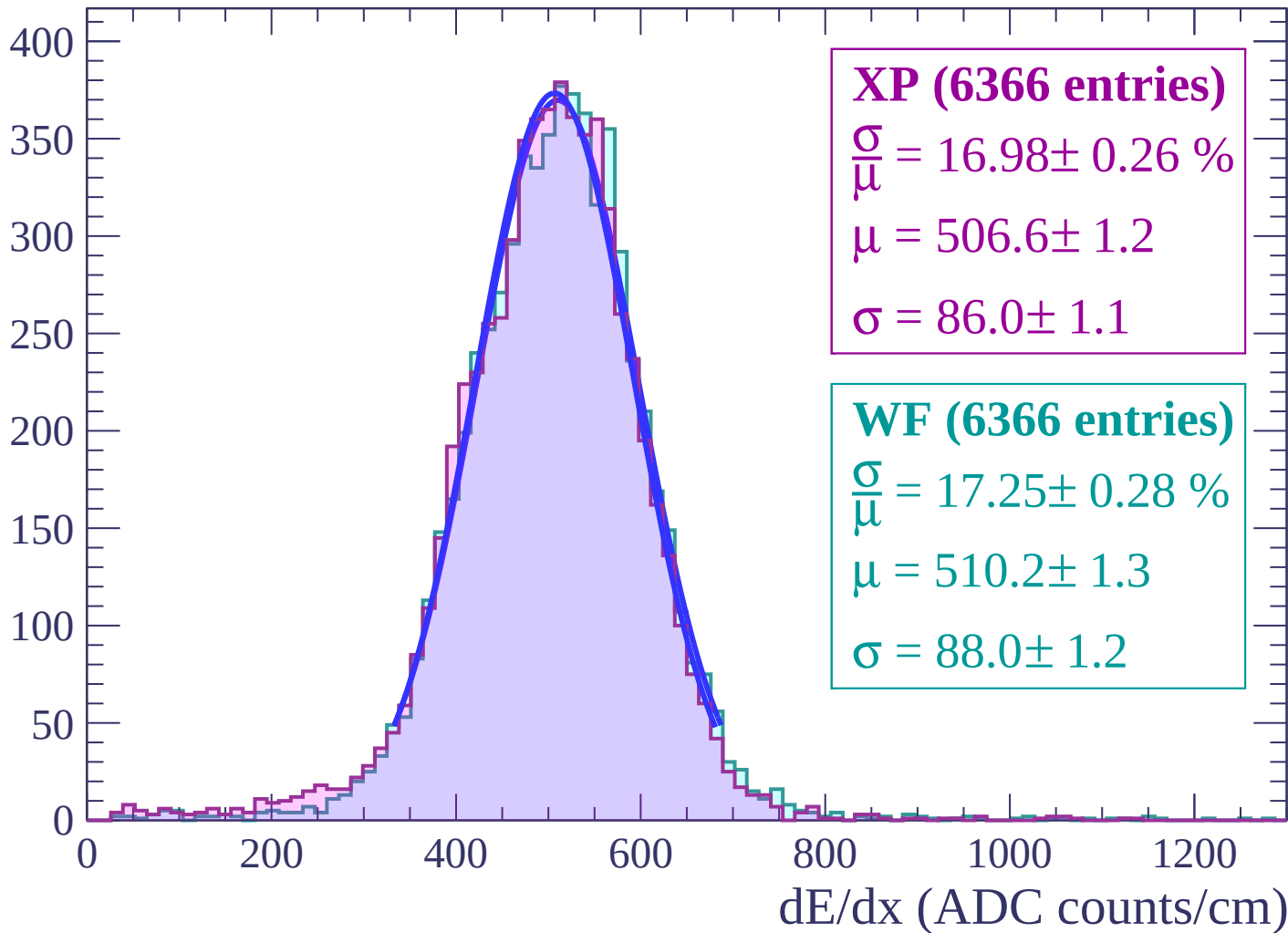
Energy loss | $-46 < \theta < -44$

Count



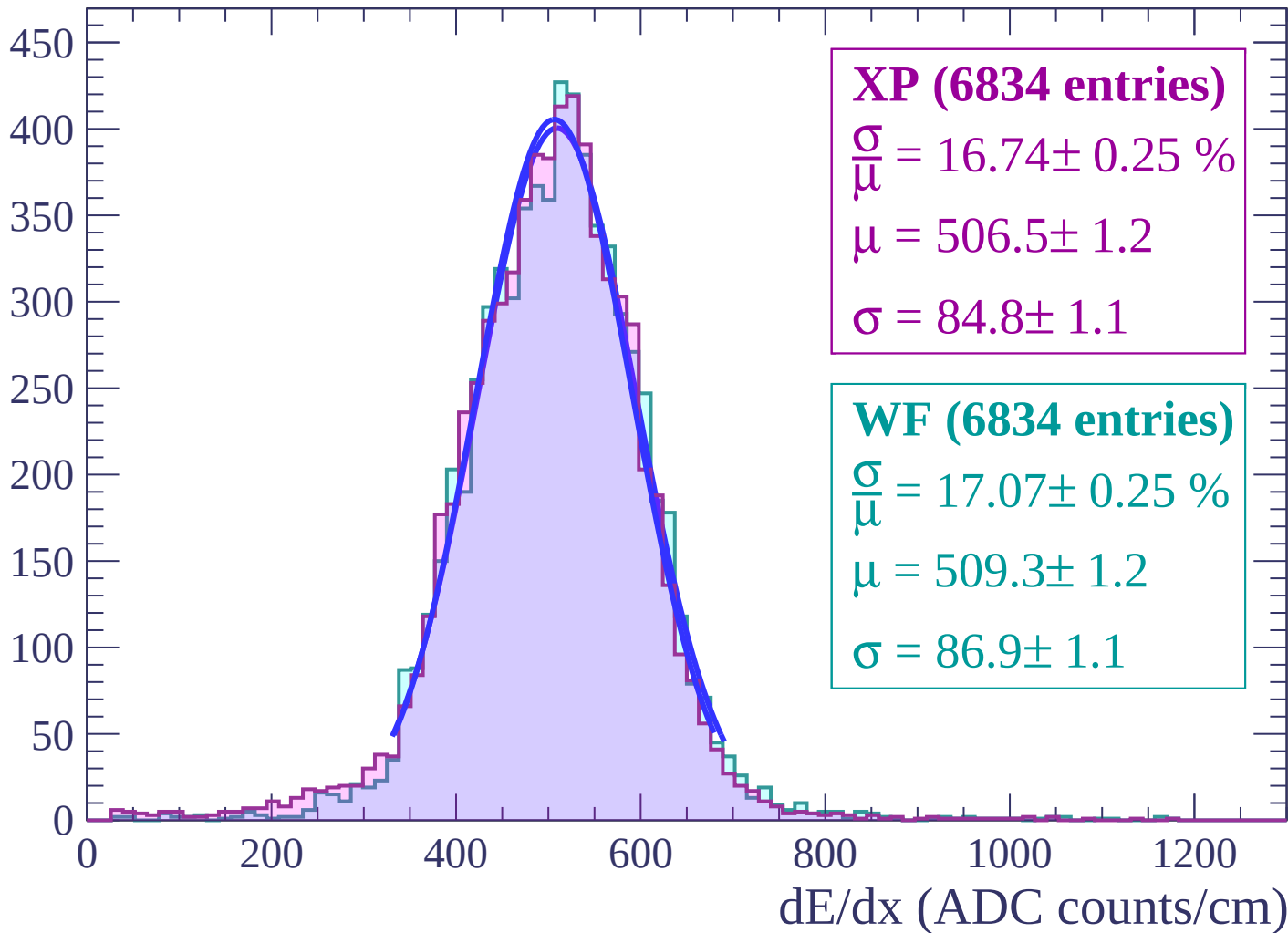
Energy loss | $-44 < \theta < -42$

Count



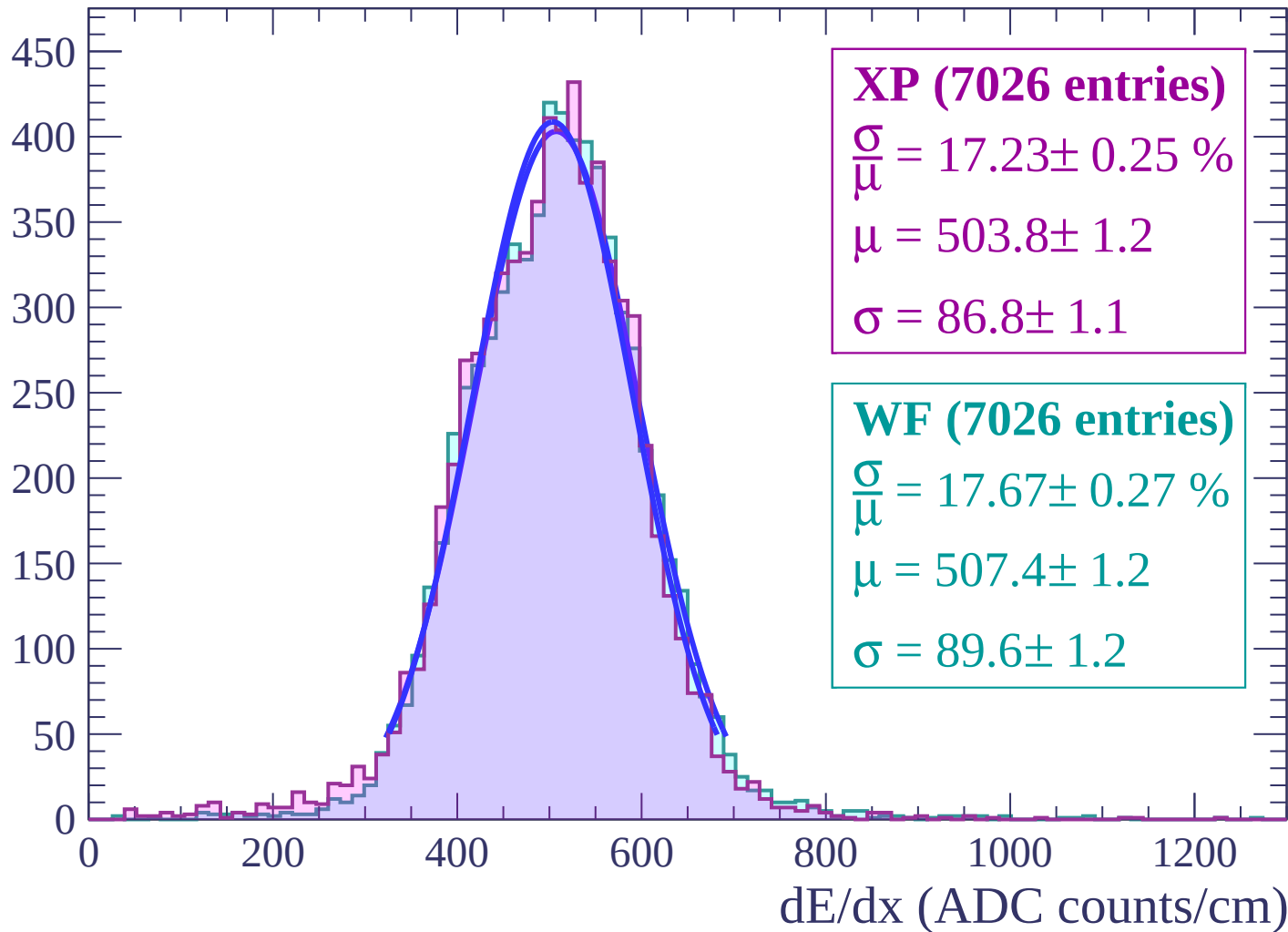
Energy loss | $-42 < \theta < -40$

Count



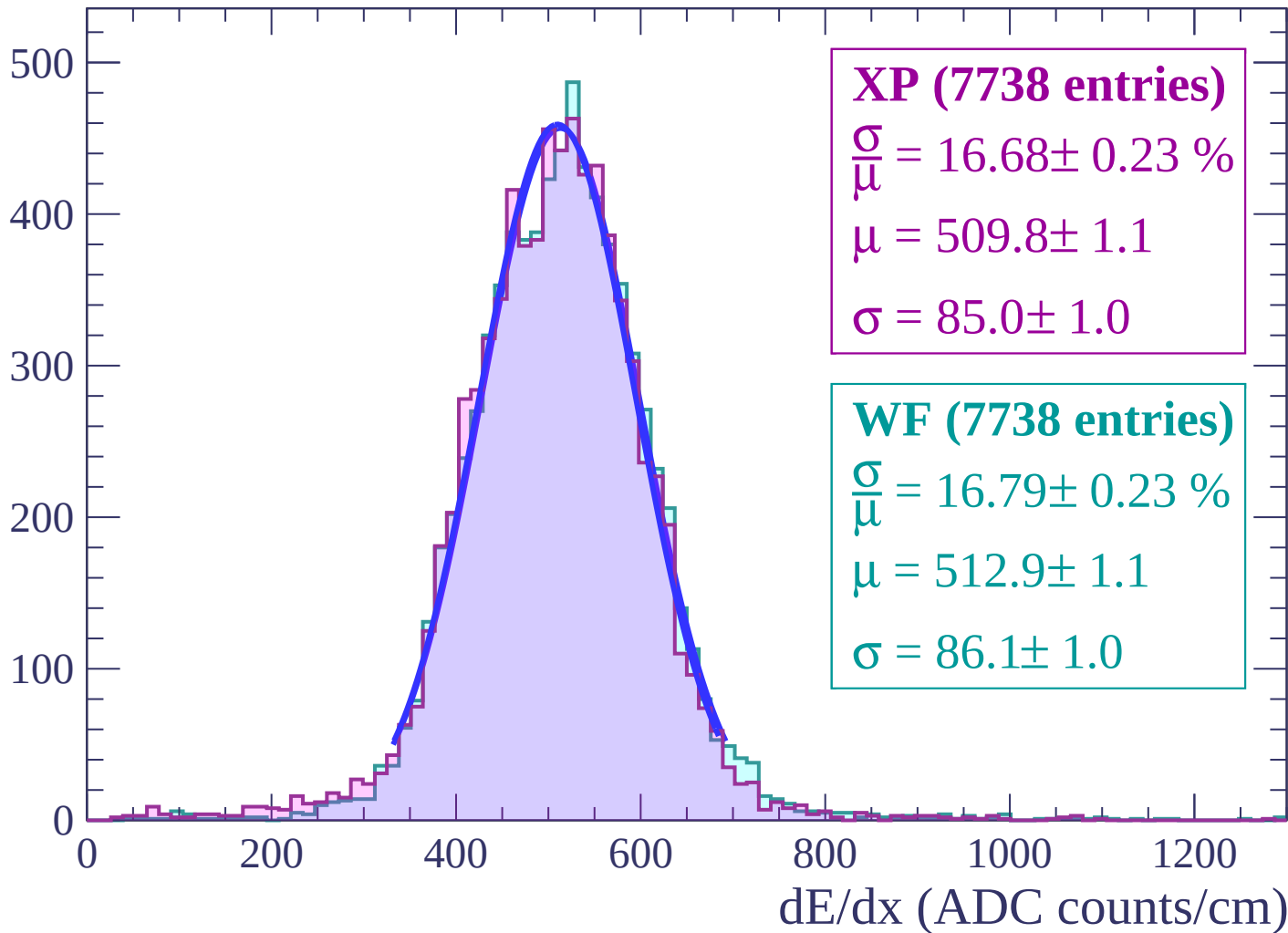
Energy loss | $-40 < \theta < -38$

Count



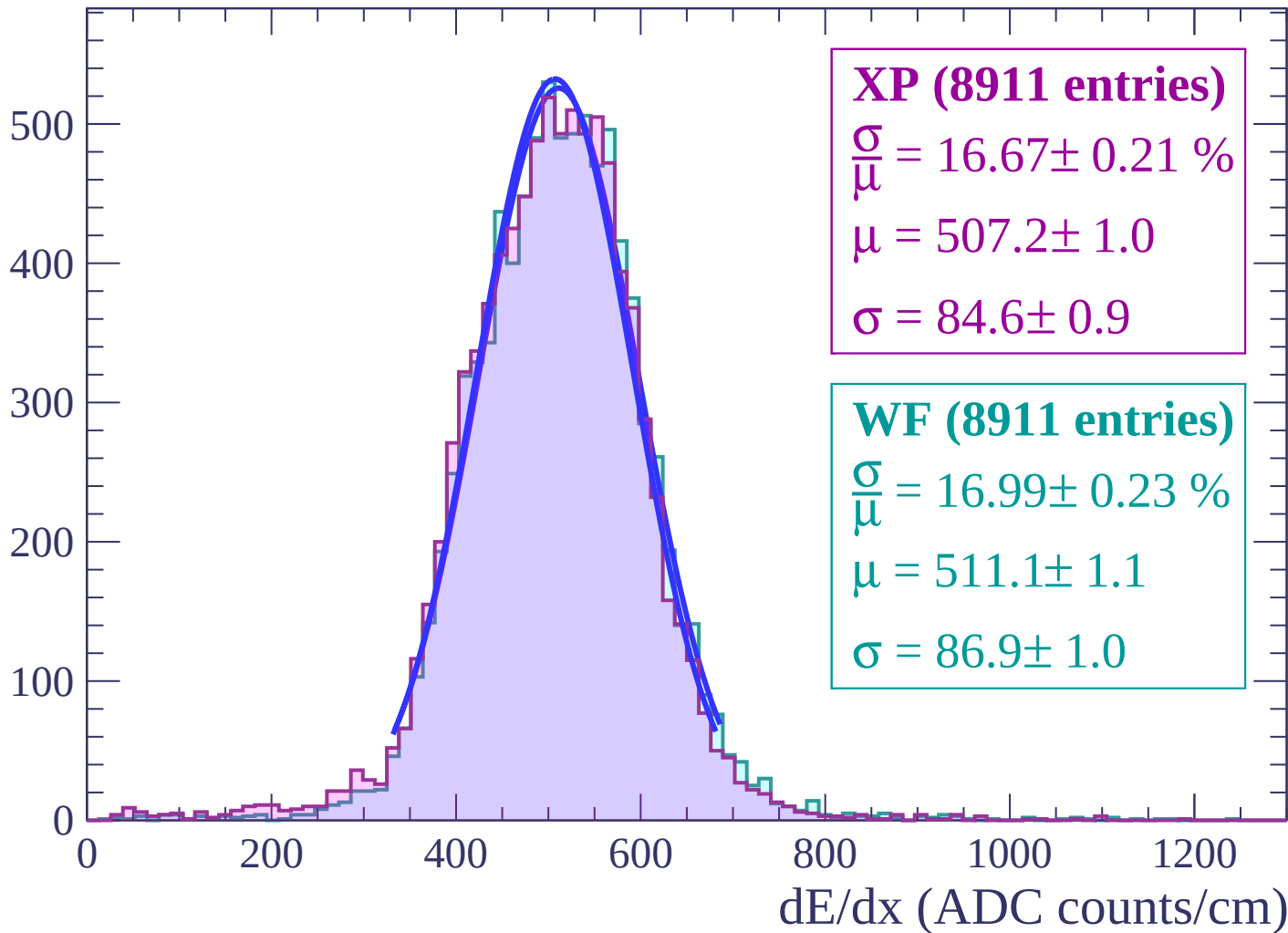
Energy loss | $-38 < \theta < -36$

Count



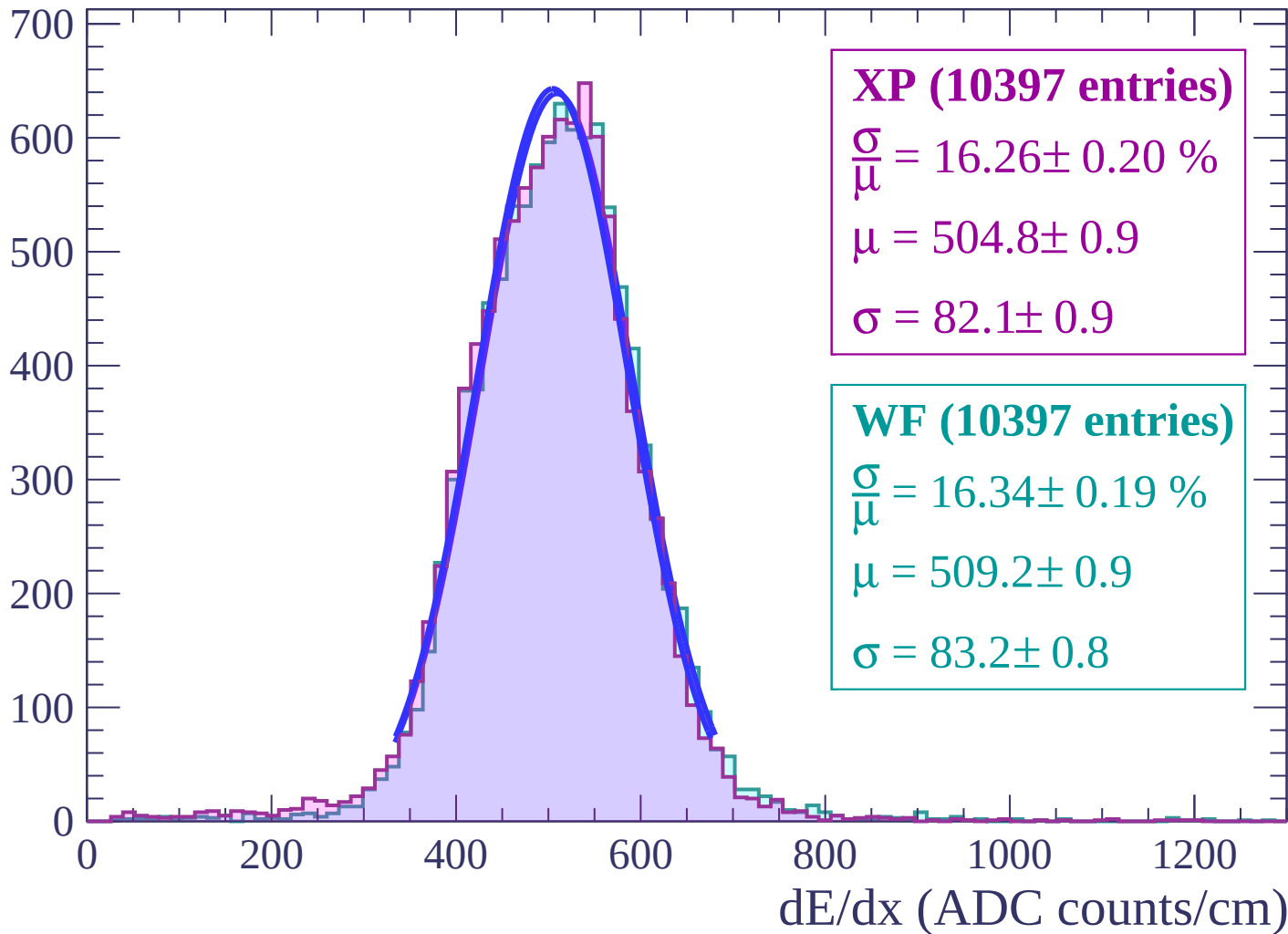
Energy loss | $-36 < \theta < -34$

Count



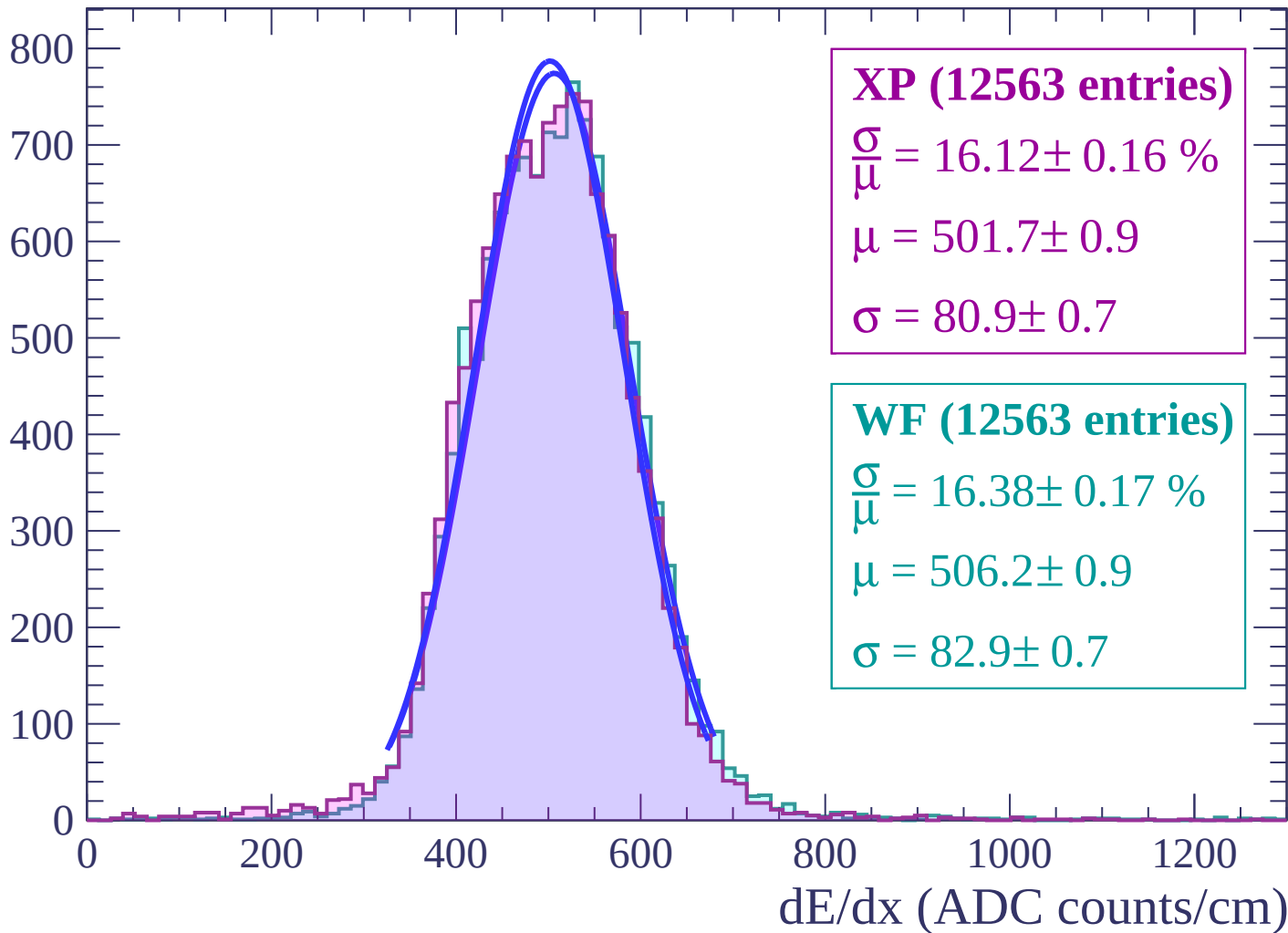
Energy loss | $-34 < \theta < -32$

Count



Energy loss | $-32 < \theta < -30$

Count



Energy loss | $-30 < \theta < -28$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (15565 entries)

$\frac{\sigma}{\mu} = 16.13 \pm 0.15 \%$

$\mu = 496.4 \pm 0.7$

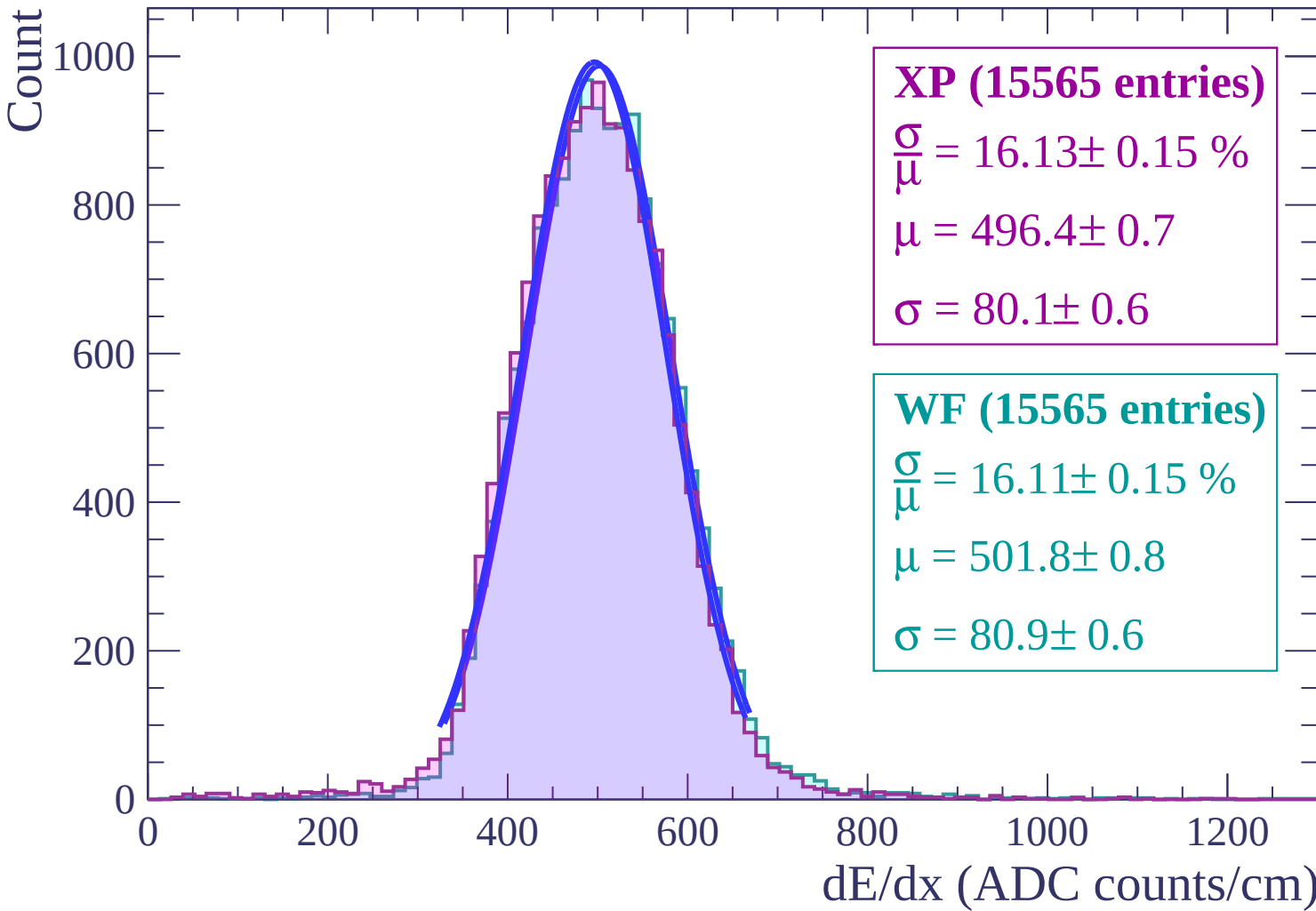
$\sigma = 80.1 \pm 0.6$

WF (15565 entries)

$\frac{\sigma}{\mu} = 16.11 \pm 0.15 \%$

$\mu = 501.8 \pm 0.8$

$\sigma = 80.9 \pm 0.6$



Energy loss | $-28 < \theta < -26$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (19278 entries)

$\frac{\sigma}{\mu} = 15.97 \pm 0.13 \%$

$\mu = 494.7 \pm 0.6$

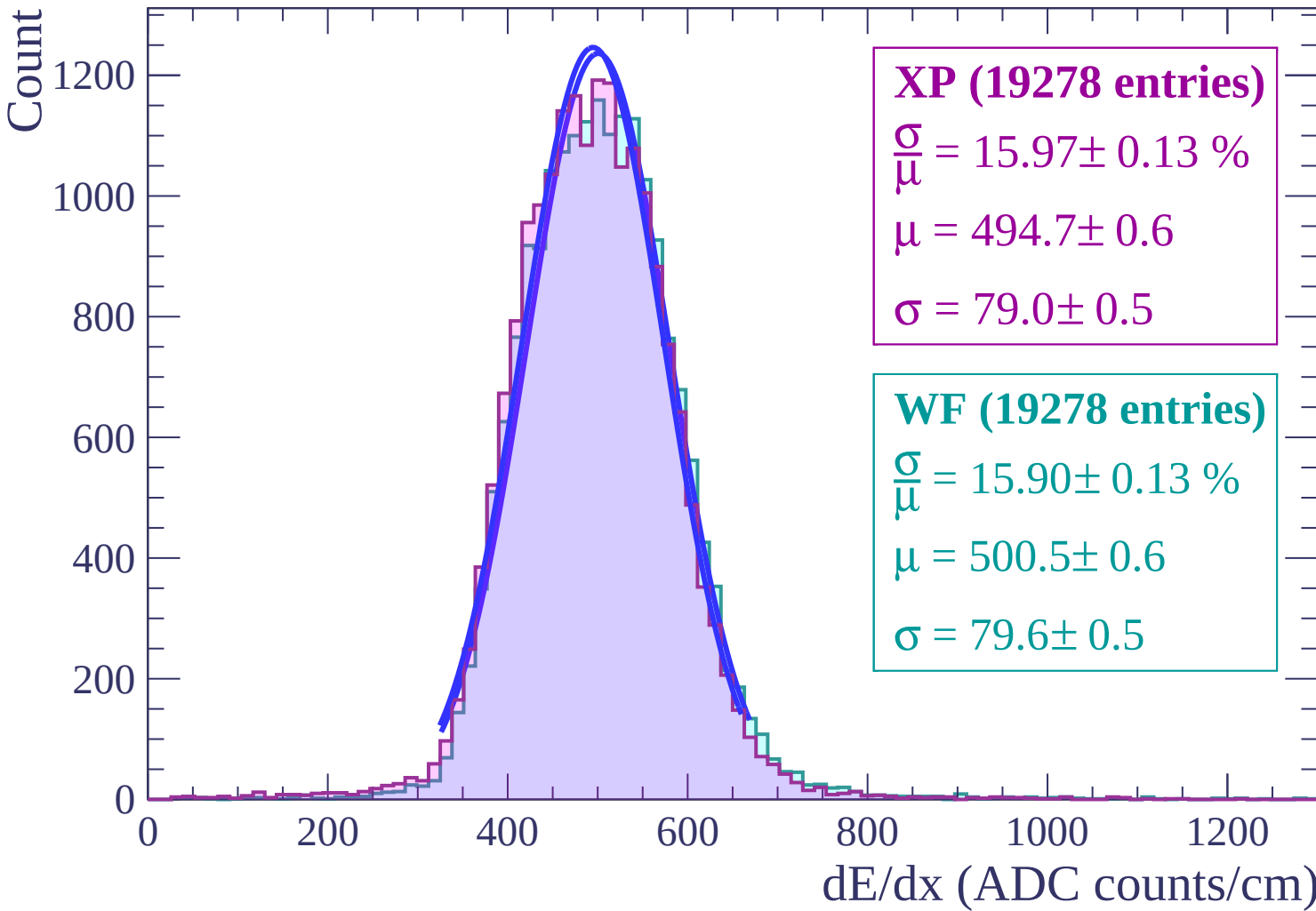
$\sigma = 79.0 \pm 0.5$

WF (19278 entries)

$\frac{\sigma}{\mu} = 15.90 \pm 0.13 \%$

$\mu = 500.5 \pm 0.6$

$\sigma = 79.6 \pm 0.5$



Energy loss | $-26 < \theta < -24$

Count

1400
1200
1000
800
600
400
200
0

XP (21927 entries)

$\frac{\sigma}{\mu} = 15.81 \pm 0.13 \%$

$\mu = 491.8 \pm 0.6$

$\sigma = 77.8 \pm 0.5$

WF (21927 entries)

$\frac{\sigma}{\mu} = 15.76 \pm 0.12 \%$

$\mu = 497.3 \pm 0.6$

$\sigma = 78.4 \pm 0.5$

0

200

400

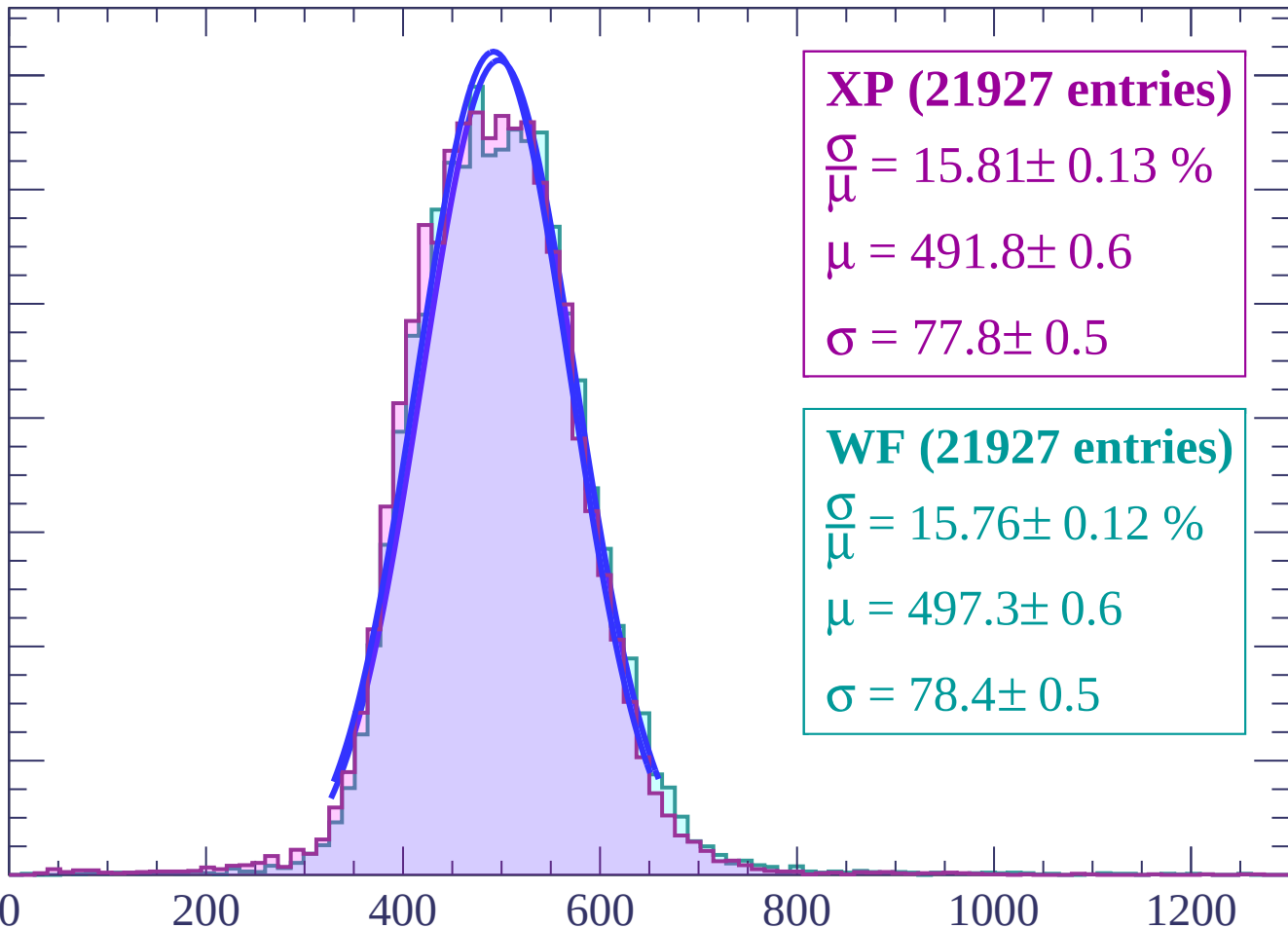
600

800

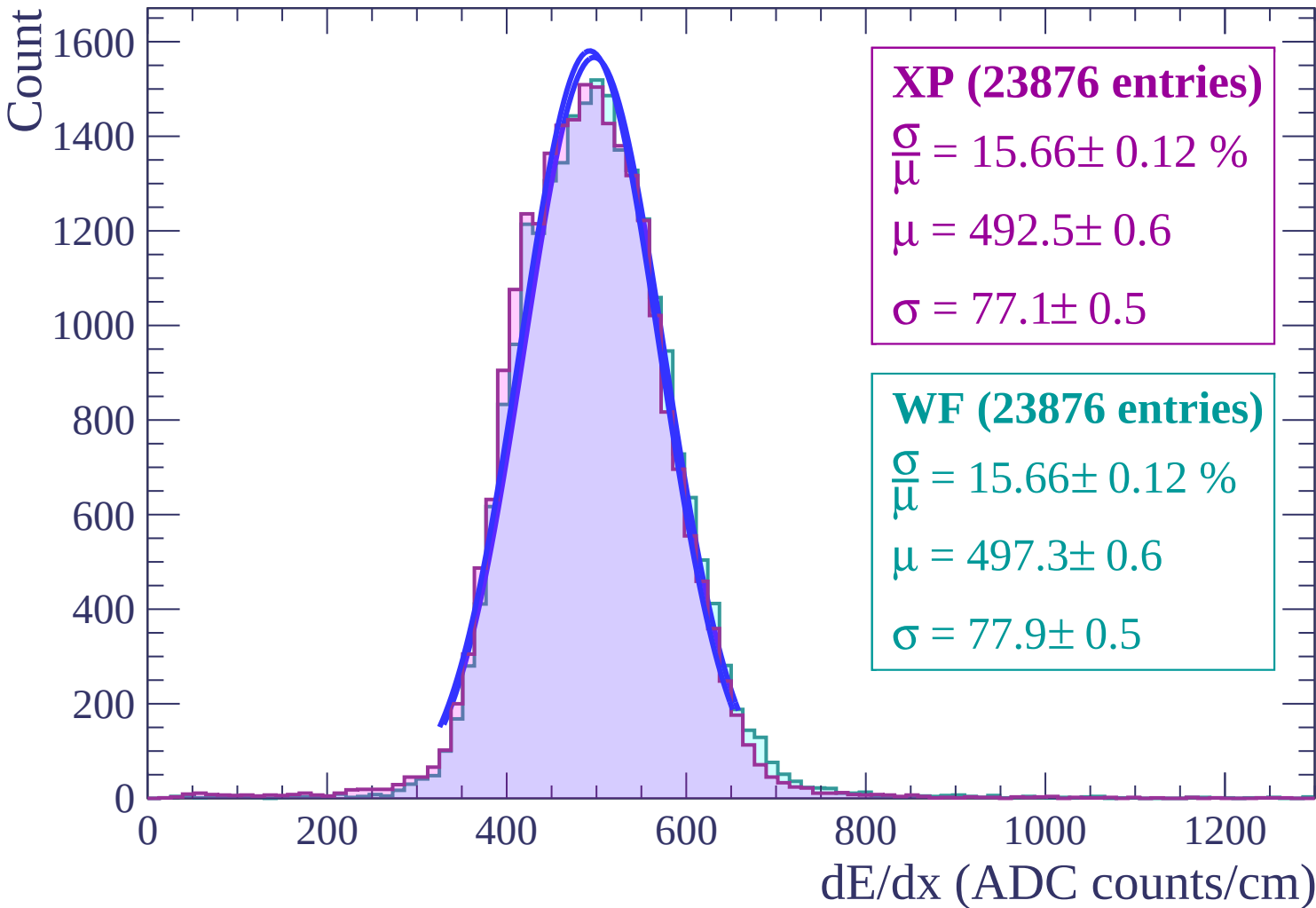
1000

1200

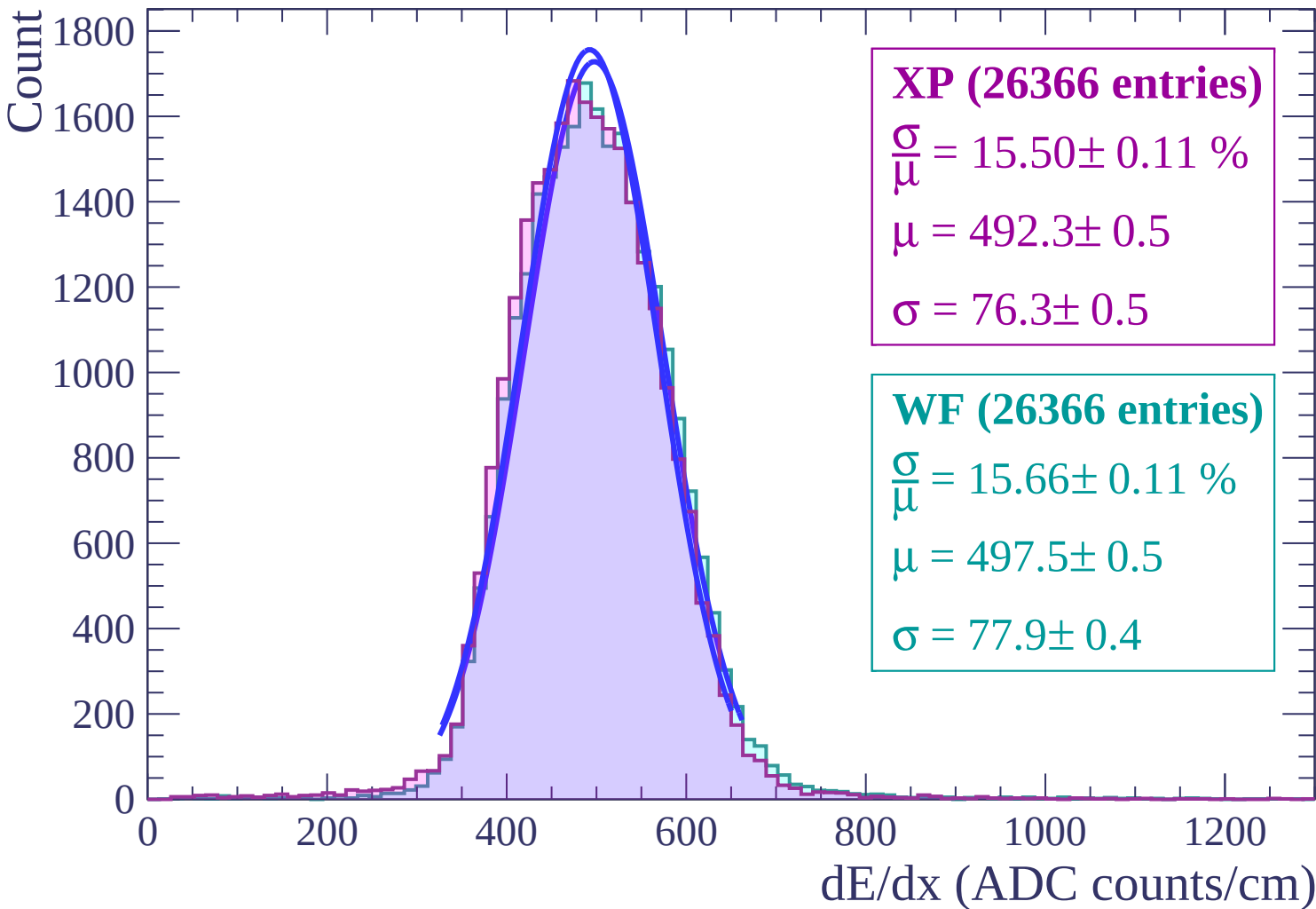
dE/dx (ADC counts/cm)



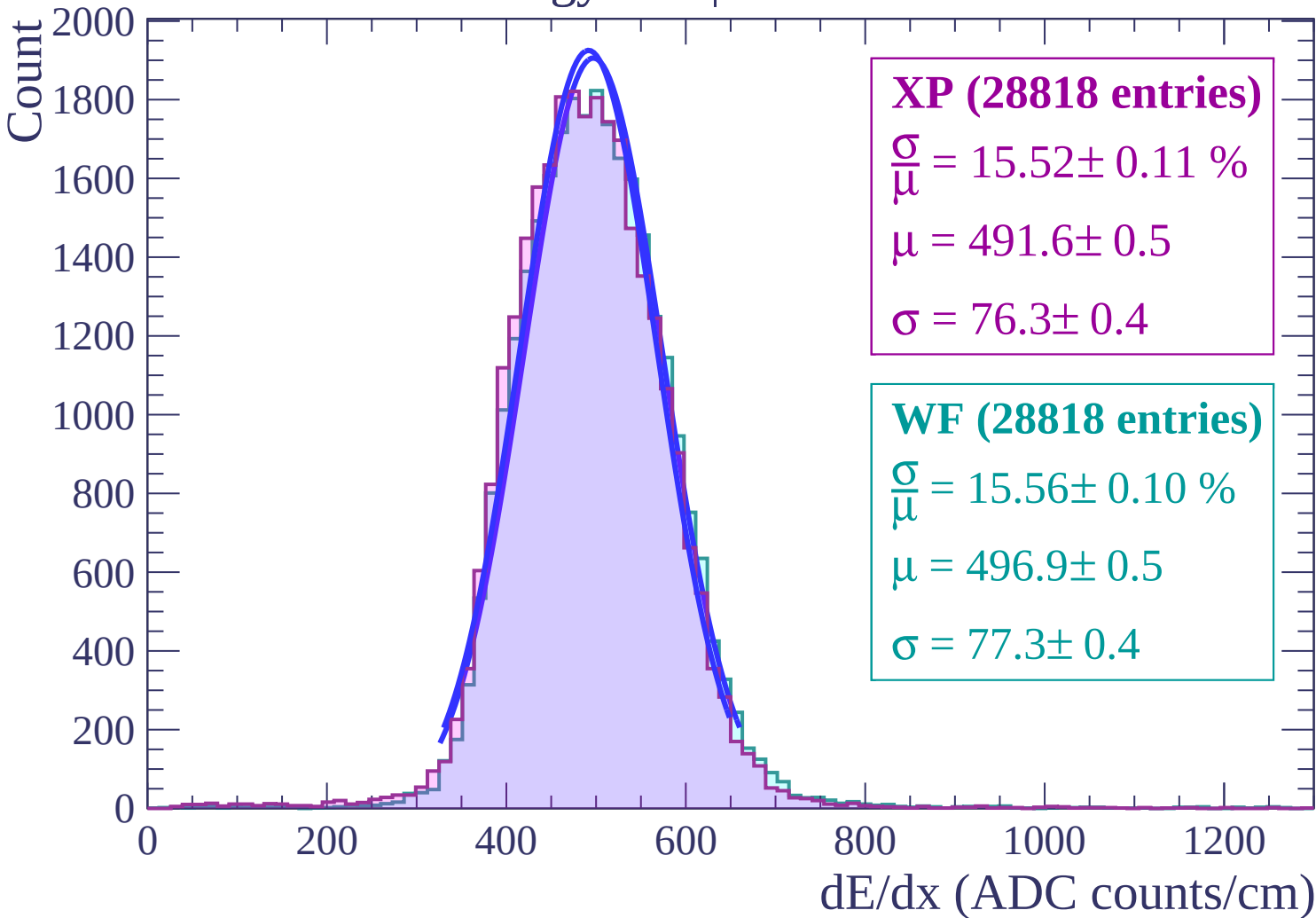
Energy loss | $-24 < \theta < -22$



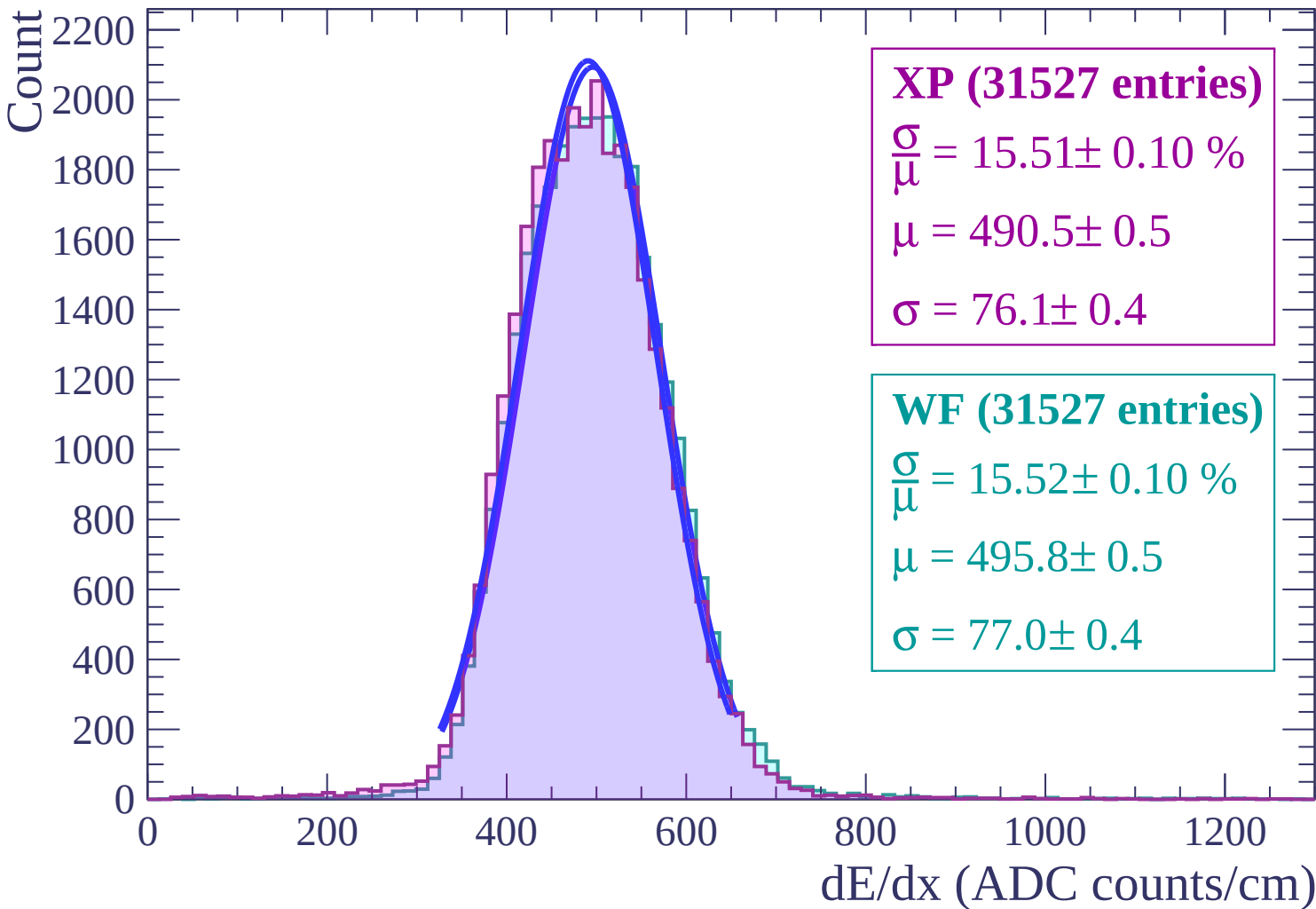
Energy loss | $-22 < \theta < -20$



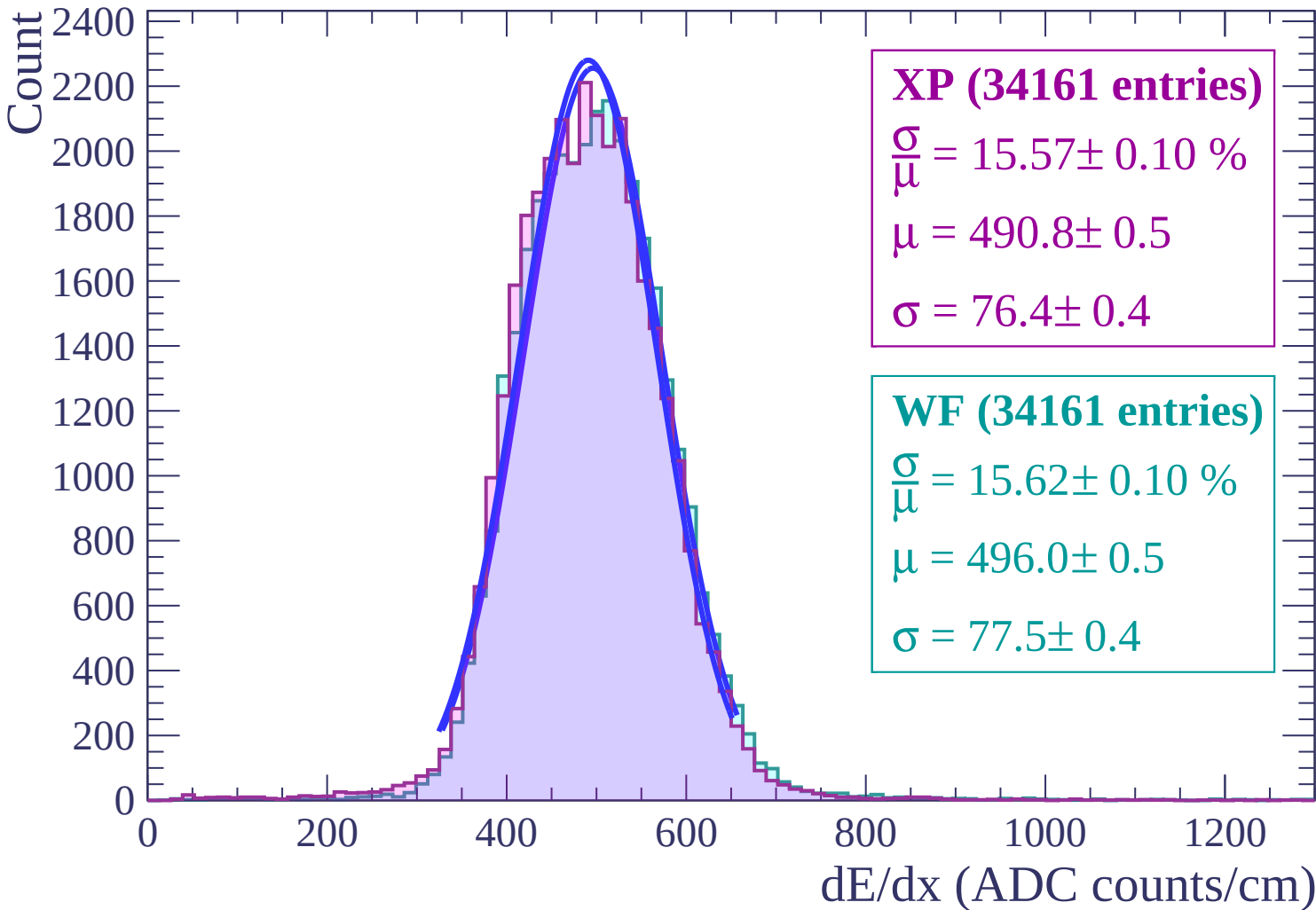
Energy loss | $-20 < \theta < -18$



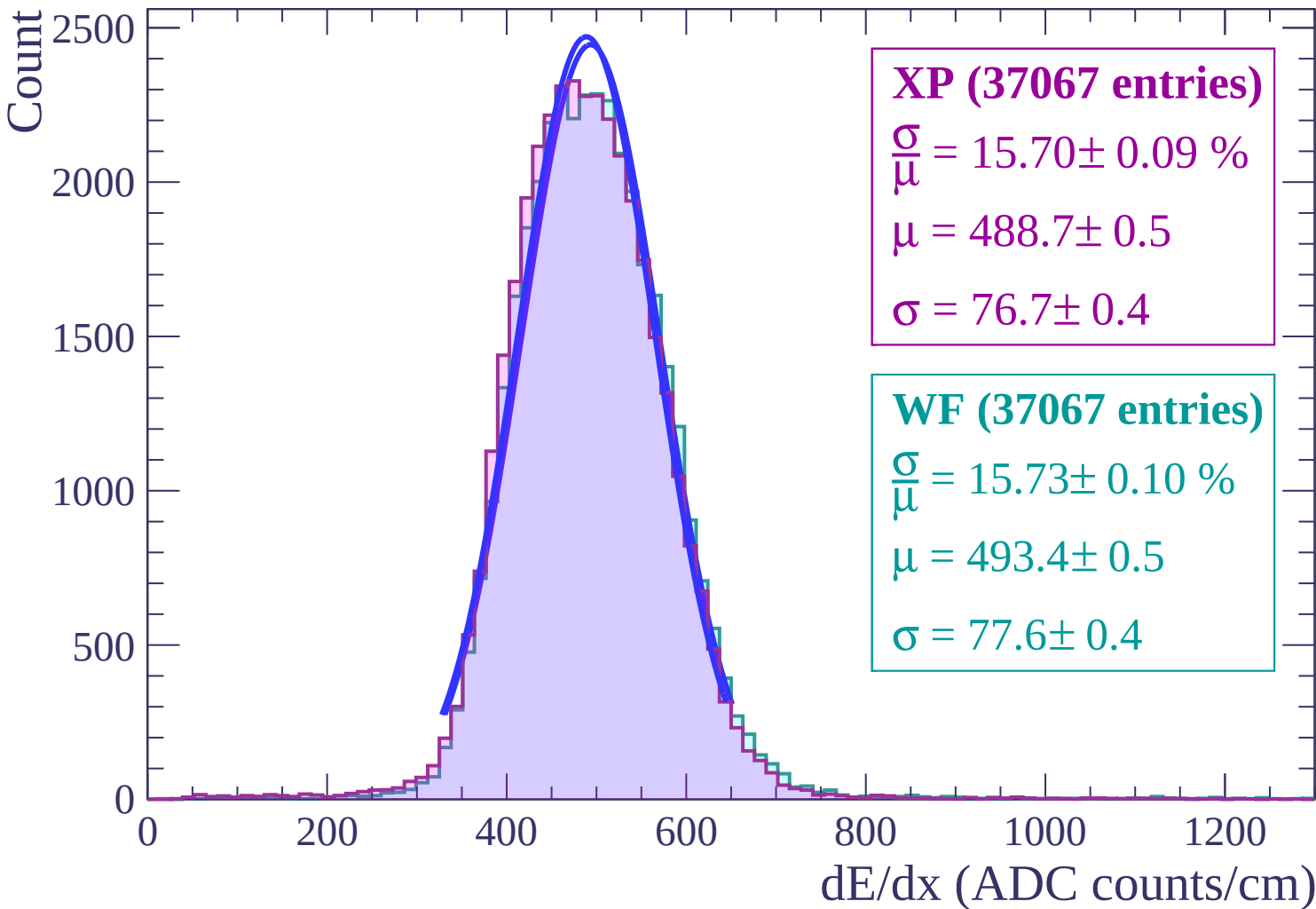
Energy loss | $-18 < \theta < -16$



Energy loss | $-16 < \theta < -14$

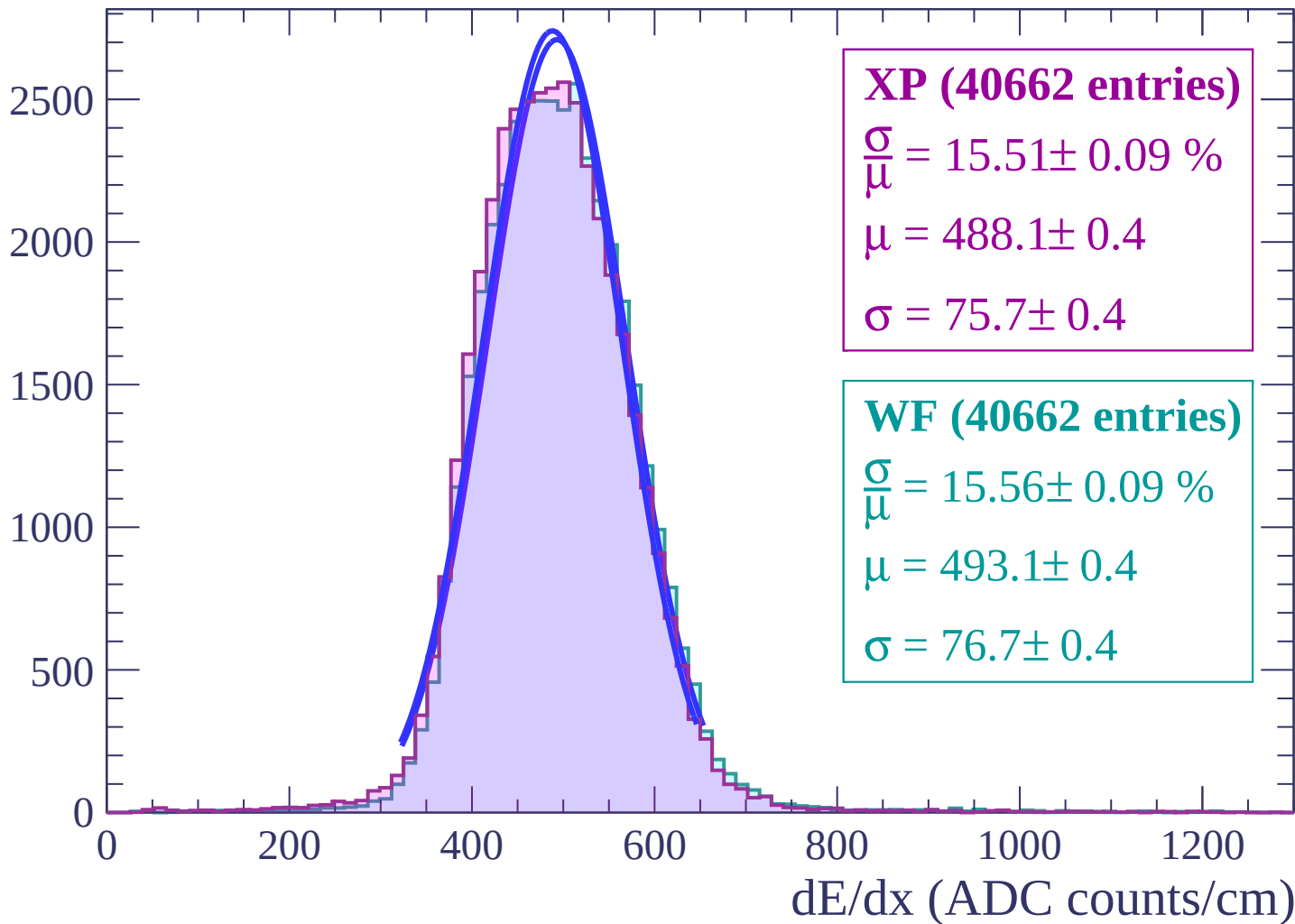


Energy loss | $-14 < \theta < -12$

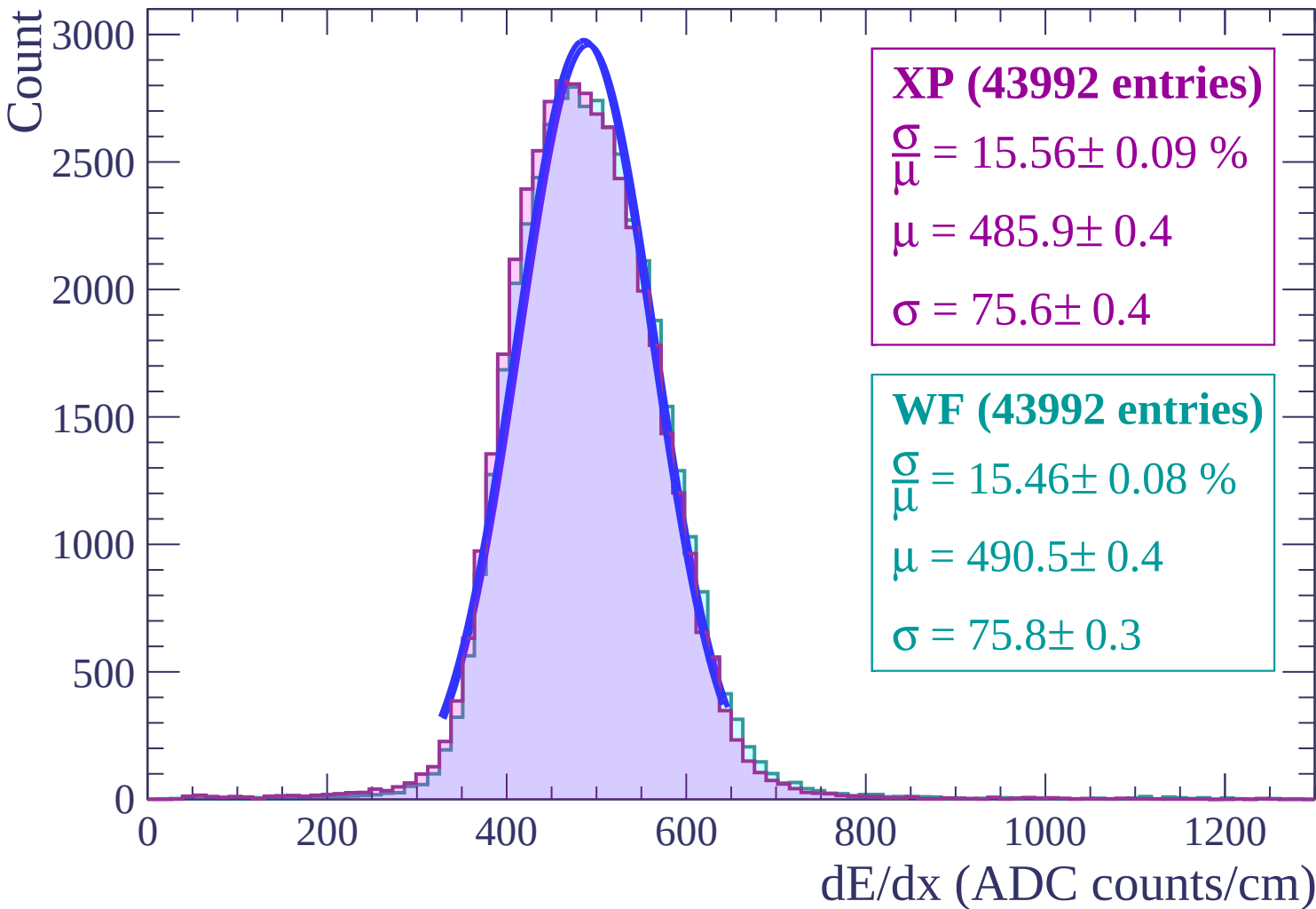


Energy loss | $-12 < \theta < -10$

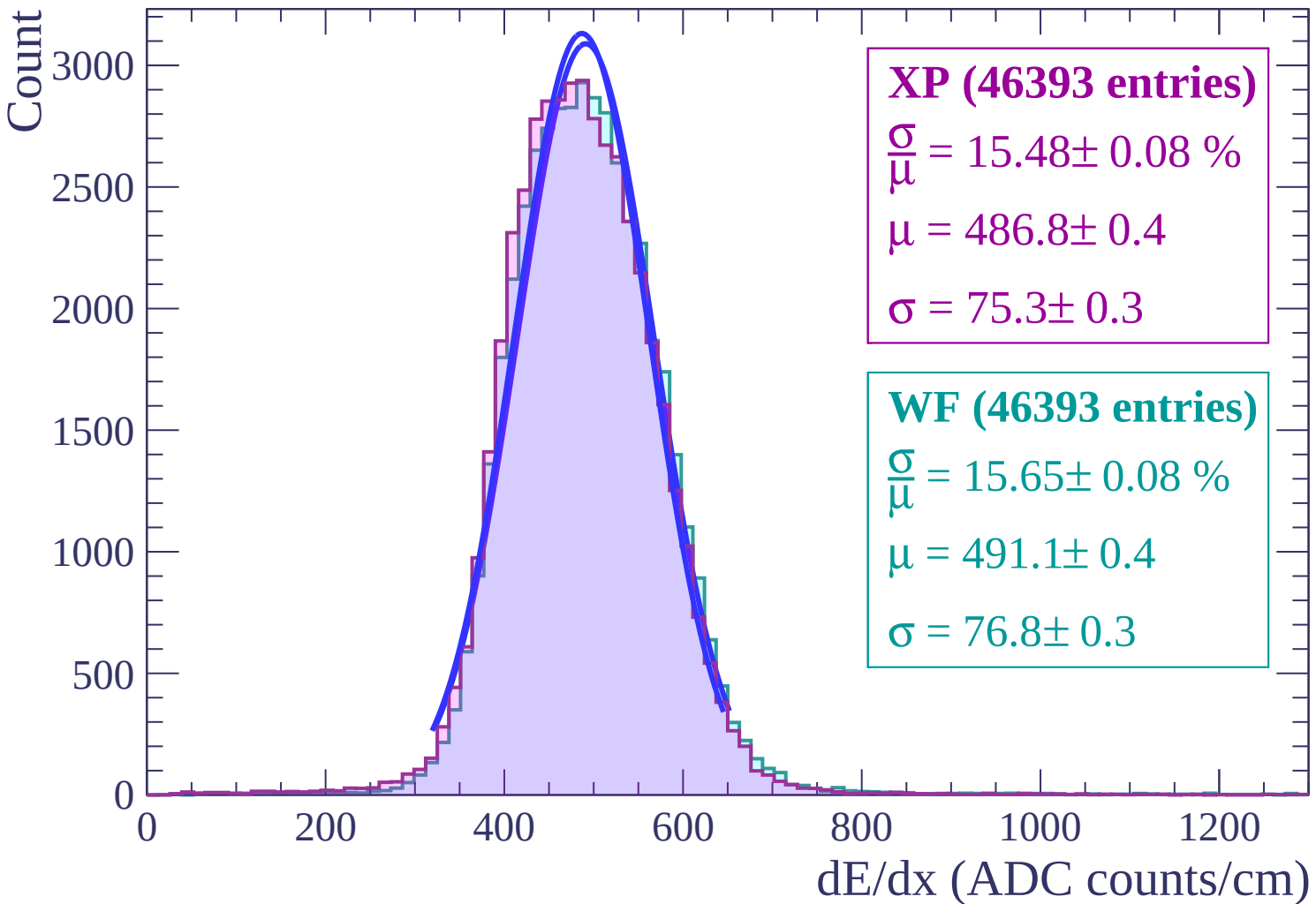
Count



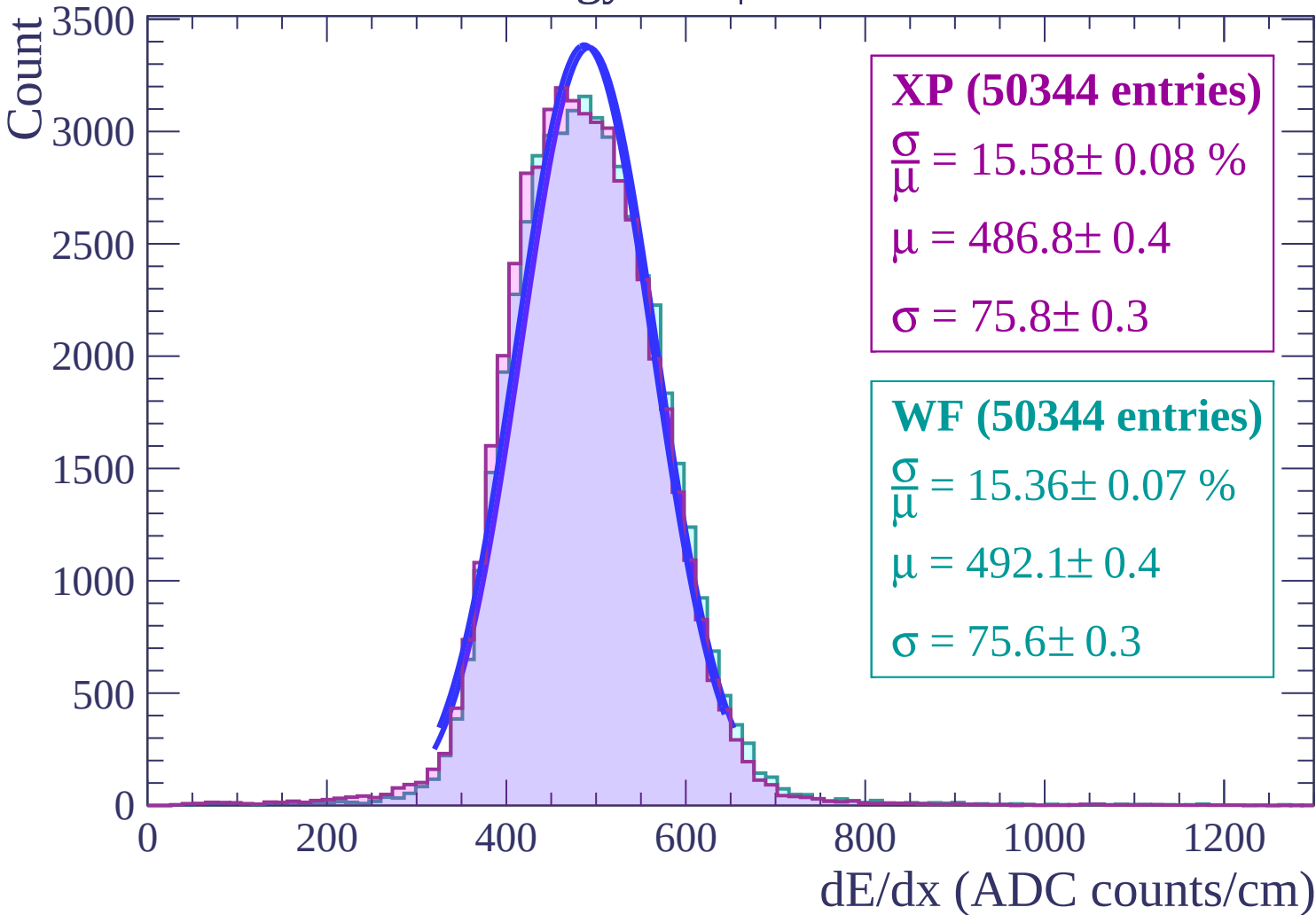
Energy loss | $-10 < \theta < -8$



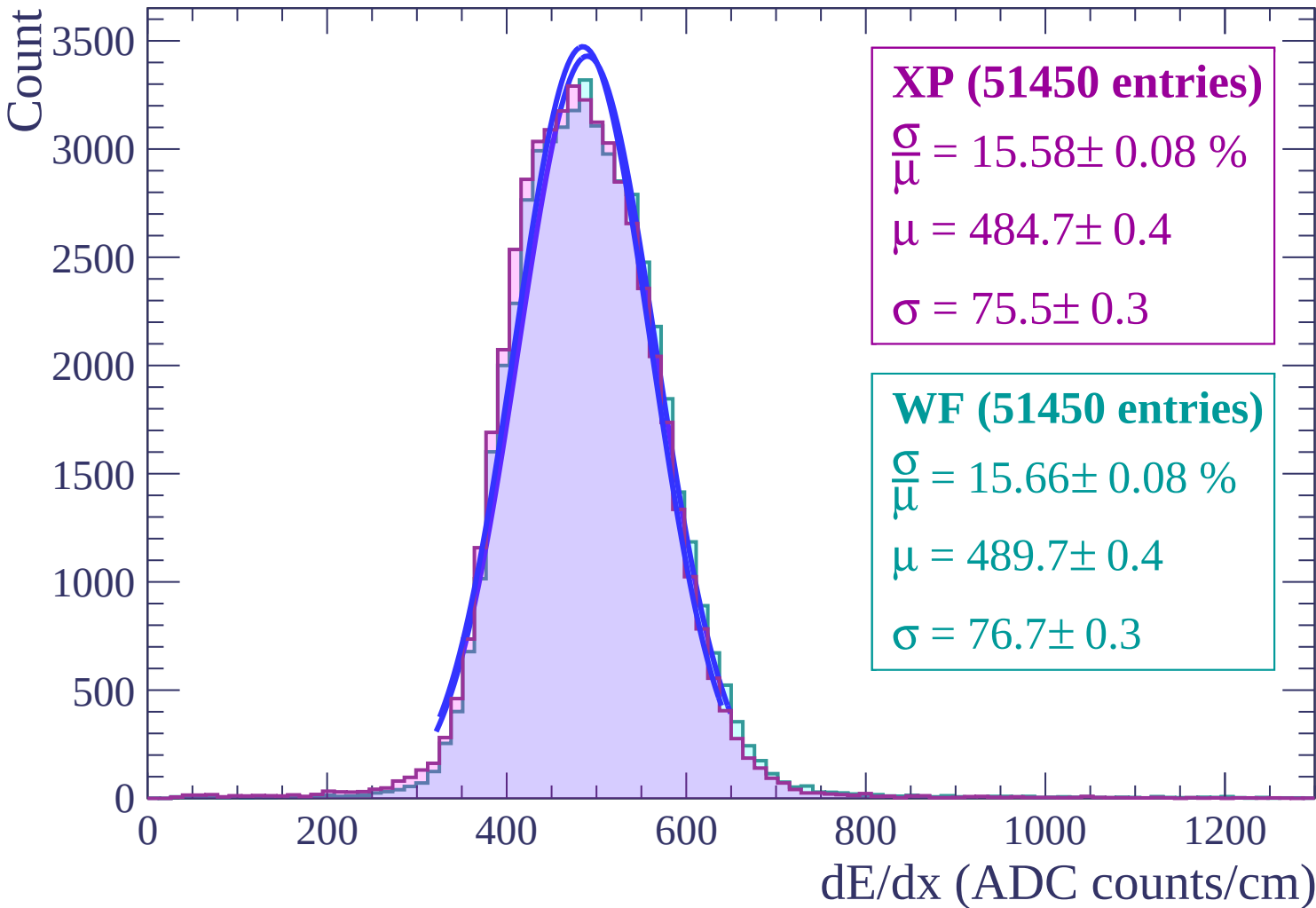
Energy loss | $-8 < \theta < -6$



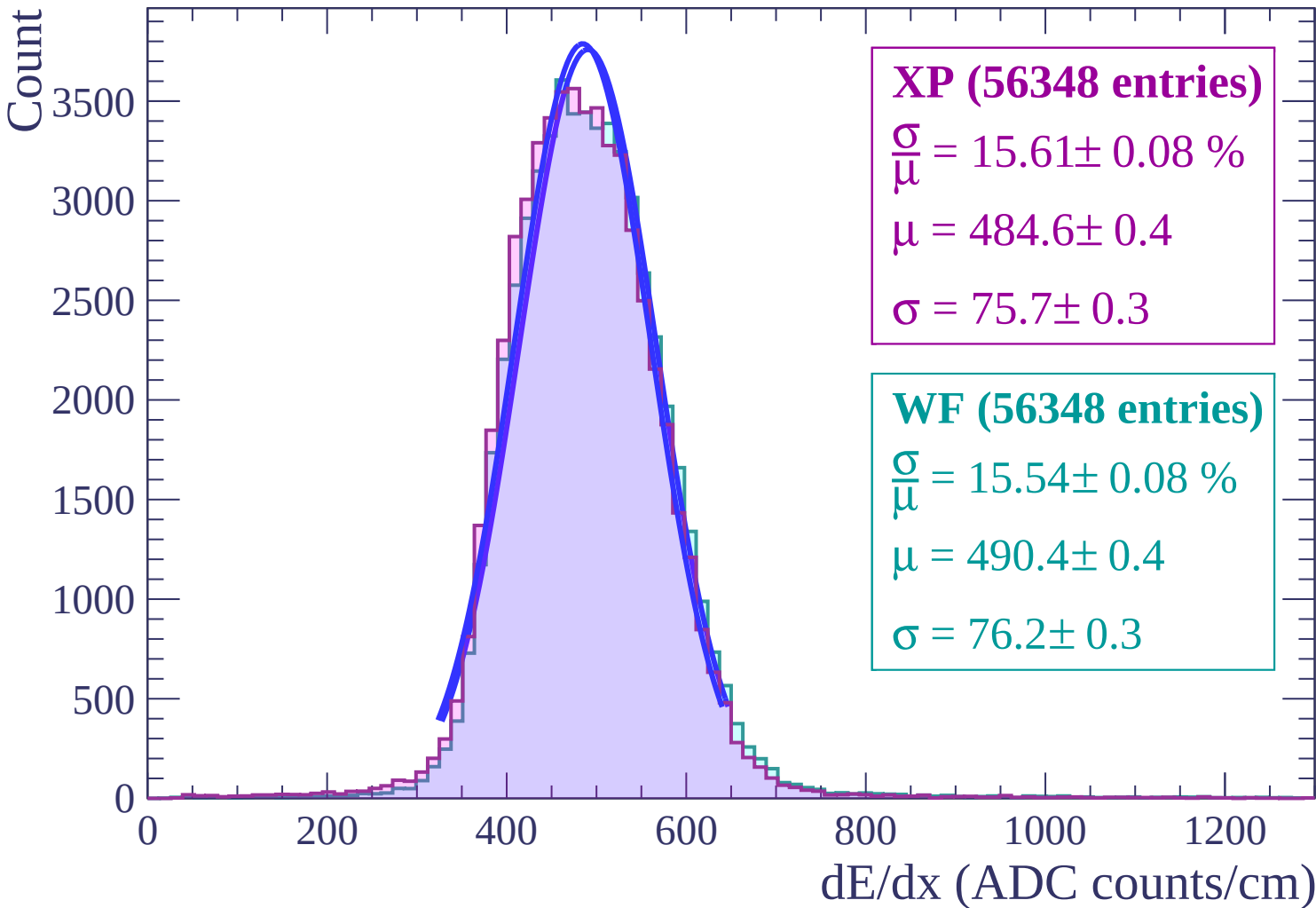
Energy loss | $-6 < \theta < -4$



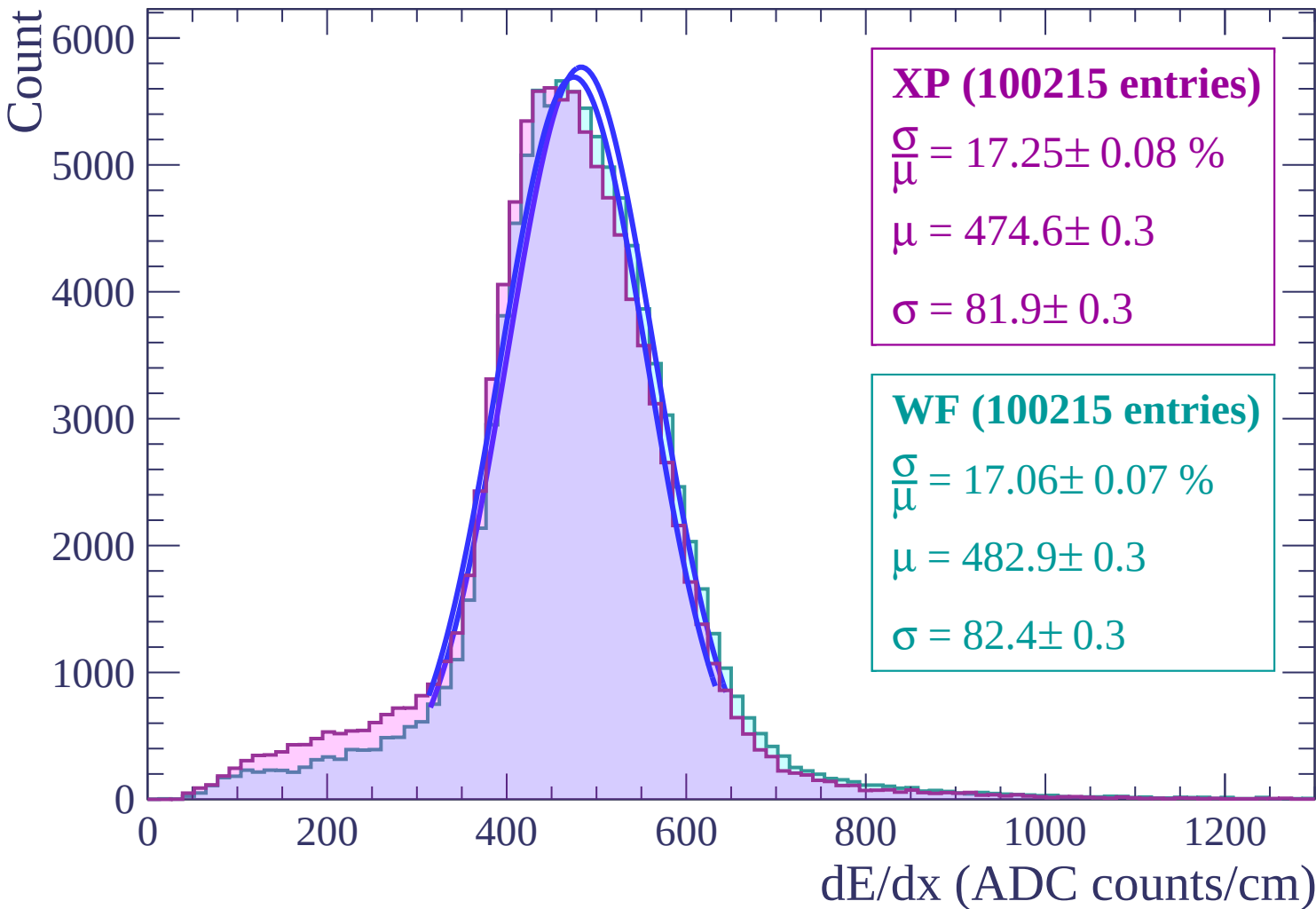
Energy loss | $-4 < \theta < -2$



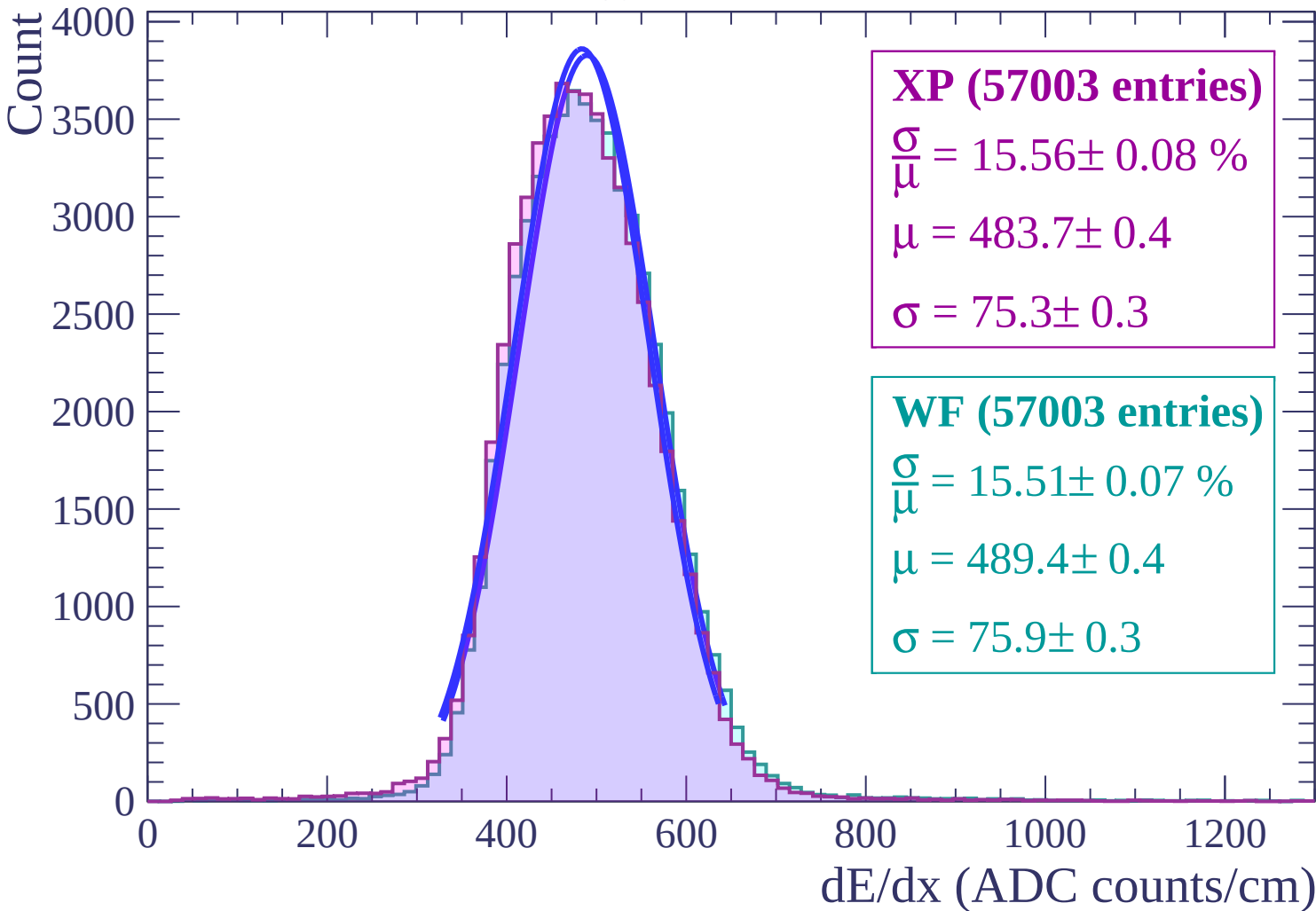
Energy loss | $-2 < \theta < 0$



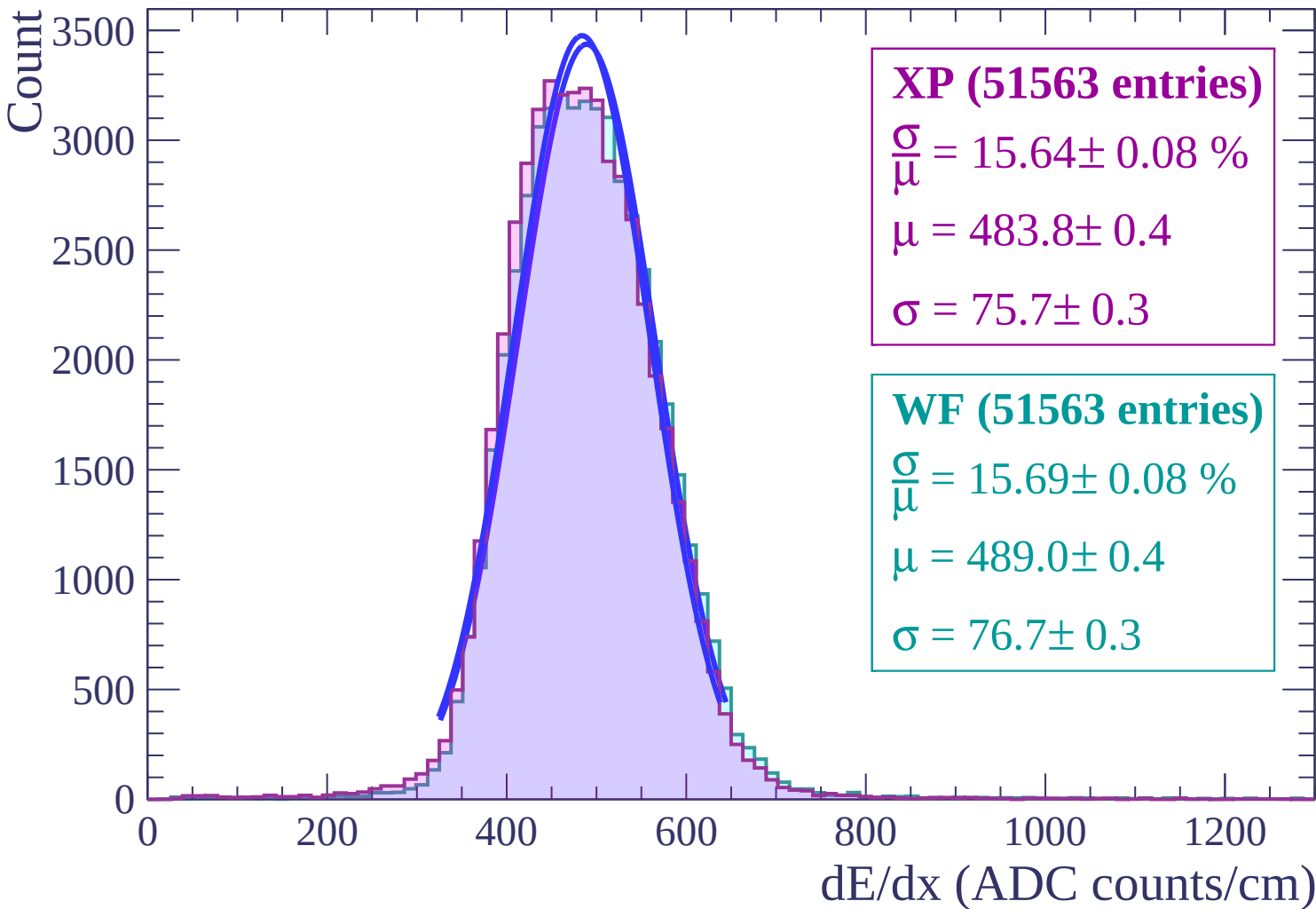
Energy loss | $0 < \theta < 2$



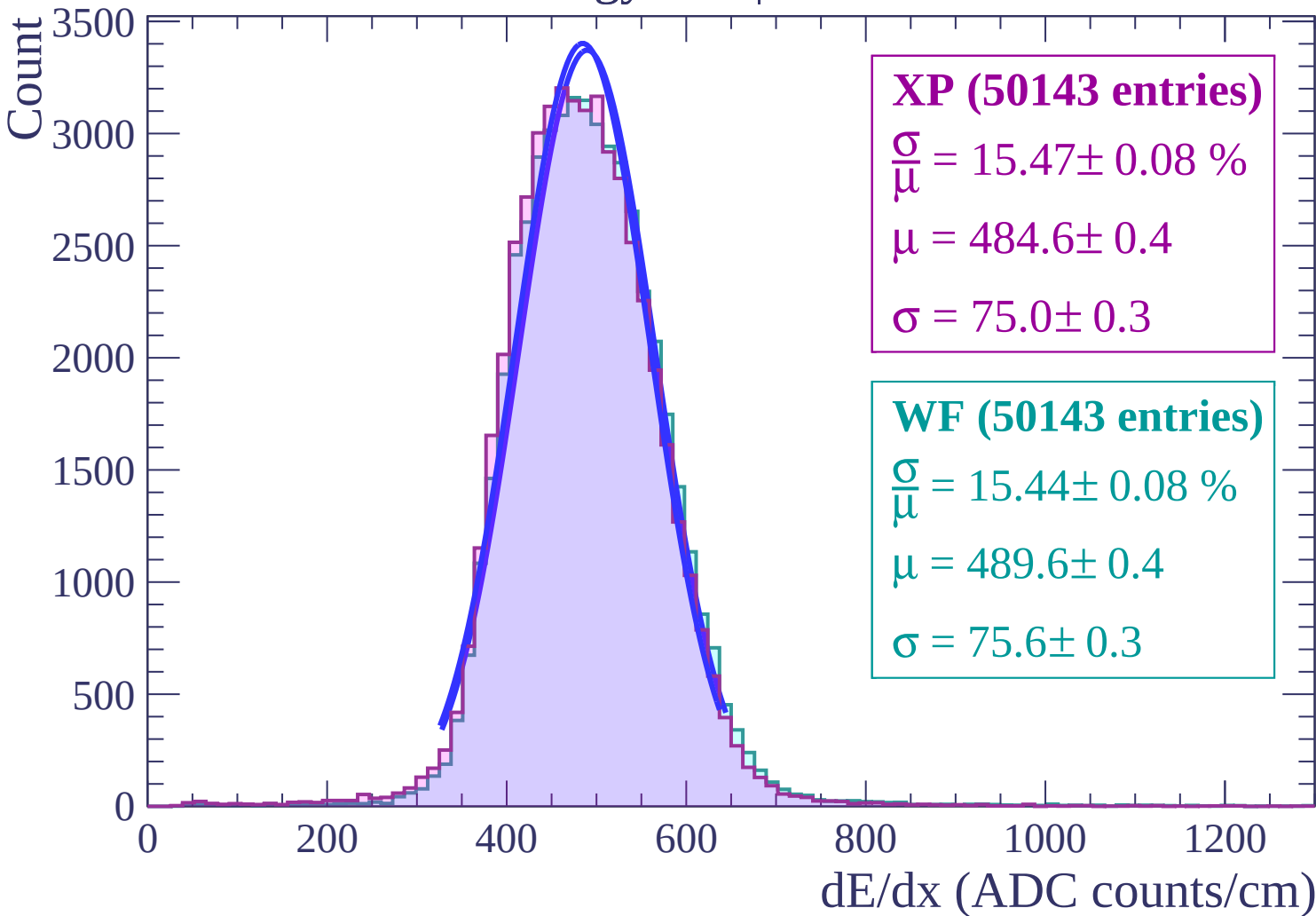
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$



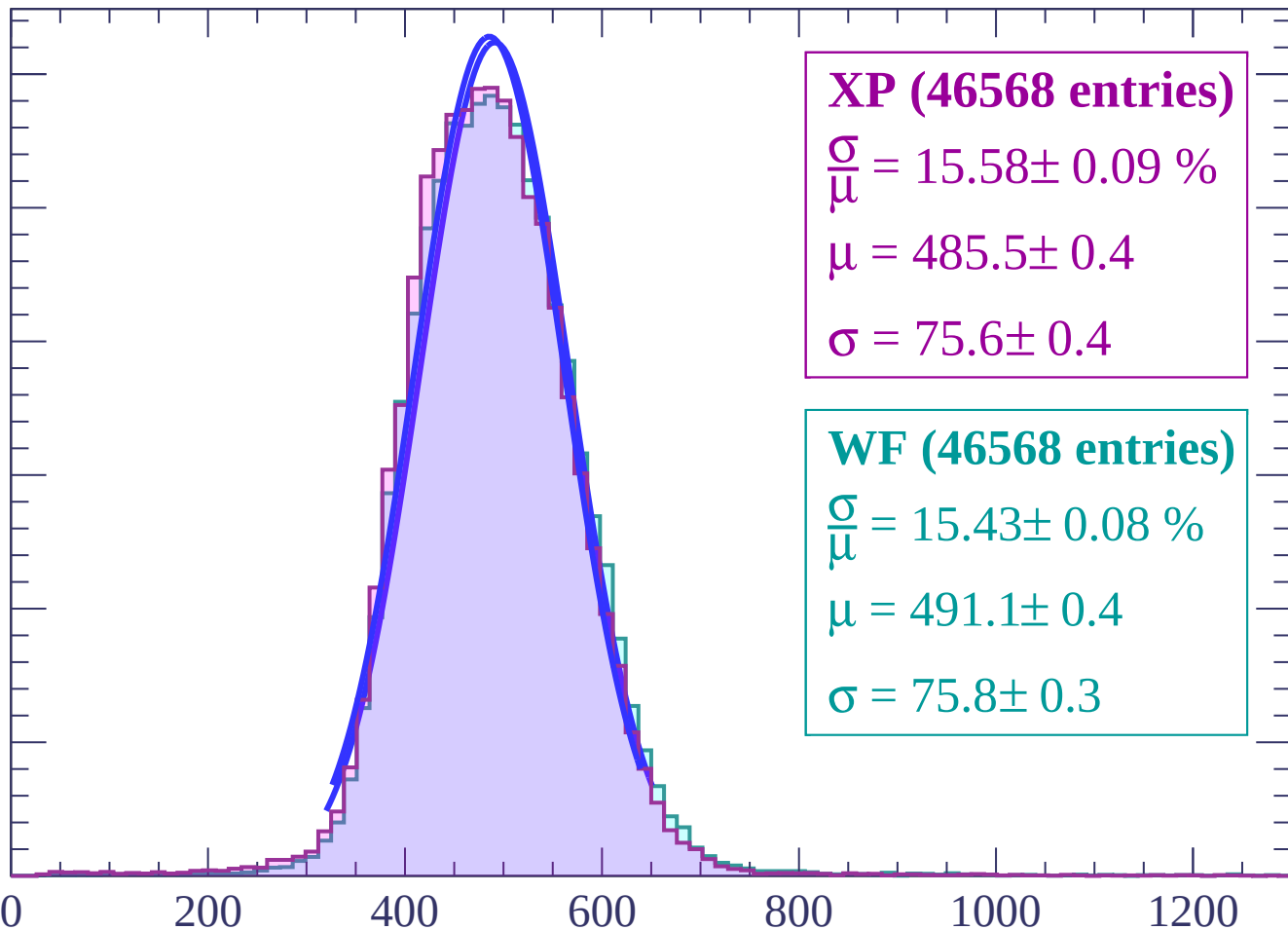
Energy loss | $6 < \theta < 8$



Energy loss | $8 < \theta < 10$

Count

3000
2500
2000
1500
1000
500
0



XP (46568 entries)

$\frac{\sigma}{\mu} = 15.58 \pm 0.09 \%$

$\mu = 485.5 \pm 0.4$

$\sigma = 75.6 \pm 0.4$

WF (46568 entries)

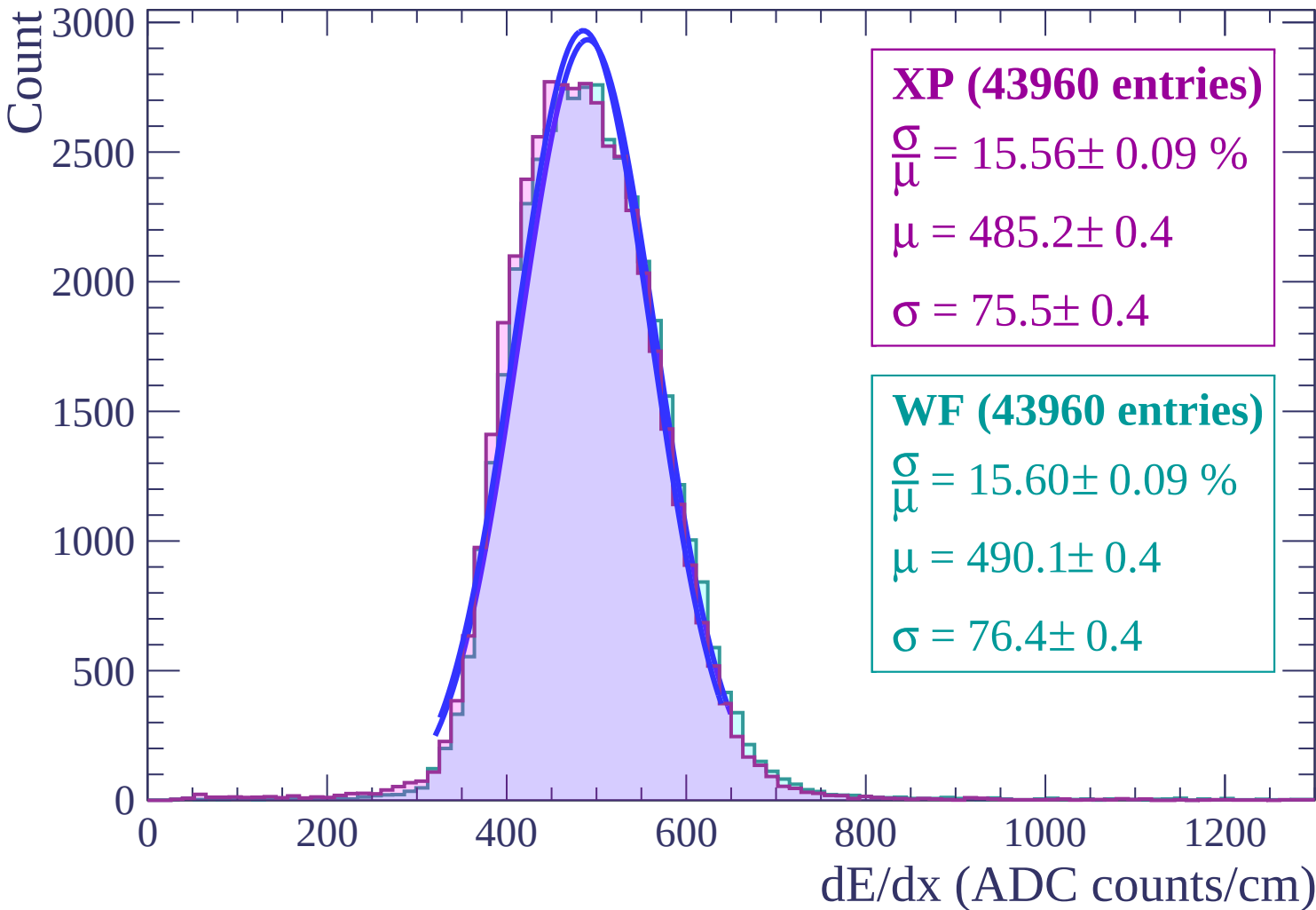
$\frac{\sigma}{\mu} = 15.43 \pm 0.08 \%$

$\mu = 491.1 \pm 0.4$

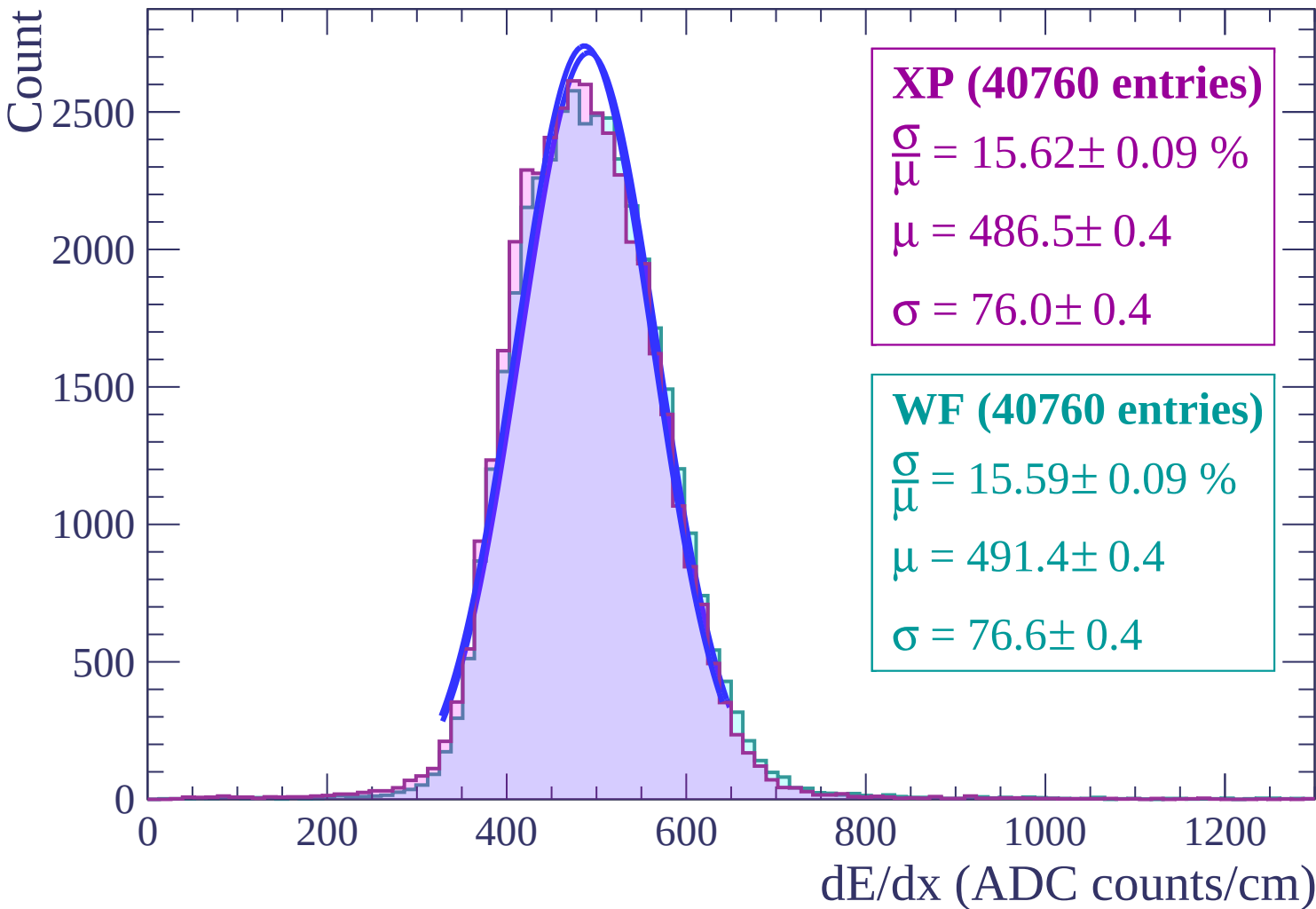
$\sigma = 75.8 \pm 0.3$

dE/dx (ADC counts/cm)

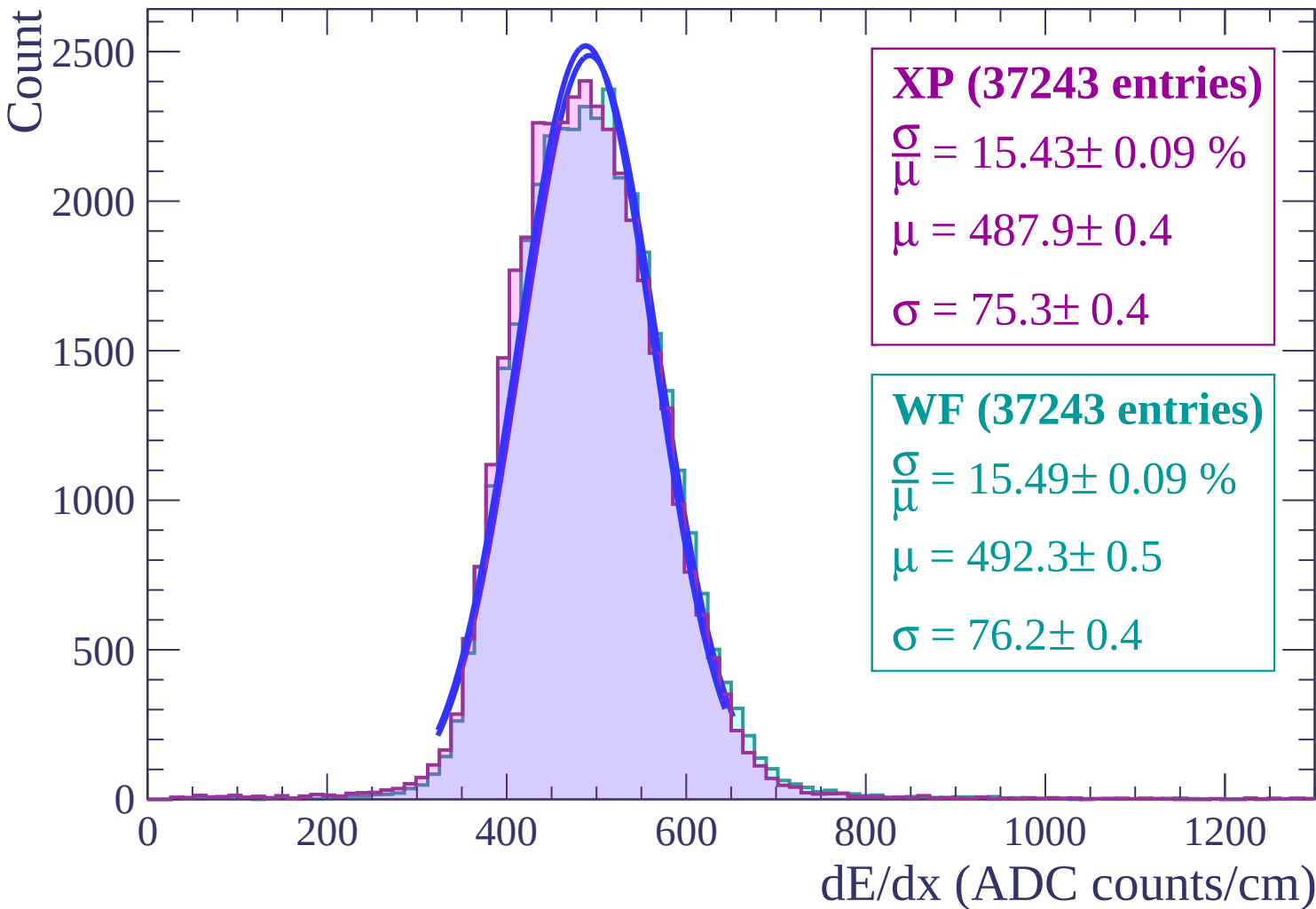
Energy loss | $10 < \theta < 12$



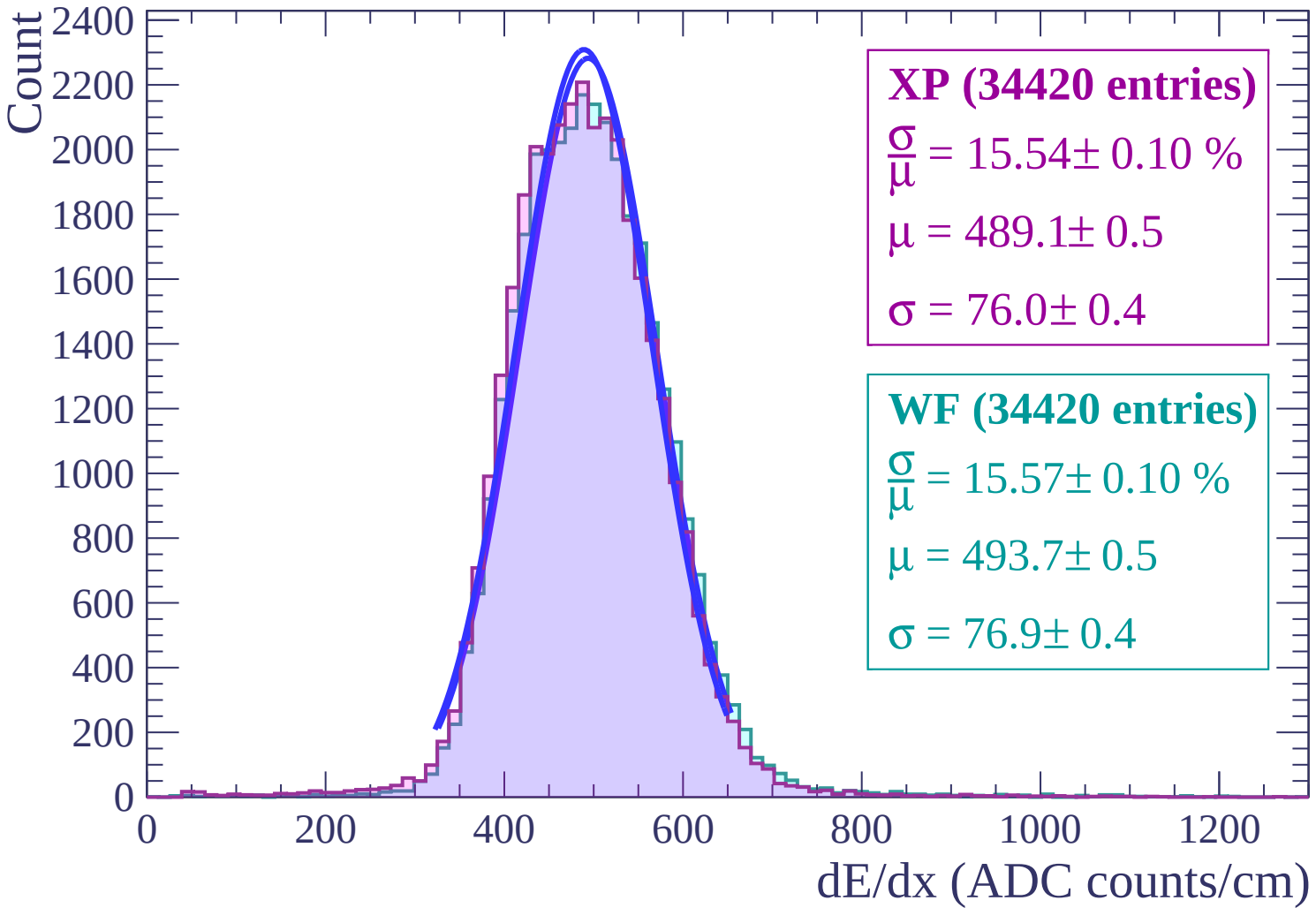
Energy loss | $12 < \theta < 14$



Energy loss | $14 < \theta < 16$

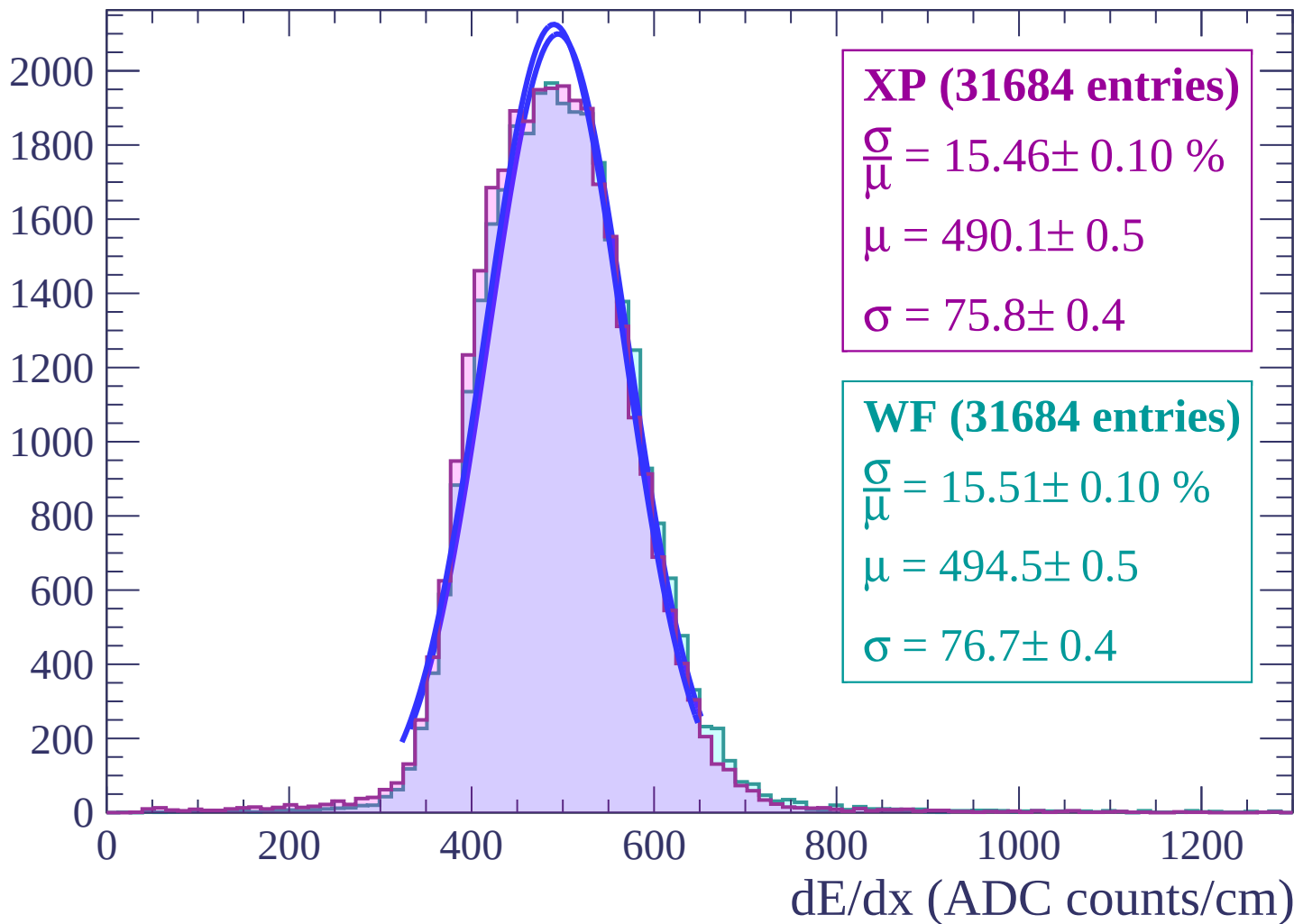


Energy loss | $16 < \theta < 18$

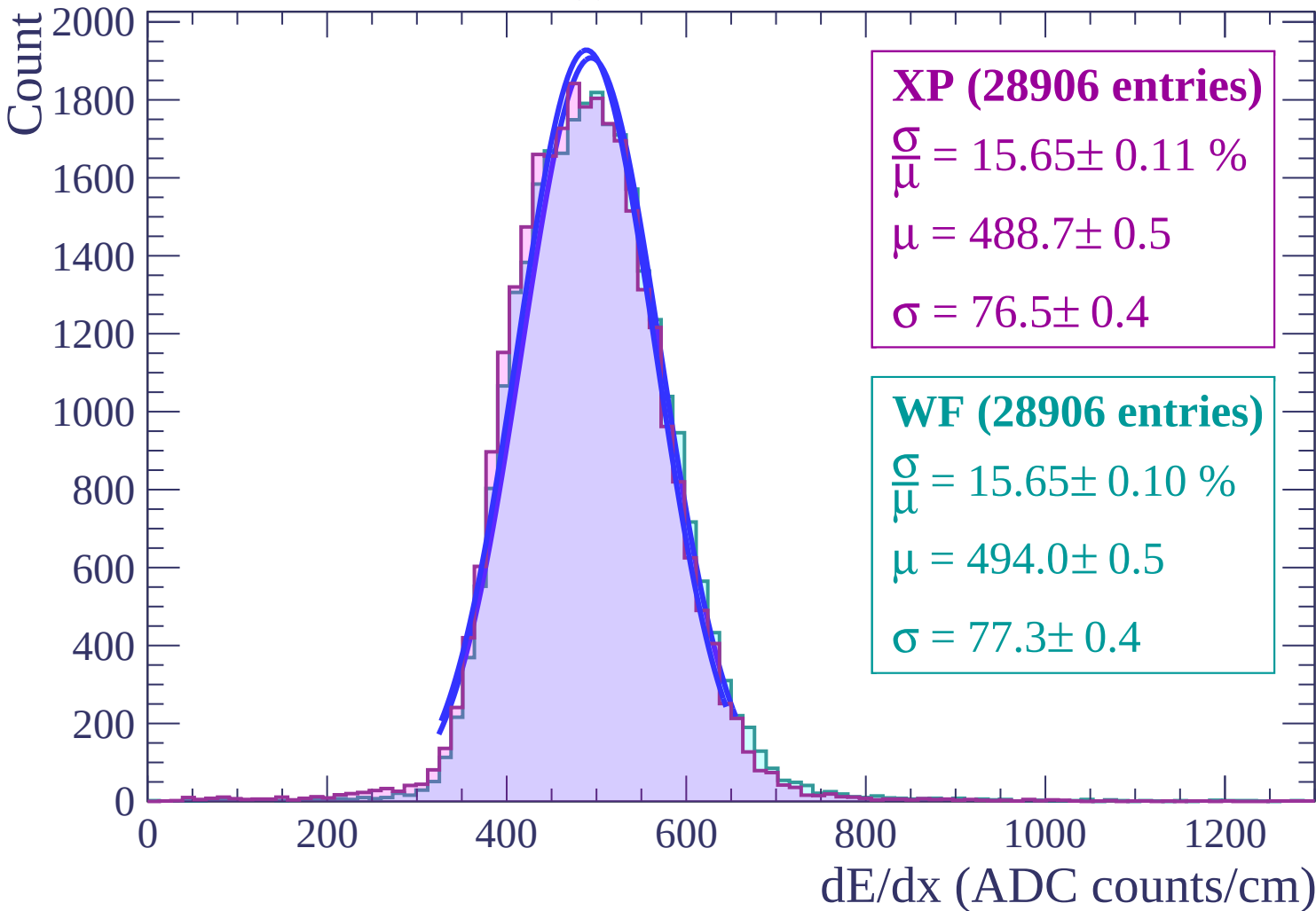


Energy loss | $18 < \theta < 20$

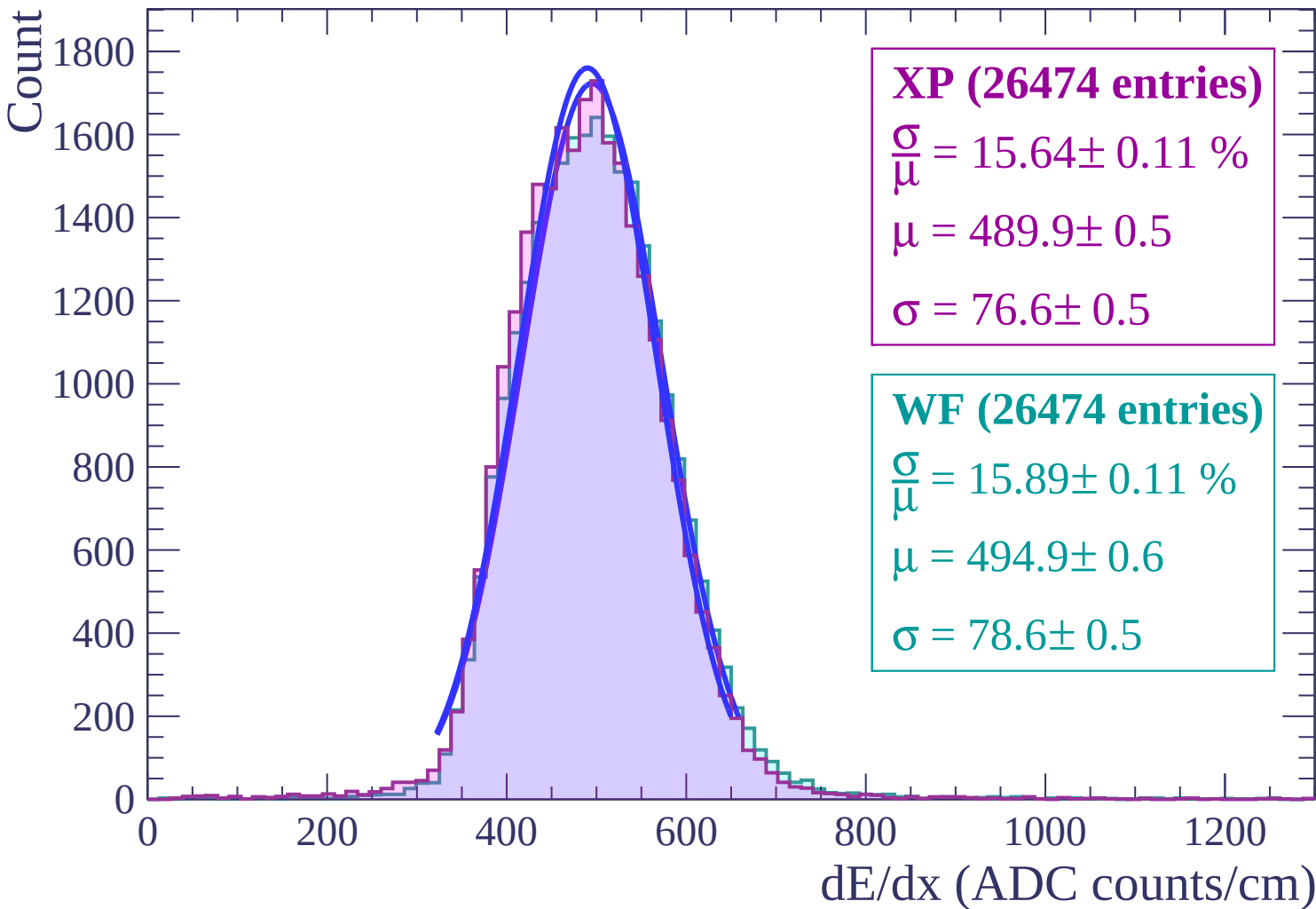
Count



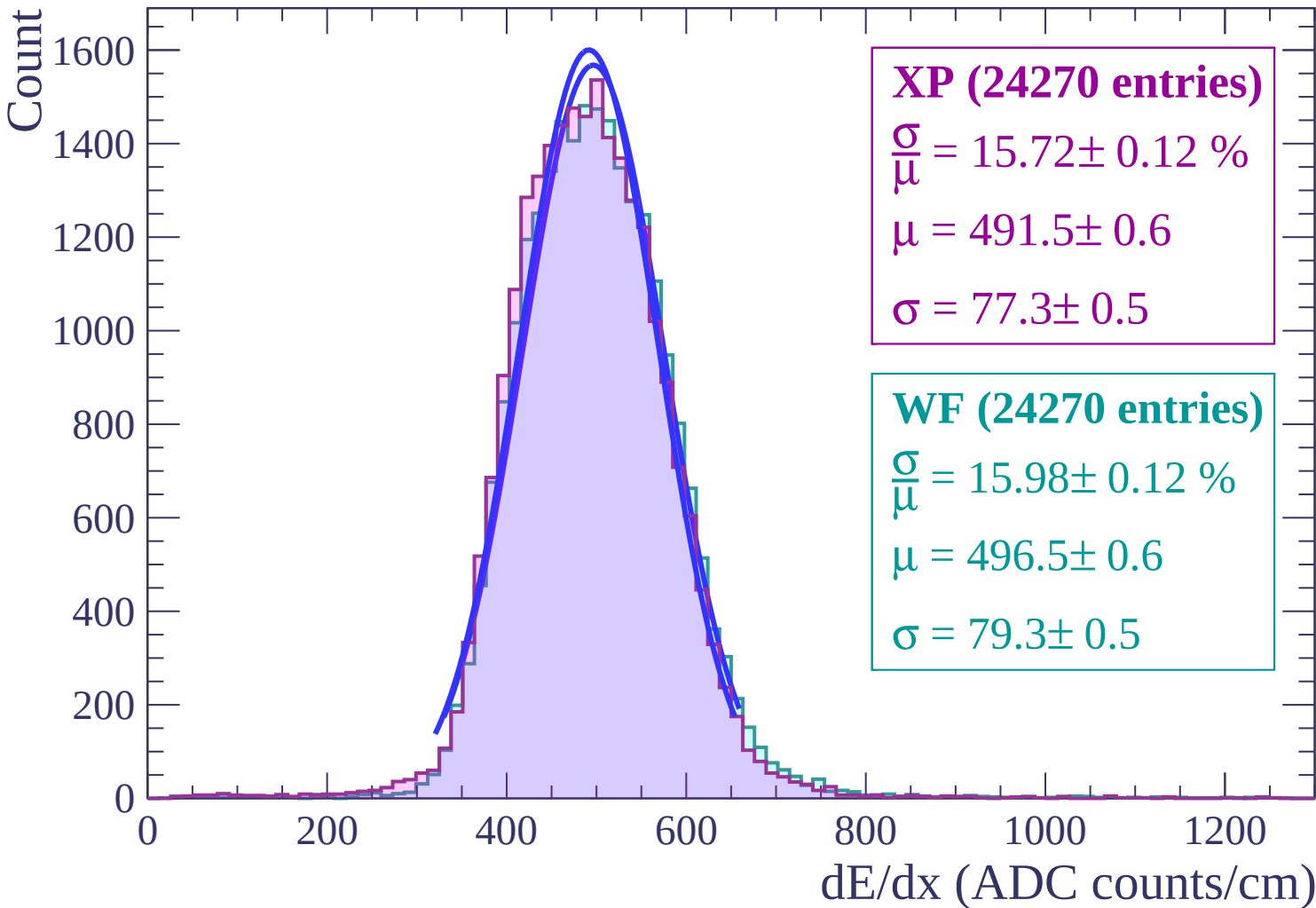
Energy loss | $20 < \theta < 22$



Energy loss | $22 < \theta < 24$



Energy loss | $24 < \theta < 26$



Energy loss | $26 < \theta < 28$

Count

1400
1200
1000
800
600
400
200
0

XP (22379 entries)

$\frac{\sigma}{\mu} = 15.97 \pm 0.12 \%$

$\mu = 492.5 \pm 0.6$

$\sigma = 78.7 \pm 0.5$

WF (22379 entries)

$\frac{\sigma}{\mu} = 16.02 \pm 0.12 \%$

$\mu = 497.3 \pm 0.6$

$\sigma = 79.7 \pm 0.5$

0

200

400

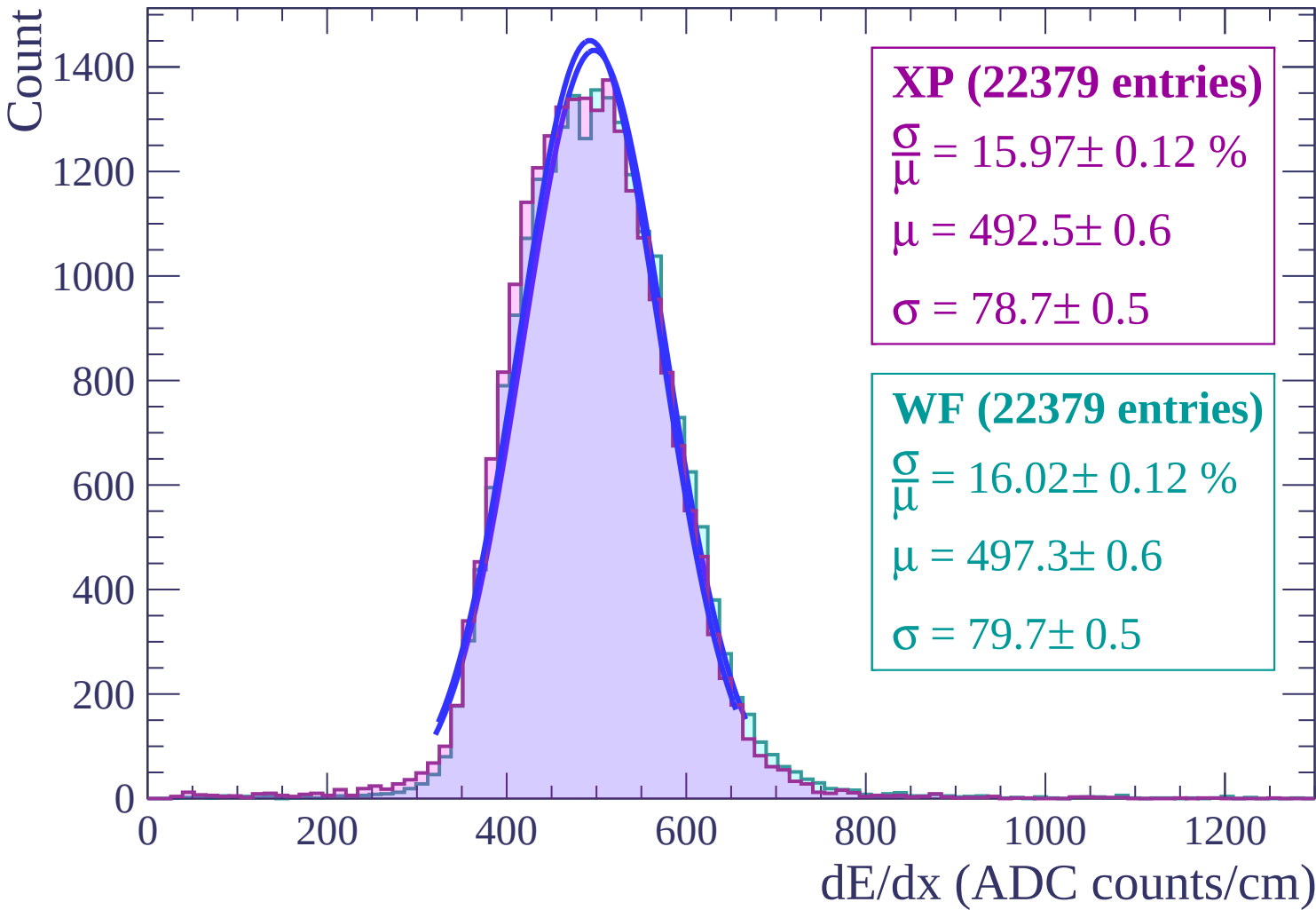
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $28 < \theta < 30$

Count

1200

1000

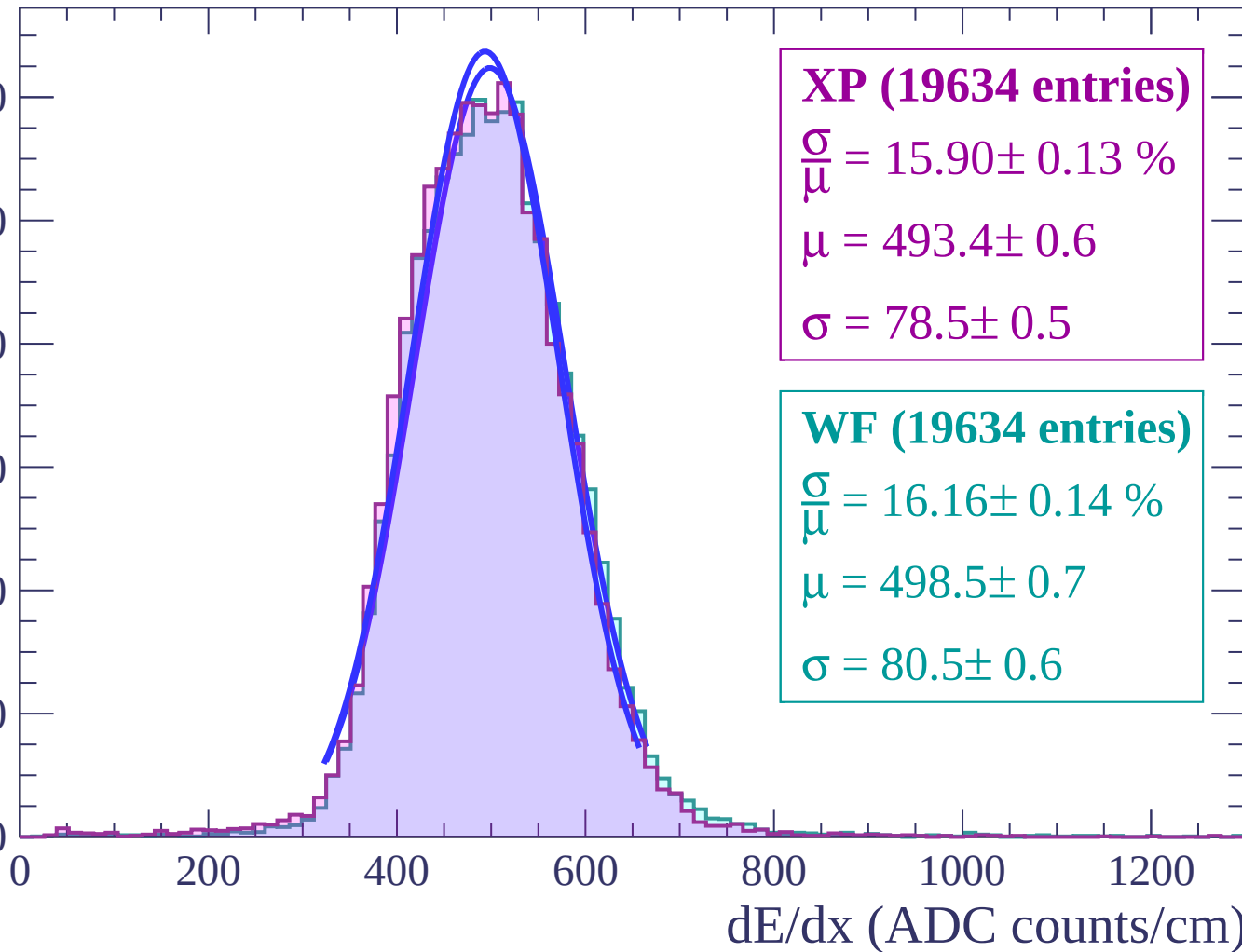
800

600

400

200

0



Energy loss | $30 < \theta < 32$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (15894 entries)

$\frac{\sigma}{\mu} = 16.05 \pm 0.13 \%$

$\mu = 497.6 \pm 0.7$

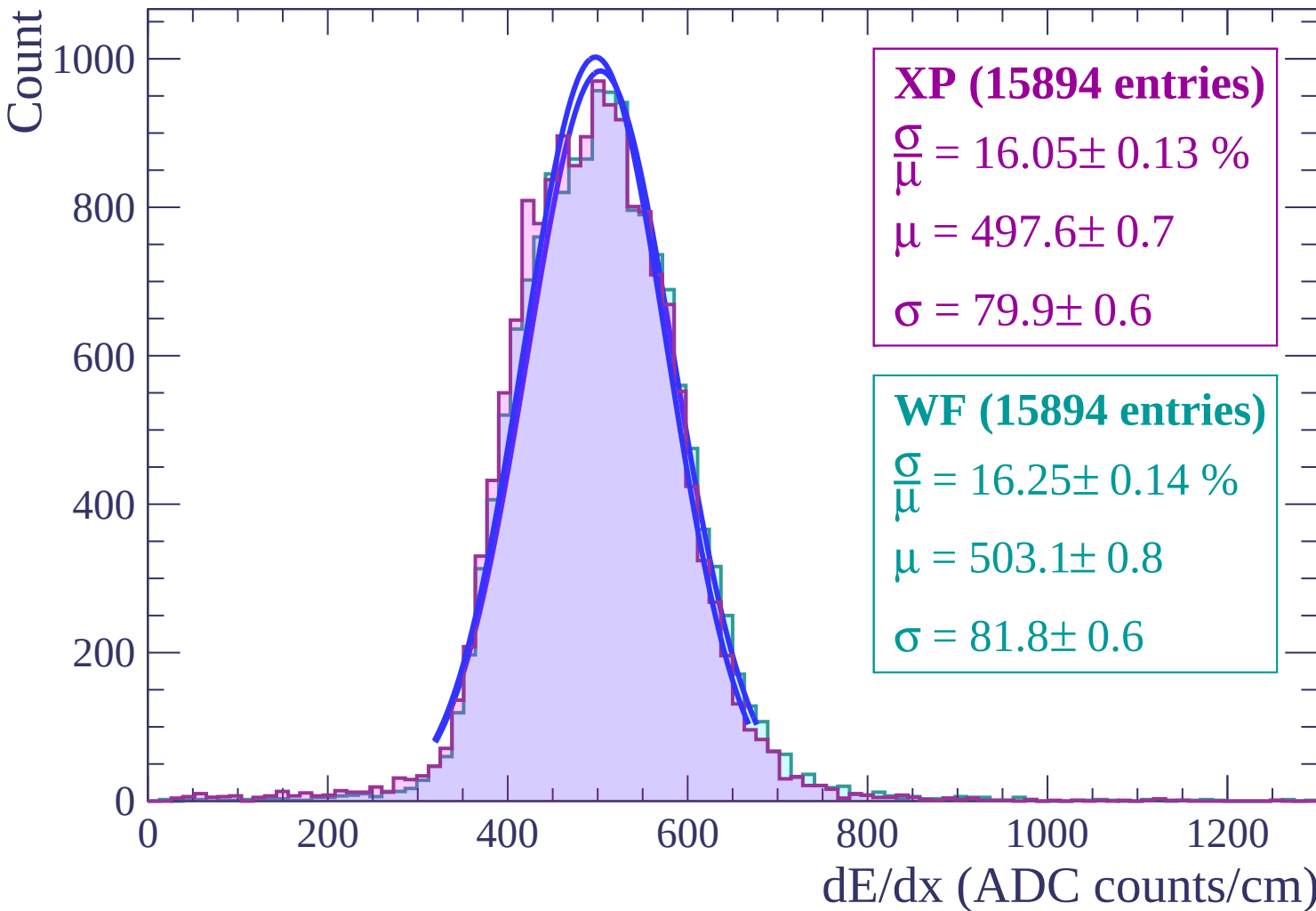
$\sigma = 79.9 \pm 0.6$

WF (15894 entries)

$\frac{\sigma}{\mu} = 16.25 \pm 0.14 \%$

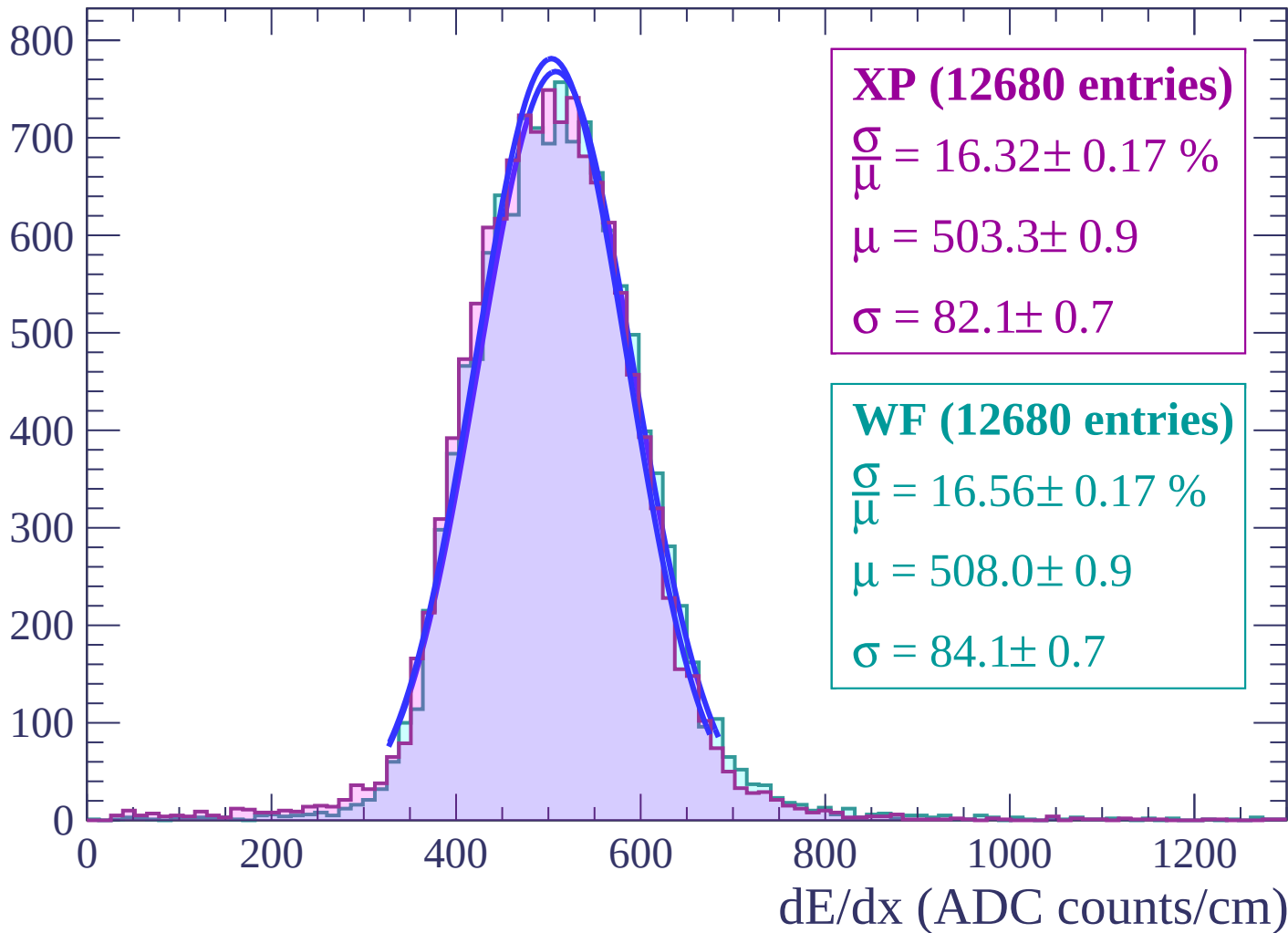
$\mu = 503.1 \pm 0.8$

$\sigma = 81.8 \pm 0.6$



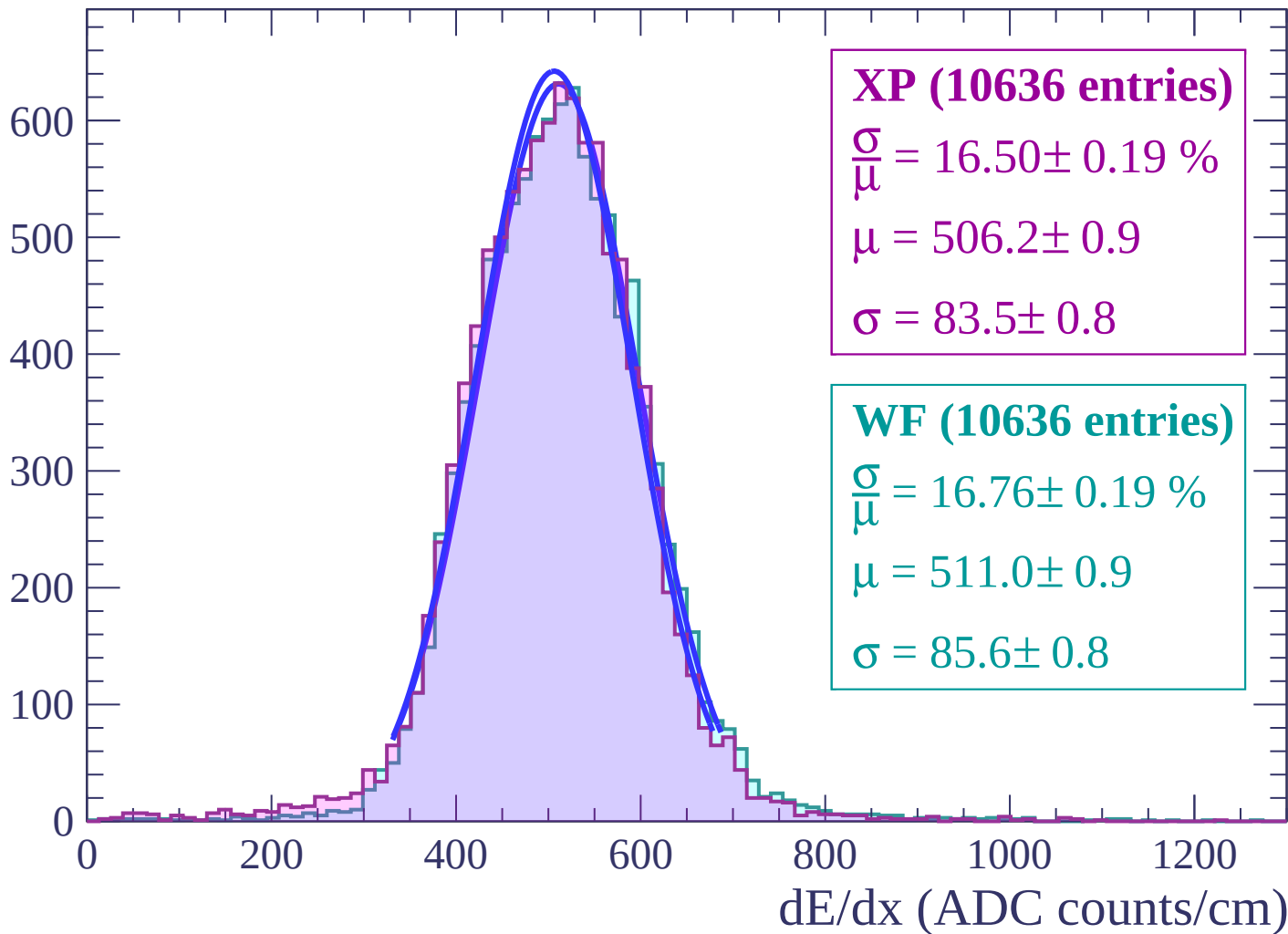
Energy loss | $32 < \theta < 34$

Count



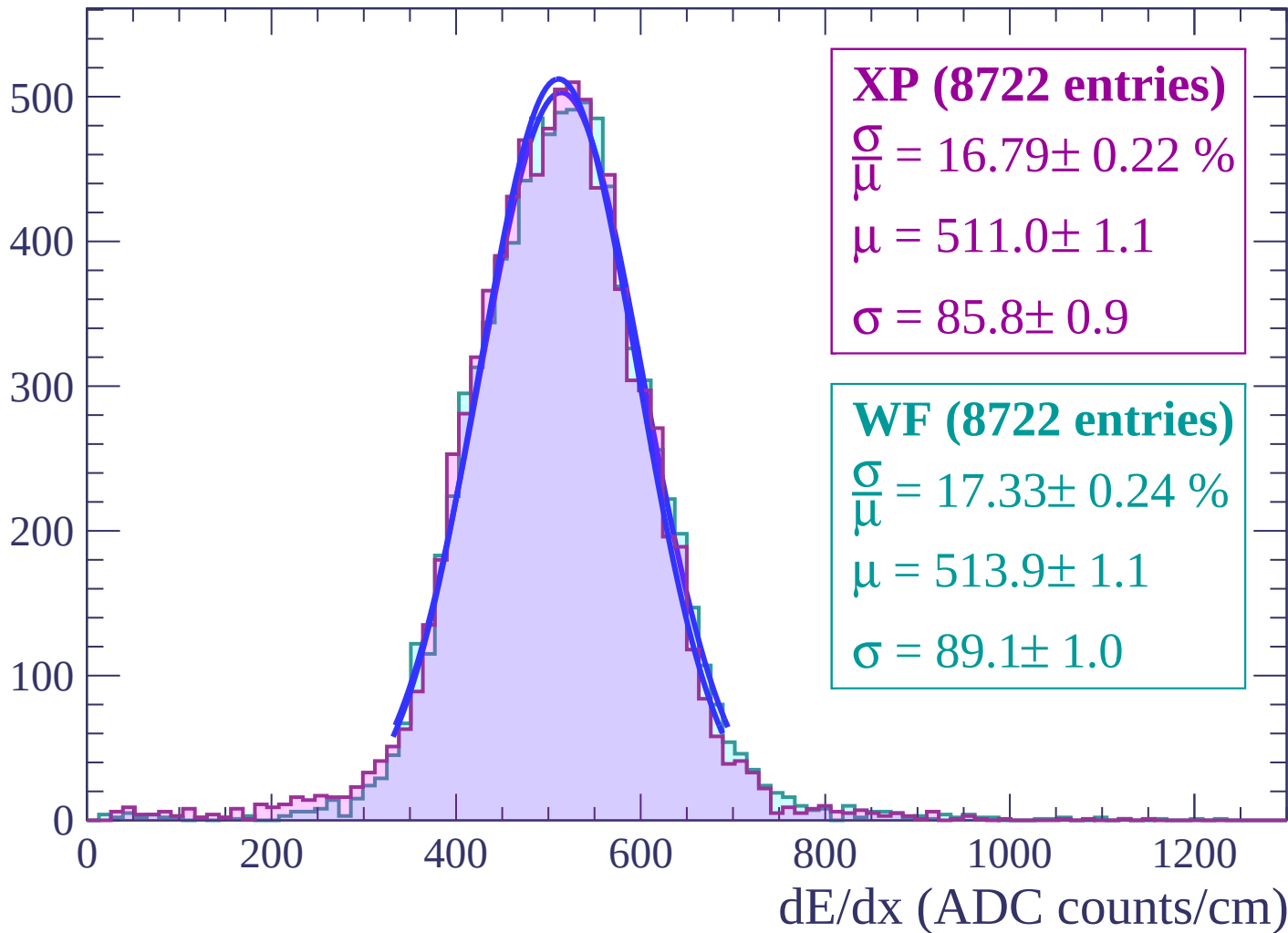
Energy loss | $34 < \theta < 36$

Count



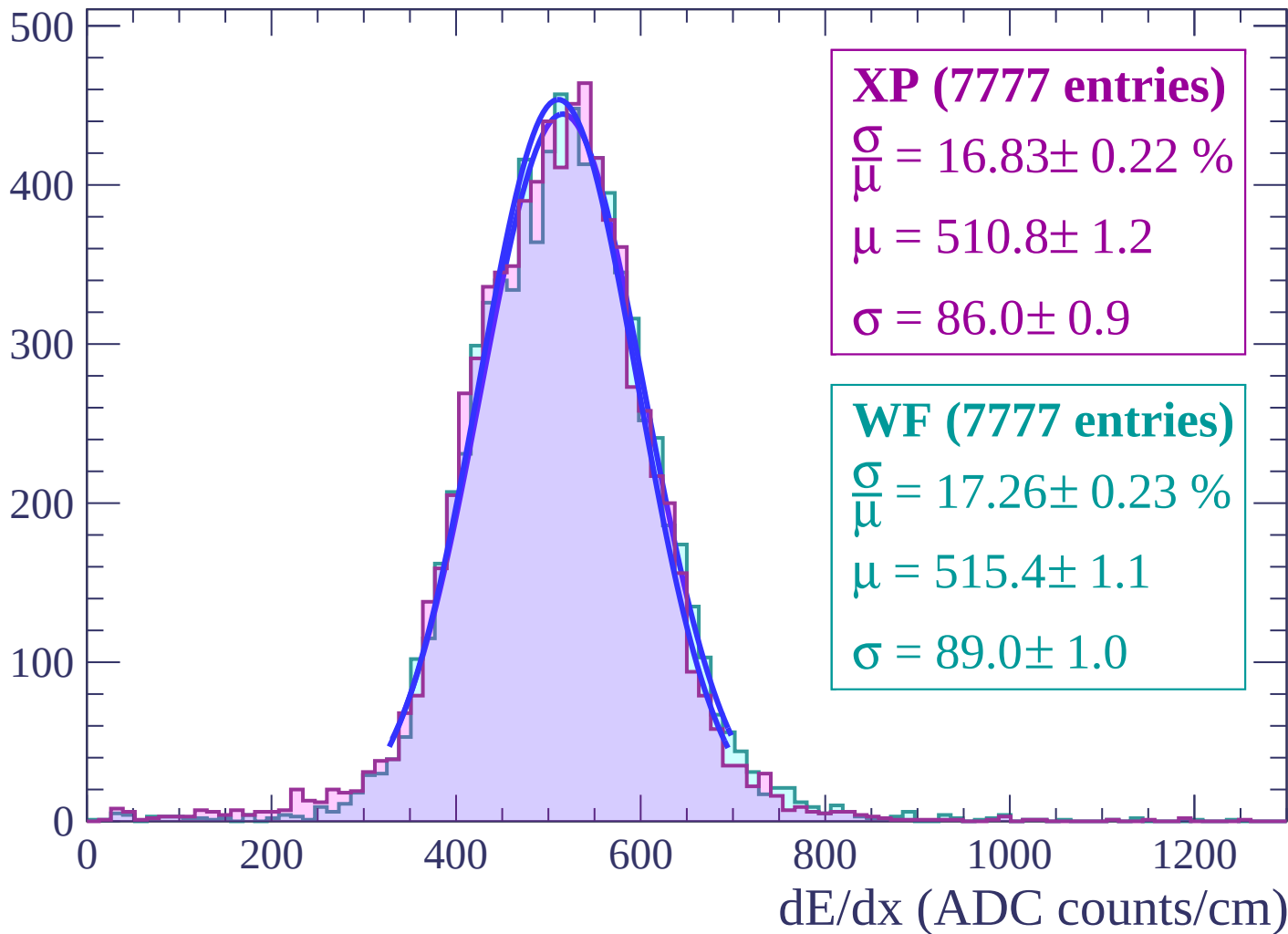
Energy loss | $36 < \theta < 38$

Count



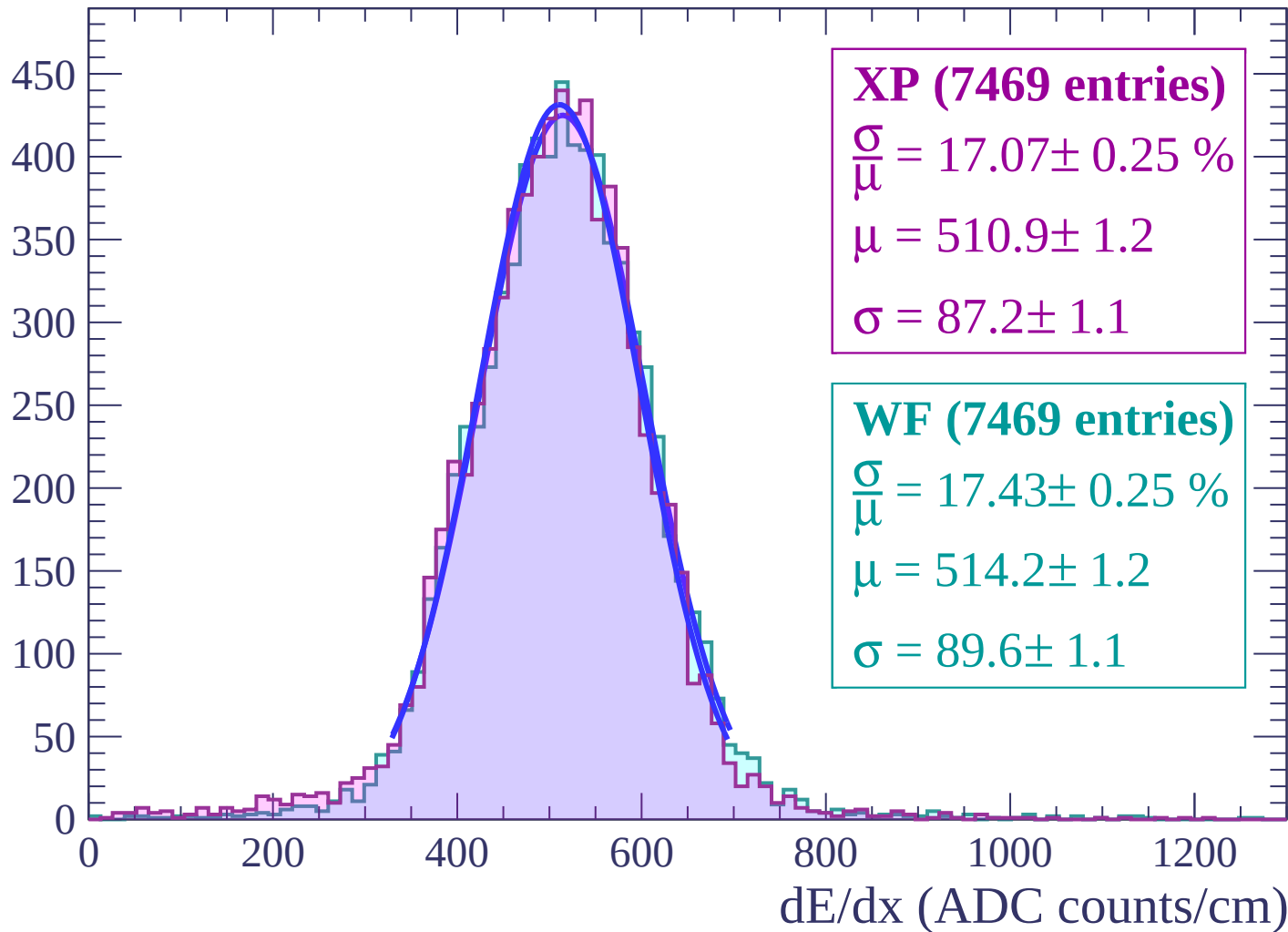
Energy loss | $38 < \theta < 40$

Count



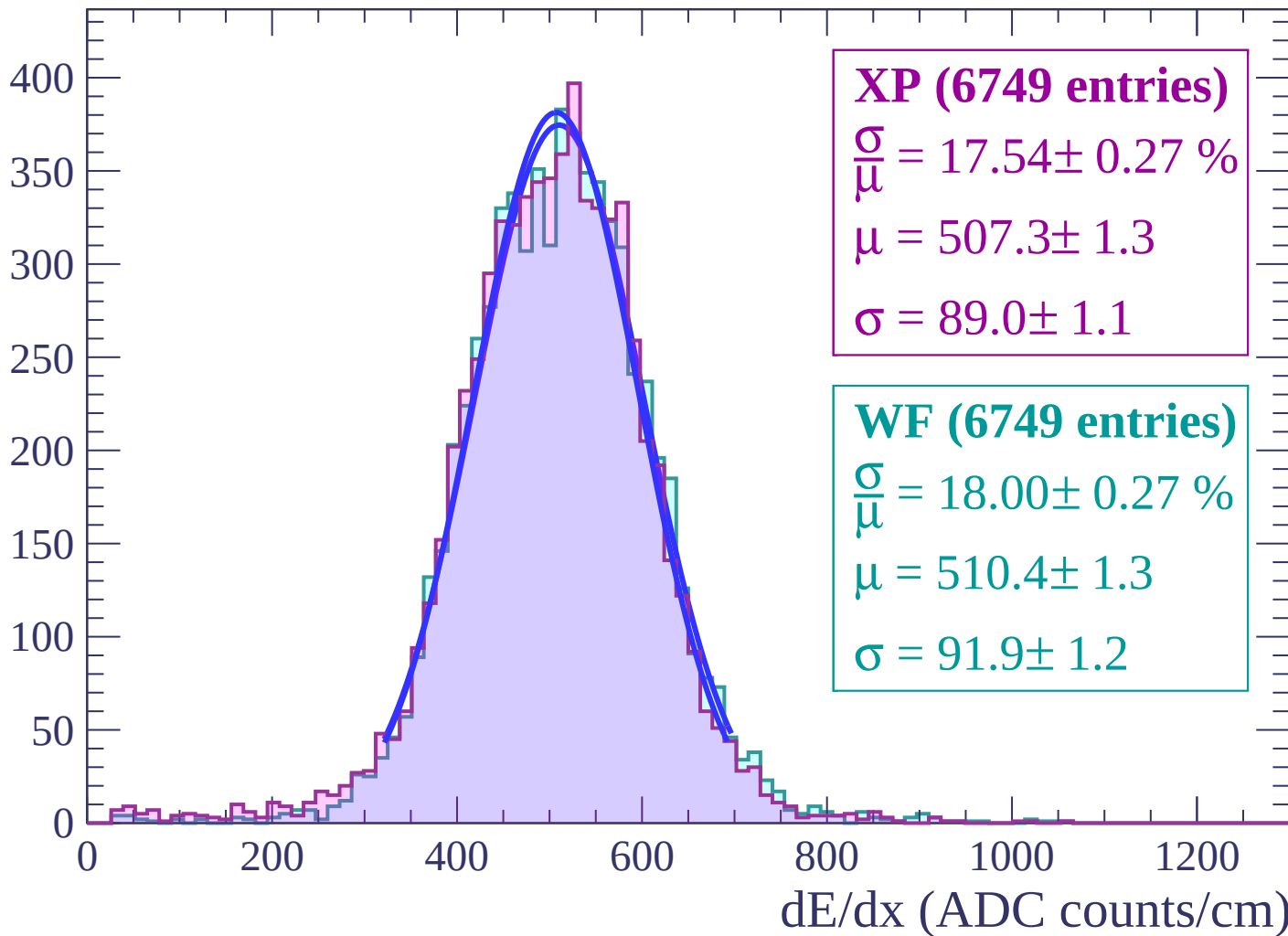
Energy loss | $40 < \theta < 42$

Count



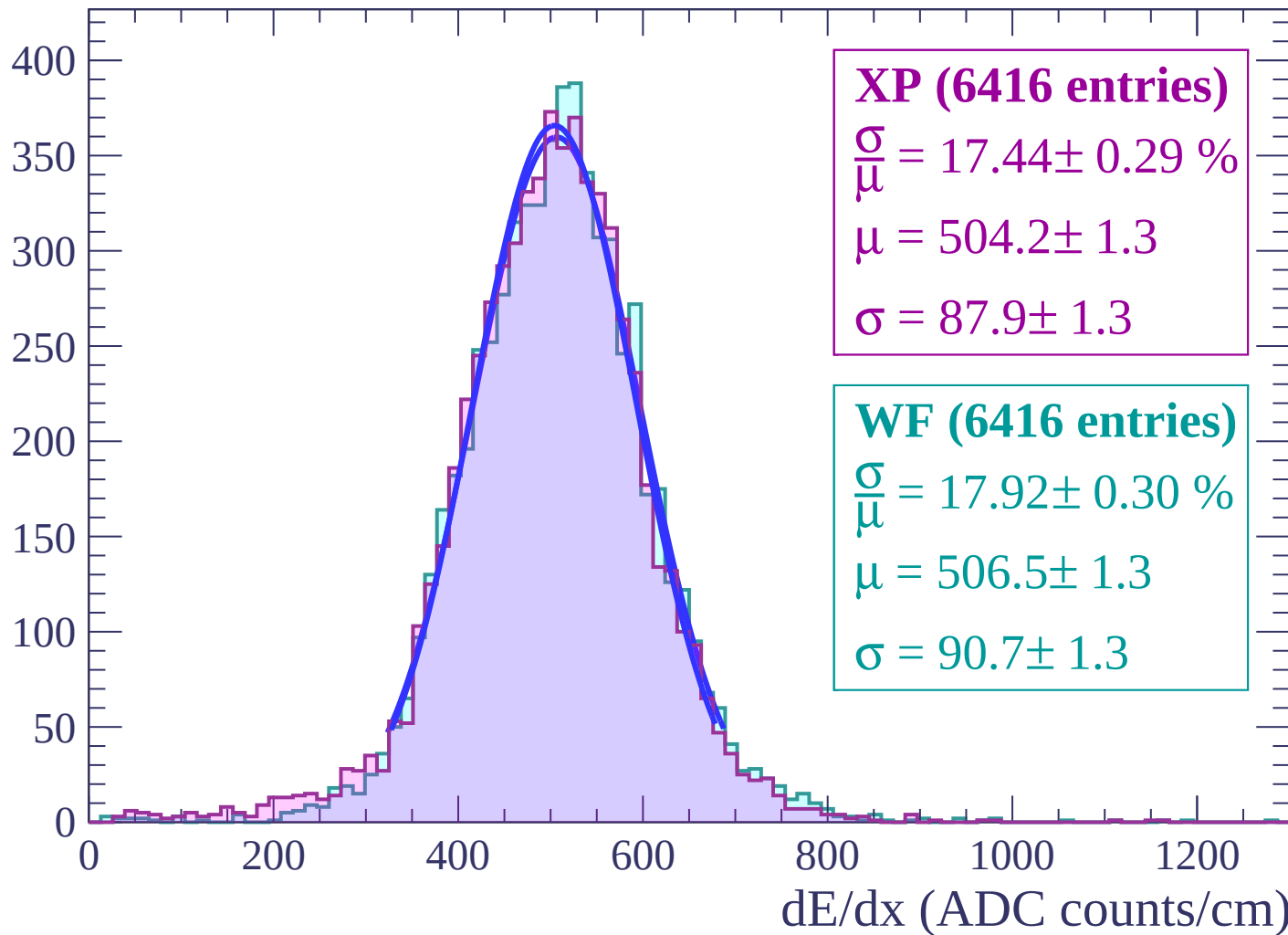
Energy loss | $42 < \theta < 44$

Count



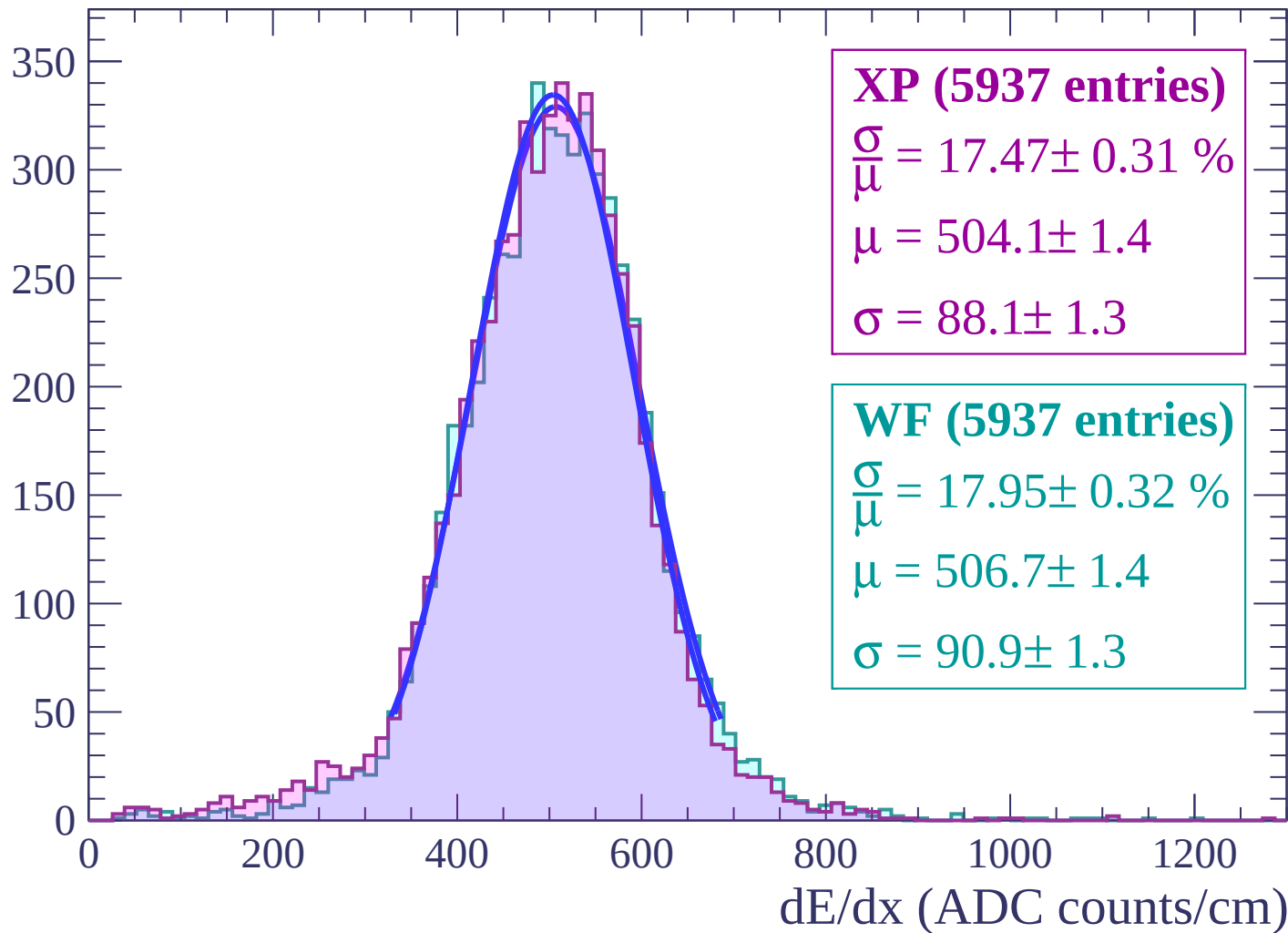
Energy loss | $44 < \theta < 46$

Count



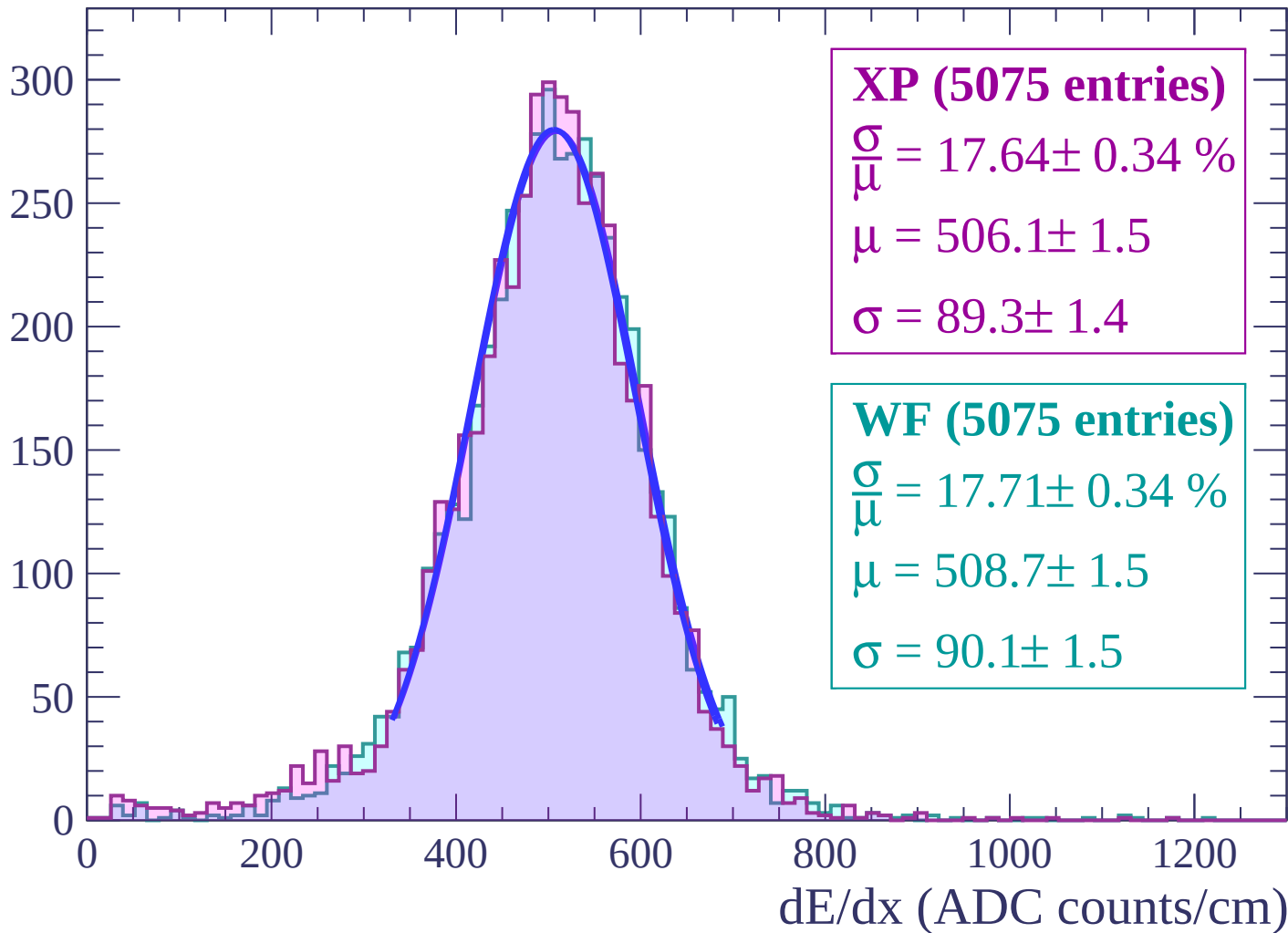
Energy loss | $46 < \theta < 48$

Count



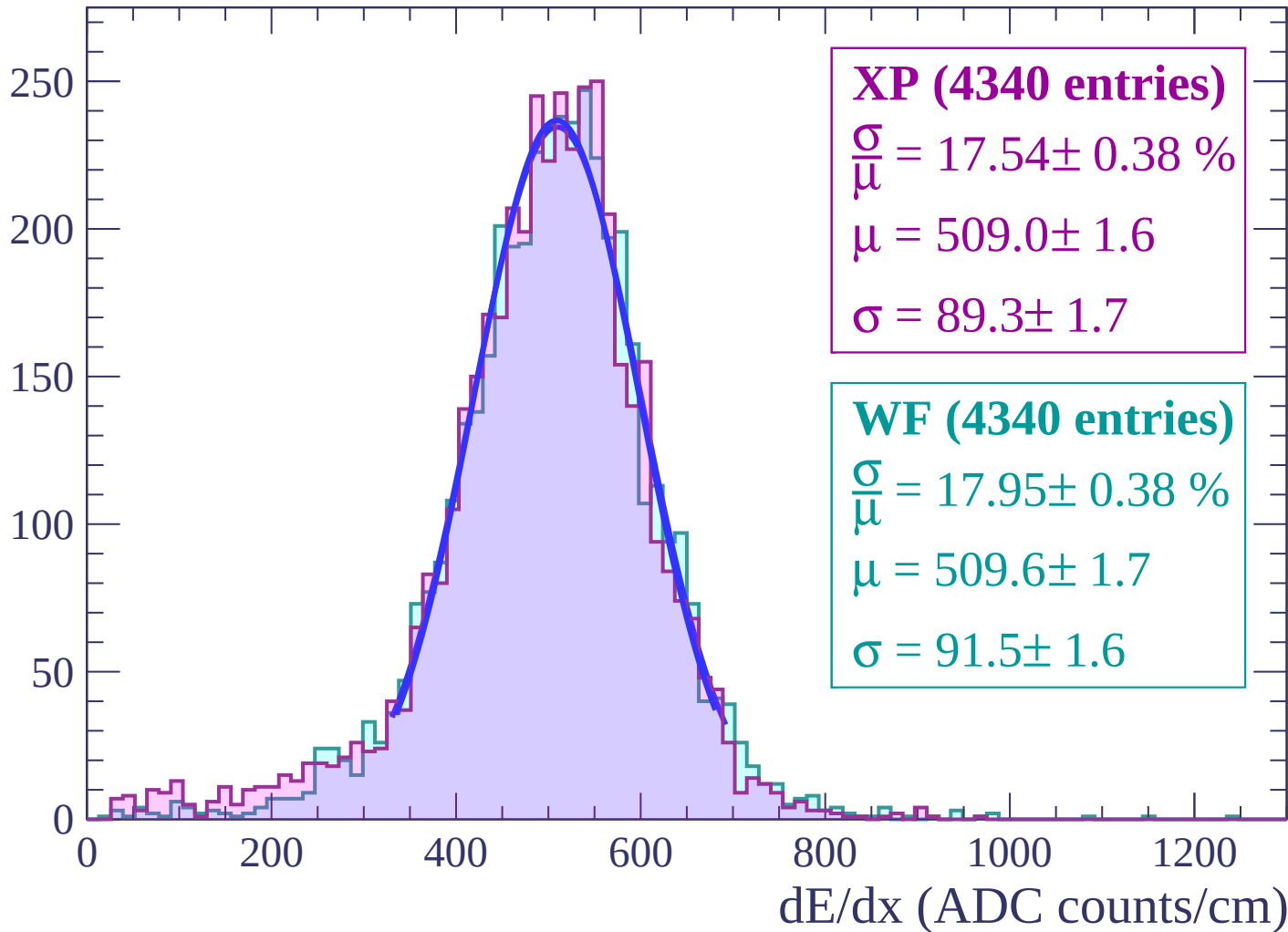
Energy loss | $48 < \theta < 50$

Count



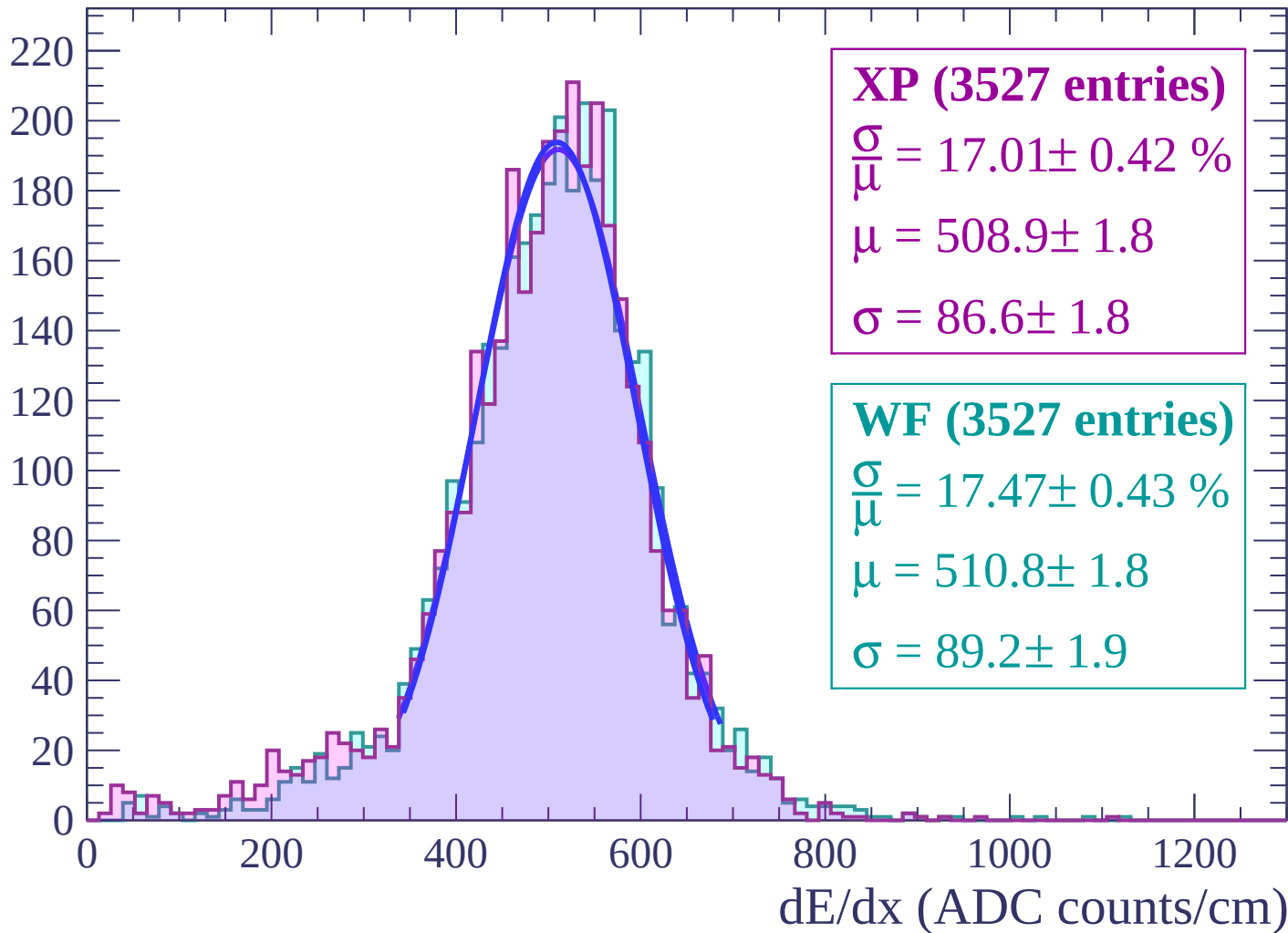
Energy loss | $50 < \theta < 52$

Count



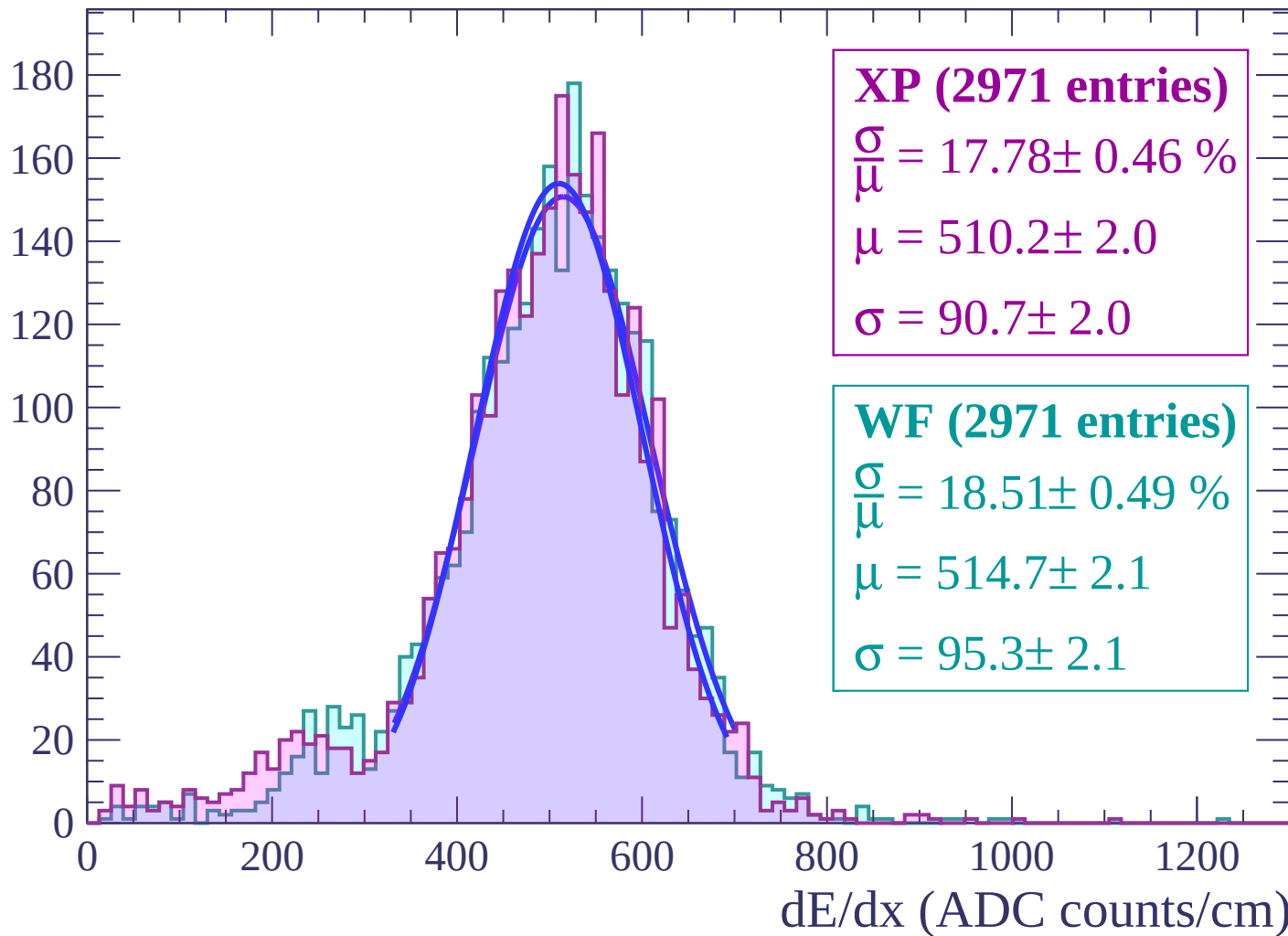
Energy loss | $52 < \theta < 54$

Count



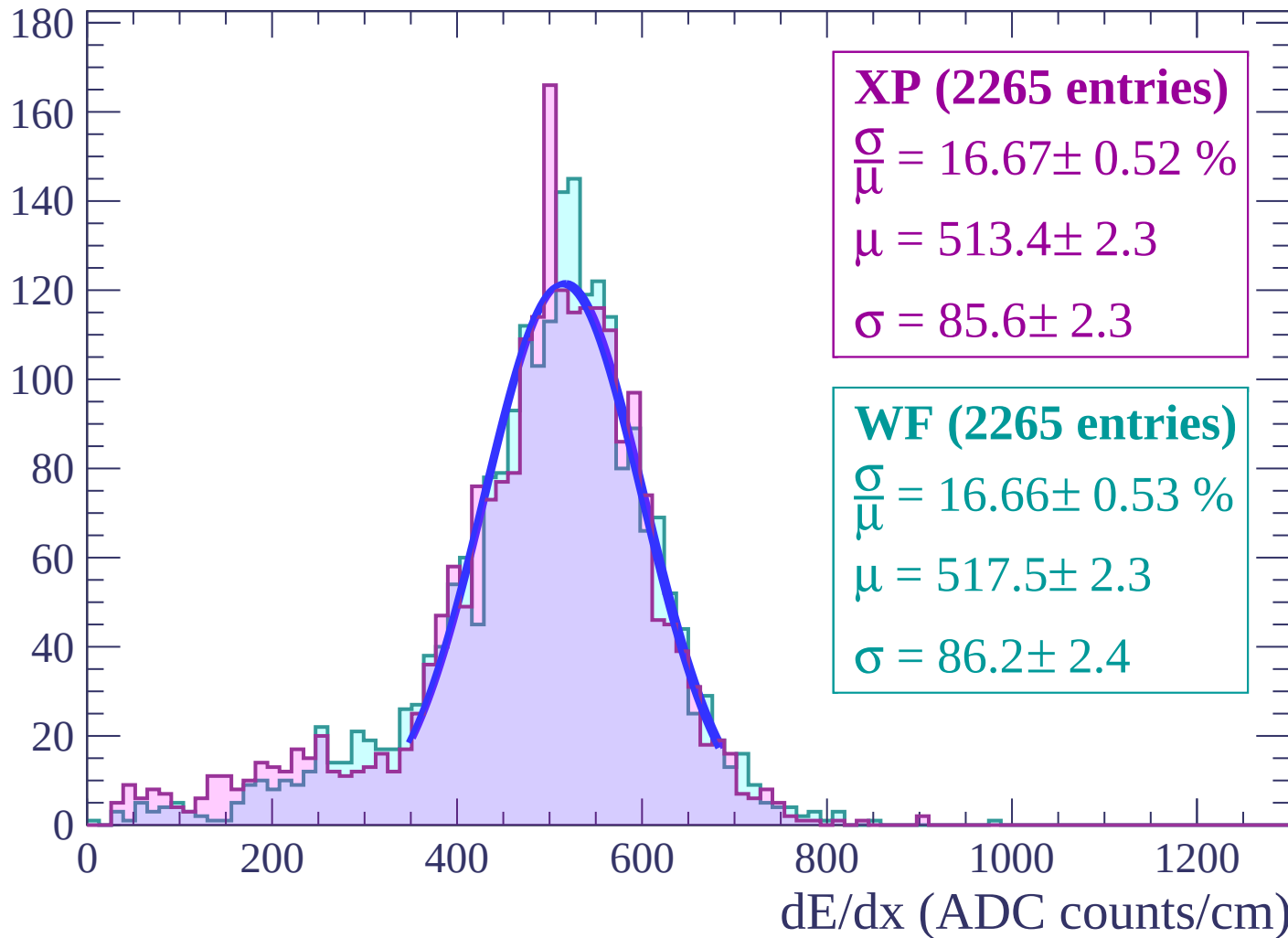
Energy loss | $54 < \theta < 56$

Count



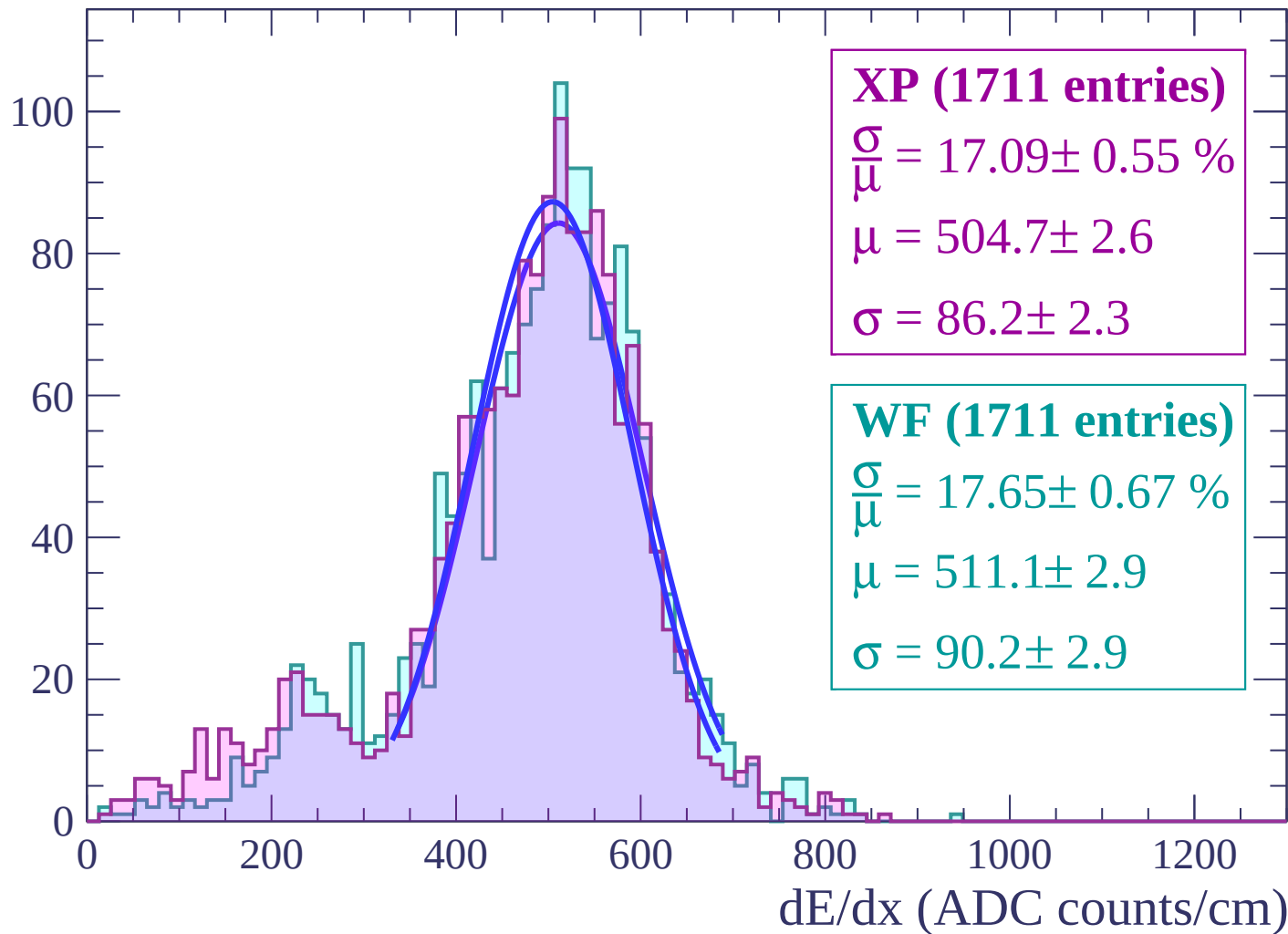
Energy loss | $56 < \theta < 58$

Count



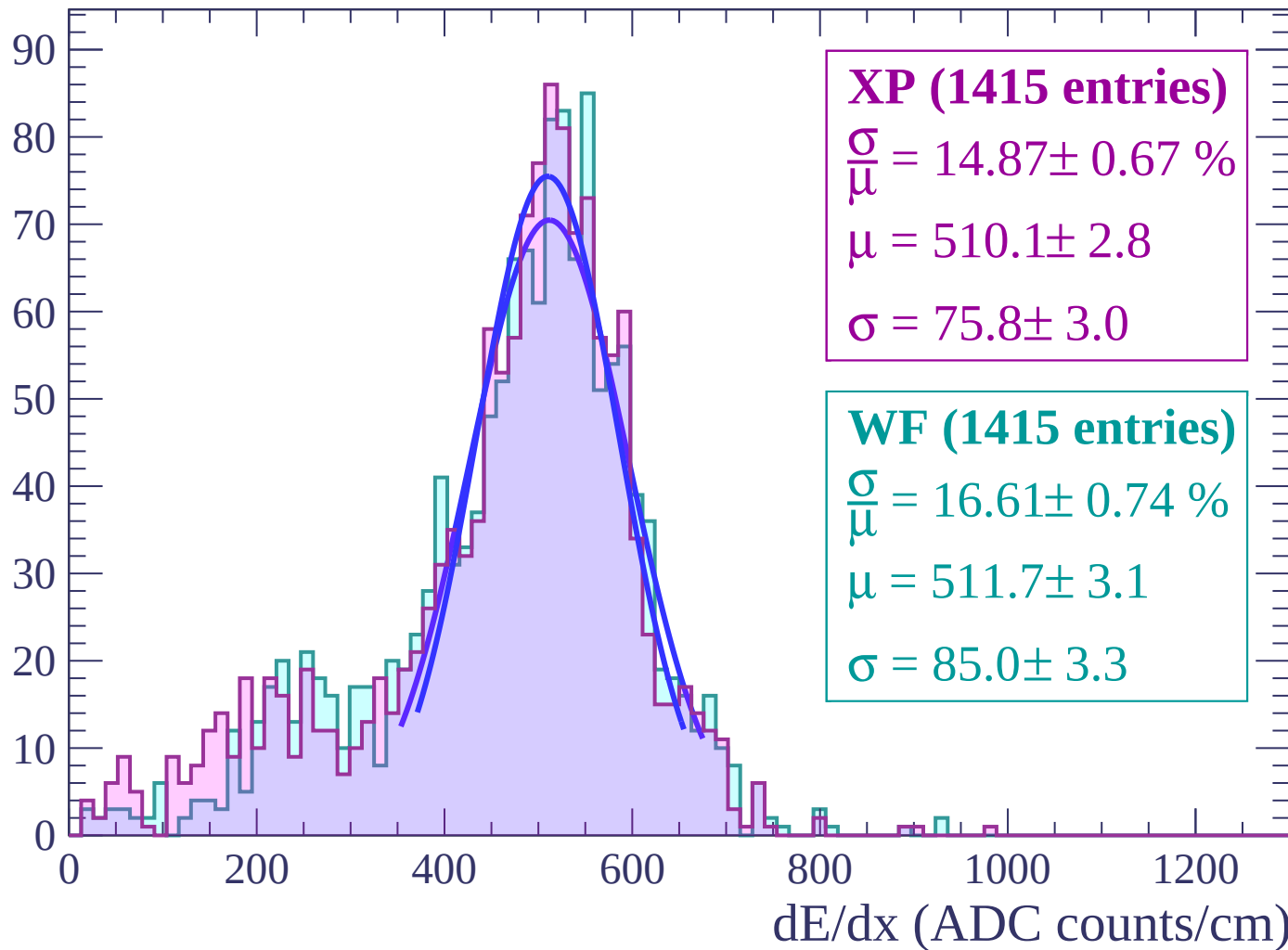
Energy loss | $58 < \theta < 60$

Count



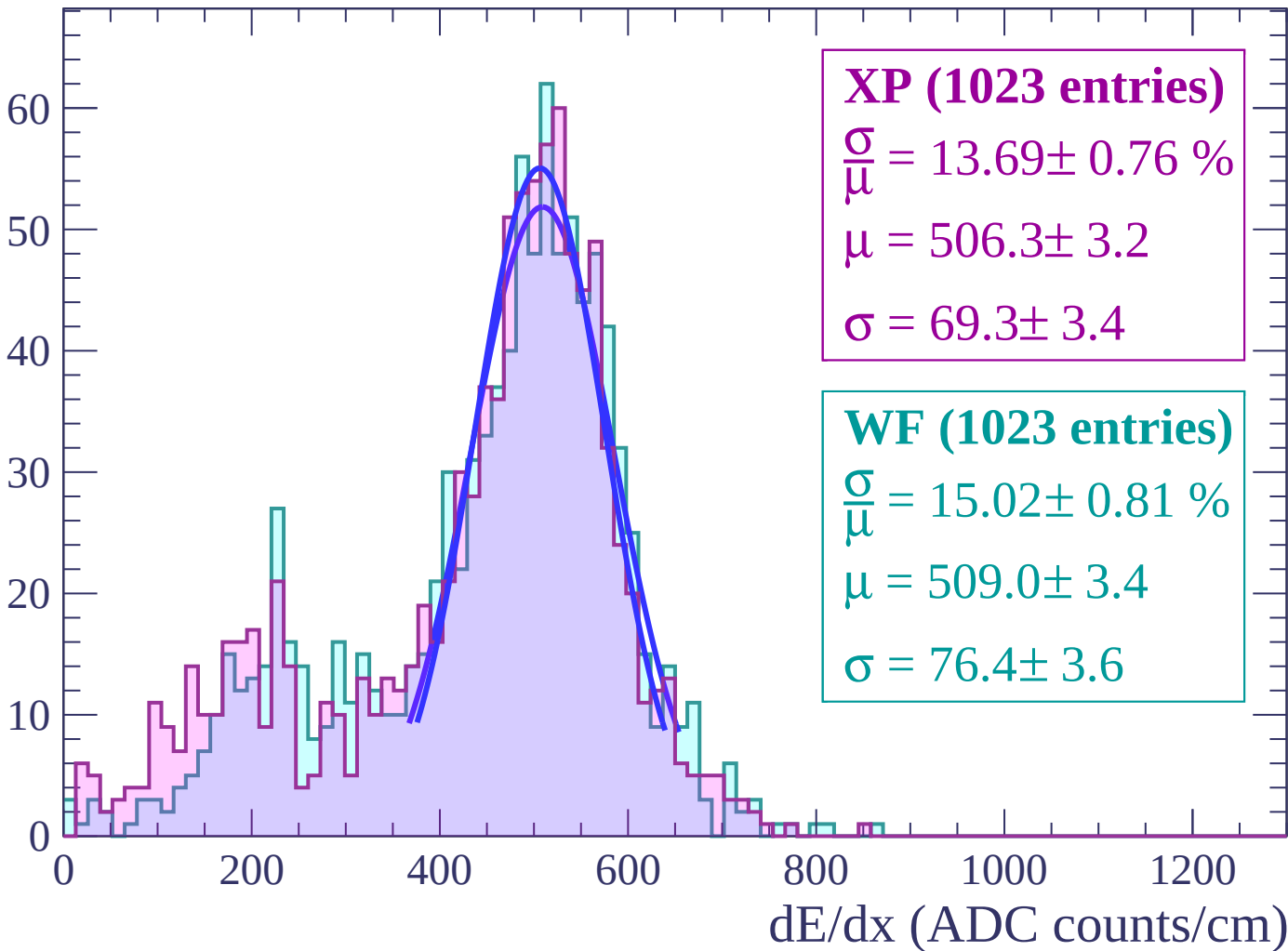
Energy loss | $60 < \theta < 62$

Count



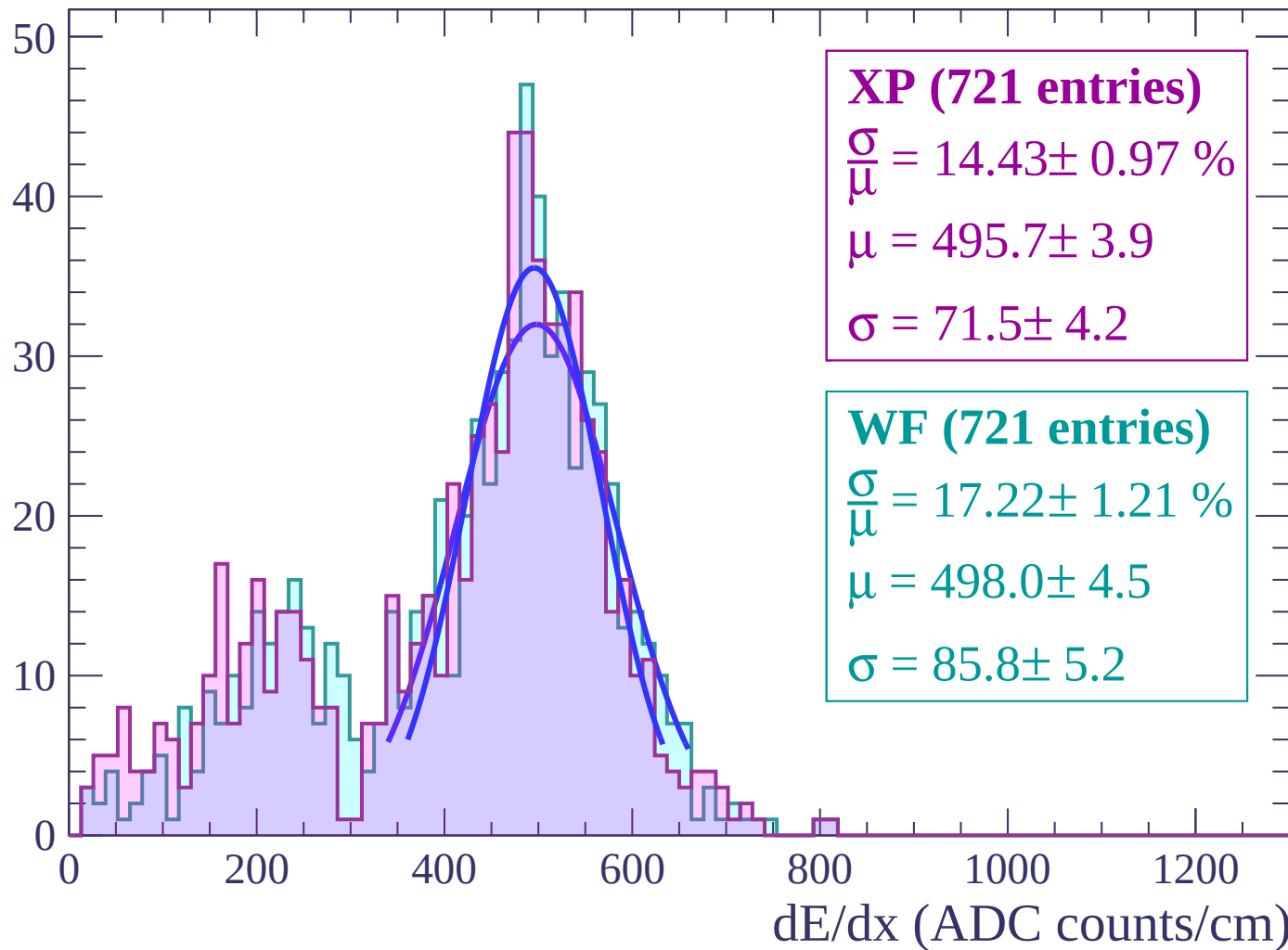
Energy loss | $62 < \theta < 64$

Count



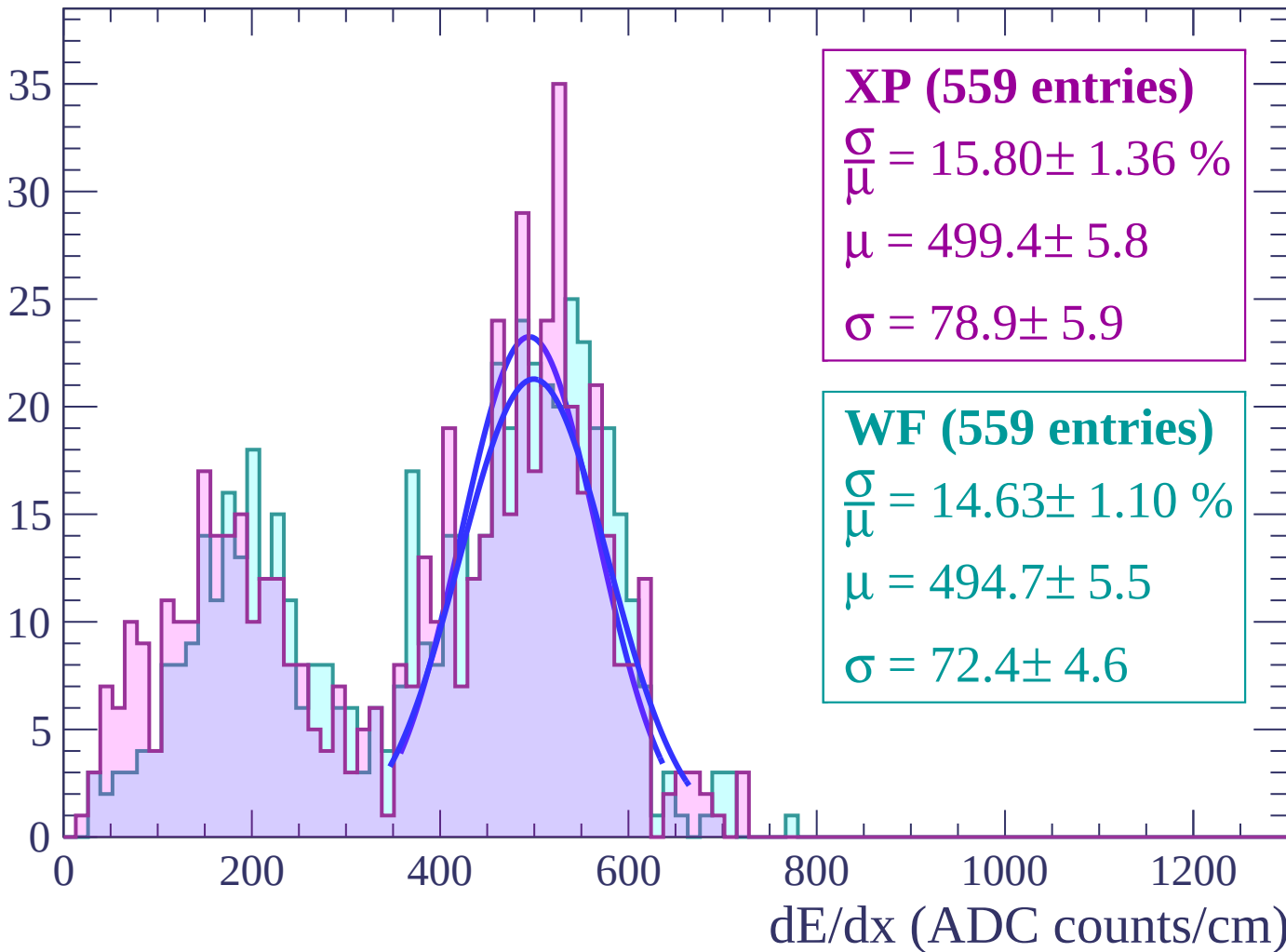
Energy loss | $64 < \theta < 66$

Count



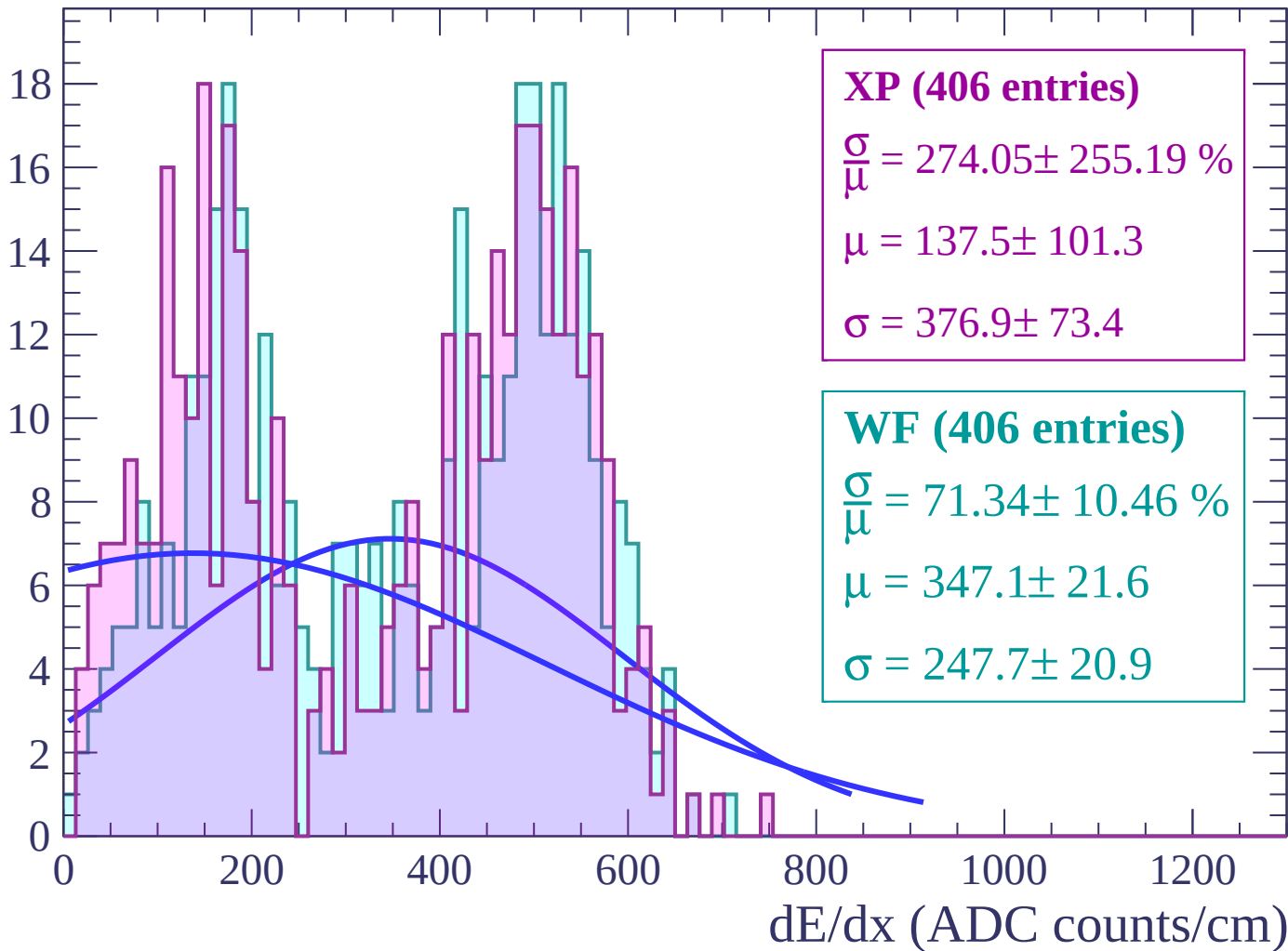
Energy loss | $66 < \theta < 68$

Count



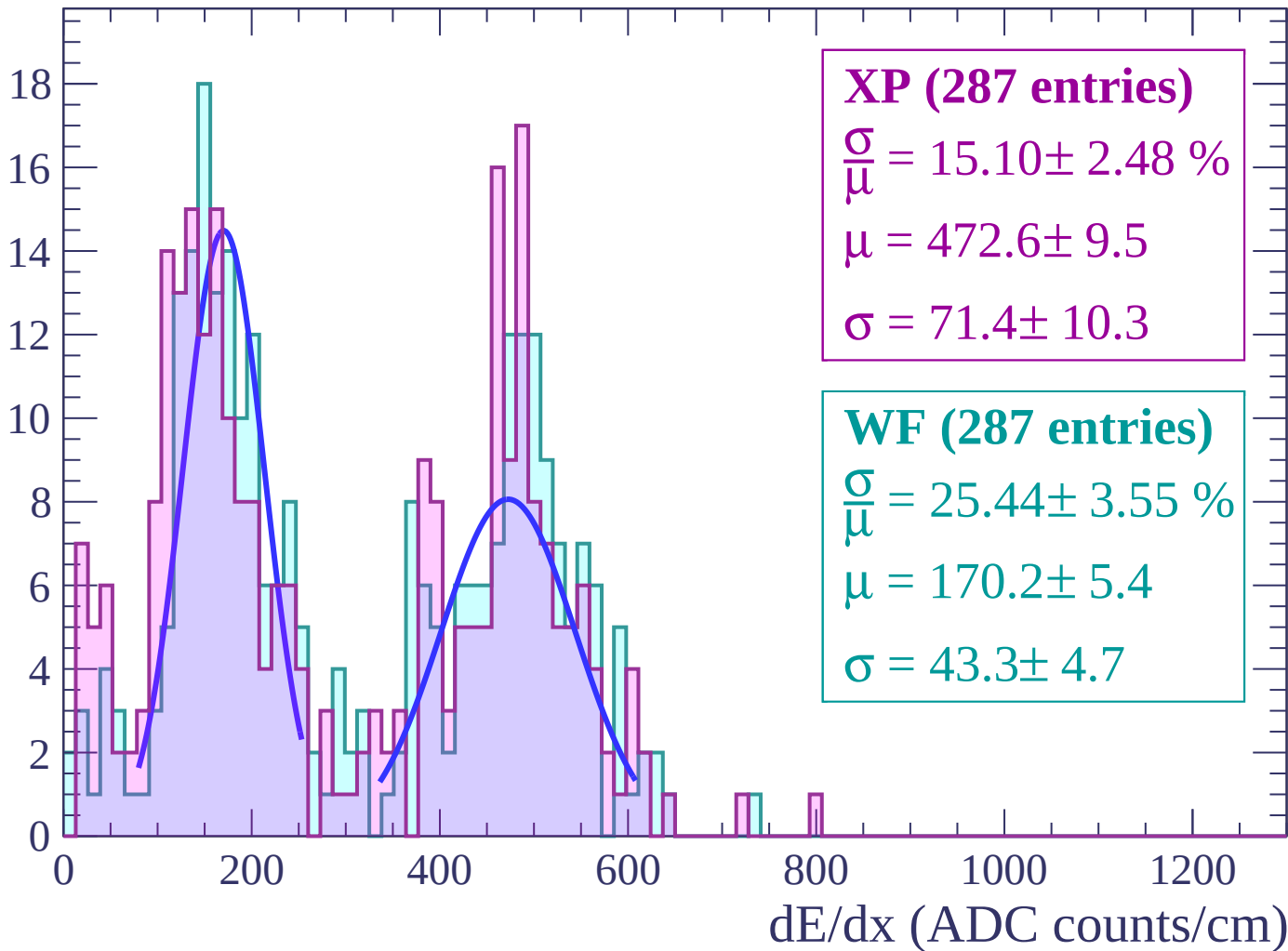
Energy loss | $68 < \theta < 70$

Count



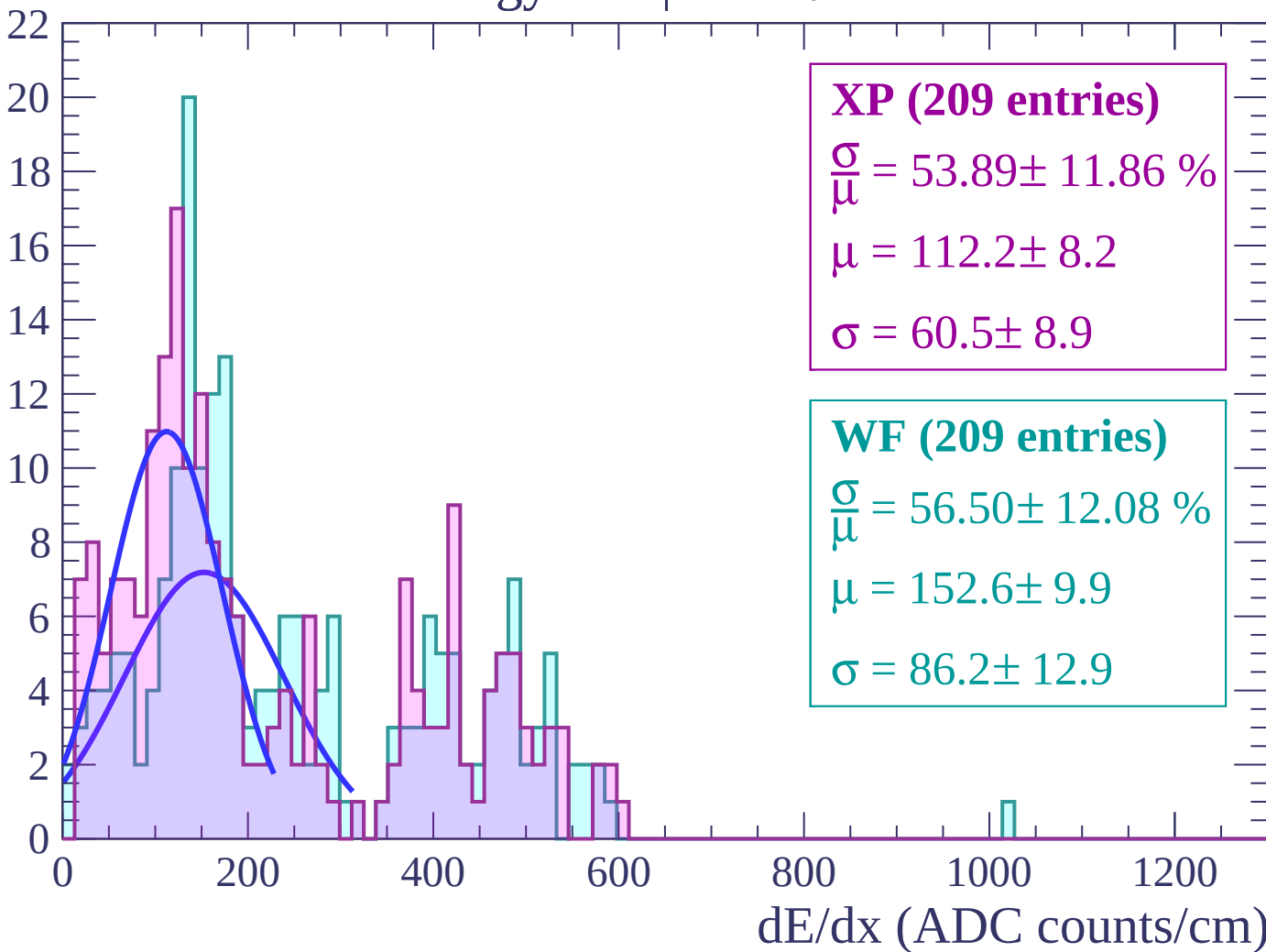
Energy loss | $70 < \theta < 72$

Count



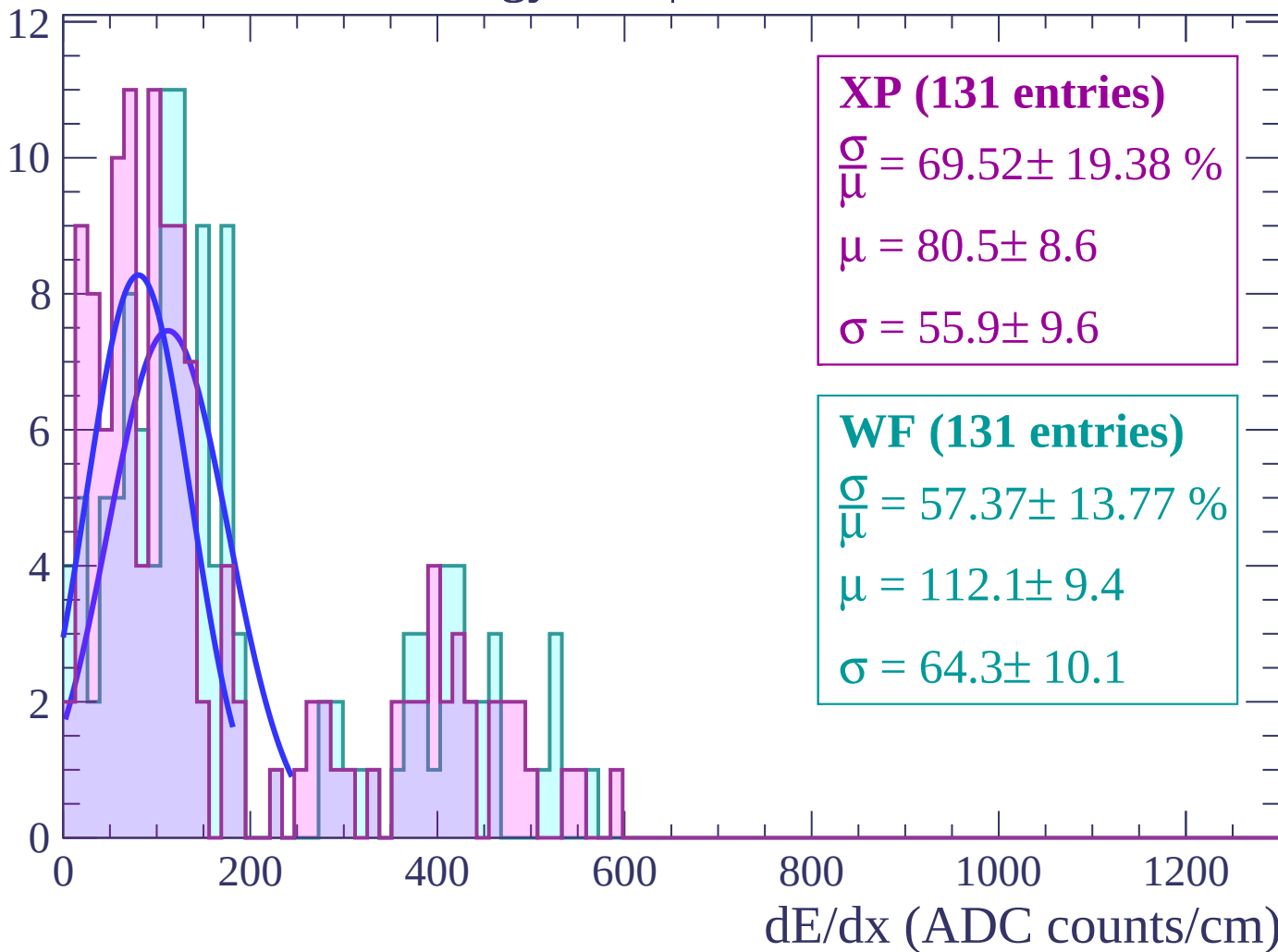
Energy loss | $72 < \theta < 74$

Count



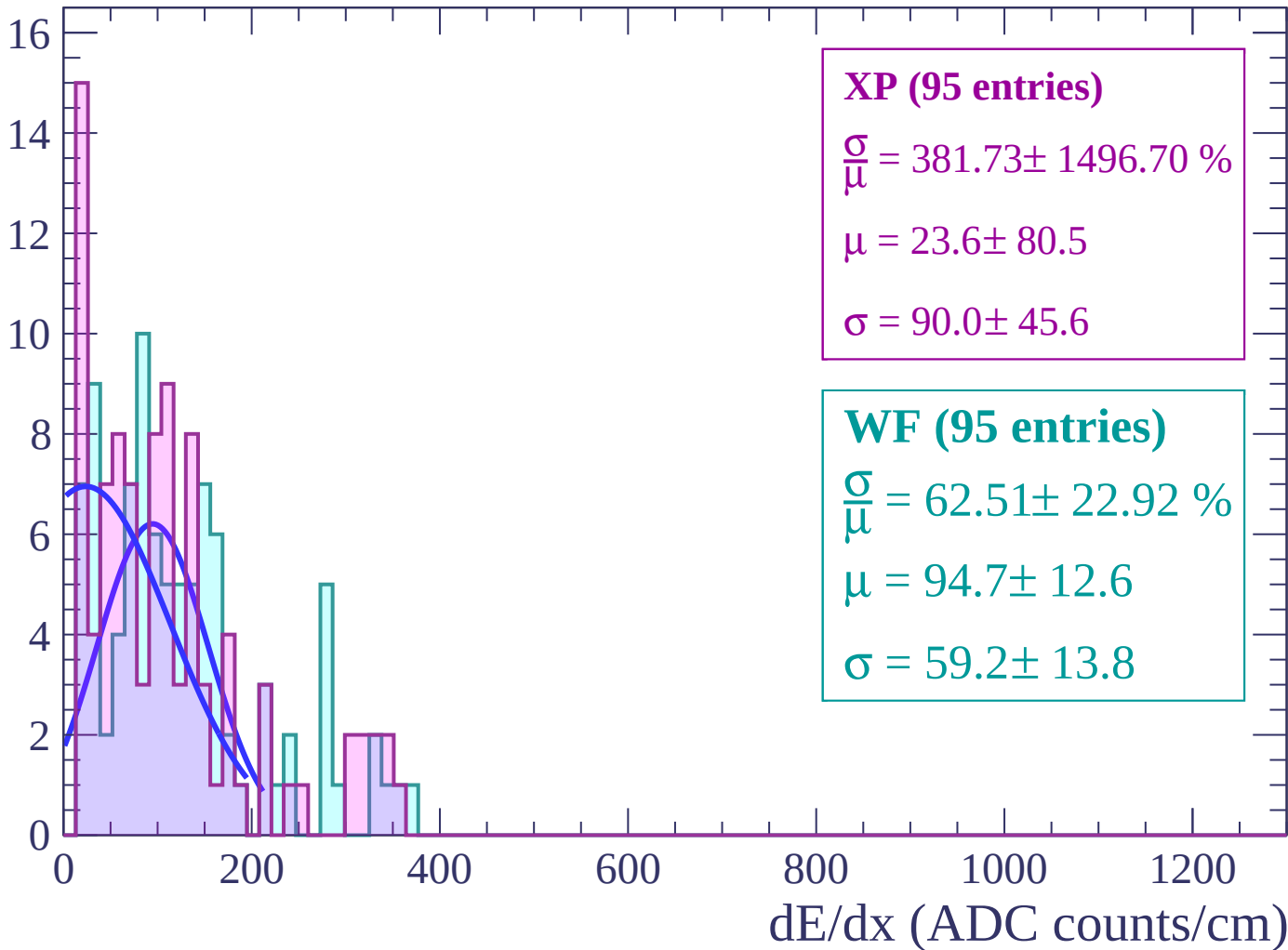
Energy loss | $74 < \theta < 76$

Count



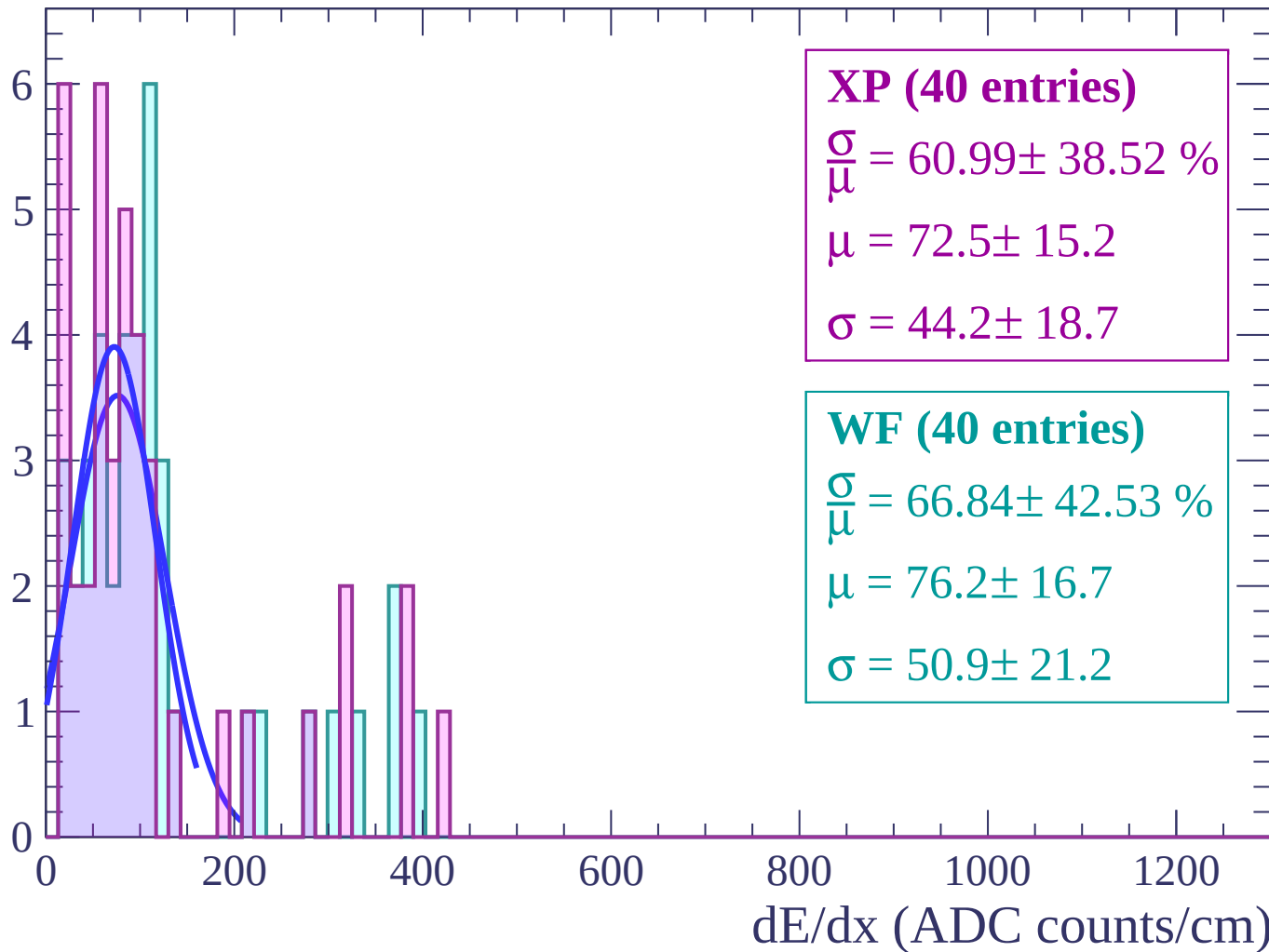
Energy loss | $76 < \theta < 78$

Count



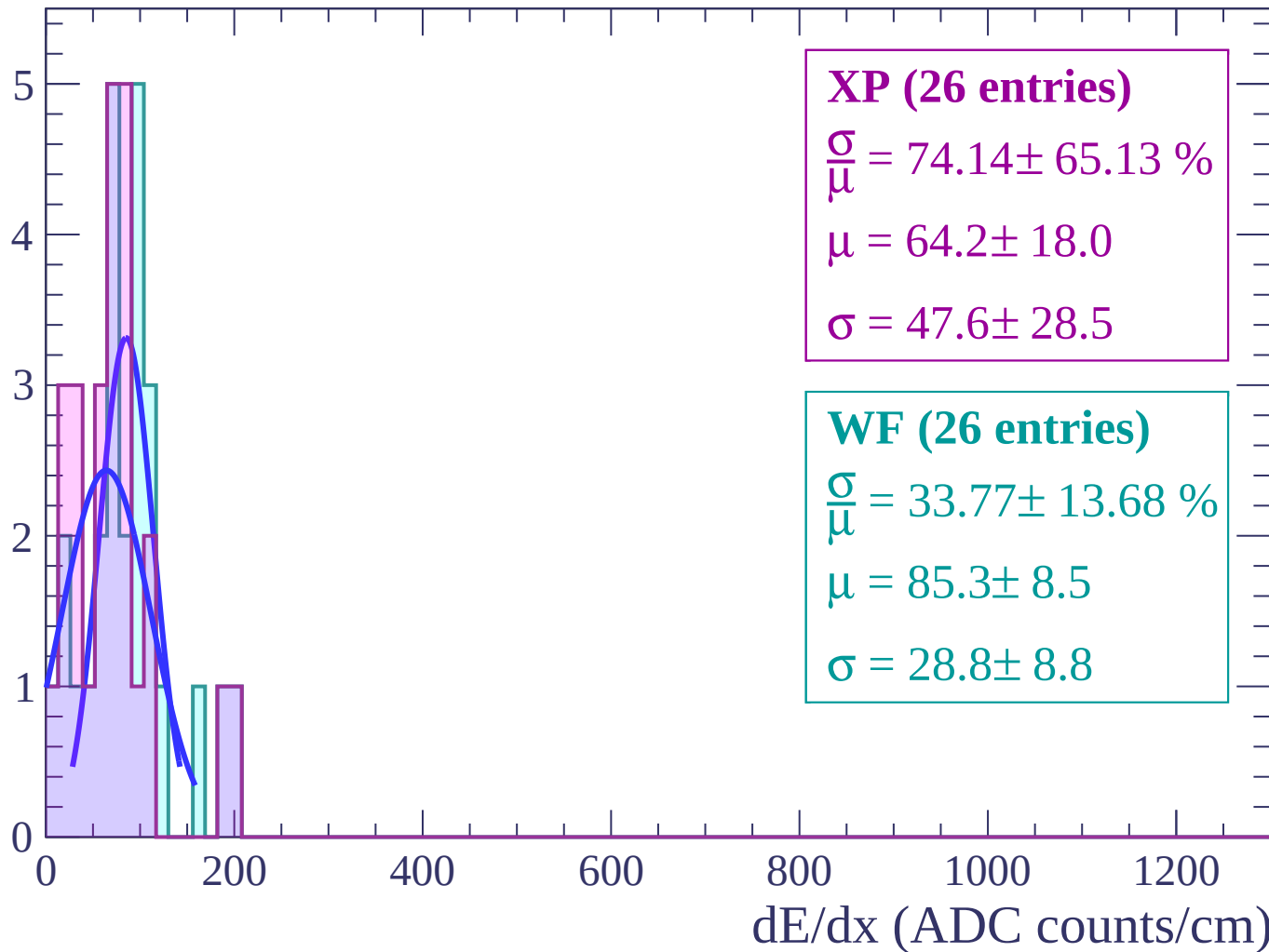
Energy loss | $78 < \theta < 80$

Count



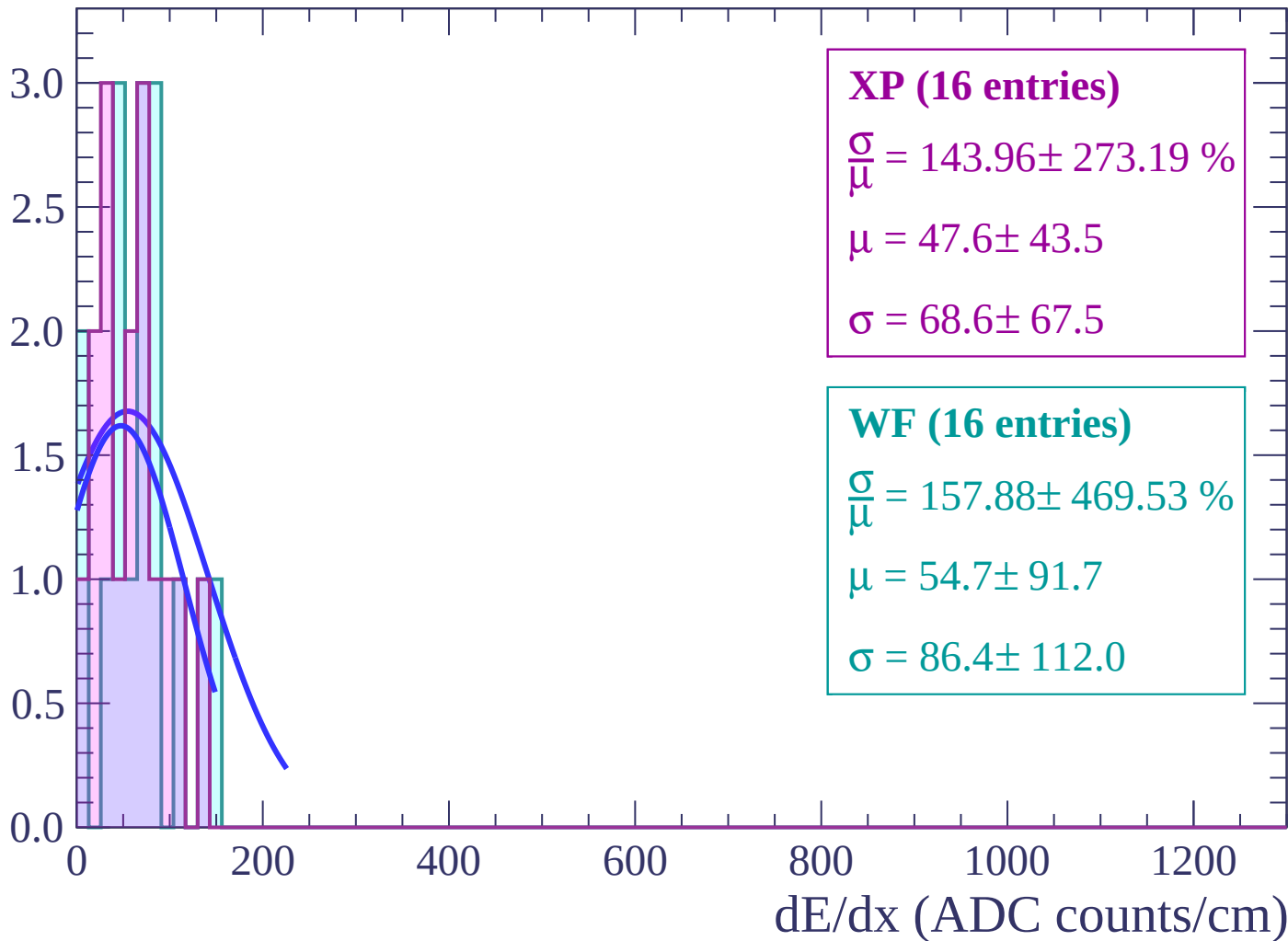
Energy loss | $80 < \theta < 82$

Count



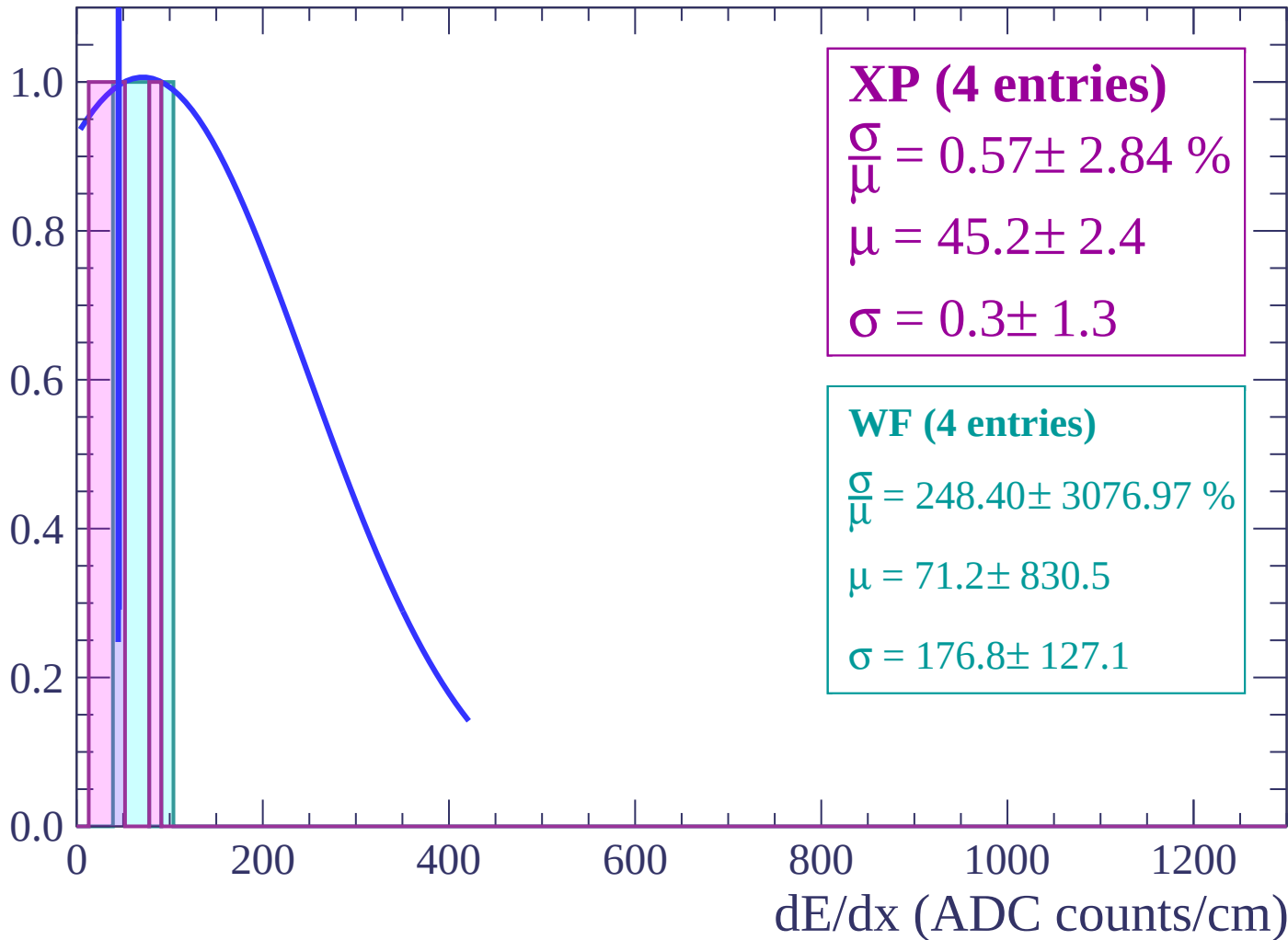
Energy loss | $82 < \theta < 84$

Count



Energy loss | $84 < \theta < 86$

Count



Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

